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MILITARY AFFILIATE RADIO SYSTEM (MARS)

Summary. This manual on methods and procedures for the operation of the Army Military Affiliate Radio System (MARS) has been revised to update the responsibilities, missions, functions, organizations, and support of the Army MARS.

Applicability. This manual applies to the Active Army, the Army National Guard, and the U.S. Army Reserve. It applies to all personnel who participate in the Army MARS Program.

Supplementation. Supplementation of this manual is authorized. If supplements are issued, Army staff agencies and major Army commands will furnish a copy of each to Commander, U.S. Army Information Systems Command, ATTN: AS-OPS- OA, Fort Huachuca, Arizona 85613-5000; other commands will furnish one copy of each to the next higher headquarters.

Suggested improvements. The proponent agency of this manual is the U.S. Army Information Systems Command (USAISC). Users are invited to send comments and suggested improvements on DA Form 2028 (Recommended Changes to Publications and Blank Forms) to the Command or Area MARS Director, who will forward them to the Commander, USAISC, ATTN: AS-OPS-OA, Fort Huachuca, Arizona 85613-5000.

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CHAPTER 1

INTRODUCTION

1-1. Purpose,

- a. This manual establishes methods and procedures for the operation of the Army Military Affiliate Radio System (MARS). It outlines responsibilities, missions, functions, organizations, and support of the Army MARS.
- b. The methods, procedures, and techniques set forth in this manual are intended as standard in-station practices and procedures for all Army MARS stations (military, club, and affiliate members). The practices and procedures outlined in this manual are designed to be compatible with the Allied Communications Publications (ACPs).
- c. Army MARS is a program conducted by the Department of the Army (DA) in which military installations, military units, military communities, clubs, and United States (U.S.) amateur licensed radio stations and operators voluntarily participate and contribute to the mission of providing auxiliary and/or emergency disaster communications on a local, national, and international basis as an adjunct to normal Army communications. MARS in this manual refers only to the Army MARS Program unless Navy- Marine Corps or Air Force is specified,

1-2. References.

Required and related publications and prescribed and referenced forms are listed in appendix A.

1-3. Explanation of abbreviations and terms.

Abbreviations and special terms used in this manual are

explained in the glossary.

1-4. Format of manual.

This manual is printed in loose-leaf format. Changes will be made on a page-for-page substitution basis. Each page is identified by manual number and appropriate change number. Changes are numbered in sequence starting with change 1. Changes must be inserted in the manual immediately upon receipt. Changes to the manual are not related; that is, change 4 does not necessarily repeat or consolidate the material in changes 1, 2, and 3. Thus, all changes are required to be posted sequentially and entered on the record of changes page for the manual to be correct.

1-5. Numbering system.

Each chapter is numbered as a separate entity. The number of each paragraph, page, illustration, and table is composed of two segments. The first segment of the number is the chapter number, the second is the sequential position of the particular paragraph, and so forth (for example, 2-4 identifies the fourth paragraph of the second chapter). This numbering system provides flexibility for changes to the manual. New material can be inserted as an additional section to a chapter. Additional chapters can be added without disturbing the numerical organization of the manual and additional material can be inserted at any place within a chapter by use of a point number. For example, two new paragraphs of material can be added to paragraph 5-8 by inserting new paragraphs 5-8.1 and 5-8.2, respectively.

CHAPTER 2

HISTORY, MISSION, AND POLICY

2 - 1 . History.

a. The Army Amateur Radio System (AARS) was founded in 1925 by the U.S. Army Signal Corps based on the need for a pool of trained radio operators in time of mobilization. This organization was a forerunner of today's Army MARS. AARS was deactivated on 7 December 1941 with the outbreak of World War II.

b. The Military Amateur Radio System was established on 26 November 1948 by authority of the Secretaries of the Army and Air Force. It was renamed Military Affiliate Radio System on 2 September 1952, to more accurately describe its nature. On 17 August 1962, the Secretary of the Navy approved the Chief of Naval Operations recommendation to organize a Navy-Marine Corps MARS Program to be implemented on 1 January 1963. DOD Directive 4650.2 formalized the composition, mission, functions, and organization of MARS and set forth policies concerning Department of Defense (DOD) support of both the MARS and Civil Amateur Radio activities within DOD.

2 - 2. Mission.

The mission of the Army MARS is to—

a. Provide DOD-sponsored emergency communications on a local, national, and international basis as an adjunct to normal communications.

b. Provide auxiliary communications for military, civil and/or disaster officials during periods of emergency.

c. Assist in effecting normal communications under emergency conditions.

d. Handle morale and quasi-official record and voice communications traffic for Armed Forces and authorized U.S. Government civilian personnel stationed throughout the world.

e. Establish programs to create civilian interest, recruit qualified volunteers, and furnish appropriate

training in military communications equipment, techniques, and procedures.

f. Initiate efforts to attain improvements in high frequency (HF) and very high frequency (VHF) radio and computer systems operating techniques and state-of-the-art technology through experimentation and testing.

g. Conduct an appropriate military or amateur radio exercise as a part of the annual celebration of Armed Forces Day in accordance with DOD Directive 4650.2.

h. Operate and maintain a MARS HF radio facility known as radio station WAR in support of Headquarters, Department of the Army (HQDA),

i. Provide a potential reserve of personnel trained in military radio communications.

2 - 3. Policy. The DA policy is—

a. To support, encourage, and cooperate in developing and promoting MARS and amateur radio activities to enhance their military and civil value.

b. To maintain liaison with Departments of the Navy and Air Force, recognized U.S. amateur radio organizations, and the Federal Communications Commission (FCC) on matters concerning Army MARS and Army amateur radio programs.

c. To establish Army MARS programs in foreign countries or territories where permitted by applicable treaties and agreements.

d. MARS will follow channels of command as prescribed in chapter 3 of this manual.

e. The Commanders, 7th Signal Command, 5th Signal Command, 1st Signal Brigade (USAISC), USAISC-Japan, and USAISC-Western Command (WESTCOM) are tasked to provide support, coordination, and execution of the Army MARS Program within their geographic areas.

CHAPTER 3

ORGANIZATION AND ADMINISTRATION

3-1. Command authority.

a. Headquarters (HQ), USAISC, a major command under command of the Chief of Staff, U.S. Army, is designated executive agency for Army MARS. The Commander, USAISC, in carrying out the MARS responsibility, appoints a Chief, Army MARS on the staff who is vested with necessary authority to carry out the DA responsibility in cognizance with DOD Directive 4650.2.

3-2. Organizational structure.

a. MARS is organized to support a worldwide auxiliary military communications system. Its composition parallels the military organizational structure and command channels in concept and operation. The authority for MARS management is vested in the Chief, Army MARS and is extended through the MARS chain of command as indicated by the organizational chart in figure 3-1.

b. Chief, Army MARS has overall responsibility for the management and direction of the MARS and direct coordination authority with command MARS directors appointed by major subordinate commands on matters pertaining to MARS administration and operation.

(1) Command MARS director, continental United States (CONUS) (7th Signal Command) is responsible for management, direction, and operation of the MARS CONUS-wide program to include Alaska, Canal Zone (Panama), Puerto Rico, and "Virgin Islands. CONUS area MARS directors are appointed by and directly responsible to the command MARS director on all matters pertaining to MARS.

(a) Eastern area MARS director has administrative and operational control over the following CONUS areas pertaining to MARS: Alabama, Connecticut, Delaware, Florida, Georgia, Kentucky, Maine, Maryland, Massachusetts, Mississippi, New Hampshire, New Jersey, New York, North Carolina, Pennsylvania, Rhode Island, South Carolina, Tennessee, Vermont, Virginia, West Virginia, Puerto Rico, and U.S. Virgin Islands.

(b) Central Area MARS director has administrative and operational control over the following CONUS areas pertaining to MARS: Arkansas,

Illinois, Indiana, Iowa, Kansas, Louisiana, Michigan, Minnesota, Missouri, Nebraska, New Mexico, Ohio, Oklahoma, Texas, Wisconsin, Canal Zone (Panama), and Central and South America.

(c) Western area MARS director has administrative and operational control over the following CONUS areas pertaining to MARS: Arizona, California, Colorado, Idaho, Montana, Nevada, North Dakota, Oregon, South Dakota, Utah, Washington, Wyoming, and Alaska.

(d) State MARS directors are appointed by the area MARS directors and, under their direction, have administrative and operational control for all matters pertaining to MARS within their State or assigned areas of responsibility.

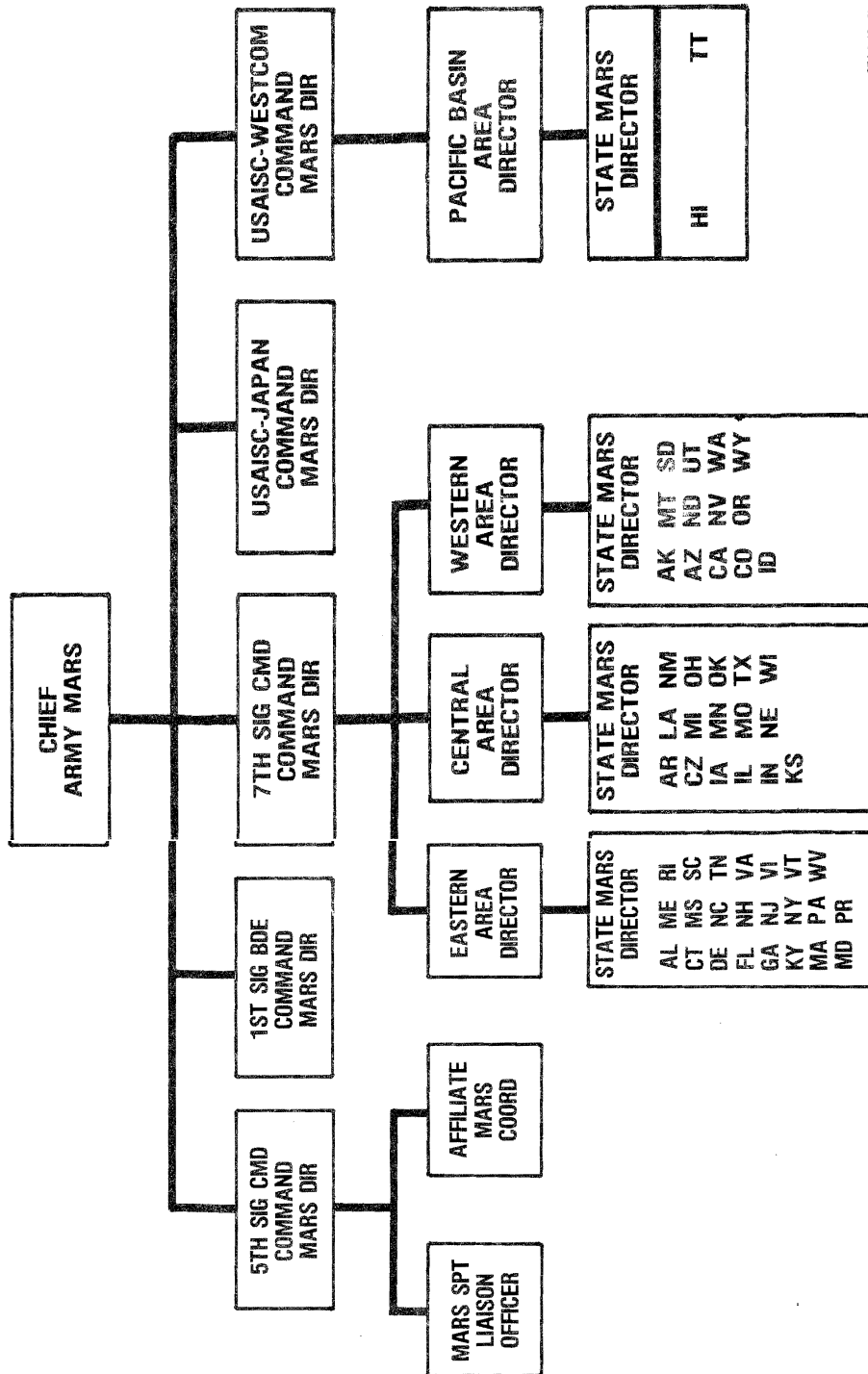
(2) Command MARS director, Pacific Basin, USAISC-WESTCOM has staff responsibility for management, direction, and operation of MARS Pacific-wide (less Japan, Okinawa, and Korea). The area MARS director, Pacific Basin, is appointed by the command MARS director, USAISC- WESTCOM, and is a part of his or her staff for program, coordination, and operation.

(a) The area MARS director, Pacific Basin, has administrative and operational control over the Pacific area extending from the Aleutian Islands to the South Polar caps, less Japan, Okinawa, and Korea.

(b) The State MARS directors are appointed by the area MARS director and under his or her direction have administrative and operational control for all matters pertaining to MARS within their State or assigned areas of responsibility.

(3) Command MARS director, Europe (5th Signal Command), is responsible for management, direction, administration, and operation of the MARS in U.S. Army Europe (USAREUR), to include the Near East and Africa. He or she will direct the establishment of a theater staff, as required, to administer both the military and the affiliate MARS network. He or she exercises direct operational coordination and technical liaison with the European MARS gateway station.

(a) MARS support liaison officer, as designated by each USAREUR major subordinate command, is responsible for coordination of MARS activities required in support of theater operations or exercises.



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Figure 3-1. MARS chain of command organizational chart.

(b) Affiliate MARS coordinator is appointed by the command MARS director and, under his or her direction, has administrative and operational control for all matters pertaining to MARS within the assigned areas of responsibility.

4) Command MARS director, USAISC Japan, is responsible for the management, direction, and operation of the MARS in Japan and Okinawa.

5) Command MARS director, Korea (1st Signal Brigade (USAISC)), is responsible for management, direction, and operation of the MARS in Korea. He or she exercises direct operational coordination and technical liaison with the Korean MARS gateway station.

3-3. Correspondence.

MARS correspondence follows the MARS chain of command. All official correspondence will be routed from the member through the State, area, and command MARS director as indicated in figure 3-1. MARS members are to correspond with the lowest echelon of the MARS organization from which an answer can be expected. Addresses for chief, Army MARS and command and area MARS directors are listed below:

- a. Chief, Army MARS
Commander
U.S. Army Information Systems Command
ATTN: AS-OPS-OA
Fort Huachuca, AZ 85613-5000
- b. Command MARS Director-CONUS
Commander
7th Signal Command
ATTN: ASN-OPS-OX
Fort Ritchie, MD
21719-5010
- c. Command MARS Director-Europe
Commander
5th Signal Command ATTN:
ASE-OP-MARS APO New York
09056
- d. Command MARS Director-Pacific Basin
Commander USAISC-WESTCOM
ATTN: ASP-OP-MARS Fort
Shafter, HI 96858-5410
- e. Command MARS Director-Japan
Commander USAISC-Japan
ATTN: ASJ-PO-OD
APO San Francisco 96343-0059
- f. Command MARS Director-Korea
Commander
USFK EUSA
ATTN: SJ-OMS
APO San Francisco 96301

- g. Area MARS Director-Eastern
Director
Eastern Area MARS ATTN: ASN-OPS-OX-MDE
Ford Meade, MD 20755-5365
- h. Area MARS Director-Central
Director
Central Area MARS
ATTN: ASN-OPS OX-SHN
Fort Sam Houston, TX 78234-5000
- i. Area MARS Director-Western
Director
Western Area MARS
ATTN: ASN-OPS-OX-PSF
Presidio of San Francisco, CA 94129-5000
- j. Area MARS Director-Pacific Basin
Commander
USAISC Signal Bn-Hawaii ATTN: ASPH-PO-
MARS Fort Shafter, HI 96858-5415

3-4. Mailing.

a. All MARS correspondence and record traffic is official communications and is authorized mailing privileges in accordance with AR 340-3.

b. The DA Official Mail Cost Control Program is to control DA official mail cost through proper and efficient use of the U.S. Postal Service (USPS). Penalty mail stamps (PMSs) are available only on order through USPS Stamp Distribution Offices (SDOs). The area or command MARS directors are designated as ordering officers for MARS. The State MARS director, or member appointed by the State MARS director, is designated as the stamp control officer for the State. The ordering officer will order through the appropriate SDO the amount of PMSs requested by the State MARS director, or stamp control officer. The SDO will ship all orders direct to the State MARS director, or stamp control officer. Each order shipped by the SDO will contain one copy of PS Form 17-G (Penalty Mail Stamp Requisition). This form will be forwarded to the area or command MARS director folded in a letter-size envelope. Orders for PMS are limited to one order per calendar quarter. State MARS directors and/or stamp control officers should submit their orders for PMS 30 days prior to the next quarter. PMS will be distributed by the State MARS directors or stamp control officers to the State MARS members in accordance with the member request.

c. State MARS directors or stamp control officers will maintain an accountable audit trail on all PMS or stamp-embossed mailing material for the State. Each MARS member will maintain an accountable audit trail on all PMS or stamp-embossed mailing material issued to them.

Upon termination from the Army MARS Program, the member will return to the State MARS director or stamp control officer all unused accountable stamps and envelopes, and so forth, along with an DA Form 5534-R (PMS Accountability Log) as shown in figure 3.2. DA Form 5534- R will be locally reproduced on 8 1/2 x 11-inch paper and is located at the back of this manual. Instructions are self-explanatory.

a. Office envelopes and postal cards that are prestamped must contain the official mail inscription "Department of the Army, Official Business, Penalty for Private Use, \$300." on the upper left corner, which will be preprinted by offset printing press. This inscription will not be typewritten, stamped, or handwritten. Official mail must contain a Government return address printed, typewritten, or hand-stamped in the upper left corner between Department of the Army and Official Business. As part of the return address the MARS member will insert only his or her call sign directly under "Penalty for Private Use, \$300."

b. Mail addressed to other MARS members or message addressees will use the appropriate area MARS director's address as the return address and mail addressed to the area MARS director will use the command MARS director's address as the return address. The return address and the to address must never be the same.

c. DA Form 5534- R will be used by State MARS directors or stamp control officers and individual members to maintain an account and audit trail of PMS and pre-stamped envelopes or postal cards issued and used. The same accountability is required by all members whether it be individual members, State MARS directors, or stamp control officers. The accountability will begin with the receipt from the SDO and end when the member submits his or her accountability log through channels to the area MARS director. DA Form 5534-R will be closed as of the last day of each calendar year with the PMS balance on hand being entered as the opening balance on the new year's log. The old year's logs will be forwarded through channels to the area or command MARS director to arrive not later than 1 March.

3-5. Forms.

a. Form authorized for use in the Army MARS are shown in appendix A.

b. State directors are authorized to use locally designed forms applicable to the local area to support preparation and submission of information required by an authorized form.

c. Recommendations for changes to DA forms authorized for use by MARS will be submitted to HQ, USAISC, ATTN: AS-OPS-OA (Chief, Army MARS), for approval and submission to HQDA.

3-6. Participation reports.

a. DA Form 5522 (Traffic and Participation Report) (RCS: ISC 65).

(1)MARS members, military installations, activities, unit, and club stations are required to submit a monthly participation report. A report is required even if a member or station had no activity for that month. Reasons for no activity will be reported. Members unable to be active may request leave of absence not to exceed 3 months. Such request must be justified and in writing. Failure to submit a report for 3 consecutive months will be considered sufficient justification for termination of membership. Military installations, activities, unit stations, and community stations will report participation through military channels to the command director.

(2) Reports will be submitted via DA Form 5522 (Traffic and Participation Report) or MARS message format shown at figure 3-3.

(3) Military installation (active, Army National Guard (ARNG), United States Army Reserve (USAR)), activity, or unit MARS stations will prepare the monthly participation report containing the information outlined in a (2) above. Reports will be forwarded through command channels to the command MARS director. An information copy will be provided to the appropriate area director. Club stations will complete DA Form 5522 in accordance with procedures outlined for affiliate members. DA Form 5522 is stocked and issued by USAISC Activity, ATTN: ASN- HUA-RAP (Post Pubs), Ft. Huachuca, AZ 85613-6000. Instructions for completing DA Form 5522 are in table 3-1

Table 3-1. Instructions for completing DA Form 5522.

COLUMN	EXPLANATION
MARS STATION	Affiliate member call sign.
NAME	Affiliate member name...
MONTH/YEAR	Self-explanatory
BLOCKS A	List hours of operation in CW nets
B.	List hours of operation in HF voice nets

NR 001

R 281500Z JUL 84

FM AAR6XX/TX

TO AAT6YY/TX

BT

1. PARTICIPATION REPORT FOR (MONTH)

A.(HOURS IN CONTINUOUS WAVE (CW) NET A1) 1 .

B.(HOURS IN VOICE NET A3) 4 .

C.(HOURS IN RADIO TELETYPEWRITER (RATT) NET F1) 1 .

D.(HOURS IN VOICE NET F3) 0 .

E.(HOURS ON THE AIR) 6 .

F.(WAR BROADCASTS COPIED) 1 .

G.(OTHER HOURS) 5 .

H.(TOTAL HOURS PARTICIPATION) 12 .

I.(TOTAL MESSAGES PROCESSED) 8 .

J. (TOTAL PHONE PATCHES COMPLETED) 0 .

K. (AREA BROADCASTS COPIED) 2 .

L. (STATE BROADCASTS COPIED) 3 .

2.NARRATIVE REMARKS (EXPLAIN OTHER HOURS SUCH AS REPAIR MARS EQUIPMENT AND PREPARED LESSON PLAN).

BT

(Information in parentheses should be transmitted.)

Figure 3-3. Sample member participation report message format.

Table 3-1. Instructions for completing DA Form 5522—continued.

COLUMN	EXPLANATION
BLOCKS C.	List hours of operation in RATT nets.
D.	List hours of operation in VHF voice nets.
E.	List total hours on the air.
F.	List 1 hour for each WAR broadcast copied.
F.	List hours of participation other than on-the-air time.
G.	List total hours claimed.
H.	List total number of messages processed.
I.	List number of phone patches completed.
J.	List number of area broadcasts copied.
L.	List number of State director broadcasts copied.
REQUESTS	List number of blank forms required.
NETS ASSIGNED	List nets member is assigned.
REMARKS	Explain other hours claimed.

DA Form 5524-R (State Consolidated Participation Report) (RCS: ISC-65). State MARS director or his or her representative will submit DA Form 5524 R monthly. Figure 3-4 shows an example of a consolidated State participation

report. DA Form 5524-R will be locally reproduced on 8-1/2 by 11-inch paper and is located at the back of this manual. Instructions for completing DA Form 5524-R are in table 3-2.

Table 3-2. Instructions for completing DA Form 5524.

COLUMN	EXPLANATION
STATE	Name of reporting state.
ZONE *	List state zone number.
MONTH	Self-explanatory.
PARTICIPATION %	List percent of membership participation.
SHEET #	Self-explanatory.
ZONE COORDINATOR/ STATE DIRECTOR	List name of Zone Coordinator/State Director submitting report.
BLOCKS A.	List hours of operation in CW nets.
B.	List hours of operation in HF voice nets.
C.	List hours of operation in RATT' nets.
D.	List hours of operation in VHF voice nets.
E.	List total hours on the air.
F.	List 1 hour for each WAR broadcast copied.
G.	List hours of participation other than on-the-air time.
H.	List total hours claimed.
I.	List total number of messages processed.
J.	List number of phone patches completed.
K.	List number of Area Broadcasts copied.
L.	List number of State Director Broadcasts copied.
REMARKS	Explain other hours claimed.

c. Quarterly area report. The area MARS director will submit a consolidated area participation report quarterly providing the information outlined in figure 3-5.

d. Command traffic and participation report (RCS: ISC-65). Command MARS director will submit a consolidated command participation report quarterly providing the information outlined in figure 3-6.

e. DA Form 5525-R (Monthly Telephone Patch Report) (RCS: ISC-65). Each phone patch station is appointed by the area MARS director. Phone patch station will report their phone patch participation

credits and number of completed phone patches on their regular report card or message through normal reporting channels, and prepare a special phone patch report and submit directly to the command or area director or the area phone patch coordinator. The report will contain the following information: country, frequency, attempted, completed. DA Form 5525-R will be locally reproduced on 8-1/2 by 11-inch paper and is located at the back of this manual. Instructions for completing DA Form 5525-R are in table 3-2.

TO: AREA MARS DIRECTOR

FM: STATE MARS DIRECTOR

DATE:

BT

1. PARTICIPATION REPORT FOR (MONTH)

A (HOURS IN CW NET A1) 1 .

B.(HOURS IN VOICE NET A3) 4 .

C. (HOURS IN RATT NET F1) 1 .

D. (HOURS IN VOICE NET F3) 0 .

E. (HOURS ON THE AIR) 6 .

F. (WAR BROADCASTS COPIED) 1 .

G. (OTHER HOURS) 5 .

H. (TOTAL HOURS PARTICIPATION) 12 .

I. (TOTAL MESSAGES PROCESSED) 8 .

J. (TOTAL PHONE PATCHES COMPLETED) 0 .

K. (AREA BROADCASTS COPIED) 2 .

L. (STATE BROADCASTS COPIED) 3 .

2. NARRATIVE REMARKS (EXPLAIN OTHER HOURS SUCH AS REPAIR MARS EQUIPMENT AND PREPARED LESSON PLAN).

BT

Figure 3-4. Example of a State consolidated participation report

Letter format

FM: Area MARS Director

TO: _____ Command MARS Director

Quarterly Participation Report for _____ Quarter Calendar Year 19_____

a. Total number of members CIV _____.

MIL STA _____

TOTAL _____

b. Total number members reporting CIV _____.

MIL STA _____

TOTAL _____

c. Total number messages processed CIV _____.

MIL STA _____

TOTAL _____

d. Total hours reported CIV _____.

MIL STA _____

TOTAL _____

e. Total WAR broadcasts copied CIV _____.

MIL STA _____

TOTAL _____

f. Total area broadcasts copied CIV _____.

MIL STA _____

TOTAL _____

Figure 3-5. Sample area MARS director's quarterly report letter format.

Total phone patches completed CIV .

MIL STA _____.

TOTAL _____

g. Applications processed:

(1) New _____.

(2) Renewed _____.

(3) Modified _____.

h. Terminations processed:

(1) Individual request _____.

(2) Nonparticipation _____.

(3) License expired _____.

(4) Deceased _____.

(5) Other _____.

j. Phone patch report broken down as follows:

(1) Country _____.

(2) Frequency _____.

(3) Attempted _____.

(4) Completed _____.

k. Narrative Report:

(1) State ALPHA, had meeting, good turnout, demonstrated emergency operation.

(2) State BRAVO, held emergency exercise, 75 members participated.

(3) State CHARLIE, no comment.

(4) State DELTA, supported Red Cross flood area.

DIRECTOR
Area MARS

Letter Format

FM: Command MARS Director TO: Chief, Army MARS

Quarterly Participation Report (RCS-AS-QPS-147) for _____ Quarter

Calendar Year 19____

- a. Total number of members,
- b. Total members reporting.
- c. Total hours reported.
- d. Total messages processed.
- e. Total phone patches completed.
- f. Total WAR broadcasts copied.
- g. Total applications processed.
- h. Total terminations processed.
- i. Phone patch report broken down as follows:
 - (1) Country _____.
 - (2) Frequency _____.
 - (3) Attempted _____.
 - (4) Completed _____.
- j. Narrative report.

Command MARS Director

Figure 3-6. Sample command director's report format.

Table 3-3. Instructions for completing DA Form 5525-R.

COLUMN	EXPLANATION
DATE	Date of report.
RCS	RCS—For use by command or area MARS directors only.
TO	Self-explanatory.
FROM	Address of reporting station.
CALL SIGN	Call sign of reporting station.
NAME	Name of affiliate member.
COUNTRY	Name of country where call was originated.
FREQUENCY	Operating frequency.
ATTEMPTED	Number of calls attempted.
COMPLETED	Number of calls completed.
REMARKS	List any information that could be useful later in reference to this report.

3-7. Participation credits.

a. All MARS members and stations are required to participate and report such activities monthly using DA Form 5522,

b. A minimum of 12 hours of participation in 3 consecutive months, consisting of at least 8 hours of HF nets is required to retain membership.

c. Participation credit will be reported on a hourly basis.

(1) Credit of 1 hour is granted for copying the WAR broadcast at the time of initial transmission from radio station WAR or initial rebroadcast by the area gateway station.

(2) Periods of on-the-air time of less than 1 hour may be accrued in quarter-hour increments and accumulated to total full-hour credit.

(3) Operation on amateur radio bands will not be reported as participation for MARS activity.

d. Credit may be given for, but not limited to, participation in the following types of activities:

(1) Operation in assigned or scheduled nets and on-the-air drills.

(2) Operation under disaster or emergency conditions. In-station support will be credited as normal net operations; deployed support will be credited from time of deployment to return to home station and will be reported as "other hours."

(3) Performing other duties and/or engaging in other MARS related activities will be reported as "other hours" and explained in the remarks section, as follows:

(a) Developing and preparing lesson plans for net use.

(b) Developing and preparing technical articles.

(c) Maintenance, installation, and repair of MARS equipment.

(d) Assistance provided to other MARS members for MARS related or public relations activities.

(e) Attending MARS conferences or meetings.

(f) In support of installation or community MARS stations.

(g) Participation in extension courses.

(h) Time used to refile MARS traffic into amateur bands.

(i) In station message processing time.

e. Participation report schedule. All MARS members and stations will submit through their normal chain of command DA Form 5522 reflecting their operation through the last day of each month. Reports not received by the schedule below will not be counted.

(1) Individual members must submit DA Form 5522 to reach the zone coordinator, State MARS Director, or appropriate participation officer not later than the 10th of the following month.

(2) Zone coordinators or participation officers will consolidate and submit DA Form 5523-R to reach the State director or his or her representative not later than the 20th of each month. Area directors will reflect participation hours on DA Form 5523-R (MARS Member Information Sheet) (RCS: ISC-65) by month and year. This information will become a permanent part of the MARS member's file. DA Form 5523-R will be locally reproduced on 8-1/2 by 11-inch paper and is located at the back of this manual. Instructions for completing DA Form 5523-R are in table 3-4.

Table 3-4. Instructions for completing DA Form 5523-R

COLUMN	EXPLANATION
OPR	List name of operator.
CALL SIGN	Operator's station call sign.
ADD	Operator's station address.
DATE	Date of report.

Table 3-4. Instructions for completing DA Form 5523-R—continued.

COLUMN	EXPLANATION
CLASS LICENSE	Amateur license class.
COMPT DATE	Date of issue.
EXP DATE	Expiration date of license.
MISSION ASSIGNMENT	List assigned mission.
PARTICIPATION	List all on-the-air time.
PHONE PATCHES	List number of phone patches attempted and number completed.
REMARKS	List any information that could be useful later in reference to this report.

(3) State directors will consolidate and submit DA Form 5523-R to reach the Area Director's office not later than the 30th of the month.

(4) Military installation, activity, or unit MARS stations (para 3-6a(3)) will submit DA Form 5522 to reach the command director's office not later than the 30th day of the month following the report month.

(5) The area directors will consolidate and submit their report to reach the command MARS director's office no later than 20 May, 20 August, 20 November, and 20 February.

(6) The command MARS directors will consolidate and submit their report to reach Chief, Army MARS no later than 30 May, 30 August, 30 November, and 2 March.

f. USAR or ARNG retirement point credit. Members of the USAR or ARNG who participate in MARS may qualify for retirement point credits as authorized in AR 140-1, AR 140-185, and NGB 640-1. Retirement point credit may be accrued on the basis of operations in a MARS supervised net using a MARS frequency and call sign at the rate of 1 point for each 2 hours of operation. Not more than 1 credit point will be accrued for any one calendar day; however, periods of operations for less than 2 hours of operation, performed on different days, may be combined to qualify for 1 retirement point. This will be recorded on DA Form 1380 (Record of Individual Performance of Reserve Duty Training) and will be authenticated as indicated below:

- 1) For USAR unit members-by the commander.
- 2) For others-by the area MARS director.

3-8. Awards.

a. In recognition of the dedicated service to the Army MARS program demonstrated by its members, a service awards program has been established wherein recognition will be given for total length of service in 5-year increments effective with accumulative service of 25 years. Any member who had interrupted service may accumulate and credit renewed service toward the 25-year requirement for the certificate. Constructive credit

for membership in the AARS will be credited.

b. To qualify for the initial (25 years) award, the member must submit a biographical outline reflecting the month and year the member first entered MARS and the month and year of each succeeding membership period. Army members who had interrupted service may accumulate membership time as credit service toward the 25-year requirement.

c. Army MARS members who have less than 25 years accumulated service but were members of the AARS between the years 1925 through 1941 may submit biographical outlines for a special award. Members who include AARS service in their biography for the 25-year award will automatically receive the AARS special award.

d. Individual members who have accumulated 25 years of service in MARS or have reached their 70th birthday with 15 years of MARS service may be placed in special status (exempt from reporting) upon request through channels to the area or command MARS director. This request must be endorsed by the State MARS director. Members placed in special status will be carried on MARS rolls in good standing without regard to amount of participation. MARS licenses must continue to be renewed when FCC amateur license is renewed.

3-9. DD Form 2350 (Military Affiliate Radio Station {MARS} Disaster Support Identification Card).

a. DD Form 2350 is an accountable and controlled card. This card will be issued and controlled by Chief, Army MARS.

b. The card will only be issued to individual members who are officially designated by the State or area MARS directors. The card will expire 2 years from date of issue or on expiration of MARS membership. When a member vacates the official position or is terminated from the program, this card must be returned through channels to Chief, Army MARS, HQ, USAISC, ATTN: AS-OPS-OA, Fort Huachuca, AZ 85613-5000.

c. State MARS directors will submit a request for DD Form 2350 to be issued to individual MARS members to the area MARS director, stating that the member is or has been appointed to a position on the state staff that deals specifically with disaster communication such as emergency coordinator, and so forth. The request will be processed by the area MARS director and forwarded through the command MARS director to Chief, Army MARS, HQ, USAISC, ATTN: AS-OPS-OA, Fort Huachuca, AZ 85613-5000, with recommendations. Chief, Army MARS will forward a completed DD Form

2350 to the individual for signature and picture (name board not required on picture) with a cover letter, instructing the MARS member to go to the nearest military post, Identification card section, to get his or her picture taken and the card laminated. If the member does not live near a military post, the member should acquire a 1- by 1-inch passport-type picture and return it with the DD Form 2350 to Chief, Army MARS for lamination. When the card has been laminated and is in the possession of the member, the receipt on the cover letter will be signed and returned to the Chief, Army MARS.

CHAPTER 4

MEMBERSHIP

4-1. Membership requirements.

a. All MARS radio stations, whether military, club, or individual, may be a member of only one service MARS. Military commands or activities will affiliate with their sponsoring service MARS unless operating in areas where there is only one service MARS authorized.

b. Membership is limited to—

(1) Individuals who are—

(a) Fourteen years of age or older. (Age may not be waived). Signature of parent or guardian is required on DD Form 630 (Application for Membership in Military Affiliate Radio Systems (MARS)) for applicants 14 to 18 years of age. DD Form 630 will be locally reproduced on 8- by 5-inch card stock, head to foot, and is located at the back of this manual. Instructions for completing DD Form 630 are self-explanatory.

(b) Citizens of the United States or aliens who have been lawfully admitted to the United States for permanent residence under the provisions of United States Code, chapter 12, title 3.

(c) Willing to operate in accordance with the regulations prescribed for participation in the Army MARS Program.

(d) In possession of a radio station capable of operating on two MARS HF frequencies and is a valid FCC and/or host country licensed amateur radio operator.

(2) Military installations, activities, and unit stations operated and maintained under the auspices of a military command.

(3) Military unit or civilian radio club stations whose custodian possesses a valid amateur radio license issued by the FCC or appropriate overseas commander.

c. Application for Army MARS membership.

(1) Applications for Army MARS affiliate membership will be submitted to the State, area, or command MARS director of the geographic area in which the applicant is located using DD Form 630. In addition to DD Form 630, an applicant will simultaneously complete and submit a DA Form 5529-R (Army MARS Questionnaire) and include a copy of the current FCC and/or host nation amateur license. No RCS is required per AR 335-15, paragraph 5-2b(4). DA Form 5529-R will be locally reproduced on 8-1/2 by 11-inch paper and is located at the back of this manual.

Instructions for completing DA Form 5529-R are self-explanatory. This form is to be filled out completely by the prospective Army MARS member and submitted with DD Form 630 to the MARS director of the area in which he or she resides.

2) Authority to approve membership applications is delegated to command or area MARS directors and will not be further delegated.

3) Upon approval, the command or area MARS director will issue the new member an Army MARS Radio Station License and Call Sign, an Army MARS Communications Practices and Procedures Manual, and other instructional and informational material.

4) Military unit or civilian radio clubs will at the time of application have a minimum of three members who have individual membership in MARS.

4-2. Membership period.

a. Membership in MARS is granted for the period covered by the member's current FCC or host nation amateur radio license.

b. Membership period for novice class license is 18 months. Membership will be terminated if the members fail to upgrade their FCC amateur licenses within this period. Members serving overseas, where upgrade facilities are not available, are automatically excluded for a period not to exceed 12 months following return to CONUS.

4-3. Modifications.

Each individual MARS affiliate member, custodian, or trustee of a MARS installation, unit, or club station will notify the area or command MARS director through channels for any of the following reasons:

a. Change of address—may be reported by letter or message.

b. Change of FCC or host nation expiration date or upgrade of amateur license—will be reported by submission of DD Form 630 and a copy of the new amateur license.

c. Change of custodian or trustee—will be accomplished by submission of DD Form 630 and a copy of the custodian's or trustee's amateur license.

4-4. Renewal.

a. The term of MARS membership is based upon the member's amateur radio license; consequently, MARS membership automatically expires when the amateur radio license expires or is revoked for any reason. To retain MARS membership, the member must immediately renew his or her MARS membership as soon as the renewed amateur radio license is received and the new expiration date is known.

b. A MARS member who has submitted a request for renewal may, while awaiting the return of the renewed license, continue to operate on MARS frequencies after his or her MARS membership has expired. The MARS membership on any individual or club station will be cancelled if a renewal request is not received by the area or overseas MARS director within 60 days after the MARS membership expiration date. A request for renewal of MARS membership will be submitted on DD Form 630 along with copies of an updated DA Form 5529-R and the new amateur radio license.

4-5. Transfer.

a. Within CONUS, Alaska, Puerto Rico, Hawaii, and the Canal Zone (Panama):

(1) Any member who transfers from one area to another will initiate a letter through channels to the MARS Director of the area the member is leaving requesting the transfer of member's MARS records to the area to which the member is relocating. The member will provide a projected date of arrival at the new location.

(2) The losing State MARS director will forward the member's records to the area MARS director who will transfer the combined member's records to the gaining area MARS director.

(3) The member must contact the gaining area MARS director during this time indicating his or her intent to remain active in the program. Failure to contact the gaining area MARS director within 60 days will be sufficient cause for termination from the program.

(4) The gaining area MARS director will retain the member's records in a suspense status for 60 days after the projected arrival date. If no contact by the end of the suspense period, the area MARS director will terminate the membership. Logistical files will be coordinated with the MARS Property Manager (MPM) in accordance with paragraph 8-9.

(5) During the transfer period and until receipt of a new MARS call sign, the member is authorized to continue using his or her current call sign.

b. Outside CONUS where MARS is not authorized.

(1) Any member who is transferred as an employee of the Federal Government (military or civilian) to a location outside CONUS where MARS is not authorized will notify the appropriate MARS director of the impending transfer date.

(2) The MARS director will drop the member from the area membership list and forward his or her State and area records to Chief, Army MARS, HQ, USAISC, ATTN: AS-OPS-OA, Fort Huachuca, AZ 85613-5000, for retention pending return to an active MARS area.

(3) Upon return to an active MARS area, it is the responsibility of the individual concerned to request MARS reinstatement from Chief, Army MARS.

(4) If the individual's MARS membership expired while he was overseas, then DD Form 630 and a copy of the individual's current amateur license must be submitted with the request for reinstatement.

c. Outside CONUS where MARS is authorized,

(1) Any member who is transferred as an employee of the Federal Government (military or civilian) to a location outside CONUS where MARS is authorized will initiate a letter through channels to the MARS director of the area the member is leaving requesting the transfer of his or her MARS records to the command MARS director of the overseas command to which he or she is being reassigned. The member will provide the area MARS director the unit to which he or she is being assigned and a projected date of arrival in the overseas command.

(2) The losing area MARS director will consolidate the State and area logistical and personnel records for review prior to forwarding to the overseas command. The area MARS director must coordinate with the MPM prior to overseas shipment of surplus property issued by the MARS equipment program.

(3) The losing area MARS director will transfer the member's consolidated records to the gaining overseas command MARS director upon notification of request to transfer from the individual and property disposition instructions from the MPM.

(4) The member must contact the gaining command MARS director during this time indicating his or her intent to remain active in the program.

(5) The gaining command MARS director will retain the member's records in a suspense status for 90 days after projected arrival date. If no contact is established by the end of the suspense period, the command MARS director will terminate the membership. Logistical files will be coordinated with the MPM in accordance with paragraph 8-9.

(6) During that part of the transfer period in which the member is in CONUS, the member is authorized to continue using his or her current call sign. Upon departure from CONUS the member is not authorized to operate in MARS until he has complied with the oversea command and host country regulations,

d. Return from overseas.

(1) Any member who is transferred as an employee of the Federal Government (military or civilian) from outside continental United States (OCONUS) to a location in CONUS will notify the command MARS director of the impending transfer and request the transfer of his or her MARS records to the area to which he or she is relocating. The member will provide a projected date of arrival at the new location.

(2) The losing command MARS director will transfer the member's records to the gaining area MARS director upon notification of request to transfer from the individual.

(3) The member must contact the gaining area MARS director during this time indicating his or her intent to remain active in the program.

(4) The gaining area MARS director will retain the member's records in suspense status for 90 days after the projected arrival date. If no contact by the end of the suspense period, the area MARS director will terminate the membership. Logistical files will be coordinated with the MPM.

(5) The gaining area MARS director will notify the gaining State MARS director of the pending arrival of a new member. After receipt of notification of intent to remain active in the program the area MARS director will provide the State MARS director with a copy of the member's records required at the State level.

(6) During the transfer period (90 days) and until receipt of a new MARS call sign, the member is authorized to continue using his or her current call sign.

4-6. Termination.

a. MARS membership may be terminated by Chief, Army MARS for the following reasons:

(1) Willful misconduct on the air.

(2) Willful misuse, sale, barter, or unauthorized disposal of U.S. Government property issued for MARS use.

(3) Failure to comply with the policies and directives of Army MARS.

b. Command or area directors are authorized to terminate MARS membership for the following reasons:

(1) Failure to maintain minimum participation requirements.

(2) Failure to submit a monthly participation report for 3 consecutive months.

(3) Failure to renew MARS license.

(4) At the request of the member (resignation).

c. State MARS directors may recommend termination of a member for any of the reasons in a and b above. State MARS directors will forward recommendations on DA Form 5531-R (MARS Membership Termination Request) to the area MARS directors for appropriate action. DA Form 5531-R will be locally reproduced on 8-1/2 by 11-inch paper and is located at the back of this manual. Instructions for completing DA Form 5531-R are self-explanatory. No RCS is required per AR 335-15, paragraph 5-2b(5). This form is to be filled out completely by the State MARS director and forwarded, with supporting documents to his or her area MARS director.

4-7. Members entering the Armed Forces.

MARS members entering the U.S. Armed Forces are eligible to receive a certificate or statement attesting to their membership. This certificate or statement may be accepted by interviewing personnel at enlistment centers and may be considered in determining assignment in the military service. MARS members enlisting in a Military Service of the United States may write to the appropriate area MARS director specifying when and where enlistment is to take place and will be furnished a MARS certificate or statement of membership.

CHAPTER 5

TRAINING

5-1. Training objectives.

The primary training objective of MARS is to develop a network of individual operators and stations capable of accomplishing the mission as outlined in paragraph 2-2. This objective can only be accomplished if every operator is trained in standardized military communication operations and traffic handling procedures. This training will enable the individual to develop the proficiency needed to establish emergency communications, and organize and conduct net operations to support emergency or contingency conditions wherever and whenever necessary. The common capability attained through training is the basis for individual pride of accomplishment and confidence in the team's ability to provide military communications. Effective military communications is the direct result of team effort.

5-2. Training program.

a. MARS directors at all levels will aid and encourage MARS members to improve military electronic

communications through on-the-air training periods, enrollment in the extension courses of the U.S. Army Signal School (U.S. Army Training Support Center, Student Service Division, ACCP. Newport News, VA 23628), and by dissemination of operational and technical information and data. DA Form 145 (Army Correspondence Course Enrollment Application) will be forwarded through channels to the command or area MARS director.

b. MARS directors will establish and implement training programs to meet the training objective. Training will be accomplished by the use of on-the-air nets, military publications, extension courses, and other MARS or local generated material.

c. Training subjects will be derived from the contents of this manual.

d. Training through experimentation with electronic equipment will be encouraged. Every effort will be made to provide for experimental purposes excess communications equipment and materials. Experimental aspects of MARS will be further encouraged by the command or area MARS directors, with demonstrations, lectures by authorities in electronic fields, and tours of unclassified installations.

Chapter 6

Call Signs

6 - 1. Call signs.

- a. MARS station call sign assignments are made by the command and area MARS directors from call sign blocks allocated by HQDA to Chief, Army MARS.
- b. A list of MARS station call sign prefixes

currently used by the U.S. Army worldwide are indicated in table 6-1. Call sign suffixes will be assigned by command or area MARS directors in alphabetical sequence beginning with the letters "AA" and continuing through "ZZ." Request for special call signs will not be honored.

Table 6-1. MARS station call sign prefixes.

COUNTRY OF AREA	MARS PREFIX
Australia	ABM9
Alaska	ALM7
Aleutian Islands	ALMS
El Salvador	ACMI
Germany (U.S. zone)	AEM1
Hawaii	ABM6
Honduras	ACM6
Japan	ABMI
Korea	ABM4
Okinawa	ABM2
Canal Zone	ACM5
Puerto Rico	ACM4
Virgin Islands	ACM3
CONUS (48 contiguous States)	AAA/AAR/AAT/AAV
Trust Territories and the Pacific less Japan, Okinawa, and Korea	ABM3

c. The use of Army MARS call signs is authorized only on Army MARS networks, except during emergencies when inter service operations are directed by service chiefs.

d. The portable suffix will not be used on MARS frequencies. Stations operating mobile on MARS frequencies are not required to give their location or indicate mobile operations unless requested to do so by the net control station (NCS).

e. Call signs assigned to military MARS stations are station call signs and will be used by all operators at that station.

f. Call signs assigned to affiliates are individual call signs based on residence location. Wherever the affiliate is, is where his or her station is located. During

emergency operations an affiliate station may assume the function of a military station and would be operated in accordance with mission assignment.

g. MARS call signs will not be used on amateur bands.

h. Use of amateur radio call signs on MARS networks is not authorized.

i. Tactical military units, under emergency conditions and when authorized by Chief, Army MARS or the command MARS director, may use tactical call signs to interface with MARS.

j. Table 6-2 provides State and regional MARS director billet call signs.

k. Table 6-3 provides special billet call signs.

Table 6-2. State and regional MARS director billet call signs.

STATE	CALL	STATE	CALL
Alabama	AAA4AL	Alaska	AAM7AK
Arizona	AAA9AZ	Arkansas	AAA4AR
California South	AAA9CS	California Central	AAA9CC
California North	AAA9CN	Colorado	AAA8CO
Connecticut	AAA1CT	Delaware	AAA3DE
Florida	AAA4FL	Georgia	AAA4GA
Hawaii	AAM6HI	Idaho	AAA0ID
Illinois	AAA5IL	Indiana	AAA5IN
Iowa	AAA7IA	Kansas	AAA7KS

Table 6-2. State and regional MARS director billet call signs—continued.

STATE	CALL	STATE	CALL
Kentucky	AAA4KY	Louisiana	AAA6LA
Maine	AAA1ME	Maryland	AAA3MD
Massachusetts	AAA1MA	Michigan	AAA5MI
Minnesota	AAA5MN	Mississippi	AAA4MS
Missouri	AAA7MO	Montana	AAA8MT
Nebraska	AAA5NE	Nevada	AAA9NV
New Jersey	AAA2NJ	New Mexico	AAA6NM
New York	AAA2NY	North Carolina	AAA4NC
North Dakota	AAA8ND	Ohio	AAA5OH
Oklahoma	AAA7OK	Oregon	AAA0OR
Pennsylvania	AAA3PA	Puerto Rico	AAM4PR
Rhode Island	AAA1RI	South Carolina	AAA4SC
South Dakota	AAA8SD	Tennessee	AAA4TN
Texas North	AAA6TN	Texas South	AAA6TS
Utah	AAA8UT	Vermont	AAA1VT
Virginia	AAA4VA	Virgin Islands	AAM3VI
Washington	AAA0WA	West Virginia	AAA3WV
Wisconsin	AAA5WI	Wyoming	AAA8WY
Trust Territories of the Pacific	AAM6TT		

Table 6-3. Special billet call signs.

CALL SIGN	STATION
AAA9CG	Commanding General, USAISC
AAA90P	Deputy Chief of Staff Operations, HQ, USAISC
AAA9A	Chief, Army MARS
AAA9B	Administrative Assistant to Chief, Army MARS
AAA9C	Operations Assistant to Chief, Army MARS
AAA9Z	Communications Clerk for Headquarters Army MARS
AAA3B	CONUS MARS Director
AAA3E	Eastern Area MARS Director
AAA6C	Central Area MARS Director
AAA9W	Western Area MARS Director
AAM6WC	USAISC-WESTCOM Command MARS Director
ABM4KOR	Korea Area MARS Director
AAA1EU	Europe Area MARS Director
ABM1JPN	Japan Area MARS Director
AAA3D	DOD-ASDCC3I)
AAA3NA	Chief, Navy/Marine Corps MARS
AAA5AF	Chief, Air Force MARS
AAA3USA	Eastern Area Gateway Station
AAA6USA	Central Area Gateway Station
AAA9USA	Western Area Gateway Station
AEM1USA	European Gateway Station
ABM4USA	Pacific Gateway Station
ABM6USA	Hawaii Gateway Station

6 – 2. Operating signals, prosigns, and prowords

Paragraph 9-8 discusses operating signals, prosigns, and prowords in detail.

CHAPTER 7

ARMY MARS NETWORKS

7-1 General

The MARS is a global network of local, State, area, continental, and intercontinental communications nets interconnected to form a worldwide traffic handling system.

7-2. Net description.

a. The following nets have been designated to service the intercontinental circuit path for Army MARS traffic. Net times are variable depending on propagation conditions.

(1) WAR/Alpha Net - meets Monday through Friday, 1600Z to 1800Z. Primary frequency 20992.5 kHz USB, alternate frequency 17501.5 kHz USB; secondary frequency 14930.0 kHz USB, alternate frequency 13479.5 kHz USB.

LOCATION	CALL SIGN	NCS	ANCS
		WAR	WAR
Lohnsfeld Ge	AEM1USA		
Fort Meade, MD	AAA3USA		AAA3USA
Fort Sam Houston Tx	AAA6USA		AAA6USA
Presido of CA	AAA9USA		AAA9USA
Hawaii	ABM6USA		
Korea	ABM4USA		

(2) Atlantic/Alpha Net - meets daily at 1300Z. Primary frequency 20975.0 kHz USB, alternate frequency 19024.0 kHz USB; secondary frequency 16041.0 kHz USB, alternate frequency 14665.0 kHz USB.

LOCATION	CALL SIGN	NCS	ANCS
Lohnsfeld, GE	AEM1USA		AEM1USA
Baumholder, GE	AEM2BPM		
Fort Meade, MD	AAA3USA	AAA3USA	
Fort McPherson, GA	AAA4USA		
Fort Sam Houston, TX	AAA6USA		
Port Bragg, NC	AAR4USC		AAR4USC

(3) Pacific/Alpha Net - meets Monday through Friday (Korea, Japan, and Okinawa time) and Sunday through Thursday (Hawaii, CONUS, and Alaska time). Zulu time is the same at 0100Z. Primary frequency 19531.0 kHz USB, alternate frequency 17498.5 kHz USB; secondary frequency 13994.5 kHz USB, alternate frequency 11990.0 kHz USB.

LOCATION	CALL SIGN	NCS	ANCS
Hawaii	ABM6USA	ABM6USA	
Presidio of SF, CA	AAA9USA		AAA9USA

Fort Ord, CA	AAR9USD	AAR9USD
Fort Lewis, WA	AAR0USB	AAR0USB
Fort Carson, CO	AAR8USC	AAR8USC
Japan	ABM1USA	
Okinawa	ABM2USA	
Korea	ABM4USA	ABM4USA
Alaska	ALM7USA	

(4) Pacific/Bravo Net - meets Monday through Friday at 0500Z • Korea, Japan, and Okinawa time and Sunday through Thursday at 0500Z Hawaii time. Primary frequency 19534.0 kHz USB, alternate frequency 17498.5 kHz USB; secondary frequency 13996.5 kHz USB, alternate frequency 11990.0 kHz USB.

LOCATION	Call Sign	NCS	ANCS
Korea	ABM4USA		ABM4USA
Hawaii	ABM6USA	ABM6USA	
Japan	ABM1USA		ABM1USA
Okinawa	ABM2USA		ABM2USA

(5) WAR broadcast information is contained in paragraph 9-19.

b. The following nets have been designed to provide CONUS a continental circuit path for MARS traffic.

(1) CONUS/Alpha Net - meets daily at 1200Z to 1600Z. Primary frequency 20105.5 kHz USB, alternate frequency 17403.5 kHz USB; secondary frequency 14939.5 kHz USB, alternate frequency 10815.0 kHz USB.

LOCATION	CALL SIGN	NCS	ANCS
Fort Meade, MD	AAA3USA		AAA3USA
Fort Sam Houston, TX	AAA6USA	AAA6USA	
Presidio of SF, CA	AAA9USA		AAA9USA

(Alternate stations designated by appropriate area director.)

(1) CONUS/Bravo Net - meets daily at 1600Z to 2200Z. Primary frequency 14936.5 kHz USB, alternate frequency 13743.0 kHz USB; secondary frequency 10815.0 kHz USB, alternate frequency 6997.5 kHz USB.

LOCATION	CALL SIGN	NCS	ANCS
Fort Meade, MD	AAA3USA		AAA3USA
Fort Sam Houston, TX	AAA6USA	AAA6USA	
Presidio of SF, CA	AAA9USA		
All Conus Military Stations			

(Alternate and other designated affiliate stations)

(2) CONUS/Charlie Net - meets daily at 2200Z to 1200Z. Primary frequency 6997.5 kHz USB (2200Z to 0459Z), alternate frequency 6825.0 kHz USB; secondary frequency 4026.5 kHz USB (0500Z to 1200Z), alternate frequency 3348.5 kHz USB.

LOCATION	CALL SIGN	NCS	ANCS
Port Meade, MD	AAA3USA		AAA3USA
Fort Sam Houston, TX	AAA6USA	AAA6USA	
Presidio of SF, CA	AAA9USA		AAA9USA

(All MARS stations.)

c. CONUS area and OCONUS command nets will be approved and NCSs will be designated by the appropriate area or command MARS director.

d. State nets will be recommended by the State directors and approved by area directors; net control stations will be designated by the State director.

e. NCSs for the WAR/Alpha Net, Atlantic/Alpha Net, Pacific/Alpha Net, Pacific/Bravo Net, and CONUS/Alpha Net will submit a monthly contact report via message to HQ, USAISC, ATTN: AS-OPS-OA, Fort Huachuca, AZ 85613-5000, with an information copy to each assigned net participant and the command or area MARS director. The report will be prepared in matrix format with assigned stations call signs across the top and the day (1 through 31) down the right side. Numbers for days the net is closed will be omitted. The letter "X" will be entered for each station that checked into the net. An NC (no contact) will be entered for each station that failed to check in. Narrative comments on net operation may be included as desired. Reports should be forwarded to arrive not later than the 15th of each month. The following requirement control symbols (RCSs) apply:

(1) Pacific/Alpha Net, RCS AS-OPS-156(1), subject: MARS Net Contact Report,

(2) Pacific/Bravo Net, RCS AS-OPS-156(2), subject: MARS Net Contact Report.

(3) Atlantic/Alpha Net, RCS-OPS-156(3), subject: MARS Net Contact Report.

(4) CONUS/Alpha Net, RCS-QPS-156(4), subject: MARS Net Contact Report.

(5) WAR/Alpha Net, RCS-OPS-156(5), subject: MARS Net Contact Report.

7-3. Message traffic.

a. Message traffic may originate anywhere within the MARS system. The path a message will follow may be between two stations (point-to-point) or passed through a series of relays and gateway stations over established radio networks.

b. In the military, communications follow the vertical structure of the chain of command. At each organizational level, radio nets are established to meet

the needs of the command. Army MARS networks are established in a similar manner.

c. As message traffic moves over prescribed networks within the military establishment, so will message traffic handled over the MARS system move over similar networks.

d. Message traffic within Army MARS is primarily for the purpose of training. It is important that all Army MARS members be familiar with the proper procedures for routing message traffic received into the MARS system.

7-4. Army MARS nets.

a. Intercontinental nets.

(1) Army MARS networks are organized and established at several levels in the chain of command. The nets are WAR/Alpha, Atlantic/Alpha, Pacific/Alpha, Pacific/Bravo, and WAR/Broadcast.

(2) The worldwide Army MARS command net (WAR/A) is composed of the HQDA command station WAR and the gateway stations worldwide; for example, Germany AEM1USA, Eastern area AAA3USA, Central area AAA6USA, Western area AAA9USA, Hawaii ABM6USA, and Korea ABM4USA.

(3) The Atlantic/Alpha net is composed of Eastern area gateway AAA3USA, Fort Meade, MD, alternate AAR4USA, Fort McPherson, GA; European Gateway AEM1USA, Lohnsfeld, Germany, alternate AEM1BPM, Baumholder, Germany; and Central Area AAA6USA, Fort Sam Houston, TX, and various alternates designated by the command or area MARS directors.

(4) The Pacific/Alpha net is composed of the gateway stations in Western Area AAA9USA, Presidio of San Francisco, CA, WESTCOM ABM6USA Hawaii, Korea ABM4USA, and their alternates Japan ABM1USA, Okinawa ABM2USA, Alaska ALM7USA, Fort Ord, CA AAR9USD, Fort Carson, CO AAR8USB, and Fort Lewis, WA AAR0USB.

(5) The Pacific/Bravo net is composed of the WESTCOM gateway ABM6USA Hawaii and alternates Korea ABM4USA, Japan ABM1USA, and Okinawa ABM2USA.

(6) The WAR/Bravo net transmits a broadcast message weekly to all Army MARS members and stations worldwide.

b. Continental nets.

(1) The CONUS nets are composed of the Central area gateway AAA6USA, Eastern area gateway AAA3USA, Western area gateway AAA9USA, their alternates, military and affiliate stations,

(2) All stations operate in accordance with the worldwide Gateway net schedule.

(3) Various area and State nets.

7-5, Net utilization.

a. The WAR/Alpha net is used to exercise the worldwide DA command net. All stations will report to the NCS WAR the status of their station; for example, equipment problems, manpower status, and traffic backlog. NCSs report this data to Chief, Army MARS, Monday through Friday.

b. The Atlantic/Alpha net is used to pass traffic between CONUS and Europe via radio or teletype and will handle high volume traffic.

c. The Pacific/Alpha and Bravo nets are used to pass traffic between CONUS and the Pacific via radio or teletype and will handle high volume traffic.

d. The WAR broadcast is designed to test the capability of Radio Station WAR and provide the Chief, Army MARS, a means of disseminating information to all members worldwide. WAR operations are defined in paragraph 9-19.

e. The CONUS nets are used to pass traffic to and from other gateway and CONUS stations. These nets will be radio or teletype, continuous wave (CW), and voice operated and will handle high volume traffic.

7-6. Gateway traffic handling procedures.

a. Each gateway station is the focal point for all MARS stations in that specific geographic area.

b. The gateway station in each area is responsible to receive traffic from their area and retransmit it to other gateway stations and receive traffic from other gateway stations and retransmit it to their area member stations.

c. All gateway stations will check all incoming traffic for completeness, errors, garbles, and so forth, before receipting for it. All discrepancies will be corrected by obtaining correct information from transmitting station before receipting for the message.

d. Garbled tapes will not be retransmitted by a gateway station. After corrections have been obtained from the transmitting station the message will be repoked to obtain a new tape for transmission.

e. All gateway stations will assign or add their station serial number to each message before retransmitting.

f. All gateway stations will expedite the process of moving their traffic to the next MARS station so that long delays and old traffic is nonexistent.

g. Net schedule in traffic handling is of utmost

importance. Schedules will sometimes be shifted due to band propagations. When this must be done, the NCS must notify all net participants of the change in order for them to follow and be of value to the net.

h. Gateway stations will not reduce traffic from other gateway stations that is addressed to their geographic area of responsibility.

i. Alternate gateway stations will be appointed. Gateway stations will coordinate with alternates and request their assistance when needed.

j. Gateway NCS will expeditiously move all traffic listed on the net. They will not leave traffic hanging.

k. MARS areas.

(1) The general breakout in (a) through (g) below shows the Army MARS areas worldwide.

(a) Eastern Area Army MARS (Gateway Station AAA3USA).

Alabama	Connecticut	Delaware	Florida
Georgia	Kentucky	Maine	Maryland
Massachusetts	Mississippi	New Hampshire	New Jersey
New York	North Carolina	Pennsylvania	Rhode Island
South Carolina	Tennessee	Vermont	Virginia
West Virginia	Puerto Rico	Virgin Islands	

(b) Central Area Army MARS (Gateway Station AAA6USA).

Arkansas	Illinois	Indiana	Iowa
Kansas	Louisiana	Michigan	Minnesota
Missouri	Nebraska	New Mexico	Ohio
Oklahoma	Texas	Wisconsin	Canal Zone
El Salvador			

(South America)

(c) Western Area Army MARS (Gateway Station AAA9USA).

Arizona	California	Colorado	Idaho
Montana	Nevada	North Dakota	Oregon
South Dakota	Utah	Washington	Wyoming
Alaska			

(d) WESTCOM Area Army MARS (Gateway Station ABM6USA). All Pacific areas between the Aleutian Islands and the South Polar Caps, less Japan, Okinawa, and Korea.

(e) Command Area Army MARS—USAISC- Japan (No Gateway Station).

Japan Okinawa

(f) Command Area Army MARS—Korea (Gateway Station ABM4USA),

(g) Europe Area Army MARS (Gateway Station AEM1USA). Area includes Europe, Near East, and Africa.

CHAPTER 8

LOGISTICS

8-1. Surplus property

a. DOD and General Service Administration (GSA) surplus property can be made available for issue to affiliate members for operational and experimental purposes in accordance with DOD Directive 4160.21-M.

b. Surplus property may be used in various ways by recipients. Equipment may be—

(1) Used as originally designed.

(2) Modified to improve its operational capability,

(3) Modified to the extent that it can be used other than originally intended.

(4) Cannibalized, when authorized by Chief, Army MARS, for parts to repair a similar item or to repair and construct different items. Remaining residue will be disposed of in accordance with paragraph 8-7c,

(5) Used in whole or part for experimental purposes when authorized by Chief, Army MARS.

c. Surplus property obtained for MARS uses remains Government property and requires care, protection, and accountability as other Government property. An audit trail of property received through the Defense Reutilization and Marketing Service (DRMS) will be maintained from point of receipt from the Defense Reutilization and Marketing Office (DRMO) through the logistics management chain to property recipient and subsequent disposal decision.

d. To maximum extent possible, surplus property will be shipped or issued directly to the ultimate user. Direct issue will be used to eliminate excessive handling and the accumulation of property at intermediate points.

e. All requests to DRMO activities and issues to affiliate members will be based on an approved requirement. No nice-to-have items will be obtained or issued and no forced-issue techniques will be used to distribute property. In no case will Government property be used as door prizes or otherwise given away for promotional purposes.

f. All Government property will be returned to Government- control upon termination of membership, whatever the reason for termination. State and area command directors will, immediately upon learning of an individual's termination, recover all Government property issued to the former member. A record of recovery or extenuating circumstances will be retained in the former member's file until the file is destroyed in accordance with paragraph 8-9c,

g. Government buildings, facilities, tools, materiel, or other Government resources will not be used to care for and/or repair surplus property which has been issued to an individual affiliate member,

h. Surplus equipment made available to the MARS equipment program is not authorized military or other Government maintenance support. No claim may be made for reimbursement for time, materiel, or funds expended on MARS issued property.

i. Equipment obtained through the DRMS for issue in the MARS equipment program will be retained and used strictly within the Army MARS Program. Equipment will not be requested for, issued, given, or loaned to table of organization and equipment or table of distribution and allowances (TOE or TDA) organizations, state or local Government organizations, or nonaffiliated individuals or institutions.

j. Equipment issued to an individual affiliate member and subsequently lost or stolen may be dropped from the accountable records after certification by the individual, review by the State director and area director, and approved by the Chief, Army MARS. The member must submit a letter to his or her State MARS director itemizing the property involved and explaining in detail the circumstances under which the event occurred. Local law enforcement authorities will be notified in case of theft. A copy of their report will be included in the member's letter. If a report has not been furnished, this fact will be indicated in the letter.

k. All acquisitions or transfers of MARS surplus or foreign excess property will be covered by requisition, issue, or transfer documents. These documents must be approved in writing by the MPM.

8-2. Authorized categories of Government surplus property.

Surplus property transferred into the Army MARS Program will be limited to communications and electronic equipment; small fixed or portable gasoline or diesel powered alternating current generating equipment; component parts and appropriate tools, test equipment, and materials directly related to MARS operational and experimental requirements. DOD Directive 4160.21-M, chapter IX. and table 8-1 of this manual contain the Federal Supply Classes (FSC) for equipment authorized to be requisitioned by Army MARS. Any requisition for an FSC not reflected on this attachment requires the approval of the Office of the Deputy Assistant Secretary of Defense for Acquisition and Logistics (DOD (A&L)), Certain FSCs in table 8-1, as annotated by subscript numerals, are restricted to issue for specific usage.

8-3. Eligibility.

- a. MARS members become eligible to receive MARS property only after active participation for 6 months.
- b. To continue eligibility to receive equipment, the minimum of 12 hours participation each 3 months must be met.

8-4. Equipment request and validation.

- a. The request for property originates with the individual MARS member. To obtain equipment through the Army MARS Equipment Program, the individual MARS member will prepare DA Form 5530-R (Army MARS Equipment Request) in 3 copies for submission to the State or area MARS director. No RCS is required per AR 335-15, paragraph 5-2b(2). The member is responsible for completion of blocks 1 through 11 of DA Form 5530-R. DA Form 5530-R will be locally reproduced on 8-1/2 by 11-inch paper and is located at

the back of this manual. Instructions for completing DA Form 5523-R are in table 8-2.

- (1) DA Form 5530-R will be forwarded to the State or area MARS director for validation of the requirement. The State MARS director will evaluate the request based on member participation during the past two full quarters. If the member has not met minimum participation requirements as set forth in AR 25-6, for the past two full quarters, the request will be rejected and returned to the member with a notation that minimum participation has not been met. If the member has met the minimum requirements for the past two full quarters, the State MARS director will validate the request and will place an alphabetic rating in the category block. The State director will retain a copy in the member's file and will forward the remaining copies to the area MARS director with comments as applicable. The alphabetic rating will be based on the following participation:

Table 8-1. Authorized FSCs for service MARS requisitions.

FSC	Material Category
5805	<u>Telephone and telegraph equipment</u>
5815	<u>Teletype and facsimile equipment</u>
5820	<u>Radio and television communications equipment, except airborne</u>
5821	<u>Radio and television equipment, airborne</u>
5830	<u>Intercommunications and public address systems, except airborne</u>
5835	<u>Sound recording and reproduction equipment</u>
5895	<u>Miscellaneous communications equipment</u>
5905	<u>Resistors</u>
5910	<u>Capacitors</u>
5915	<u>Filters and networks</u>
5920	<u>Fuses and lightning arrestors</u>
5925	<u>Circuit breakers</u>
5930	<u>Switches</u>
5935	<u>Connectors, electrical</u>
5940	<u>Lugs, terminals, and terminal strips</u>
5945	<u>Relays and solenoids</u>
5950	<u>Coils and transformers</u>
5955	<u>Piezoelectronic crystals</u>
5960	<u>Electronic tubes and associated hardware</u>
5961	<u>Semiconductor devices and associated hardware</u>
5962	<u>Microcircuits, electronics</u>
5965	<u>Headsets, handsets, microphones, and speakers</u>
5970	<u>Electrical insulators and insulating material</u>
5975	<u>Electrical supplies and hardware</u>
5977	<u>Electrical contact brushes and electrodes</u>
5985	<u>Antennas, waveguides, and related equipment</u>
5990	<u>Synchros and resolvers</u>
5995	<u>Cable, cord, and wire assemblies; communications equipment</u>

Table 8—1. Authorized FSCs for service MARS requisitions—continued.

FSC	Material Category
5999	Miscellaneous electrical and electronic components
6105 (1)	Motors, electrical
6110 (1)	Electrical control equipment
6115 (2)	Generators, generator sets electrical (10kW or less only)
6125	Converters, electrical rotating
6130	Converters, electrical nonrotating
6135	Batteries, primary
6140	Batteries, secondary
6145	Wire and cable electrical
6150 (1)	Miscellaneous electrical power and distribution equipment
6625	Electrical and electronic properties measuring and testing instruments
6940	Communications training devices (code practice equipment only)
7021 (1)	Automatic data processing, processing unit computer, digital
7110 (3)	Office furniture (desk, chair, and file cabinet only)
7125 (3)	Cabinets, lockers, bins, and shelving
7430 (3)	Typewriters and office type composing machines
7530	Stationing and record forms
7610 (1)	Office supplies

NOTES:

1 Restricted communications-electronics (C-E) related equipment.

2 Restricted to C-E related equipment (15Kw or less).

3 Clearly ascertained to meet MARS requirements (for example, MARS officers' files, technical library shelves, and so forth).

Table 8-2. Instructions for completing DA Form 5530-R.

ITEM NO	TITLE	INSTRUCTIONS
1	Member's Name	Self-explanatory.
2	Call Sign	Affiliate member call sign.
3	Street	Self-explanatory.
4	City	Self-explanatory.
5	State	Self-explanatory.
6	Zip Code	Self-explanatory.
7	Phone Work or home	phone number including area code.
8	Nomenclature	A detailed description, including model of item being requested.
9	National Stock Number	Enter stock number if available or Federal Supply Class.
10	Number required	Quantity being requested.
11	Description and justification	This block may be used to further describe the item being requested. Justification must be complete and in detail as to why the item is required and how the member will use the requested equipment.
12	State Director	a. Must approve or disapprove (use remarks block below to explain why request was disapproved or approved, if applicable). b. Signature required. c. Date. Self-explanatory. d. Category. See paragraph 8-4a(1)(a) for alpha ratings.
13	Area Director	a. Must approve or disapprove (use remarks block below to explain why request was disapproved or approved, if applicable). b. Signature required. c. Date. Self-explanatory. d. Category. See paragraph 8-4a(1)(a) for alpha ratings.
14	Command MARS Director	a. Must approve or disapprove (use remarks block below to explain why request was disapproved or approved, if applicable). b. Signature required. c. Date. Self-explanatory. d. Category. See paragraph 8-4a(1)(a) for alpha ratings.
15	Chief, Army MARS	a. Must approve or disapprove (use remarks block below to explain why request was disapproved or approved, if applicable). b. Signature required. c. Date. Self-explanatory. d. Category. See paragraph 8-4a(1)(a) for alpha ratings.
16	Remarks	
17	MARS Property Manager	Name of MARS Property Manager.

Table 8-2. Instructions for completing DA Form 5530-R—continued.

ITEM NO	TITLE	INSTRUCTIONS
18	Date received	Date approved request was received by the MPM.
19	Date added to requirements list	The MPM will enter date that the requested item was added to the requirements list.
20	Disposition	The MPM will annotate what action took place on the approved requested item.

- (a) Over 75 hours on-the-air net time per quarter.
- (b) From 25 to 75 hours on-the-air net time per quarter.
- (c) Minimum of 12 to 25 hours' participation per quarter.

(2) The State and area MARS directors will compare the member's request with the list of equipment previously issued to that member. In order to maintain equitable distribution of property among the entire membership, requests for issue of duplicate items are not normally approved. If a member requests an item similar to one previously issued, special justification for one of the following situations must be provided:

- (a) Equipment will replace an unserviceable like item. The request for issue will be accompanied by a request for disposition of the unserviceable item.
- (b) Equipment will replace a like item because of loss or theft. The certification required by paragraph 8-1j will accompany a request of this type.
- (c) Equipment will significantly upgrade and improve the member's communications capability.

b. Requests validated by the area and the command MARS director will be forwarded to the MPM.

c. Requests will be retained by the MPM until review and approval by the Chief, Army MARS.

d. HQ, USAISC will designate an Equipment Review Board consisting of Chief, Army MARS as the chairman, a representative from the Deputy Chief of Staff for Logistics (DCSLOG), and the MPM as the recorder. This board will review all equipment requests and recommend final approval or disapproval. Those items approved will be placed on the equipment requirements list for subsequent screening at DRMO.

8-5. Screening and freezing.

a. The MPM will review DRMS, excess personal property lists (EPPLs), and the GSA regional catalog. Requests for property on these lists will be—

- (1) Based on requests from the members which have

been approved by the Chief, Army MARS. .

- (2) Limited to equipment in the FSCs listed in table 8-1.

b. MARS is authorized to screen and freeze DOD excess personal property being processed for reutilization. The FSCs listed in table 8-1 identify the types of surplus property that Army MARS is authorized to screen and freeze.

(1) Surplus property may be screened by volunteer MARS members who have been designated in writing by the MPM.

(2) Freeze authority is delegated by Chief, Army MARS to the MPM and OCONUS command MARS directors. This authority will not be further delegated.

(3) Based on the recommendation of the State MARS director and area or command MARS director, the MPM will appoint volunteer MARS members as screeners and provide them with GSA Form 2946 (Screener's Identification Card) which will be an accountable item. Only two volunteer screeners per DRMO will be nominated by the State or area director. A letter of appointment (fig 8-1) will be addressed and forwarded with DA Form 1687 (Notice of Delegation of Authority- Receipt for Supplies) to the commander or chief of the appropriate disposal facility. A copy of the letter will also be furnished to the individual screener and the State and area MARS directors.

c. Information concerning available nonreportable surplus property may be obtained by the authorized screener through visits to property disposal activities in their respective areas. Requests for this property will be subjected to the same limitations prescribed above for property on Defense Logistics Services Center (DLSC) or GSA lists.

d. Screeners will screen only for equipment included in the equipment requirements list as furnished by the MPM. When a screener locates Items listed on the equipment requirements list, the screener will fill in and attach DA Form 5532 (Screener's Tag) to the selected equipment. This form is used to identify MARS equipment. Instructions for completing DA Form 5532 are in table 8-3. DPDS Form 103 (Screener's Tally



DEPARTMENT OF THE ARMY

UNITED STATES ARMY INFORMATION SYSTEMS COMMAND
FORT HUACHUCA, ARIZONA 85613-5000

(Office Symbol)

SUBJECT; Authorized MARS Screeners Chief,

Reutilization and Marketing Office

1. This letter is to inform your activity that the following individuals are appointed as authorized MARS Screeners for excess/surplus equipment.

REPLY TO ATTENTION OF

(Name}

(SSN)

(Call Sign)

(Name)

(SSN)

(Call Sign)

2. Point of contact is _____

JOHN J. JONES

MARS Property Manager

Figure 8-1. Letter of appointment.

Request to Freeze Excess/Surplus Property) will be completed and may contain any number of items. The proponent for DPDS Form 103 is Defense Logistics Agency (DLA). DPDS Form 103 can be obtained from General Services Administration, 525 Market Street, San Francisco, CA 94104. It will be mailed without delay to the MPM or OCONUS command director who will telephone the disposal facility and freeze items involved. The DPDS Form 103 will be marked "Approved For Freeze," annotated with the name of the DRMO representative and date contacted. It will be signed by the MPM or OCONUS command director and mailed to the disposal facility to confirm the telephonic action. When

justified, the screener may expedite the freeze process by providing the information from DPDS Form 103 to the MPM or the OCONUS director via MARS priority message or telephone. In addition to the information from blocks 1 through 10 of DPDS Form 103, the complete mailing address and telephone number of the disposal facility where the item is located will be furnished, Instructions for completing DPDS Form 103 are in table 8-4, To enable the MPM or the OCONUS director to identify the specific requirement for the item, the screener should also provide the date and number of the item from the equipment requirements list used during screening.

Table 8-3. Instructions for completing DA Form 5532.

ITEM NO	TITLE	INSTRUCTIONS
1)	Date	Self-explanatory.
2)	Phone	Work or home phone number including area code of MARS member who was designated by the MPM.
3)	Name	Screener's name.
4)	Signature	Designated screener's signature.

Table 8-4. Instructions for completing DPDS Form. 103.

ITEM NO	TITLE	INSTRUCTIONS
1	National Stock Number (NSN)	Enter NSN if available or Federal Supply Class.
2	Unit of Issue	
3	Quantity	Self-explanatory.
4	Turn-in document	Obtained from DRMO, DD Form 1348-1, blocks 30 through 43.
5	Storage location	Obtained from DRMO, DD Form 1348-1, block 1.
6	Demi! Code	Obtained from DRMO, DD Form 1348-1, block V.
7	GSA condition code	Obtained from DRMO, DD Form 1348-1, block P.
8	Unit price	Obtained from DRMO, DD Form 1348-1, blocks 74 through 80.
9	Extended price	Obtained from DRMO, DD Form 1348-1, block E.
10	Noun description	Obtained from DRMO, DD Form 1348-1, block X.

8 - 6. Property accountability.

a. The MPM is responsible for maintaining property control records to ensure an audit trail for equipment from the time it is received from DRMO or GSA until turned in to DRMO or until property control has been dropped as authorized by Chief, Army MARS, in accordance with DOD Directive 4160.21-M, chapters XIV and XVI. Issue, turn-in, transfer, or disposal of MARS property will be accomplished only when authorized by a DD Form 1348-1 (DOD Single Line Item Release/Receipt Document) issued by the MARS MPM. These documents will include a voucher number consisting of the DOD activity address code (DOI)AAC) of the MPM, the Julian date, and a four-digit serial number, the last three digits of which will be constructed sequentially starting with 001 each Julian date. The first digit may be used to identify types of transactions, separate blocks of numbers, or for any other application useful to the MPM. The letter "K" plus 3

numbers will be used when the item is found in the EPPL. The letter "L" plus 3 numbers will be used when the item is found through the local screeners.

b. Each of these vouchers will be recorded sequentially on a DA Form 2064 (Document, Register for Supply Actions), The DA Form 2064 will identify the type of transaction (issue, turn- in, transfer, donation, disposal, and so forth), and the call sign(s) of the affected member(s). A copy of each completed property voucher will be retained permanently in the individual files of the member(s) gaining or losing the property. A copy of the initial, incomplete voucher will be retained in the losing member's file until the transaction has been completed, all signatures have been obtained, and the original copy of the completed document has been returned to the MPM. Maintenance of a sequential document file is not mandatory. Document registers and members' files will be retained permanently, providing an audit trail to documents. All supply transactions will be entered on DA Form 2064.

c. Maintenance of the MARS property book will be accomplished as follows:

(1) All nonexpendable Government property used in the MARS program, and not authorized by TOE or TDA, will be accounted for in an authorized MARS property book maintained by the MPM. The MPM will be appointed, in writing, as the accountable officer for the MARS property book. The appointing document will become a permanent part of the property book records.

(2) Property book records, as a minimum, will consist of the following:

(a) DA Form 3328 (Property Record), maintained in accordance with paragraph 8-6e,

(b) A separate property record for each individual affiliate member containing, as a minimum, a list of the member's privately owned equipment used in the MARS program. Additionally, correspondence and documentation pertaining to Government property requested or received by the member will be maintained in these files (para 8-7h).

(c) A document register maintained in accordance with paragraph 8-6b and DA Form 2064. Transactions for nonexpendable and expendable items will be listed separately, using the first digit of the document serial number to establish separate blocks of numbers. Documents for expendable items will be filed in the same fashion as documents for nonexpendable material, but will not be posted to property book pages.

(d) A due-in suspense file of copies of DD Forms 1348-1 submitted to DRMO or GSA for items frozen for MARS use for which issue transactions have not been completed.

(e) Second destination transportation funds records.

(f) General correspondence files essential to the functions of the MPM, maintained in accordance with AR 340-18,

(3) Property book pages will be maintained as follows:

(a) Nomenclature, national stock number (NSN), or manufacturer's part number, initial acquisition cost, date, document numbers, total balance onhand, unit of issue, recoverability, and security codes will be recorded in ink on the front of the page.

(b) Recipients of the item will be recorded in pencil on the reverse of the page, showing call sign and quantity issued. Serial numbers, if applicable, will be recorded in ink. Gains, losses, and transfers between members will be recorded as they occur.

(c) Each affiliate member will conduct an annual inventory of all MARS property for which the affiliate member is responsible. The MPM will prepare and forward the inventory listing in quadruplicate. Upon

receipt by the member, the listing will be compared with property on hand. Any changes, additions, or deletions will be noted on the listing and explained. All copies will be signed and copy 1 will be returned to the MPM; copy 2 sent to the State MARS director; copy 3 to the area MARS director; and copy 4 retained by the member. Discrepancies will be reconciled by the MPM in accordance with paragraph 8-lj,

d. The MPM will develop a detailed standing operating procedure (SOP) for preparing, maintaining, and processing all documents and records referenced in this chapter.

8-7. Equipment issues.

a. All property requested from DRMO activities and issued through MARS channels will be documented by use of DD Form 1348-1 issued by the MPM. Only one line item, sufficiently identified by NSN, noun, model number, quantity, and condition, will be listed per DD Form 1348-1.

b. The following notation will be recorded on each DD Form 1348-1 used to issue serviceable equipment to an affiliate member: "NOTICE: The equipment listed on this document is subject to recall by the Government. The equipment may be used in its original configuration or modified to meet MARS communications requirements; however the equipment will not be cannibalized. The recipient agrees, by signature hereto, to keep the equipment in as good condition as received subject to fair wear and tear. The Government will assume no responsibility for maintenance or repair of the equipment." Serviceable equipment is DOD surplus or foreign excess personal property received from the DRMO in supply condition codes A through D and disposal condition codes 1 through 5 (tables 8-5 and 8-6).

c. Equipment issued to an affiliate member for cannibalization will be made using DD Form 1348-1 with the notation: "NOTICE: Signature of addressee on this document indicates receipt of property only. This property may be cannibalized for repair parts. Use of the equipment for other than MARS purposes is prohibited. Residue after cannibalization will be disposed of as directed by Chief, Army MARS." Only unserviceable surplus or foreign excess personal property received from DRMO in supply condition codes E through H, with disposal condition codes of 9, S, or X will be Equipment requisitioned for cannibalization is not issued for cannibalization (tables 8-5 and 8-6). authorized shipment at Government expense.

Table 8-5. Supply condition codes.

Code	Title	Definition
A	Serviceable (Issuable Without Qualification)	New, used, repaired, or reconditioned material which is serviceable and issuable to all customers without limitation or restriction. Includes material with more than 6 months' shelf life remaining.
B	Serviceable (Issuable Without Qualification)	New, used, repaired, or reconditioned material which is serviceable and issuable for its intended purpose but which is restricted from issue to specific units, activities, or geographical areas by reason of its limited usefulness or short service life expectancy. Includes material with 3 through 6 months' shelf life remaining.
C	Serviceable (Priority Issue)	Items which are serviceable and issuable to selected customers, but which must be issued before conditions A and B material to avoid loss as a usable asset. Includes material less than 3 months' shelf life remaining.
D	Serviceable (Test/Modification)	Serviceable material which requires test, alteration, modification, conversion, or disassembly. (This does not include items which must be inspected or tested immediately prior to issue.)
E	Unserviceable (Limited Restoration)	Material which involves only limited expense or effort to restore to serviceable condition and which is accomplished in the storage activity where the stock is located.
F	Unserviceable (Reparable)	Economically reparable material which requires repair, overhaul, or reconditioning (includes reparable items which are radioactively contaminated).
G	Unserviceable (Incomplete)	Material requiring parts or components to complete the end item prior to issue.
H	Unserviceable (Condemned)	Material which has been determined to be unserviceable and does not meet repair criteria (includes condemned items which are radioactively contaminated).
S	Unserviceable (Scrap)	Material that has no value except for its basic material content. No stock will be recorded as onhand in condition code S. This code is used only on transactions involving shipments to DPDOs. Material will not be transferred to condition code S prior to turn-in to property disposal officers (PDOs) if material is recorded in condition codes A through H at the time material is determined excess. Material identified by NSN will not be identified by this condition code.

Table 8-6. Disposition condition codes.

Code	Title	Definition
1	Unused-good	Unused property that is usable without repairs and identical or interchangeable with new items from normal supply source.
2	Unused-fair	Unused property that is usable without repairs, but is deteriorated or damaged to the extent that utility is somewhat impaired.
3	Unused-poor	Unused property that is usable without repairs, but is considerably deteriorated or damaged. Enough utility remains to classify the property better than salvage.
4	Unused-good	Used property that is usable without repairs and most of its useful life remains.
5	Used-fair	Used property that is usable without repairs, but is somewhat worn or deteriorated and may soon require repairs
6	Used-poor	Used property that may be used without repairs, but is considerably worn or deteriorated to the degree that remaining utility is limited or major repairs will soon be required.
7	Repairs required-good	Required repairs are minor and should not exceed 15 percent of original acquisition cost.
8	Repairs required-fair	Required repairs are considerable and are estimated to range from 16 percent to 40 percent of original acquisition cost.
9	Repairs required-poor	Required repairs are major because the property is badly damaged, worn, or deteriorated, and are estimated to range from 41 percent to 65 percent of original acquisition cost.
X	Salvage	Property has some value in excess of its basic material content, but repair or rehabilitation to use for the originally intended purpose is clearly impractical. Repair for any use would exceed 65 percent of the original acquisition cost.
S	Scrap	Material that has no value except for its basic material

8-9. Maintenance of property records and files.

Records and files will be maintained as prescribed by AR 340-18 with the following exceptions:

a. The MPM is responsible for maintaining the permanent property record on each affiliate member. This record will not be destroyed.

b. State, area, or command directors will also maintain copies of all equipment issue and turn-in documents for members in their jurisdiction. These documents are furnished by the MPM. A separate file will be maintained for each member. The files will be retained at State, area, or command level until transfer or termination of affiliate member. This will follow the member upon transfer to another state or MARS area,

c. When membership terminates and the member's equipment has been redistributed, recalled, or dropped from control, equipment files maintained by the MPM will be permanently retained. Member files maintained by State, area, or directors will be destroyed 1 year after termination by burning or shredding.

8-10. Member transfers.

When a MARS member transfers from one area to another, the member will notify the losing State MARS director of the move. The losing State MARS director will review the member's property file to determine if there is any MARS property that requires recovery or redistribution prior to the member's departure. Issued equipment will not be moved from one MARS area to another without written approval of the State and command or area MARS directions. A copy of the approval letter will be provided to the MPM by the losing area director. The losing State and command or area MARS directors will transfer the individual's property file to the gaining State and command or area MARS directors. If approval is not granted, the MPM will process the transfer or turn-in documents in accordance with paragraph 8-8.

8-11. Gifts and donations.

a. In deciding to receive a gift, the recipient of an offer must consider the propriety of the gift as well as the relationship between the intrinsic value of the gift to the Army MARS Program and the cost of acceptance and maintenance. Gifts will be categorized as conditional or unconditional.

b. Conditional gifts require the approval of the Secretary of the Army. Normally gifts with a value of more than \$1,000 will be treated as conditional gifts. Conditional gifts will not be accepted by any MARS member for the Army MARS Program.

c. Normally a gift with a value of \$1,000 or less and

whose acceptance and maintenance entail negligible cost will be treated as an unconditional gift and processed as follows:

(1) When consistent with the intent of the donor, unconditional gifts of tangible personal property made directly to the Army MARS Program will be treated as gifts to the Army MARS Equipment Program and will become U.S. Government property. Property acquired under the provision of this paragraph will be accounted for as prescribed in this chapter.

(2) No arrangements will be made to grant special privileges or concessions to a donor, DA personnel and affiliate MARS members will not solicit gifts from potential donors. They may, however, in response to an appropriate inquiry, inform potential donors of needs of the service.

d. An affiliate MARS member or MARS official receiving the offer of an unconditional gift will advise the MPM, through channels, of the nomenclature, serial number, estimated value, and recommended disposition. The MPM will prepare a DD Form 1348-1 to the donor recognizing receipt of the gift and a DD Form 1348-1 authorizing issue.

e. Gifts donated for use as door prizes at meetings and conventions are considered conditional gifts and cannot be accepted by the Army MARS Equipment Program. This type gift is considered a gift between the donor and the recipient and will not be entered into nor accounted for by the Army MARS Equipment Program.

8-12. Labeling MARS equipment.

All MARS equipment belonging to the U.S. Government and being used by MARS affiliate members will be labeled by using a stick-on MARS Property Tag. These tags will be provided by Chief, Army MARS. Labeling U.S. Government MARS property will facilitate the identification of items during annual inventory and enable family or friends to distinguish U.S. Government MARS equipment from personally owned equipment.

8-13. Reports.

a. Area directors (CONUS) and command directors (OCONUS) will submit a monthly report to the MPM reflecting the following:

- (1) Number of members at beginning of month.
- (2) New members by name, call sign, and mailing address.
- (3) Members terminated by name and call sign.
- (4) Number of members at end of month,

b. Reports will be submitted as of the last day of each month and forwarded to the MPM to arrive no later than the 15th of each month.

CHAPTER 9

OPERATIONS

9-1. General.

a. The ability to maintain efficient communications under all conditions depends on adherence to standardized operating procedures. It is only through a thorough working knowledge of these procedures, followed by constant training and application of the principles involved, that proficiency can be attained. MARS procedures are developed from military ACP for standardization of operation.

b. The instructions contained herein are sufficiently comprehensive to govern MARS operations during day-to-day and/or emergency conditions. A degree of flexibility is provided and, at the same time, basic communications principles have been preserved. Instances may arise in which a specific operating requirement is not completely covered by these instructions. Under these circumstances, common sense and initiative must prevail.

c. The instructions in this chapter are devoted to those basic procedures which apply equally to all means of communications covered in succeeding chapters.

d. Reliability, security, and speed are the three fundamentals of military communications. Reliability is always paramount. It must never be diminished nor sacrificed to meet the conflicting demands of security, speed, or convenience. The relationship between reliability and speed is by nature elastic and must be balanced according to specific existing conditions.

e. Appendix H provides the emission designations and frequency tolerances applicable to MARS.

9-2. Circuit discipline.

a. Adherence to prescribed procedures is mandatory.

b. All transmissions will be concise without sacrificing clarity.

c. The following practices are forbidden.

(1) Violation of radio silence.

(2) Transmitting in a directed net without permission.

(3) Exceeding the specific power ratings placed on individual MARS frequencies.

(4) Use of other than authorized prosigns, prowords, or operating signals.

(5) Use of profane, indecent, or obscene language.

(6) Activation of a MARS station in an area of military operations without the military commander's authorization.

(7) Transmission of third-party traffic from an area of military operations without the appropriate military or host nation approval.

d. The following practices are to be avoided:

(1) Use of excessive transmitting power when lower power will suffice.

(2) Excessive time consumed in on-the-air tuning, changing frequency, or adjusting equipment.

(3) Transmitting at speeds beyond the capability of receiving operators.

(4) Testing or tuning on the air on other than assigned frequencies and net.

9-3. Definition or types of radio nets.

a. A radio net is an organization of radio stations capable of direct communications on a common frequency.

b. A directed net is one in which it is necessary to obtain permission from the NCS before transmitting to other stations in the net. MARS nets will be directed unless otherwise indicated by the NCS. Only those stations which are assigned may enter a directed net. Stations not assigned to a specific directed net must call the NCS for permission to operate in that net. A directed net may be operated in a formal or informal format.

(1) Formal operations will be conducted the same as a directed net. The NCS may declare a net informal after all traffic has been handled. Any new stations entering the net or any net station listing traffic will automatically convert net back to formal operations.

(2) Informal net time is that period of net operation authorized by the NCS during which stations may transmit informal comments. Operators may be addressed personally during informal periods only. Informal net operations will be directed.

c. A free net is a net in which any station may communicate with any other station in the same net without first obtaining permission from the NCS. The NCS may declare a free net only after all traffic has been cleared and training has been conducted. Free net operations are not directed and do not relieve the NCS of the authority and responsibility for circuit discipline.

d. The command or area MARS director may approve the establishment of the following common types of MARS nets;

(1) Administrative net. A net primarily established for administrative purposes linking any level of authority with subordinates and such other stations as may be specifically designated.

(2) Technical net. A net primarily established for the exchange of technical information between members and stations experimenting with advanced technology in the communications field.

(3) Command net. A net approved and established by the command or area MARS director for control, management, and coordination of the Army MARS Program.

(4) Radio wire integration (RWI) net. A net established to combine radio and wire communications. (Phone patch nets are RWI nets.)

(5) Traffic net. A net primarily established to handle record message traffic.

(6) Training net. A net primarily established to promote technical and procedural training in matters pertaining to MARS communications. Training nets will be directed.

(7) Emergency or disaster net. A net established to provide emergency communications support of military and civil relief operations in natural or manmade disasters. During emergencies or disasters, real or simulated, all operations by MARS member stations will be on MARS frequencies and the net directed by an NCS.

9-4. Operational control.

a. NCS. A station in a radio net designated to control traffic and responsible for circuit discipline within a given net.

b. Responsibilities of NCSs:

(1) Open and close net at exact scheduled time.

(2) Verify the identify of reporting stations through authentication as deemed necessary.

(3) Limit transmissions to essential length.

(4) Monitor for operation within authorized frequency tolerance and for spurious and substandard emission.

(5) Expedite the clearing of all traffic in the net.

(6) Prescribe speed of transmissions,

(7) Resolve disputes incident to traffic handling and net procedure.

(8) Monitor net operations and take appropriate action to correct procedural discrepancies.

(9) Conduct training discussions in MARS procedures.

(10) Maintain a net operating log on DA Form 5527-K (Net Control Station Report (RCS: ISC-66) and submit such net reports as required by State MARS director or higher authority, DA Form 5527 R will be locally reproduced on 8-1/2 by 11-inch paper and is located at the back of this manual. Instructions for completing DA Form 5527-R are in table 9-1 and on reverse side of form.

(11) Initiate and terminate radio silence as required.

c. Authority of NCS. The authority of the NCS extends only to the operation of the net and does not allow jurisdiction over local administration of individual stations within the net. However, within its scope of authority, decisions of the NCS are final.

d. Responsibility and authority of alternate net control station (ANCS). To ensure continuity of control, an ANCS will be appointed. The ANCS will aid the NCS in the operation of the net, reporting any errors or omissions in the callup or traffic reports, acting as a relay station, noting inaccurate net procedures, and aiding the NCS in procedural training discussions. In the absence of both the NCS and ANCS, any station in the net may act as interim NCS. That station will normally be the first station present in the pre-established net station reporting order. If the NCS or ANCS is not heard by 1 minute past the opening net time, the first station in the reporting order or the first station to call will assume control of the net until relieved by the regular assigned NCS or ANCS.

Table 9-1. Instructions for completing DA Form 5527-R.

ITEM	INSTRUCTIONS
NET DESIGNATOR	List the designator assigned to the net.
NET CALL	List the net call assigned to the net,
NCS	List call sign of net control station,
ANCS	List call sign of the alternate net control station.
TIME	List the time the net starts and stops. Use Zulu time.
DAY	List the day of the week the net is called.
FREQUENCY & EMISSION	List the frequency and emission of net operations.
TRNG OFF	List the call sign of the training officer.
MONTH	Indicate the month of the year.

Table 9-1. Instructions for completing DA Form 5527-R—continued.

ITEM	INSTRUCTIONS
STATION	List the call signs of the member of entering the net.
DATE	Indicate various dates stations participated in net.
TOTAL	Indicate total hours of participation for the month.
REMARKS, TRAFFIC, ETC.	List any special notes or remarks.
COMMENTS	List any comments or other information that may be useful.
SIGNATURE	Net control station will sign here.
DATE	Enter the submission date here.

9-5. Radio station WAR.

a. WAR is the army command radio station for the Army MARS worldwide.

b. WAR functions as NCS in support of DA operations, field exercises, command post exercises, disaster operations (real or simulated) when Army MARS has been activated in support of such operational missions.

9-6. MARS frequencies.

a. MARS radio frequency assignments are made by DA for CONUS use and by the theater commander in oversea areas. Since many MARS frequencies are near the frequencies of other services, caution must be used by all MARS stations to ensure compliance with time and frequency assignments, authorized emissions and input power limitations, and to maintain accuracy of frequency within equipment design limits and prescribed tolerances.

b. All MARS members and custodians will limit the operation of their station on MARS frequencies to the following:

(1) Authorized and organized MARS operations on a specific frequency at a specific time.

(2) When an emergency situation exists, real or simulated, such as floods, severe storms, or any situation that may endanger human life and property.

c. Chief, Army MARS requires that each MARS member or station will be assigned to operation on a minimum of two different frequencies (one must be HF) consistent with capabilities of the station concerned. MARS directors will plan their nets accordingly. MARS operations and planning are considerably enhanced by stations capable of operating on numerous MARS frequencies.

d. The use of MARS frequencies in areas outside the jurisdiction of the U.S. Government is governed by the regulations of the host government concerned. Members traveling in these areas are responsible for gaining operating clearance for MARS use, from the command

or areas MARS director.

e. MARS members will only use call signs assigned by command or area MARS directors on MARS frequencies. During emergencies, other agency assigned calls may be authorized. Call signs will normally be sent in their entirety, especially during initial calling and answering.

9-7. MARS net call signs.

a. Net call signs will be assigned by the command or area and State MARS director at the time an NCS is appointed. **Net** calls may be based on the call sign of the NCS,

(1) To form a net call, a slant (/) bar and a single letter will be added to suffix of the call sign of the NCS. Letters will begin with "A" and followed by succeeding single letters of the alphabet to denote additional nets under the control of the same NCS.

(2) The suffix "E" will be reserved to denote the net call of an emergency net.

b. Net designators assigned to nets by the command or area MARS director provide a more permanent administrative identification of a net than the net call. (Net designators should not be confused with net calls, even though they may be identical.) Such designators when assigned are used for permanent administrative identification of the net according to geographic coverage or mission and are not to be substituted for or used in addition to the net call in on-the-air operations.

9-8. Operating signals, prosigns, and prowords.

a. Operating signals consisting of the Q and Z signals, with amplifying information as required, will be used in accordance with instructions contained in appendix B. Q and Z signals should not be used in radiotelephone procedures; instead, the operating signals will be conveyed by concise phrases. When it is necessary to relay operating signals over voice circuits, they will be transmitted by their phonetic equivalent,

b. Prosigns are one or more letters, characters, or combinations thereof used to facilitate communications by conveying, in a condensed form,

certain frequently used orders, instructions, requests, and information relating to communications,

c. Prowords are pronounceable words or phrases which have been assigned meanings for the purpose of expediting message handling on nets where radiotelephone procedure is used. In no case will a proword or a combination of prowords be substituted for the textual component of a message.

d. Appendix C gives a list of prosigns and prowords which are authorized for use on military circuits. An overscore (a line over two or more letters) indicates that the letters overscored are to be transmitted as a single character; Uiat is, without a pause between letters (such as BT).

9-9 Phonetic alphabet.

a. When necessary to identify any letter of the alphabet, the standard phonetic alphabet shown in table 9- 2 will be used. Difficult words or groups may be spelled using the phonetic alphabet preceded by the proword "I SPELL." If the operator can pronounce the word to be spelled, he or she will do so before and after the spelling to identify the word. Where a text is composed of pronounceable words, they will be spoken as such.

Examples:

"CATENARY - I SPELL, CHARLIE ALFA TANGO ECHO NOVEMBER ALPHA ROMEO YANKEE - CATENARY"

"RENDEZVOUS POINT IS UNIMAK - I SPELL, UNIFORM NOVEMBER INDIA MIKE ALFA KILO"

(The encrypted group LUXOW will be pronounced individually as letters, and counted as one group.)

LUXOW - "LIMA UNIFORM XRAY OSCAR WHISKEY"

(1) Encrypted or enciphered groups will always be transmitted by the phonetic equivalents of the individual letters without using the proword "I SPELL."

(2) A single letter will be identified phonetically preceded by the proword "INITIAL."

(3) The phonetic alphabet in table 9-2 is the only alphabet authorized for use in MARS operations.

9-10. Pronunciation of numerals.

a. To distinguish between numerals and words similarly pronounced, the proword "FIGURE(S)" must precede such numbers. Numbers will be transmitted digit by digit except that exact multiples of hundreds and thousands are spoken as such. The proword "FIGURE(S)" will be used whenever numerals are to be transmitted except in conjunction with the prowords "NUMBER," "TIME," "GROUPS," and in the phonetic pronunciation of a call sign. The numerals pronunciation chart is also contained in table 9-2. The decimal point is to be spoken as "DAY-SEE-MAL." Dates will be spoken digit by digit, with the months in full. Roman numerals will be transmitted as the corresponding Arabic numeral preceded by the proword "ROMAN."

Examples:

123.4 is to be spoken as "WUN TOO THUH-REE DAY-SEE-MALL FO-WER"

"20 AUG" is spoken as "TOO ZE-RO AUGUST"

"VII" is spoken as "ROMAN SEVEN"

b. Abbreviations in the text are transmitted as follows:

(1) Initials used alone or in conjunction with short titles will be spoken phonetically.

(2) Personal initials will be spoken phonetically preceded by the proword "INITIALS."

(3) Abbreviations frequently used in normal speech may be used in the same manner when transmitted by voice.

Examples:

"PARAGRAPH A" will be spoken as "PARAGRAPH INITIAL ALFA"

"ACP" will be spoken as I spell "ALPHA CHARLIE PA PA"

"G. M. SMITH" will be spoken as "INITIAL GOLF INITIAL MIKE SMITH"

"NATO" may be spoken as "NATO" "CONUS" may be spoken as "CONUS" "U. S. ARMY" may be spoken as "U.S. ARMY"

c. Miscellaneous abbreviations are in appendix D. International prefixes are in appendix E.

Table 9-2. Phonetic equivalents.

LETTER	WORD	*PRONUNCIATION OF LETTERS
A	ALFA	ALFAH
B	BRAVO	BRAH VOH
C	CHARLIE	CHAR LEE
D	DELTA :	DELL TAH
E	ECHO	ECK OH
F	FOXTROT	FOKSTROT
G	GOLF	GOLF

Table 9-2. Phonetic equivalents—continued

LETTER	WORD	PRONUNCIATION OF LETTERS
H	HOTEL	HOH TELL
I	INDIA	<u>IN</u> DEE AH r
J	JULIETT	<u>JEU</u> LEE ETT
K	KILO	<u>KEY</u> LOH
L	LIMA	<u>LEE</u> MAH
M	MIKE	MIKE
N	NOVEMBER	NO <u>VEM</u> BER
O	OSCAR	<u>OSS</u> CAH
P	PAPA	PAH <u>PAH</u>
Q	QUEBEC	KEY <u>BECK</u>
R	ROMEO	<u>ROW</u> ME OH
S	SIERRA	SEE <u>AIR</u> RAH
T	TANGO	<u>TANG</u> GO
U	UNIFORM	<u>YOU</u> NEE FORM
V	VICTOR	<u>VIK</u> TAH
W	WHISKEY	<u>WISS</u> KEY
X	X-RAY	<u>ECKS</u> RAY
Y	YANKEE	<u>YANG</u> KEY
Z	ZULU	<u>ZOO</u> LOO
NUMBER		PRONUNCIATION OF NUMERALS
0		<u>ZEE</u> ROW
1		WUN
2		TOO
3		THUH <u>REE</u>
4		<u>FO</u> WER
5		FI YIV
6		SIX
7		<u>SE</u> VEN
8		ATE
9		<u>NIN</u> ER

"The underlined portion denotes the accented syllable(s).

9-11. Official time.

a. The 24-hour clock systemj will be used in expressing time within MARS. Time is expressed by means of a six-figure group of numerals (for date-time groups), the first two numerals expressing the day of the month and the next four the clock time, followed by the time zone suffix "Z" to denote Greenwich mean time (G.m.t.) or universal coordinated time (u.c.t.). Standard time zones and time conversion charts are at appendix F. In stating time for checks or schedules, the two-digit date will not be used. When a message is referred to by its time of origin, the method of expressing such time will be exactly as in the original message.

b. MARS stations will be responsible for keeping correct time. Synchronizing procedures will be as follows:

(1) Radiotelegraph procedures: Request: INT ZUA meaning "Request timing signal now (or at . . .)" Reply: ZUA 1630Z meaning "Timing signal will be transmitted at 1630Z." The numerals indicating time will be followed by a 5-second dash which terminates exactly at the time indicated.

(2) Radiotelephone procedures: Request:

"Request time check at. . . ." Reply: "Time Check at 1630Z 15 seconds; 10 seconds, 5, 4, 3, 2, 1 time 1630Z"

(3) Radio Teletype procedure same as Radiotelephone.

9-12 Interservice operations

a. The MARS service chiefs have authorized interservice operations for the transfer of narrative traffic between MARS services.

b. Interservice operation is intended to augment each service's capability and open other outlets for record traffic delivery. To accomplish this for Army MARS, the State director will—

(1) Coordinate with other service MARS representatives to establish schedules and procedures.

(2) Determine from present net schedules which frequencies will work best with the other service for split frequency operations,

(3) Appoint specific stations to be part of the interservice operations

(4) Obtain participation and traffic information from appointed participants to include hours and message count for interservice operation.

(5) Consolidate member reports and submit totals in the narrative statement of their monthly- report.

(6) Ensure that split-frequency operation is used. Army MARS will transmit only on Army authorized frequencies and receive on the other service frequency.

9-13. Radio station log

a. DA Form 4341 (Radio Station Log) (fig 9-2) will be maintained by all stations to record MARS operational information and to provide data for the monthly MARS Traffic and Participation Report (DA Form 5522). Instructions for completing DA Form 4341 are in table 9-3.

b. Radio station logs will be retained for a minimum of 1 year.

Table 9-3, Instructions for completing DA Form 4341.

ITEM	INSTRUCTIONS
DATE TIME (2)	Time of opening the net or station.
STATION CALLED	Call sign of the station calling.
CALLED BY	Call sign of the station called
FREQUENCY	List the frequency used.
POWER AND EMISSION	List the power and emission used.
MESSAGES SENT	List the number of messages sent.
MESSAGES RECEIVED	List the number of messages received.
TIME END CONTACT	Time of closing the net or station.
REMARKS	List any remarks or notes here
OPR SIGN	Operators should sign their initial here.

9-14. Message files.

a. The maintenance of a complete and accurate message file aids in orderly handling of messages, in answering services which pertain to such messages, and in the compilation of traffic reports.

b. A consecutive numbering system for each calendar month ensures proper Filing and handling of traffic. DA Form 4016 (Telecommunications Center Originating Message Register) (fig 9-2) will be used for this purpose. Instructions for completing DA Form 4016 are in table 9-4.

Table 9-4, Instructions for completing DA Form 4016.

ITEM	INSTRUCTIONS
STATION SERIAL NO	List the station serial number (SSN) of each message beginning with 001 each day and continuing in sequence through the day.
OFFICE SYMBOL	List your MARS call sign.
DATE-TIME GROUP	List the date and time of the message.
PRECEDENCE	List the priority of the message.
CLASSIFICATION	List the classification of the message.
GROUPS OR CARDS	List the group count of the message if the groups have been counted.
TIME OF FILING	List the time the message was filed.
LOG SIGN	The logging operator will sign his or her initials here.
TIME OF XMSN	List the day and time the message is transmitted.
HANDLING TIME	Subtract the file time from the transmit time and list the amount of time here.
FILE SIGN	The transmitting operator initials here.
REMARKS	List the call sign of the station that the message was sent to or pertinent information,
NOTE; Use this form for both origination and receipt of narrative traffic.	

c. Message files will be retained for a minimum of 90 days after the date of the last entry.

9-15. Phone patches.

a. No Army MARS member or station will use military frequencies to originate or terminate phone patches for any exchange of information or conversation

unless authorized by an appropriate command or area MARS director.

b. Phone patch operations or RWI may be used as part of any planned drill or emergency training exercise for training purposes. DA Form 4342 (Radio Wire Integration Log) (fig 9- 3) will be used for this operation. Instructions for completing DA Form 4342 are in table 9-5.

For use of this form, see FM 11-490-7; the proponent agency is USA/SC

[illegible]DA FORM 4016
1 APR 73

REPLACES DA FORM 11-189, 1 OCT 69, WHICH IS OBSOLETE

Figure 9-2. Sample of DA Form 4016.

☆ U. S. GPO: 1976-0-665-697/236

For use of this form, see FM 11-490-7; the proponent agency is USAIS/SC

[illegible]

Table 9-5. Instructions for completing DA Form 4342.

ITEM	INSTRUCTIONS
STATION TO STATION	Identify type of call.
PERSON TO PERSON	Identify type of call.
AREA CODE	List area code of call.
TELEPHONE NUMBER	List telephone number to call.
FROM	Name and rank of calling party.
TO INFORMATION	Name of called party or other information.
FREQUENCY	List the operating frequency.
OVERSEAS STATION	Call sign of the MARS station placing the call.
TIME (GMT)	Time the call starts.
REMARKS	Time the call stops and any other information that could be useful in reference to the telephone call.
later	
DATE	Indicate the date the calls are made.

c. During actual emergencies, disasters, and/or military operations, phone patch (KWI) may be employed by military or civil emergency disaster officials.

d. Phone patch stations that have been appointed will submit DA Form 5525-R to the area MARS director monthly as stated in paragraph 3-6e.

e. Phone patch stations will implement MINIMIZE when applicable,

9-16. MINIMIZE procedures.

a. MINIMIZE applies to MARS, as it does to active Army. The reason for MINIMIZE is to control the volume of traffic entering telecommunications systems during periods of emergencies. Notification of MINIMIZE affecting MARS will be originated by the Chief, Army MARS.

b. Telecommunications traffic is defined as information transmitted, emitted, or received in the form of messages (signs, signals, writing, images, and sounds) of any nature by wire, radio, visual, or other electromagnetic systems. When an actual or simulated emergency arises, or is anticipated, it is necessary to reduce the volume of traffic ordinarily transmitted over military telecommunications networks.

c. During a MINIMIZE condition, traffic of any precedence may be considered essential when it must be transmitted electrically for the command or activity concerned to avoid a serious detrimental impact on mission accomplishment or to ensure safety of life. Originators of traffic must apply the following test during periods of MINIMIZE:

(1) Will immediate mission or operation be adversely affected if proposed traffic is not electrically transmitted?

(2) Is the message or call in question of vital and urgent importance or within a category specifically exempted from MINIMIZE by the imposing authority?

d. Only messages or calls which pass this test are authorized during periods of MINIMIZE. Whenever MINIMIZE is imposed worldwide or for specified areas, MARS messages already in the pipeline of the system will be processed onward to destinations. New MARS messages will not be accepted unless they meet the criteria of c above.

e. MINIMIZE may be imposed by commanders at any level within their areas of responsibility. The authority to impose MINIMIZE upon all users of DOD telecommunications networks simultaneously is limited to the Joint Chiefs of Staff (JCS). When MINIMIZE is imposed, a Chief, Army MARS message will be transmitted to all gateway stations and to Radio Station WAR for relay advising MARS stations of MINIMIZE, indicating the area of restriction and the type of restriction.

f. When MINIMIZE is imposed, MARS stations will not accept any message traffic or telephone patching intended for an area in which MINIMIZE is imposed unless the message or call meets the test above. MARS nets will not close down during MINIMIZE but will be on listening watch to handle essential traffic.

9-17. Handling of distress calls.

a. Any operator who hears a distress call should answer the call promptly and take whatever steps might be necessary to give assistance (see e below).

b. Operators not located close to the sources of immediate relief should continue to monitor.

c. Whether they answer or not, every operator hearing a distress call or monitoring distress traffic should copy everything and retain the copy in his or her log, including calls of stations answering the call and handling the traffic.

d. Generally, once a distress call has been answered by a station which is in a position to send aid, other stations should confine themselves to monitoring.

RADIO WIRE INTERGRATION LOG

OPERATOR
SGT DOE

ALL CALLS
COLLECT

DATE 19 Nov 85

PAGE 1 OF 1 PAGES

[illegible]

SAMPLE

Figure 9-3. Sample DA Form 4342.

e. A station which answers a distress call and handles the distress message should try to maintain radio contact with the station in trouble as long as possible. He or she should try to keep the distressed station advised, if possible, as to the progress of relief measures and should be ready to handle any further traffic which might originate with the distressed station. At this stage of the situation, traffic originated by the station in distress has absolute precedence over all other communications. Messages to persons involved in the disaster take precedence over all other traffic except traffic coming out of the distress zone. While rescue or relief is actually in progress, only communications of immediate precedence should be handled, whether or not a directed net has been established. Priority traffic can wait until the immediate danger to human life or safety has cleared. Local emergency SOPs will furnish agency's contact numbers for assistance,

a. If an emergency net has been set up to handle the distress communications, the station in distress should identify some station he or she can hear well to assume direction of the circuit so that a net control with a good signal can dominate the frequency and relay, as necessary,

b. If the situation is likely to generate much traffic, or if it is likely to continue for some time, regular emergency operations should be implemented,

c. If a distress call is determined to be false, stations responding to the distress call will report all details such as date, time, frequency, call sign, and other available factors to the area MARS director immediately. Area MARS directors will initiate corrective action and report such infractions through command channels to Chief, Army MARS, HQ, USAISC. ATTN: AS-OPS -OA, Fort Huachuca, AZ 85613-5000.

9-18. Break-In procedures.

a. Break-in is the method whereby a receiving station may interrupt a transmission to request the transmitting station to wait, shift frequency, repeat, and so forth. This procedure will not be used to obtain repetitions when more than one station is receiving a message.

b. Break-in will not be used by a station not directly concerned with the transmission of a message unless that station has a message or information of an immediate or urgent nature for the NCS or other net station. When break-in is used by such a station, a full or abbreviated call will

follow the prosign "BT" proword "BREAK,"

9-19. WAR broadcasts.

a. WAR broadcasts originate and are sent by the Chief, Army MARS. They contain general interest information, announcements of special events, and instructions in MARS operating procedures. Each MARS member should copy these broadcasts or obtain them through local rebroadcast or other sources. Direct copy of these broadcasts also provide the MARS member a chance to improve their CW copying speed and test radioteletype (RATT) reception capability.

b. WAR broadcasts are transmitted at 25, 20, and 15 words per minute (WPM) (CW), at 0130Z and 0430Z Tuesday (Monday eastern standard time (e.s.t.)) on 3347, 5217, 6997.5, and 14405 Khz. The text is repeated by RATT 60 WPM (170 cycle and 850 cycle shift) at 0130Z and 0430Z Wednesday (Tuesday e.s.t.) on 5217, 6997.5, and 14405 Khz. No MARS station will transmit on the above frequencies during the scheduled WAR broadcast.

9-20. Jamming.

a. Where deliberate jamming of a frequency is suspected, ignore the jammer and make no reference on frequency to the jamming. Continue working as best possible, with repeats or fills as needed. To recognize the jamming gives the jammer the recognition he or she seeks. Note any peculiarity and strength of the jammer.

b. On VHF nets, especially when a repeater is used, attempt to identify the jammer by listening on the repeater input frequency while attempting to determine direction of interfering signal. Do not discuss direction finding (D/F) activities regarding the jammer on the air on any frequency,

c. Submit information derived to the NCS for compilation into an interference report.

9-21. Interference report.

An interference report should be reported on DA Form 5528-R (Interference Report) (RCS: ISC-67) when adequate data can be ascertained regarding interference. DA Form 5528-R will be locally reproduced on 8-1/2 by 11-inch paper and is located at the back of this manual. Instructions for completing DA Form 5528-R are in table 9-6 and on reverse side of form.

Table 9-6. Instructions for completing DA Form 5528-R.

ITEM	INSTRUCTIONS
1,2,3. and 4	Indicate the time the interference started and stopped.
5	Indicate your operating frequency.
6	List the type of equipment you used to measure the frequency of your signal and interfering signal.
7	Indicate the type of emission of the interfering signal.
8	If possible, identify the source or origin of the interfering signal.
S	List type of information heard that will help identify the interfering signals.
10a	If heard, list the call sign of the interfering station.
b	If possible, identify the geographical location of the interfering signal.
c	List the frequency of the interfering signal.
11	Did you make contact with the interfering station?
12	Was the interfering station in contact with another station? If so, list the other station call sign if heard.
13	Was the interfering station involved with an organized net?
14a	Identify any special characteristics heard on CW signals.
b	Identify any special voice characteristics heard.
15	Indicate the bandwidth of the interfering signal.
16	List (in this block and on additional paper) any information not specifically covered in blocks above. List anything that may help officials to absolve any frequency discrepancies that may exist

NOTE: Each member submitting an interference report will list his or her name, address, call sign, date, and date of interference. Complete each of the following items as accurately as possible—the better the information submitted, the better chance someone will have getting the problems corrected.

9-22. Authentication procedures and system.

a. Authentication systems are used to maintain and protect the reliability of military communications systems by preventing the introduction of fraudulent messages or signals intended to cause erroneous actions, damage, confusion, or the disclosure of important information.

b. In actual practice, authentication systems are appropriately classified and safeguarded in order to prevent disclosure to unauthorized individuals. The contents of the various systems are changed frequently to prolong their usefulness. Security of authentication systems is enhanced by random selection of test elements each time authentication is employed.

c. Instructions concerning the method of selection or application of elements in authentication systems will not be discussed on the air. Such instructions will accompany the authentication publication. On-the-air instructions will include only general information indicated by appropriate operating signals.

(1) Authentication within MARS is employed in two ways. One method is to establish the authenticity of a communications station; the other is to ensure the validity of a message. Authentication in MARS will be used to authenticate stations only. Authentication systems published by MARS are unclassified and are used for training purposes only.

(a) The ability to authenticate rapidly and accurately is of the great importance in military operations. As NCS should ensure that proficiency in authentication procedures is maintained through conduct of training at periodic intervals simulating the situations listed in (2) below.

(b) Under normal training conditions, authentication

of stations is not to become a permanent part of MARS net opening and closing. Training in authentication will normally be reserved until all messages other than drill messages have been cleared.

(2) Station authentication is employed between stations to guarantee their authenticity. Station authentication is used under the following circumstances:

(a) Whenever fraudulent transmission or operation is suspected in a radio net.

(b) When transmitting initial and amplifying reports concerning an emergency .

(c) When opening or closing a radio net.

(d) When instructions are issued to change radio frequency and call signs and when initial contact is made after such change.

(e) When a station reports into a radio net.

(3) Test elements of authentication, in the form of a challenge, are transmitted after the call as follows:

CW: INT ZNB.....K.

VOICE: WHAT IS THE AUTHENTICATION OFOVER.

(a) The reply consists of the authenticator as follows:

CW: ZNB.....K.

VOICE: AUTHENTICATION IS.....OVER.

(b) The reply may include a counter challenge as follows:

CW: ZNB.....INT ZNB.....K.

VOICE: AUTHENTICATION IS.....WHAT IS AUTHENTICATION OF.....OVER.

(c) The reply to a counter challenge would be ZNB.....AR or AUTHENTICATION IS....OUT.

The transmission of AR or OUT does not preclude

further transmission if the authentication is incorrect.

(d) No further transmission is necessary if the authentication is correct. If the authentication is incorrect, the proword "INCORRECT" followed by a new challenge will be transmitted in voice procedure. ZND (with numeral, if applicable) followed by a new challenge will be transmitted in CW.

(e) The phonetic alphabet (table 9-2) will be used when transmitting elements and authenticators by voice.

(2) Suggested methods for station authentication are as follows:

(a) The NCS may challenge, in turn, each station in the reporting order replying to a counter challenge from

only the first or from all stations in the reporting order.

(b) The NCS may challenge the first station in the reporting order who will reply and then challenge the second station in the reporting order followed by succeeding challenges and answers until all stations in the reporting order have replied to the preceding station and challenged the following station.

(c) Stations may authenticate before receiving or transmitting a message.

(3) The operating signals shown in table 9-7 will be used in CW authentication. The literal definitions of these signals will be used in voice authentication. The method of determining an authentication from an authenticator is shown at table 9-8.

Table 9-7. Operating signals.

Operating Signal	Question	Answer, Advice, or Order
ZNB	What is the authentication of.....(1. Message....	Authentication of.....is) 1. Message.....; 2. Last
	2. Last transmission.	transmission).
ZNC		All transmissions will be authenticated...(1. On all
		circuits; 2. On this circuit; 3. On frequency.
ZND		You are using authenticator incorrectly.....(1. Verify
		authenticator system key; (2. Check authentication of your last
		transmission). (6) Instructions for use of authentication table.
		Two test elements will always be used in forming an
		authenticator.

Table 9-8. MARS authentication table.

A	BFIWZ	EHOXC	QYDJU	ASMRK	GLTPVN
B	TYJVA	XNGOS	FLRCI	WBEZM	QUKHDP
C	WKRBI	UPAHN	DZYFQ	ELXCS	JGMVOT
D	RIYTE	ADOZL	GBMXN	VPKFQ	WCSJHU
E	FAMVC	YZTLW	IOBQS	PDENR	XHUGKJ
F	YEMPT	FBRHI	WAUOK	LCXSD	ZJNQGV
G	UZKIB	RLEYG	OCSWD	NXAJT	PVHFQM
H	HFAEZ	QDIYR	KUCMB	XGWLW	NTJSPO
I	PHAGM	IQBXN	VZDSO	CFRKU	YWLTJE
J	DNZHI	XIYAK	LEWGT	OVBPF	CSQMUR
K	MSXQD	FVPTY	HJNLE	KZUGC	RAOIWB
L	PUCOR	VEJFH	XTKGW	QSBDM	LZANIY
M	EVPNC	OAQJT	MWFLR	GUHDI	SBZXYK
N	SCUYF	NGZDM	QIVTE	HKPWJ	LOBRXA
O	JBFSW	CKVEM	OYQZH	TINUP	GDRLAX
P	PODSF	LJCUA	RKNBV	ZMQYG	EXIWITH
Q	GWHXJ	QYDFC	AREUZ	KTOIL	VMPBNS
R	LGBUH	MFCRZ	XJAIY	SNDVO	TPWKEQ
S	QHLAV	PKBES	ZTXCJ	YOIMW	DNFURG
T	CTXOK	SJPNU	GLAHQ	BYVEZ	IRDFMW

Table 9-8. MARS authentication table—continued.

U	ZJUNR	OHMTB	DPGEW	FICQX	AKVYSL
V	XTWJN	VLQSD	UMKPC	RFGOE	HZAYBI
W	IQDMW	UNXRP	ESOVA	JLCTH	BKFZGY
X	VLGAS	KMWBN	XPJYR	HTEIC	ZDOFQU
Y	NOQDG	PWKVF	TRAMJ	YIBZU	CXHELS
Z	KHTLZ	VSWMQ	GPHEU	BYDIN	OFAXJC

(a) Using as an authenticator the two test elements "L" and "G," locate the first test element "L" in the left column (row designators of the authentication table) and proceed across the row designated by the first test element until the second test element G is located. The letter immediately to the right of the second test

element is the authentication. In this case the letter "W" is the authentication.

(b) Table 9-8 is a sample of procedures used for authentication. Instructions for use of table 9-8 will vary with the specific systems issued. Operators must familiarize themselves prior to use.

CHAPTER 10

MESSAGES

10-1. Spurious messages.

No person will knowingly or willfully originate, accept for transmission or transmit, deliver or cause to be delivered, spurious or misleading messages to any persons.

10-2. Originator's authority and responsibility.

a. The originator of a message is the person or command by whose authority a message is sent and who determines what to say and to whom. The originating station is the MARS station that first enters that message into the MARS traffic channels.

b. The originator is responsible for—

(1) Determining whether a message is necessary.

(2) Ensuring proper use of the prescribed message form.

(3) Assigning the proper precedence.

(4) Drafting the message properly; for example, keeping the text brief by using concise words and phrases, authorized abbreviations, American radio relay league (ARRL) numbered radiograms, and the necessary punctuation. Originating station will assist the message originator in message preparation. Appendix G discusses the use of ARRL numbered radiograms.

10-3. Privacy of messages.

a. All messages (classified or unclassified) will be treated with privacy. No message or part thereof will be discussed with any unauthorized person nor will its contents be revealed except by the originator or addressee, or as properly authorized. Even the fact of the receipt or existence of a message will not be divulged except to proper authorities.

b. False or forged messages. No MARS member will knowingly transmit or cause to be transmitted, or deliver or cause to be delivered, a false or forged message.

10-4. Messages authorized for transmission.

The following types of messages are authorized for transmission over MARS networks in the following recommended order.

a. Official military originated messages.

b. Civil emergency traffic.

c. Official traffic such as American Red Cross.

d. Health and welfare traffic to and from emergency areas, chaplain, recreational services, and so forth.

e. Personal messages.

(1) Personal messages, including third party messages or phone patches not of a business nature, are permitted in MARS provided such a message would not normally be sent by available commercial means.

(2) When personal messages are accepted for transmission via MARS circuits, MARS members (stations) will advise the originators of the various aspects of good message writing; for example, conciseness of text, precedence, use of punctuation, abbreviations, and use of ARRL numbered radiograms. Maximum length of a MARS plain text message is usually 30 words. Such messages, when accepted, will be given the same care and consideration as official messages.

f. Administrative traffic pertaining to MARS.

g. Exercise traffic (drill messages). These messages will use the words "MARS Exercise" as the first two and last two words of text.

10-5. Messages not authorized.

a. Messages relating to legislative matters will not be accepted.

b. Messages of a derogatory nature will not be accepted for transmission by MARS.

c. A message containing initial notification of the death of any individual will be accepted for transmission via MARS only if addressed to the Area American Red Cross field director, the chaplain, or commanding officer of the area or unit to which the individual is assigned. Under no circumstances will any MARS stations deliver such a message directly to the individual concerned. A message pertaining to a death other than an initial report may be handled without restriction. In this case, the operator will query the originator whether the message is initial notification or is a followup.

d. Messages clearly of business nature (commercial enterprise).

10 - 6. Types of messages.

Military messages are prepared in single address, multiple address, book, or general message form.

a. Single address. A message which has only one addressee.

b. Multiple address. A message which has two or more addressees and is of such nature that the drafter considers that each addressee must know of all other recipients.

c. Book. A message of such nature that it is not mandatory that each addressee know who the other recipients are.

d. General. A message is designed to meet the

requirement for the dissemination of information or instructions to a wide standard distribution. They are assigned an identifying title and usually a sequential number; for example, ALCONCEN 25/84 (All concerned number 25 calendar year 84).

10-7. Forms of messages.

Messages transmitted over military radio circuits are prepared in plaindress or abbreviated plain- dress form as shown in table 10-1.

Table 10-1. Plaindress or abbreviated plaindress message format.

PARTS	COMPONENTS	ELEMENTS	FORMAT LINE	CONTENTS
			1	Not used.
	Procedure	Called station(s)	2	Station(s) called, prosign "XMT" and exempted call signs.
		Calling station	3	Prosign "DE" and call sign of station calling: serial number. Prosign "NR" followed by a number or a combination of letters and numbers.
		Transmission instructions	4	Prosigns "F," "G," "T": operating signals; call signs, address groups, plain language address designators, address indicating groups (AIGs), routing indicators.
	Preamble	Precedence: DTG; message instructions	5	Precedence prosign, Data and time expressed in digits and Zulu suffix followed by month indicated by first 3 letters, the year indicated by the last 2 digits; and operating signals and Prosign "IX."
	Address	Originator's prosign; originator	6	Prosign "FM," Originators designator (call sign address group, plain language address designator, routing indicator) or call sign, and two-letter State abbreviation or geographical location of originating station when different from originator.
		Action addressee prosign; action addressee(s)	7	Prosign "TO." Action addressees' designators (call signs, address groups, plain language address designators, AIGs, routing indicators).
		Information addressee prosign; information addressee(s)	8	Prosign "INFO." Information addressees' designators (call signs, address groups, plain language address designators, routing indicators).
		Exempted addressee prosign; exempted addressee(s)	9	Prosign "XMT." Exempted addressees' designators (call signs, address groups, plain language address designators).
		Accounting information group count	10	Accounting symbol, Prosign "GR" (number of groups) or prosign "GRNC,"
	SEPARATION		11	Prosign BT.
T E X T	Text	Subject matter	12	UNCLAS, or appropriate classification, prosign "SVC," internal instructions and appropriate textual matter, including signature, if appropriate.
E	SEPARATION		13	Prosign "BT."
N	Procedure	DTG	14	Hours and minutes expressed in digits and Zulu suffix.
D I		Final instructions	15	Prosigns "SB," "B," "C," "IMI," "ZNC," "IX" operating signals, station designators.
N G		Ending sign	16	Prosign "K," "AR."

a. Plaindress. A plaindress message is one in which the originator and the addressee designators are indicated externally of the text.

(1) A plaindress message contains all the components as shown in the basic format and must always include the precedence and date-time group (DTG).

(2) When the call serves as the address, lines 6, 7, 8, and 9 may be omitted.

(3) The precedence and DTG must be included.

b. Abbreviated plaindress. An abbreviated plaindress form message uses the call to represent the addressee and the originator; therefore, the address component will be omitted, and any or all of the following elements: station serial number, precedence, DTG, and group count.

c. Service message. A service message is one between communications personnel pertaining to any phase of traffic handling, communication facilities or circuit conditions. Service messages are indicated by the prosign "SVC" or proword "service." Service messages are prepared and transmitted in plaindress or abbreviated plaindress form. Generally, service messages concern messages originated at, destined to, or refiled by a station and will be assigned a precedence equal to that of the message to which they refer. Message reference in service messages should be sufficiently complete to ensure positive identification. The precedence, DTG, and textual reference may be used as necessary.

d. Abbreviated service message. An abbreviated

service message is one in which the text contains only prosigns, operating signals, identification of messages, parts of messages, and amplifying data as necessary. It may be originated by operators and may contain any of the components shown in the basic message format except that:

(1) All message parts and a majority of the components and elements have a standard arrangement or sequential order of appearance.

(2) The basic message format is the basis for the procedure contained in this manual.

(3) In table 10-1, format lines 2, 3, 4, 14, 15, and 16 identify the procedural portion of the basic message format as designated for radiotelegraph operation. The format lines 5 through 13 are the nonchangeable components of the basic message format. All format lines do not necessarily appear in every message; however, when used, the-; will be in the order shown.

(4) In table 10-2, format lines 2, 3, 4, 14, 15, and 16 identify the procedural portion of the basic message format as designated for radiotelephone operation. Lines 5 through 13 are the nonchangeable elements of the basic format. All format lines do not necessarily appear in every message, however, when used they will be in the order indicated.

(5) In table 10-3, format lines 2 through 5, 15, 16 identify the procedural positions of the basic message format as designated for RATT operation. All format lines do not appear in every message; however, when used they will be in the order indicated.

Table 10-2. Procedural portion for radiotelephone operation.

PARTS	COMPONENTS	ELEMENTS	FORMAT LINE	CONTENTS
H	Procedure	Called station's(a)	1	Not used.
			2	Station(s) called, proword "EXEMPT," exempted calls.
E		Calling station	3	Proword "THIS IS"—station calling.
			4	Prosign "NR" followed by a number or a combination of letters and numbers.
		b.Message follows		Proword "MESSAGE."
		c. Transmission identifications		Proword "NUMBER" and SSN.
A		b.Transmission instructions		Prowords "RELAY TO;" "READ BACK DO NOT ANSWER;" "WORDS TWICE;" operating signals; call signs; plain language designators.
D				
I	Preamble	a. Precedence; DTG; message instructions	5	Precedence designation; proword "TIME;" date and time expressed in digits and Zulu suffix; and operating signals.
	Address	a.Originator's sign; originator	6	Proword "FROM." Originator's address designator.
N		b.Action addressee sign	7	Proword "TO." Action addressee designator.
		c.Information addressee sign; information addressee	8	Proword "INFO." Information addressee designators.
G				

Table 10-2. Procedural portion for radiotelephone operation—continued.

PARTS	COMPONENTS	ELEMENTS	FORMAT LINE	CONTENTS
		d. Exempted addressee sign; exempted address	9	Proword "EXEMPT." Exempted addressee designators.
	Prefix	a. Accounting information, group count	10	Accounting symbol; group count; proword groups or (group no count).
SEPARATION			11	Proword "BREAK."
T E X T	Test	a. Subject matter	12	Clear, unclassified, Proword "SERVICE," and/or internal instructions as appropriate; thoughts or ideas as expressed by the originator.
SEPARATION			13	Proword "BREAK."
E N D I N G	Procedure	a.DTG	14	Proword "TIME." Hours and minutes expressed in in digits and Zulu suffix, when appropriate. Prowords
		b.Final instructions	15	"WAIT," "CORRECTION," "AUTHENTICATE," "MORE TO FOLLOW;" station designators.
		c.Ending sign	16	Prowords "OVER" or "OUT".

Table 10-3. Procedural portion for radioteletype operation.

PARTS	COMPONENTS	FORMAT LINE	ELEMENTS	CONTENTS	EXPLANATION
H E A D I N G	PROCEDURE	1	Handling instructions	Transmission Identification; security warning prosign (when used): pilot—pilot contains repeated precedence prosign**; routing indicators); prosigns, operating signalsand address designations*!*!* as required.	Always contains transmissionidentification (which included the "Start of Message Indicator"when necessary); also contains pi- lot(s) as required to convey specific message handling instructions.
		2	Called stations(s)	Repeated precedence prosign**;routing indicator {s) of station(s)responsible for delivery or refile.	Basic routing line. If message isdual precedence, only the higher precedence is shown in this line.
		3	Calling station and filing time	Prosign "DE": routing indicator of station preparing message for transmission; station serial num- ber; filing time; date separated by - slant from hour and minute expressed in digits followed by zone	Filing time is the date and time the message was filed with the communication center.
		4	Transmission instructions	SUIiiX. Security warning operating signal (when used); prosign "T";other operating signals: special operating group(s) designator(s) routing indicator(s)	Indicates specific transmission responsibility not apparent in other components of the message heading. Not to be used unless necessary. Plain language address designators are not permitted in messages.
H E A D I N G	PREAMBLE	5	Precedence:DTG; message instructions	Precedence prosign(s) DTG and zone suffix (Z indicat (ing G.m.t.); operating signal(s).	In the case of dual precedence, both prosigns are shown separated by a space. Operating signals are used only when required to convey message handling instructions.
		6	Originator	Prosign "FM"; originator's designation	Message originator is indicated plain language routing indicator, address group or call sign.
		7	Action	Prosign "TO"; routing indicators); operating signal(s); address designator(s)	Action address(es) is indicated by plain language, routing indicators) address group(s), or call sign(s). In the case of multiple address messages, when addressees are listed individually each address designation will be on a

Table 10-3. Procedural portion for radioteletype operation—continued.

PARTS	COMPONENTS	FORMAT LINE	ELEMENTS	CONTENTS	EXPLANATION
H E A	ADDRESS	8	Information addressee	Prosign "INFO" routing indicators); operating signal(s); address designator(s).	separate line and may be preceded either by the operating signal "ZEN" (meaning delivered by other means) or by the routine indicator of the station responsible for delivery. Such use is mandatory on all joint and combined addresses. Same as for line 7 except that line 8 pertains to information addressee. A collective address designation or an AIG in format line 7 may include information addresses.
D I N G		9	Exempted	Prosign "XMT;" address designator(s).	Used only when a collective address designation is used inline 7 or 8, or an AIG is used in line 7, and an indication of the addressee exempted from the collective address or AIG is required.
G		10	Accounting group count.	Accounting symbol (when required); group count prosign "GR;" group count.	The group count prosign and group count will be used only when the text consists of countable encrypted groups.
SEPARATION T E X T		11		Prosign "BT."	
		12	Classification; internal instructions; thought or idea expressed by originator (in that order).		See ACP 121-series.
SEPARATION E N D I N G	PROCEDURE	13		Prosign "BT."	
		14	Confirmation.		Not used in tape relay operation.
		15	Correction.	Prosign "C"; other prosigns; operating signals and plain language as required.	
		16	End of message functions.	2CR, 8 LF, 4N's, 12 LTRS,	The 4N's in this sequence are the end of message indicator.

""Included only when required for clarity.

**If message is dual precedence, only the higher precedence is shown in this line.

* "Plain language designators are not permitted in codress messages.

10 - 8. Message diagram—radiotelegraph, radiotelephone, or radioteletype
In table 10-4, it should be noted that every element is

indicated in order of appearance in the graph, radiotelephone, or radioteletype. message, but the contents of the various elements are not necessarily as they will appear.

Table 10-4. Message Format

FORMAT LINE	RADIOTELEGRAPH; RADIOTELETYPE	RADIOTELEPHONE
1	Not used	Not used
2	AAM1USA 11 XMT 11 AAM5USA	AAM1USA EXAMPT AAM5USA
3	DE AAM2USA NR48 11	THIS IS AAM2USA MESSAGE NUMBER 48
4	T 11 AAM4USA 11,	RELAY TO AAM4USA
5	R II 251020Z APR 77 ZEX II	ROUTINE TIME 251021 ZULU APRIL 77
6	FM AAM3USA II	FROM AAM3USA
7	TO AAM1USA II	TO AAM1USA
8	INFO WAR II	INFO WAR
9	XMT AAM6USA	EXEMPT AAM6USA
10	GR OR (GRNC)	GROUP or (GROUPS NOT COUNTED)

Table 10-4. Message Format—continued

FORMAT LINE	RADIOTELEGRAPH/RADIOTELETYPE	RADIOTELEPHONE
11	BT	BREAK
12	(TEXT)	(TEXT)
13	BT	BREAK
14	(1020Z)	(TIME 1020 ZULU)
15	B (AS) (C) (IMI)	MORE TO FOLLOW (WAIT) (CORRECT) (SAY AGAIN)
16	K (AR)	OVER (OUT)

10 - 9. Message preparation—radiotelegraph or radioteletype.

a. Use of separative sign (prosign II). The separative sign (prosign II), written as a short dash, is used to avoid mistakes in reception which might occur if letters or figures of adjacent groups are run together. The separative sign is used only in the heading of a message, format lines 2 through 10 as follows:

(1) Before and after all prosigns in the procedure and preamble components of the heading, except prosigns "DE," "AA," and "NR." The separative sign after a prosign is omitted when that prosign is followed by the prosign "BT."

(2) To separate each element of the address component; for example, preceding the prosigns "FM," "TO," "INFO," and "XMT."

(3) To separate, in the preamble components, the DTG from the message instructions.

(4) Between the call and the beginning of the repetition of a message repeated back.

(5) To separate call signs belonging to adjacent multiple transmission instructions as in c(3)c below.

(6) To separate portions of an abbreviated service message,

b. The call (format lines 2-3). The call is a means of establishing communications. In the call, the prosign "XMT" is used to indicate that stations whose designations follow are not required to respond. This is followed by the prosign "DE" and the designator of the calling station. The prosign "DE" is used only in this call and means, "THIS TRANSMISSION IS FROM THE STATION WHOSE DESIGNATOR FOLLOWS." The transmission identifier prosign "NR" may be employed to provide a means of assisting the receiving station in ascertaining that it has received all messages sent to it by a particular transmitting station.

EXAMPLE: (Station to Station):

AAR1USA DE AAR2USA NR71
(Exempting one station) AAR1USA
II XMT II AAR3USA DE
AAR2USA NR72

c. Transmission instructions (format line 4). This line may contain the prosigns "F," "G," and "T," operating signals, call signs, address groups, and plain language address designators.

(1) Prosign "G" (REPEAT BACK).

a) The prosign "G," meaning "REPEAT BACK THE ENTIRE TRANSMISSION," is used by the transmitting station to ensure that the receiving station received the message as transmitted. It may be ordered by the originator or the transmitting operator and may apply to the station called, all stations called, or stations designated before the prosign "G" in the transmission instructions. The instruction to repeat back applies to the whole transmission.

EXAMPLE 1: (Station called to repeat back).

AAR2USA DE WAR II G BT
TEXT BT
(AAR2USA complies as follows)
WAR DE AAR2USA II
BT TEXT BT K

EXAMPLE 2: (The instructions to repeat back given to one station of a collective or multiple call)

WAR/A DE WAR II AAR3USA
II G
BT TEXT BT K
(AAR3USA complies)
WAR DE AAR2USA II
WAR/A DE WAR II AAR3USA
II G
BT TEXT BT

b) When the prosign "G" is used with collective or multiple calls, designated station repeat back, stations receiving are to wait until the completion of the repeat back regardless of answering sequence

EXAMPLE: (After AAR3USA repeats back, in example 2 above, the response is given by WAR)

AAR3USA DE WAR C AR
Other stations then receipt:
AAR2USA R AR
AAR4USA R AR
AAR5USA R AR
AAR6USA R AR
AAR9USA R AR

(2) DTG. The DTG is expressed in digits, Zulu suffix, month abbreviated, and year expressed by two digits. An abbreviated plaindress message may carry no DTG or the DTG may be replaced by a DTG transmitted in line 14.

EXAMPLE: AAR2USA DE WAR NR5 II P II
211635Z JUN 84 II

(3) Message instructions—use of operating signals. The message instructions are normally ordered by communications centers or originators to express handling and delivery instructions and follow the DTG. Message instructions are to remain unchanged.

EXAMPLE 1: AAR4USA transmits a suspected duplicate message to: AAR9USA
AAR9USA DE AAR4USA NR9 II R
II 071754Z MAY 84 ZFD* II FM
WAR II TO AAR9USA INFO
AAR6USA BT TEXT BT K

*ZFD means "THIS MESSAGE IS A SUSPECTED DUPLICATE."

EXAMPLE 2: (WAR transmits a drill message to AAR2USA)
AAR2USA DE WAR NR321 II P II
071513Z APR 84
II ZEJ*
BT

*ZEJ means "EXERCISE (DRILL) MESSAGE."

e. Expressing the address (format lines 6, 7, 8, and 9).

(1) Use of prosigns "FM," "TO," "INFO," and "XMT." The prosigns "FM," "TO," "INFO," and "XMT" are used to indicate the originator, action addressees, information addressees, and exempted addressees as required.

EXAMPLE 1: (Action addressees only):
AAR2USA DE WAR NR322 II R II
081517Z APR 84 II FM WAR II
TO AAR2USA II AAR3USA
BT TEXT BT K

EXAMPLE 2: (Action and information addressees):
AAR2USA DE WAR NR 323 II R II
081520Z APR 84 II FM WAR II TO
AAR2USA II INFO AAR3USA II
AAR4USA BT TEXT BT K

EXAMPLE 3: (Action, information, and exempted addressees): WAR/A DE WAR

NR342 II R II

081523Z APR 84 II FM WAR II TO
AAR2USA II INFO AAR3USA II XMT AAR4USA II
AAR5USA BT TEXT BT K EXAMPLE 4: (Information addressees only): AAR2USA DE WAR NR325 II R II
081613Z APR 84 II *ZEX II FM WAR II INFO
AAR2USA BT TEXT BT K *ZEX means "THIS IS A BOOK MESSAGE AND MAY BE DELIVERED AS A SINGLE ADDRESS MESSAGE TO ADDRESSEES FOR WHOM YOU ARE RESPONSIBLE."

(2) Call serving as the address. In plaindress messages when the originator is in direct communication with the addressees, the call will serve as the address.

EXAMPLE 1: (When all addressees are action):
AAR2USA II AAR3USA II
AAR4USA DE WAR NR 326 II R
II 082343Z APR II
BT TEXT BT K

EXAMPLE 2: (When there are both action and information addressees. The information addressees must be indicated additionally in the transmission instructions by use of the operating signal "ZFH2," followed by the designation of the information addressees.)
AAR2USA II AAR3USA II
AAR4USA DE WAR NR327 II
ZFH2 AAR3USA II R II 011645Z
APR 84
BT TEXT BT K

EXAMPLE 3: (When all addresses are information. This is indicated by the inclusion of the operating signal "ZFH2" in the transmission instructions with no address designations following.)
AAR2USA II AAR3USA II
AAR4USA DE WAR NR 328 II
ZFH2*
II R II 081535Z APR 84
BT TEXT BT K

*ZFH2 means "THIS MESSAGE IS PASSED TO YOU FOR INFORMATION."

f. Prefix (prosign "GR," "GRNC") (format line 10).

(1) Prosign "GR" followed by numerals means "THIS MESSAGE CONTAINS THE NUMBER OF GROUPS INDICATED". It must be included in the prefix of all messages containing countable groups.

EXAMPLE; (AAA4USA transmits a message of 8 groups to AAR9USA);
AAR9USA DE AAA4USA NR 417 II
R II 300240Z APR 84
GR8
BT
KAHO TUON GREU AHID XOVO
DEAK FOLB DUTA BT K

(2) Prosign "GRNC" means "THE GROUPS IN THIS MESSAGE HAVE NOT BEEN COUNTED." This prosign is included in the prefix if it is necessary to indicate that groups have not been counted.

g. DTG (format line 14). This line when used will contain the time group expressed in hours, minutes, plus the Zulu suffix. Used only in abbreviated plaindress messages,

h. Final instructions (format line 15). This line is identified by the appearance of the prosigns "AS," "B," "C," "IMI," operating signals, and station designators as required.

i. Transmission ending (prosign "K," "AR") (format line 16). Every transmission will end with the prosign "K" (invitation to transmit) or the prosign "AR" (end of transmission).

10-10. Message diagram radiotelephone.

In table 10-2, it should be noted that every element is indicated in the order of appearance in the message, but the contents of the various elements are not necessarily indicated as they will appear.

10-11. Message preparation radiotelephone.

a. Each message prepared for transmission by radiotelephone will be in plaindress or abbreviated plaindress as described in paragraph 10-7.

b. Operating signals are not designed for radiotelephone transmissions. In radiotelephone procedures, the operating information will normally be conveyed in concise phrases. Where there are language difficulties, it will be permissible to use operating signals instead of standard radiotelephone phrases. In such instances, operating signals are transmitted by using the authorized phonetic equivalents.

c. The call (format lines 2-3).

(1) The call of a message serves to identify the stations between which that particular message is being transmitted. It may also serve as the address of the message when the designators of the originator and the addressees are the same as the call signs of the stations in communication with each other on the same net. In the call, the proword "EXEMPT" is used to indicate that station, whose designation follows is not required to respond. The proword "FROM" is used only in the call and means "THIS TRANSMISSION IS FROM THE STATION WHOSE DESIGNATOR FOLLOWS."

(2) Message. The proword "MESSAGE" may be transmitted immediately following the call to indicate that a message which requires recording is about to follow.

(3) Transmission identification. The transmission identification will be the last element of format lines 2 and 3 and will consist of the station serial number (SSN) preceded by the proword "NUMBER."

EXAMPLE: (Station to station):

ALFA ALFA ROMEO TWO
UNIFORM SIERRA ALFA THIS IS
ALFA ALFA ROMEO SIX
UNIFORM SIERRA ALFA
MESSAGE NUMBER TWO
SEVEN

(Exempting one station) ALFA ALFA
ROMEO TWO UNIFORM SIERRA
ALFA
EXEMPT
ALFA ALFA ROMEO THREE
UNIFORM SIERRA ALFA THIS IS
ALFA ALFA ROMEO FOUR
UNIFORM SIERRA ALFA
MESSAGE NUMBER SIX SEVEN

d. Transmission instructions (format line 4). This line contains the transmission instructions which may consist of the prowords "RELAY(TO)," "WORDS TWICE," or "READ BACK." The use of these prowords is explained in paragraphs 14-12, 14-13, and 14-18.

e. Precedence, DTG, and message instructions (format line 5).

(1) Precedence. The appropriate precedence designation, proword "PRIORITY" or "ROUTINE," is transmitted as the first element of format line 5. In the case of dual precedence messages, the higher precedence designation will be transmitted first.

(2) DTG. The proword "TIME" followed by the DTG and Zulu suffix is transmitted immediately after the precedence designation.

(3) Message instructions, These should not normally be required on radiotelephone messages. When included, they will consist of concise instructions which will remain with the message to the station destination.

EXAMPLE: ALFA ALFA ROMEO THREE

UNIFORM SIERRA ALFA THIS IS
ALFA ALFA ROMEO FIVE
UNIFORM SIERRA ALFA
PRIORITY

TIME ZERO SEVEN ONE SIX
THREE ZERO ZULU MAY EIGHT
FOUR

THIS MESSAGE IS A
SUSPECTED DUPLICATE

f. Expressing the address (format lines 6, 7, 8, and 9). These lines form the address of the message and are recognized by the prowords "FROM," "TO," "INFO," and "EXEMPT," respectively,

EXAMPLE A: (Plaindress message heading showing all possible elements of the address components)

ALFA ALFA ROMEO FIVE
UNIFORM SIERRA ALFA THIS IS
ALFA ALFA ROMEO NINER
UNIFORM SIERRA ALFA
MESSAGE NUMBER TWO
SEVEN PRIORITY
TIME ONE TWO ONE SIX THREE
ZERO ZULU MAY EIGHT FOUR,
FROM ALFA ALFA ROMEO TWO
UNIFORM SIERRA ALFA
TO ALFA ALFA ROMEO FIVE
UNIFORM SIERRA ALFA
INFO ALFA ALFA ROMEO SIX
UNIFORM SIERRA ALFA
EXEMPT ALFA ALFA. ROMEO
THREE UNIFORM SIERRA ALFA

EXAMPLE B: (Abbreviated plaindress message showing three elements in the address *component* assuming abbreviated call signs are used.)
ROMEO FIVE UNIFORM: SIERRA
ALPHA THIS IS
ROMEO NINER UNIFORM

SIERRA ALPHA

FROM ALPHA ALPHA ROMEO
TWO UNIFORM SIERRA ALPHA
TO ALPHA ALPHA ROMEO FIVE
UNIFORM SIERRA ALPHA
INFO ALPHA ALPHA ROMEO
SIX UNIFORM SIERRA ALPHA

EXAMPLE C: (Abbreviated plaindress message

with call serving as the address,
action addressees only assuming
abbreviated call signs are used.)

ROMEO FIVE UNIFORM SIERRA
ALPHA THIS IS
ROMEO NINER UNIFORM
SIERRA ALPHA PRIORITY
TIME ONE TWO ONE SIX THREE
ONE ZULU APRIL EIGHT FOUR

EXAMPLE D: (Abbreviated plaindress message

with the call sign serving as the
address, action, and information
addressees assuming abbreviated call
signs are used.)

ROMEO SIX UNIFORM SIERRA
ALPHA INFO ROMEO FOUR
UNIFORM SIERRA ALPHA
THIS IS

ROMEO THREE UNIFORM
SIERRA ALPHA

(BREAK)

(TEXT)

(BREAK)

TIME ONE TWO ONE EIGHT
TWO FOUR ZULU

g. Proword groups, groups no count (format line 10).

(1) Group count. Radiotelephone messages (plain text) are usually short (normally not exceeding 30 words). A group count is required and will be preceded by the proword "GROUPS" which will normally appear in the message prefix.

h. DTG (format line 14). This line is used only in abbreviated plaindress messages when a DTG is omitted in line 5. It consists of the proword "TIME" followed by the DTG plus the Zulu suffix, when appropriate.

i. Final instructions (format line 15). This line contains any final instructions necessary. It is identified by the prowords "WAIT," "CORRECTION," or "MORE TO FOLLOW."

j. Ending sign (prowords over or out) (format line 16).

(1) Every transmission will end with either the proword "OVER" or "OUT," except that the proword "OVER" may be omitted when two stations are in continuous communication with each other on a net not shared with a third station.

(2) In all transmissions where the proword "DO NOT ANSWER" is used, the transmission will be ended with the proword "OUT."

10-12. Service messages.

a. General. A service message is a message between stations that pertains to any phase of traffic handling, communications facilities, or circuit conditions. Service messages must be clearly worded to prevent misinterpretation and maximum use made of operating signals, prosigns, and authorized abbreviations. In radiotelephone operations, appropriate prowords will be used. These messages are used, for example, to correct message handling discrepancies, report delivery of a message, explain traffic handling delays, and assist generally in processing and handling traffic. They may also deal with traffic loads of unusual nature, traffic which requires special handling, the adjustment of procedural discrepancies, or changes in available facilities.

b. Procedures.

(1) Service messages are identified by the abbreviation "SVC" which is always used in the message text following the classification or the unclassified abbreviation "UNCLAS."

(2) Service messages are assigned the same classification as the message being serviced if the textual content is quoted or revealed by the service message.

(3) A service message normally will be assigned the precedence of the message being serviced. This may be modified by circumstances; for example, if a message has already been excessively delayed and must be held for service action, it may be necessary to assign priority precedence to the service message, even though the message being serviced is only ROUTINE precedence. Conversely, a service message reporting an after-the-fact action can be assigned a lower precedence.

(4) The text of the service message will refer to the SSN, time of file (TOF), and DTG of the message being serviced and will indicate the required action by operating signal or plain language.

EXAMPLES: (1) Request for repeat of entire message.

AAR1USA DE AAR2USA NR269
021015Z MAY 84 BT

SVC ZES2 NR 67 020537Z MAY
84
BT

(2) Reply to request for repeat of an entire message.

AAR2USA DE AAR1USA ZUI
AAR2USA 021015Z MAY 84
ZDK AAR2USA
R 020537Z MAY 84 FM AAR1USA
TO AAR2USA INFO AAR3USA
BT
TEXT
BT
021530Z
K

(3) Request for a repeat of a portion of a plain text message.

AAR3USA DE AAR4USA NR29
030518Z JUN 84
SVC INT ZDK AAR3USA
022025Z JUN 84
EXAMPLE TO REQUEST
BT

(4) Reply to request for repeat of a portion of a plain text message

AAR4USA DE AAR3USA NR87
030711Z JUN 84
SVC ZUI AAR4USA NR29 ZDK
AAR3USA NR43 022025Z JUN 84
EXAMPLE TO REQUEST PD
EXAMPLE OF A SERVICE
MESSAGE REPLYING TO A
REQUEST
BT
K

10-13. Counting and checking groups.

a. The prosign "GR" followed by numerals means "THIS MESSAGE CONTAINS THE NUMBER OF GROUPS INDICATED." It must be included in the prefix of all messages containing countable groups.

b. The prosign "GRNC" means "THE GROUPS IN THIS MESSAGE HAVE NOT BEEN COUNTED." This prosign is included in the prefix if it is necessary to indicate that groups have not been counted.

c. When counting groups (plain text or coded) all characters between space and space will be counted as one group.

EXAMPLE 1: NR 84
R211200Z JUL 84
FM AA6USA/TX

TO AAR6XX/TX
GR 6
BT
THIS IS HOW YOU COUNT
GROUPS.
BT
EXAMPLE 2: NR 85
R221200Z JUL 84
FM AA6USA/TX TO
AAR6XX/TX
GR 10
BT
DIGITS 4 - 1 000 A AND
AAA6USA/A WOULD BE COUNTED
AS ONE.

10-14. Book messages.

Book messages are prepared in the same manner as multiple address messages with the exception that the operating signal "ZEX" (and "ZEZ" if appropriate) will appear in the message instructions. In order to reduce transmission time and enhance security, the address designations not required in a transmission may be segregated and eliminated at origination as well as relay stations, thus address designations not pertaining to a given transmission can be deleted from the address component of book messages.

CHAPTER 11

DRAFTING MESSAGE TEXT

11-1. General.

a. It takes thought to originate a good message. What may seem simple and appropriate to the originator may be interpreted in an entirely different light by the addressee if proper care is not exercised. Vague and ambiguous text can be misinterpreted, resulting in an exchange of additional explanatory messages.

b. The need for brevity in message preparation is stressed. To avoid misinterpretation and further explanatory messages, the message must state exactly what is meant. Consistent with this action, all unnecessary words are to be eliminated. Commonly used conjunctions, prepositions, and articles (for example, and, but, for, in, on, and the) are to be eliminated unless essential to the meaning.

c. Abbreviations and short titles are used in messages in order to shorten the text, thereby saving transmission time.

11-2. Punctuation.

Punctuation is not used unless necessary to the sense of the message. When it is essential to employ punctuation, abbreviations and symbols as shown in table 11-1 are authorized.

Table 11-1. Punctuation abbreviations and symbols.

Punctuation	Abbreviation	Symbol
Question mark	QUES	?
Hyphen or dash		-
Colon	CLN	:
Comma	CMM	,
Decimal point	PT	.
Paragraph	PARA	
Period	PD	.
Parentheses	PAREN	()
Quotation marks	QUOTE/UNQUOTE	"
Semicolon	SMCLN	;
Slant bar	SLANT	/

11-3. Phonetic equivalents.

When it is necessary to include isolated letters in messages, the authorized phonetic equivalents (table 9-1) should be used. It is not advisable to employ the phonetic equivalent for single-letter words when names are to be transferred (for example, J.C. Porter or John Cook Porter vice Juliett Charlie Porter); when the actual

word might better be used (for example, 26 degrees West instead of 26 degrees Whiskey); when the abbreviation is readily recognizable and authorized (for example, USA, IJSN, USAF, USAISC).

11-4. Numbers.

a. Numbers may be written as digits or spelled out. When spelled out, they are expressed in words for each digit; except numerals 10 through 20 may be expressed as one word, and exact multiples of thousands are one word when the word "THOUSAND" is used. For example, 246.8 is written TWO FOUR SIX PT EIGHT and is un mistakeable, but when spoken, as TWO FORTY SIX PT EIGHT could be interpreted to mean 2406.8.

b. The figure 1500 would be written ONE FIVE HUNDRED. Cardinal numbers rather than ordinal numbers will be employed whenever practicable; for example, FIVE TWO instead of 52d, ROMAN NUMERALS will be expressed as cardinal numbers preceded by the word ROMAN to avoid misunderstanding; for example, IX written as ROMAN 9. Fractions such as one-half will be written as 1/2.

EXAMPLE 1: Voice "ROMAN NINER"

EXAMPLE 2: CW "ROMAN 9"

EXAMPLE 3: Voice "WUN SLANT TWO"

EXAMPLE 4: CW "1 fraction Bar (_ .. _)2"

11-5. Expression date and time in the text.

a. Date. When it is necessary to indicate a date alone in a message, it will be expressed by one or two figures indicating the date of the month followed by the first three letters of the name of the month, and the last two figures of the year.

EXAMPLE: 9 Oct or 9 Oct 84,

b. Time. Times in message text will be expressed as four figures followed by a time zone letter when it is known to be understood by all addressees, or the word "local" (for example, e.s.t, or P.d.t (Pacific dayline time)). The first pair of figures indicates the hour on a 24-hour clock. The second pair indicates the minutes past the hours.

EXAMPLE: 1535T. Where practicable, the time 2400 should be avoided unless it is necessary to indicate this particular instant of time; instead use 2359 or 0001.

c. Date and time together. When it is necessary to indicate date and time together, the DTG will be expressed as six figures followed by the zone letter. The first pair of digits indicates the date, the second pair indicates the hour (24-hour clock) and the third pair indicates the minutes past the hour. The DTG of a message will always be expressed in this manner followed by the time suffix letter "Z."

EXAMPLE: 271630Z Jun 84 represents 1630 G.m.t, on 27 June 1984.

11-6. Repetition.

a. A word may be repeated to prevent errors. It is not to be repeated solely for the purpose of emphasis. An example where repetition serves a legitimate purpose is "MIYAZAKI I SAY AGAIN MIYAZAKI" (to minimize the possibility of mistaken identity or incorrect spelling).

b. Letters and numbers should not be repeated for emphasis, but should be spelled out.

11-7, Message references.

a. References in message texts will be avoided unless essential. When used, the following provisions and instructions apply. References will normally consist of

<i>COLUMN</i>	<i>EXPLANATION</i>
NR	Message number.
PREC	Message precedence.
DATE-TIME	Message DTG.
FROM	Name of message originator.
ORIGINATING STATION	Originating station call sign.
TO	Name of message addressee.
STREET ADDRESS	Self-explanatory.
CITY AND STATE	Self-explanatory.
ZIP CODE	Self-explanatory.
TELEPHONE NUMBER	Self-explanatory.
TEXT	Text of message.
SIGNED	Name of message originator (transmitted).
NAME AND ADDRESS OF SENDER	Complete name and address and telephone number of message originator (not transmitted)

b. Determining precedence.

(1) Responsibility. The assignment of precedence to a message is the responsibility of the message originator. The precedence assigned to a message by the originator does not necessarily indicate the action to be assigned to the reply. Such instructions, when necessary, will be included in the text. Misuse of the precedence system defeats its purpose which is to ensure rapid handling of message traffic over available facilities with a minimum of backlog and delay resulting from competing priorities.

(2) Significance. Precedence categories are used to specify the relative order in which messages are to be handled. These categories include—

(a) To the originator: The required speed of delivery of the message to the addressee.

(b) To communication personnel: The relative order of message processing, transmission, and delivery.

YRMSG (your message), MYMSG (my message), YRLTR (your letter), MYLTR (my letter), followed by the DTG of the message or the serial number of a letter, order, or other document. Use of the date of such correspondence, in addition to the serial number, often is advisable, especially when it is not of recent issue or when the subject cannot be identified readily from the content of the message.

b. When reference is made to a DTG or originator's reference number, the month will be abbreviated using the first three letters and the year by the last two digits.

EXAMPLE: MYMSG 151517Z JUL 84, YRMSG 021345Z JUL 84.

11-8. Message preparation.

a. Standardization. To achieve standardization, all messages prepared for transmission follow a prescribed format, DA Form 5526-R (MARS Radiogram) is the prescribed form for drafting messages. It serves as a guide for preparation of record traffic for transmission. DA Form 5526-R will be locally reproduced on 8-1/2 by 11-inch paper and is located at the back of this manual. Instructions for completing DA Form 5526-R are in table 11-1.

(3) Precedence categories.

(a) Flash (Z), Immediate (O), Priority (P), and Routine (R).

(b) Flash (Z) and Immediate (O) messages are reserved for military traffic of highest urgency. Flash precedence will not be used on any MARS originated traffic.

(c) Immediate precedence is the highest precedence authorized for use by MARS during emergency conditions. Day-to-day operations will be limited to Priority and Routine.

(d) Priority is reserved for messages which must have precedence over Routine traffic. This is the highest precedence which may be assigned to messages of an administrative nature. Priority messages are processed, transmitted, and delivered in order received and ahead of all messages of lower precedence. Routine messages

being transmitted should not be interrupted unless they are extremely long.

(e) Routine is the precedence used for all types of messages which are not of sufficient urgency to justify a higher precedence but must be delivered to the addressee without delay. Routine precedence will be assigned to MARS administrative messages which require normal handling. Routine messages are processed, transmitted, and delivered in the order received and after all messages of higher precedence.

(4) Dual precedence. Multiple address messages having both action and information addressees may be assigned a single precedence in which it indicates the precedence for all addressees or they may be assigned two precedences, one precedence for all action addressees and a lower precedence for all information addressees. Dual precedence is indicated in the headings of plaindress messages by including the two precedence prosigns in the preamble and indicating the status (action or information) or the addressees in the address component.

c. FROM line.

(1) Military to military. When the message is directed to military addressees, the FROM line will consist of the command title and location of the commander in conjunction with the office symbol. In the following examples the official designation of originators and addressees are deemed to be self-sufficient and except where absolutely essential, personal names of originators or addressees or members of their staffs will not be included in the text of the message. Authorized abbreviations may be used. Army Post Office (APO) numbers or other coded distribution lists will not be used as an originator's address or as part of an address.

EXAMPLE 1: FM CDRUSAISC FR
HUACHUCA AZ //AS-OPS-CM//
TO CDRUSAISC FR MEADE
ME //ASNA-MDE//

EXAMPLE 2: FM CDRUSAISC FT MEADE
MD //ASN-OP-OX-MDE//
TO AAA6USA

EXAMPLE 3: FM AAA9USA CA
TO AAA6USA TX

EXAMPLE 4: FM AAA9USA CA

TO CDRUSAISC FT SAM
HOUSTON TX //ASN-OP-OX-
SHN//

(2) From military to nonmilitary. When the message is directed to a nonmilitary addressee, the originator's designation will consist of the full title and address of the commander. Short titles or abbreviated titles will not be used.

EXAMPLE 1: FROM: COMMANDER UNITED
STATES ARMY
INFORMATION SYSTEMS
COMMAND FORT HUACHUCA
ARIZONA

EXAMPLE 2: FROM: COMMANDER UNITED
STATES ARMY
INFORMATION SYSTEMS
COMMAND FORT WIEADE
MARYLAND

d. TO line.

(1) To a military addressee. Messages will be addressed to the commander or head of the command, installation, or agency for whom the message is intended. Messages for the attention of a specific person or unit will be addressed to the commander or head of the organization and will contain internal delivery instructions.

EXAMPLE 1: TO CDRUSAISC FT HUACHUCA
AZ //AS-OPS-OA//

EXAMPLE 2: TO CDRUSAISC FT MEADE MD
//ASN-OP-OX-MDE

(2) When a message is to be transmitted or delivered to nonmilitary addressees, names of cities will be spelled out in full; names of states or countries or lands will be abbreviated.

EXAMPLE: TO MRS ROBERT ROE 123 WEST
STREET ALBANY NY 12345
PHONE 518-XXX XXXX

e. Third-party messages.

(1) In drafting third-party messages, the FROM and TO lines will contain complete name (rank or rating), city, state, county, or land address, zip code or APO of Fleet Post Office (FPO), and telephone number when available.

EXAMPLE: FROM SP5 JAMES ROE
HH BTRY 123 FA BN
APO NY 09000

TO PHONE EXTENSION 6789
TO MRS SANDY ROE 456
WEST STREET
BALTIMORE MD
21212
PHONE 301-543-6789

INFO line. Prepared in the same
manner as the TO line. XMT line.
Prepared in the same manner as the
TO line.

(2) Oversea message originators. Where
military security precludes disclosure of military unit
name and/or location, the PSOM line need only show
country and MARS gateway station call.

EXAMPLE: FM SGT RICHARD RAL
PIRMESANS GE/AEM3USA GE

f. Message identification. The DTG, Zulu time,
month and year, combined with the abbreviated title of

the originating command, agency, or
installation and the drafter's office symbol will be the
only means of identifying Army messages.

EXAMPLE: MY AS-OF-OX-MDE 180739Z JUL 84

g. Classification line. Messages addressed military-
to-military or other Government departments or agencies
will carry a classification line. A separate subject line
will be inserted in the message text following the
internal instructions and preceding any references to
previous communications. Military-to-nonmilitary and
third-party traffic will not carry a subject line.

h. Body of text. The body of the text must be clear
and concise. Eliminate as many words as possible
without making the meaning vague or misleading. Digits
and authorized punctuation will be used. Conjunctions,
prepositions, and articles should not be used unless
essential to the meaning.

CHAPTER 12

OPERATING METHODS AND REFILE INSTRUCTIONS

12-1. Principal operating methods.

The use of the three principal operating methods available for passing messages from one station to another are determined by operational requirements.

a. Receipt method. The receipt method requires the receiving station to give a receipt for each message to the transmitting station, thus there is certainty of reception. The receipt method is normally used when sending messages-

- (1) Between fixed stations.
- (2) From a mobile to a fixed station.
- (3) Between mobile stations.

b. Broadcast method. Broadcasts are used to disseminate information and are made on specific frequencies and at specified times. Messages transmitted by this method are considered received when transmitted. The reliability of this method can be increased appreciably by assigning a DTG and sequential broadcast number, and by monitoring off the air to ensure quality of transmission.

(1) Broadcasts not specifically addressed, (a) Prior to commencing a broadcast, a 5-minute series of test calls will be transmitted. The test call consists of a series of three V's repeated three times; the prosign "DE," and the station call letter repeated three times (call tape for 5 minutes maximum}. At the appropriate time transmit broadcast.

EXAMPLE: VW VVV VVV DE WAR WAR NR 60
R 1G1520Z APR 84
FM
TO
GR NO BT
TEXT BT AR

(b) The text may contain the message number, date, DTG, or other identifying or specific information as deemed necessary. The broadcast will be concluded by transmitting the general call "CQ" three times, prosign "DE," station call repeated three times, followed by the operating signal "QRU" and the prosign "AR."

EXAMPLE: CQ CQ CQ DE WAR WAR WAR QRU
AR

- (2) Retransmission.

In net operations, the retransmission of a broadcast not specifically addressed will require a new heading in plaindress or abbreviated form.

EXAMPLE: AAA9USA/A DE AAA9USA

NR 60 II R II 101520Z APR 84 II FM
WAR II
TO ALL MARS MEMBERS AND
STATIONS WORLDWIDE
GR NC
BT
TEXT
BT (K) (AR)

Transmission ending with the prosign "K" requires the NCS or designated station to receipt for the broadcast.

c. Intercept method. The intercept method is one in which one transmitting station sends a message to a second station; the latter station obtains the necessary repetitions to ensure correct reception and, if directed by the transmitting station or if prescribed by the operating agency, repeats back the message. Messages thus transmitted are actually intended for third stations which copy the message but do not use their transmitters directly in connection with the intercept method transmission. Messages transmitted by the "I" method will normally be addressed to the intercepted station. The address of plaindress messages will be expressed in full. Serial numbers will be used for all messages and messages will be transmitted in strict sequence of serial numbers.

EXAMPLE: AAA1USA DE AAA2USA NR17
II I II G II
R II 121314Z APR 84 II
FM WAR II TO AAA9USA
GR NC
BT
TEXT BT K

AAA1USA repeats back the complete message obtaining a receipt or completing necessary fills as required.

Because the message was addressed to AAA9USA and transmitted by the INTERCEPT method, AAA1USA would take no action on the message.

12-2, Reliability

a. Reliability is paramount in military communications, being more important than speed. Accuracy of transmission directly affects reliability. Accurate forming and spacing of international Morse code characters and proper enunciation enhance reliability and increase the speed of net operations.

b. Operators will transmit messages exactly as written. Prosigns, operating signals, or abbreviations will not be substituted for the text as written by the originator. Numerals in DTGs, serial numbers, call signs and those used in operating signals and prosigns will be written and transmitted as digits (CW and RATT) and pronounced in voice.

c. An operator who receipts for a message is responsible for further relay or delivery of that message. Any long delay or nondelivery of a message will be reported to the originator.

12-3, Speed of transmission.

Speed of transmission is achieved by accurate sending and by adherence to procedures. Operators will transmit only as fast as the receiving operator can record the message. In net operation the speed of the slowest operator will govern the speed of all stations in the net. This rule may be suspended by an NCS when practical reasons dictate higher speeds.

12-4, Means of operation.

a. Messages are transmitted over MASS circuits employing radiotelegraph (CW using international Morse code or teletypewriter) and radiotelephone,

b. Messages which have been received in the basic format by one means will require reprocessing before they are forwarded by another means; for example, a message received by international Morse code and/or teletypewriter will have to be reprocessed before transmitting in the voice mode.

c. MARS is basically a relay system and the major portion of messages entering the system may pass through several MARS stations before delivery to the addressee.

d. In the movement of messages from originator to addressee, one means of transmission should not be used to the exclusion of the other means. Volume traffic is best moved over MARS between major relay stations by means of radioteletype writer. MARS members should use initiative in employing CW telegraphy when circuit conditions are such as to render voice or radioteletypewriter transmissions difficult. A message transmitted at five or ten WPM and receipted for is better than delaying a message for hours because of the

inability to copy through noise and atmospherics.

12-5, Mars Network

MARS radio net-works are organized along command lines into a series of interconnecting networks operating under a designated NCS. These networks consist of—

a. A HQDA MARS net, comprised of the headquarters command station WAR, oversea and CONUS gateway stations, and alternate gateway stations providing radio communications between geographical areas to supplement command and emergency communication channels as required.

b. An area command MARS net comprised of the area gateway station and subordinate affiliate or military MARS stations representing each state and/or military installation within the area. These stations provide routine communications and communication needs based on requirements of military and civilian community disaster plans.

c. State, district, or local MARS nets to provide routine communication support and relay facilities between military and civilian community, initial support at scenes of disaster, and continuing support at disaster scenes to base and/or mobile facilities as required.

12-6, Traffic flow

a. Messages entering the Army MARS networks will be moved through the interconnecting networks through a system of relays. During the movement of the message through the interconnecting network, it may be necessary for the message to be refiled from one system to another. Traffic refiled may require minor changes in the basic message format.

b. Each station in relay sequence will receipt for a message and assign a new station serial number before retransmitting. Receipts will be exchanged only between stations transmitting and receiving the message. No MARS station will receipt for messages which have been intercepted and for which that station was not the designated receiving station unless authorized to do so by the designated receiving station.

12-7 Refile

a. REFILE IS THE reprocessing of messages into appropriate format for transfer to or from another system or means of communication. Traffic that is transferred from RATT to CW and voice nets constitutes refile since some minor changes may be necessary in message format to adapt it to another means of transmission.

b. MARS members will refile messages into the amateur system only when MARS facilities are not available to deliver the message to its final destination. MARS messages should not be refiled into the amateur system merely to provide training in refile procedures. The refile processes required to transfer a message from or into the amateur system are the responsibility of the MARS member since the transfer always occurs at a MARS station. Therefore, any improperly constructed or mishandled refile message which appears in either system can initially be charged to the MARS member who processed the message for refile. Messages can be accurately and efficiently refiled only if MARS members study, understand, and properly apply the detailed refile procedures contained in the following paragraphs.

c. Procedure for refiling a MARS message into the amateur system. The MARS station refiling a MARS message into the amateur system will make the necessary changes to the text and heading as follows:

(1) Delete all the heading before the address.

(2) Delete the entire group count before the first break.

(3) Delete the abbreviation "SGD" (signed) between the last word of the text and the signature and insert a BREAK sign.

(4) Add a new heading including all of the following in sequence:

(a) New message number.

(b) Amateur call sign of the refiling MARS station.

(c) The sign "CK" and the number of groups excluding the signature.

(d) The place of origin.

(e) The words "VIA MARS/MARS" originating station call and geographic location.

(f) The originator's time, month, and day.

(5) The address remains unchanged. The text remains unchanged except as indicated in (3) above. A MARS message will not be released into the amateur system until all changes listed above have been completed.

EXAMPLE 1: A message to be refiled from MARS to amateur.

(Received via MARS)

DE AEM1US NR 55

R 13140Z APR 84

FM AJ5AB FRENCH MOROCCO

TO MRS JOHN DOE

123 SPRING DR

KANSAS CITY KA

66122

PHONE: 816 555-1212

GR NC

BT

ARRIVED SAFE AND WELL PACKAGE
AND LETTER RECEIVED ON ARRIVAL
HAVE MAILED LONG LETTER LOVE
SGD JACK BT

EXAMPLE 2: (Refiled to amateur)

NR 75 K4USA CK15 AJ5SB FRENCH
MOROCCO VIA MARS

AEM1USA/GE

1430 13 APR 84

TO MRS JOHN DOE

123 SPRING DR

KANSAS CITY KA

66122

BT

PHONE: 816 555-1212

ARRIVED SAFE AND WELL PACKAGE
AND LETTER RECEIVED ON
ARRIVAL HAVE MAILED LONG
LETTER LOVE

BT JACK BT

(Required changes have been made. The message can be sent via amateur.)

d. Procedure for refiling an amateur message into the MARS system. An amateur message will not be accepted for refile unless it contains all the following:

(1) Message number.

(2) Call sign of the amateur station of origin.

(3) Check (number of text, groups) (CK).

(4) Place of origin.

(5) Time of origin (or file time), date, or no file time (NFT) date. (NOTE: If NFT and date is included, time of refile will become the file time.)

(6) Complete address.

(7) Complete text.

(8) Signature.

e. To convert (refile) the amateur message into a normal form MARS message, the following changes will be made to the amateur message.

(1) Delete the amateur message number and check (CK).

(2) Delete the BREAK (BT) between the last word of the text and the signature and insert the word "SGD."

(3) To make a normal form MARS heading, add the following in sequence:

(a) Message number of the refiling MARS station.

(b) Precedence.

(c) DTG (date and time of the refiling MARS station).

(d) Insert abbreviation "FM" before the
amateur call sign of the station of origin.

EXAMPLE: Message refiled from amateur to MARS.

(Received via Amateur)

NR 56 W3USA CK18 FT MEADE MD

1510 (or NFT) OCT 16

TO JOHN ADAMS

324 FERNDALE AVE

ATLANTA GA 00000 BT

YOUR LETTER RECEIVED TODAY

IF LEAVE GRANTED WILL ARRIVE

YOUR HOME SUNDAY IF NOT

WILL NOTIFY YOU LATER BT

DAVE BT

(Refiled to MARS)

AAA4USA DE WAR NR 79

R 162020Z OCT 84

FM W3USA FT MEADE MD

1510 (or NFT) OCT 16

TO JOHN ADAMS

324 FERNDAL AVE

ATLANTA GA 00000

GR NC

BT

YOUR LETTER RECEIVED TODAY

IF LEAVE GRANTED WILL ARRIVE

YOUR HOME SUNDAY IF NOT

WILL NOTIFY YOU LATER SGD

DAVE

BT

CHAPTER 13

RADIOTELEGRAPH PROCEDURES

13-1. General.

This chapter contains specific instructions for the conduct of communications employing CW telegraphy. Development and use of other high-speed communications equipment have not decreased the need for well-trained radiotelegraph operators who can copy through natural and man-made interference and who are capable of detecting deceptive practices. The capability to maintain communications using radiotelegraph is a vital and basic requirement of the MARS. There is conclusive evidence of situations where CW operation provided the only usable method of communicating. All MARS members must recognize the continuing requirements to establish and maintain the highest proficiency in radiotelegraph net operation.

13-2. Continuous wave net operations.

a. Network activities, for example, message exchange, contacts and transmissions are controlled (directed) by an NCS. MARS nets are directed unless otherwise indicated. A directed net may be designated by use of the operating signal "ZKB" meaning "IT IS NECESSARY TO OBTAIN THE PERMISSION OF THE CONTROLLING STATION (NET CONTROL STATION) BEFORE TRANSMITTING MESSAGES." In these circumstances, the NCS may also indicate a specific sequence of answering. The signal "ZKB" may be preceded by ZKA meaning "I AM NET CONTROL STATION."

b. NCS may lift the restrictions of a directed net by use of the operating signal "ZUG" meaning "NEGATIVE" in conjunction with the "ZKB" signal,

EXAMPLE 1: (Making a FREE NET a DIRECTED NET)
AAA3USA/A DE AAA3USA ZKA
ZKB K AR

EXAMPLE 2: (Making a DIRECTED NET a FREE NET)
AAA3USA/A DE AAA3USA ZUG
ZKB K AR

c. Occasions may arise when, after a station has been given permission to transmit in a directed net and commences the transmission, another station may break in with higher precedence traffic. Should this happen, the original transmitting station is not to resume

transmission of the message until directed to do so by the NCS.

13-3. Circuit discipline.

Adherence to prescribed procedure is mandatory. Unauthorized departures from or variations to prescribed procedure often create confusion and reduce reliability and speed. The basic rules essential to circuit discipline and good transmission practices are outlined in paragraph 9-2. If the procedure prescribed herein does not cover a specific operating requirement, initiative and common sense should prevail.

13-4. International Morse code.

a. Transmissions over military telegraphic systems use the international Morse code. The characters used are as follows:

LETTERS

A .-	F ..-	K -.-	P .--.	U ..-
B-...	G --.	L .-..	Q--.-	V...-
C -.-	H	M -	R.-.	W .--
D-..	I ..	N -.	S ...	X -..-
E .	J .---	O----	T -	Y -..-
				Z --..

FIGURES

1 .----	4 ...-	7 --...	0-----	-
2 ..---	5	8 ---..		
3...--	6 -....	9-----		

(NOTE: The figure zero is to be written "0." When this figure occurs in the text of messages which consists entirely of figures or in weather messages which use the number code, it may be transmitted as "T" (-). On all other occasions the figure 0 is to be transmitted as "-----" Transmission of 0 as the letter "O" is expressly forbidden.)

PUNCTUATION AND OTHER SIGNS

BLOCK (capital letters or underline) ...-

FRACTION BAR (slant) -.-.

PERIOD (full stop) .-.-

HYPHEN or DASH -....-

PARENTHESIS (left hand bracket) -.-.

(right hand bracket) -.-.-

COMMA --..--

QUESTION MARK ..--.. in MARS (..--..) is a
prosign defined as "say again (repeat)."

COLON ---...

APOSTROPHE .-----.

13-5. Transmission techniques and speeds.

a. Each character will be transmitted clearly and distinctly. The speed of transmission will be governed by the prevailing conditions and the capability of the receiving operators.

b. Accuracy in transmission is far more important than speed. The difference in time required to send a message at 18 WPM and that required to transmit it at 25 WPM is slight. Even this slight gain in time may be negated by any added time required for repetitions.

c. The speed at which the receiving operator can copy without having to obtain repetitions is the speed at which the transmitting operator will transmit. When transmitting to more than one station in a net, the governing speed of the transmitting operator is that of the slowest receiving operator. The speed of transmitting message headings on manually operated circuits should be slower than the speed of the transmission of texts. If advisable, the NCS should prescribe the speed of transmission on a circuit.

13-6. Call signs.

It is mandatory that call signs be phonetically expressed when employed by stations calling, entering, reentering, joining, or rejoining nets. Call signs should be expressed phonetically when difficult operating conditions may result in confusion or mistaken identity. At all other times, phonetic expression of call signs is not required. Full call signs are to be used when first establishing a net, when reporting into a previously established net, and in the transmission instructions and address components when a message is required to be relayed to a station on a different net. Provided no confusion can arise, call signs may be abbreviated when authorized by the NCS. Abbreviated call signs are developed by dropping the first two letters of the MARS call.

EXAMPLE: "AAV7US" becomes "V7US" in abbreviated form

13-7. Calling or answering continuous wave net.

a. The call. The preliminary call transmitted to establish communications may be a single, net, or multiple call as outlined in (I)(a) through (I)(c) below.

Under difficult conditions the call signs may be transmitted twice. Two or more call signs transmitted as part of the call or appearing in message components will be arranged in alphanumerical order. When net calls are used, the prosign "XMT" may be used to indicate stations that are not required to respond. Before transmitting, an operator will listen long enough to ensure that transmission will not cause interference to other transmissions in progress.

1) The call will consist of the call sign(s) of stations called, the prosign "DE," and the call sign of the calling station. The call may contain:

(a) Single call sign. Identifies a single station.

EXAMPLE: (Single) AAA2USA DE AAA4USA K

(b) Multiple (collective) call sign. Identifies a predetermined group of stations.

EXAMPLE: (Multiple or collective) AAA2USA II
AAA6USA DE AAA4USA K

(c) Net call sign. Identifies all stations on that net.

EXAMPLE: (Net) AAA USA / A DE AAA1USA
K

2) The call may be of two types:

(a) Single call. Whereby only one call sign precedes the prosign "DE." This may be an individual, net, or collective call sign.

EXAMPLE: (Single) AAA2USA DE AAA4USA K
(Net) AAA2USA / A DE
AAA4USA K

(Multiple or collective) AAA2U3A II
AAA6USA DE AAA4USA K

(b) Multiple call. Whereby two or more call signs precede the prosign "DE." These may be individual, net, and/or collective call signs.

b. Answering.

(1) To answer a preliminary call, stations will transmit the identification of the calling station, the prosign "DE," the identification of the answering station, and the prosign "K."

EXAMPLE: AAA4USA DE AAA2USA K

(2) After good communication has been established, an answer may consist of the prosign "DE," the identification of the answering station, and the prosign "K." This is an abbreviated answer.

EXAMPLE: DE AAA2USA K

(3) Stations will respond to a preliminary call in the order called, alphabetical or numerical sequence, or the order directed by the NCS.

(4) If any station fails to answer in proper sequence, the next station waits 5 seconds and answers. The station which fails to answer in proper order must wait until all other stations have answered or have had time to answer.

(5) If a station called fails to answer, the NCS will wait 5 seconds after the last response received, then initiate a new preliminary call specifically to that station.

c. Unknown station - prosign "AA." When a station hears a call without being certain that the call is intended for it, it will not answer until the call has been repeated and is understood. When a station is called, but is uncertain of the call sign of the called station, it will answer immediately by transmitting the prosign "AA" followed by the prosign "DE," its own call sign, and prosign "K."

d. Test signals. When it is necessary for a station to initiate test signals for adjustment of a transmitter before making a call, adjusting frequency measuring equipment, or adjusting a receiver, such signals will consist of not more than three series of V's followed by a 10-second dash, followed by the call sign of the station and the prosign "K" or "AR." Annotate equipment dial readings for future tuning assistance. A second transmission may be made if necessary.

e. Signal strength and intelligibility.

(1) A station assumes it has an intelligibility of "good" unless otherwise notified. Signal strength and intelligibility reports will not be exchanged unless communications are unsatisfactory. Intelligibility is the ease with which the incoming signals can be received. This depends on the relative strength of the desired versus the undesired signal (for example, interference, static, inherent receiver noises, and so forth) and on the capabilities of the receiving operator.

(2) Signal strength and intelligibility are indicated by means of the operating signals "QSA" and "QRK" (see table B-2 for meanings).

f. Transmitting a message.

(1) Messages will be transmitted exactly as written. Abbreviations will not be substituted for plain language or plain language substituted for abbreviations.

(2) Numerals in DTGs, serial numbers, call signs, and those used with operating signals and prosigns will be written and transmitted as digits,

(3) Separation of parts. The prosign "BT" is to be transmitted immediately following the last element of the heading and of the text of all messages, except abbreviated service messages in which a DTG is not used.

g. Pause in transmission (prosign "AS"),

(1) The prosign "AS" made during a transmission and without an ending sign indicates a short pause.

EXAMPLE: AAA2USA DE WAR NR 1 II R II
101214Z APR 84

BT

ARRIVED SAFE AND WELL

AS

When ready to resume, WAR completes the transmission commencing with a repetition of the last work, group, or prosign transmitted.

(2) The prosign "AS" followed by prosign "AR" means "YOU ARE TO WAIT," or "I AM OBLIGED TO WAIT," as applicable.

EXAMPLE: AAA4USA DE WAR NR1 II P II
121520Z APR 84

BT

ARRIVED SAFE AND AS AR

(3) A station having received the prosign "AS" will wait for the prosign "K" before transmitting unless in the meantime the station has been given a message or higher precedence to transmit or if it appears the station has been overlooked.

(4) A station resuming transmission of a message after a pause as in (2) above will, after the call, identify the message, transmit the prosign "AA," the last word, group or prosign transmitted, the separative sign, and resume transmission of the original message.

EXAMPLE: AAA4USA DE WAS NR 1 121520Z
APR 84 II AA AND II WELL
PACKAGE RECEIVED BT K

h. Transmission of messages containing more than 100 groups (prosign "B").

(1) When a group count is used, messages containing more than 100 groups will be transmitted in groups of 100 with only the last portion containing additional groups. A receipt will be given for each portion,

EXAMPLE: (WAR transmitting a message containing 160 groups to AAA5USA, stops after transmitting the 100th group, indicates there is more to follow and requests a receipt for the portion transmitted.)

AAA5USA DE WAR NRI II R II
151530Z APR 84

GR160

BT

(FIRST 100 GROUPS)

II

B 100 K

AAA5USA having received the portion, transmits:

WAR DE AAA5USA R K

WAR resumes transmission after
a short pause.
AAA5USA DE WAR 101 II
(101 to 160 inclusive)
BT K

(2) In the above example, should AAA5USA have required any repetitions, they would have been asked for and given prior to receipting for the portion.

(3) Messages without group counts will be transmitted as in the example above but without prosign "GR NC." A new portion will start with the last word or group of the previous portion.

i. Transmitting messages in strings (prosign "B").

(1) When radio communication is good, frequently it facilitates the handling of traffic for one station to send several messages to another station without interruption. The receiving station will indicate the number of messages to be transmitted in a given sequence.

EXAMPLE: (WAR has 10 messages for AAA6USA.
WAR transmits): AAA6USA DE WAR
ZBO 6P 4R K
AAA6USA transmits:
WAR DE AAA5USA QSG5 K

(2) When messages are to be sent in sequence, the transmitting station will indicate immediately after the prosign "B" at the end of message in the sequence, the precedence prosign of the message to follow. Upon transmitting the last message in a sequence, a receipt is required before continuing with another sequence. Therefore, the last message of each sequence will be terminated with the prosigns "B" and "K," meaning "THERE IS MORE TO FOLLOW, RECEIPT FOR WHAT I HAVE SENT."

EXAMPLE: (WAR transmits the first of a string of
five messages): AAA9USA DE WAR
NR 1 II P II
160237Z APR 84
GR23
BT
TEXT
BT
B K

(NOTE: There is a short pause to allow
any station to break in to transmit traffic
of a higher precedence. If no station
interrupts, WAR proceeds):
AAA9USA DE WAR NR5 II P II
160242Z APR 84 II
FM WAR

WAR terminates the ending of the fifth
message.

BT B K

AAA9USA requests the needed
repetitions if break-in procedure has not
been employed. Otherwise AAA9USA
receipts for the string:

WAR DE AAA9USA R K

j. Repetition of difficult portions (prosign "IMI"). In
the text of a plain language message, difficult portions,
at the discretion of the transmitting operator, may be
repeated. For this purpose, the prosign "IMI" is used,
and means "I AM GOING TO REPEAT THE
DIFFICULT PORTION JUST TRANSMITTED."

EXAMPLE: AAA4USA DE AAA6USA NR 11 II R II
171951Z APR 84

BT

UNCLAS YOUR 121212Z.

REPLACEMENT PART NO.
23A/MBS/M/47 IMI 23A/MB6/
M/47 being dispatched

BT K

k. Errors during transmission and cancelling
transmission (prosign "EEEEEEEE").

(1) The error prosign will be of not less than eight
E's. It should not be unduly prolonged. It may be used to
correct errors and to cancel transmission in progress.

(2) Correction of errors. If an error is made during
transmission, the error prosign will be transmitted,
followed by the repetition of the last word or group
correctly transmitted.

EXAMPLE: AAA9USA DE WAR NR12 II T

II R II 171231Z APR 84 II

FM WAR II

TO AAA9USA II

INFO AAA9USA EEEEEEEEE INFO
AAA6USA

BT

UNCLAS MY 160603Z DETAILS

WILX EEEEEEEEE DETAILS WILL
FOLLOW BT K

(3) Cancelling transmissions in progress. The error
prosign when used for cancellation of transmission may
be used only for the transmission in progress. Cancellation
of messages already receipted for must be effected by means
of a separate message. When used, the prosign will be
followed by the prosign "AR," which means "THIS
TRANSMISSION IS IN ERROR, DISREGARD IT."

EXAMPLE: AAA3USA DE AAA9USA NR 13 II R II
222327Z APR 84 BT UNCLAS RETURN TO
EEEEEEEE AR

1. Receipts (prosign "R").

(1) The prosign "R" will be used to indicate receipt of messages or portions thereof,

(2) After receiving instructions, information or an inquiry prosign "K" means "I HAVE RECEIVED YOUR LAST TRANSMISSION."

EXAMPLE: AAA6USA DE WAR R AR

(3) After a call, the prosign "R" preceded by prosign "INT" means "HAVE YOU RECEIVED MY LAST TRANSMISSION OR HAVE YOU RECEIVED ____?"

EXAMPLE: AAA3USA DE AAA5USA NR 2 INT
R K
or
AAA3USA DE AAA5USA NR 2
INT R AAA9USA
121753Z APR 84
K

13—8. Indicating precedence in a preliminary call.

a. Precedence may be indicated in a preliminary call as follows:

PROSIGN	MEANING
Z	I have FLASH traffic to transmit.
O	I have OPERATIONAL IMMEDIATE traffic to transmit
P	I have priority traffic to transmit.
R	I have routine traffic to transmit.

EXAMPLE: (AAA2USA tells AAA4USA that he or she has priority traffic to transmit.)
AAA4USA DE AAA2USA P K

b. To indicate a number of messages of one or more precedences in a preliminary call, the operating signal "ZBO" may be used.

EXAMPLE 1. (Free or directed net)
AAA3USA DE AAA9USA ZBO
3P II 2R K

EXAMPLE 2: (Directed net. AAA2USA tells AAA4USA he or she has two priority messages for AAA3USA.)
AAA4USA DE AAA2USA ZBO 2P
AAA3USA K

EXAMPLE 3: (Directed net. AAA2USA tells AAA4USA that he or she has seven routine messages for AAA4USA and two priority and two routine messages for AAA5USA.)
AAA4USA DE AAA2USA ZBO
7R II 2P 2R AAA3USA K

13-9, Establishment and operation of a MARS continuous wave net.

a. Establishment and operation.

(1) In both MARS and tactical operations the establishment of an efficient communications system is dependent upon an understanding and the proper execution of certain procedures by net participants. Prior to the establishment of a MARS net or prior to the assignment of a station to a MARS net, the Chief, Army MARS, command or area MARS directors, or the State MARS director will inform each participant of the assigned net designator, call sign, net control and alternate net control stations, operating frequency, power, emission, allowable frequency tolerance, date and time of net schedule and operating procedures in this manual.

(2) The following rules will govern the operation of a CW net on MARS frequencies:

(a) Net operations will always be directed unless indicated otherwise by the NCS.

(b) Net stations will answer the net call in the predetermined alphanumeric position occupied on the roster or indicated callup order of net stations. The normal alphanumeric order may be modified by the NCS to meet local requirements.

(c) Each station will list their traffic in the initial transmission when entering the net.

(d) The NCS will transmit one net call for net stations and thereafter will provide several 5-second unannounced pauses during a net period for unreported stations, after a message transmission has been completed and before a new message transmission is directed. Stations waiting to enter the net may do so during these pauses.

(e) In the CW net operation break-in procedure, a station will not break-in unless it has a message of higher precedence than the message being transmitted or has some information of immediate or urgent nature for the NCS or other station.

b. Tuning stations on a net.

1) Ordering and controlling netting is a responsibility of the NCS.

2) The operating signals "ZRF," "ZRC," "ZGE," and "ZRA" are used for tuning (see table B-1 for meanings).

3) The NCS is responsible for having its transmitter accurately tuned to the assigned frequency. Each subordinate station must tune to the NCS even if the NCS is off frequency. If this is the case, the subordinate station(s) should inform the NCS, using appropriate operating signals.

(4) Prior to opening a net, preliminary adjustments to transmitter and receiver will be made, using a dummy load on the assigned frequency. Then, at the appropriate time, the receiver can be zero beat with the NCS and the transmitter tuned to the receiver in a minimum amount of time on the assigned frequency. Care should be taken that emission standards are observed; for example, power and spurious emissions are minimized.

EXAMPLE: (AAA5USA, the NCS, wishes to tune the net)

AA5USA/A AAA5USA/A DE
AAA5USA AAA5USA ZRF II
ZRC2 II AAA5USA AAA5USA
(repeated for 20 seconds, then
depress key for 10 seconds to permit
zero beat) II ZRC1 AR. After a 50-
second pause to allow stations to
tune their equipment: AAA5USA/A
DE AAA5USA ZGE2 K

Station answers:

AAA5USA DE AAA2USA II
AAA2USA K
AAA5USA DE AAA3USA II
AAA3USA K
AAA5USA DE AAA4USA II
AAA4USA K

All prove to be on the correct
frequency except AAA3USA who is
1 kHz low. AAA5USA
transmits:

AAA3USA DE AAA5USA ZRA3 1
K

AAA3USA receipts, readjusts his or
her transmitter and asks how the
frequency checks. AAA5USA DE
AAA3USA INT ZRA K

AAA5USA replies giving further
tuning instructions if necessary for
frequency correction.

(5) Any station desiring to readjust the transmitting system during net operation requires NCS permission.

c. Opening a CW net.

(1) The following example of a MARS CW net operation will be used as a guide for all MARS nets. It will be followed as closely as possible. This theoretical net operation is based upon the following:

(a) Net call—WAR/A.

(b) NCS WAR.

(c) Alternate NCS AAA3USA.

(d) Alphabetical or numerical roster or reporting order of net stations, AAA2USA AAA3USA AAA4USA AAA5USA AAA6USA AAA9USA

(2) An example of opening net call is shown below.

WAR/A WAR/A WAR/A DE WAR WAR
WAR

ZRF II ZRC2 (10 second dash) ZRC1 AR
Meaning, "I AM ABOUT TO SEND A TUNING
SIGNAL ON MY PRESENT FREQUENCY:
ZERO BEAT WITH MY TRANSMITTER (10-
second dash to zero beat); TUNE AND/OR
ADJUST YOUR EQUIPMENT."

This transmission will be followed by a 50-
second clear period for adjusting of equipment.

WAR/A DE WAR K

(3) An example of answering calls by stations entering net is shown below.

DE AAA2USA ZBO IP K

DE AAA3USA ZBO 1R AAA6USA K

AAA4USA fails to answer immediately.

AAA5USA waits 5 seconds and transmits:

DE AAA5USA ZBO 1R II IP

AAA2USA K

DE AAA6USA ZBO 1R K

DE AAA9USA QRU K

AAA4USA having missed his place in the
callup waits until the roster has been
completed and then transmits:

WAR DE AAA4USA QRU K

If AAA4USA had not responded, WAR
would have initiated a single call to
AAA4USA.

(4) The NCS acknowledges the transmissions of the net stations.

WAR/A DE WAR II R II AAA2USA IP II
AAA3USA 1R AAA6USA II AAA4USA QRU
II AAA5USA 1R II IP AAA2USA II AAA6USA
1R II AAA9USA QRU AR WAR then pauses 5
seconds to allow any other stations to call and to
allow the ANCS or other net stations to inform
the NCS of any errors or omissions in the callup
or traffic reports.

(5) WAR, not hearing any station call, proceeds to
direct the clearance of traffic in order of precedence
highest to lowest. WAR transmits:

AAA2USA DE WAR ZOG P K

(Meaning, "TRANSMIT YOUR MESSAGE
TO THIS STATION.")

AAA2USA hearing WAR's transmission,
transmits:

WAR DE AAA2USA NR1 II P 112025Z
 APR 84 II FM AAA2USA NJ II
 TO WAR GR10 BT TEXT BT K
 WAR having received the message
 transmits;
 AAA2USA DE WAR R NR1 AR
 WAR pause 5 seconds and not hearing
 anything, transmits;
 AAA5U3A DE WAR ZOG 1P AAA2USA
 AR
 AAA2USA HEARING WAR's
 transmission, transmits:
 AAA5USA DE AAA2USA K
 If AAA2USA had not called AAA5USA,
 AAA5USA would have had to call
 AAA2USA to see if AAA2USA was
 ready to copy his or her message.
 AAA5USA receiving AAA2USA
 invitation, transmits:
 AAA2USA DE AAA5USA NR5 P II
 012345Z
 APR 84 II FM BT TEXT. .
 BT K
 AAR2USA after requesting and receiving
 missed words, transmits:
 AAA5USA DE AAA2USA R NR 5 AR
 After a 5-second pause WAR transmits:
 AAA6USA DE WAR ZOG I R K
 (Meaning, "TRANSMIT YOUR
 MESSAGE TO THIS STATION")
 AAA6USA transmits:
 WAR DE AAA6USA NR7 R II 120456Z
 APR 84 II FM BT TEXT. .
 BT
 Upon completion of AAA6USA
 transmission, WAR transmits:
 AAA6USA DE WAR R NR7 AR
 After a 5-second pause WAR proceeds:
 AAA3USA DE WAR ZOG 1 R
 AAA6USA AR
 AAA6USA hearing WAR transmits:
 AAA3USA DE AAA6USA K
 AAA3USA hearing AAA6USA's
 invitation transmits:
 AAA6USA DE AAA3USA NR 4 R II
 050607Z
 APR 84 FM BT TEXT BT K
 AAA8USA having received the message
 transmits:
 AAA3USA DE AAA6USA R NR4 AR
 WAR pauses for 5 seconds and
 AAA9USA having received an original
 priority message for transmission
 transmits:
 WAR DE AAA9USA ZBO 1 P
 AAA6USA K

WAR transmits:
 AAA9USA DE WAR ZOG 1 P
 AAA6USA
 AR
 AAA6USA hearing the last two
 transmissions, transmits:
 AAA9USA DE AAA6USA K
 AAA9USA hearing AAA6USA
 invitation, transmits;
 AAA6USA DE AAA9USA NR8 P II
 121450Z
 APR 84 II FM BT K
 AAA6USA having requested and received
 missed words transmits:
 AAA9USA DE AAA6USA R NR8 AR
 WAR pauses 5 seconds then proceeds:
 AAA5USA DE WAR ZOG 1 R
 AAA2USA AR
 AAA2USA hearing WAR transmits:
 AAA5USA DE AAA2USA K
 AAA5USA hearing AAA2USA's
 invitation transmits:
 AAA2USA DE AAA5USA NR3 R II
 120340Z
 APR 84 II FM BT T E X T.
 . B T K
 AAA2USA having received the message
 transmits:

AAA5USA DE AAA2USA R NR3 AR Traffic is
 now cleared and the NCS will proceed to other business.

(6) In response to the initial net call, stations
 reporting into the net will indicate traffic or no traffic.
 Stations entering a net already in progress will include a
 traffic report at the time they enter the net. However,
 when an NCS desires to obtain information about the
 traffic to be transmitted, the operating signal "INT
 ZBO," meaning "OF WHAT PRECEDENCE AND FOR
 WHOM ARE YOUR MESSAGES?," will be
 transmitted. The station(s) called will answer using the
 operating signal "ZBO" meaning, "HAVE (or
 .has). . . , . . . (number followed by precedence
 prosign of each precedence message(s)) for you (or for
)."

EXAMPLE: (WAR transmits):

 AAA4USA DE WAR INT ZBO K
 AAA4USA has two priority messages for
 WAR, two routine messages for
 AAA5IJS, and one routine for New
 York City. AAA4USA transmits):
 DE AAA4USA ZBO 2P II 2R
 AAA5USA II IE NEW YORK CITY K
 (If WAR could not hear AAA4USA and
 requested AAA2USA to relay
 AAA4USA traffic report, WAR would
 transmit):

AAA2USA DE WAR INT ZOD
 AAA4USA K
 (AAA2USA having heard AAA4USA
 respond to WAR original request
 transmits):
 DE AAA2USA II AAA4USA ZBO 2P II
 2R AAA5USA II 1R NEW YORK CITY
 K (WAR transmits); DW WAR R AR

d. Closing the net. At the end of the time allotted
 for the net, WAR will transmit:
 WAR DE WAR QRU ZKJ 1/2 AR
 When the net is closed at the end of the allocated period,
 no further transmissions, including on the air tuning, will
 be made on the frequency.

13-10, Procedure signs (prosigns),

a. Examples of prosigns used in radiotelegraph are
 shown in (1) through (3) below.

(1) If the delay will be a few seconds only, the
 prosign "AS" will be made after the identity of the
 responding station. This transmission must then be
 followed with a few seconds by a normal response.

EXAMPLE: AAA4USA DE AAA2USA AS after a
 few seconds DE AAA2USA K

(2) The prosign "AS" followed by prosign "AR"
 means "YOU ARE TO WAIT" or "I AM OBLIGED TO
 WAIT." But not over 1 minute. It applies only to and
 between the stations designated in the call.

EXAMPLE: AAA4USA DE AAA2USA AS AR

(3) The prosign "AS" followed by a numeral and the
 prosign "AR." indicates number of minutes of "WAIT."

EXAMPLE: AAA4USA DE AAA2USA AS 5 AR

b. Words twice (OPERATING SIGNAL QSZ). During
 periods of poor or difficult reception conditions the
 receiving operator may request to have messages sent
 "WORDS TWICE." This is accomplished using the
 operating signal "QSZ."

EXAMPLE: AAA2USA DE WAR QSZ K
 AAA2USA transmits:
 WAR WAR DE DE AAA2USA
 AAA2USA NR 3 NR 3 II RR II
 141418Z 141418Z JUL JUL 84 84 BT
 BT
 UNCLAS UNCLAS YOUR
 YOUR 131313Z 131313Z APR APR 84
 84 APPROVED APPROVED BT BT K
 K

c. During transmission corrections are made in

accordance with paragraph 13-7j. After transmission and
 before receipting for a message, the receiving station
 operator should check for a complete and accurate copy.
 If incomplete copy, either (1) or (2) below should be
 accomplished.

(1) Request repetitions of the whole message or
 portions thereof by means of prosign "IMI," with
 identifying data as shown in paragraph 13-11.

(2) Assume the following message has been
 transmitted:

AAA2USA DE WAR NR 8 II P II
 231426Z AUG 84 II FM AAA3USA II
 TO AAA4USA II
 INFO AAA5USA
 BT
 MY 222300Z AUG 84 PROCEED ON
 ASSIGNED MOVEMENTS. MAKE
 REPORTS OF CIRCUIT 03A.
 BT K

EXAMPLE 1: (AAA2USA missed the entire message.
 He or she transmits):

WAR DE AAA2USA IMI K
 (WAR replies)
 AAA2USA DE WAR (transmits entire
 message as previously transmitted)
 EXAMPLE 2: (AAA2USA misses the
 action addressee and the word "circuit."
 He or she transmits): WAR DE
 AAA2USA IMI TO TO INFO II WB 03A
 K
 (WAR replies)
 AAA2USA DE WAR TO TO INFO TO
 AAA4USA II INFO II WB 03A II
 CIRCUIT K (AAA2USA then receipts
 for the message.)

EXAMPLE 3: (AAA2USA questions his or her
 doubtful receipt of the word "proceed."
 He or she transmits); WAR DE
 AAA2USA INT WA 84
 II PROCEED K II

(WAR replies)
 AAA2USA DE WAR C K (AAA2USA
 then receipts for the message.)

d. Requests for repetition after receipt. After receipt
 has been obtained for a message, all requests for
 repetition must be in the form of a new message. When
 in direct communication, this may be accomplished by
 use of an abbreviated service message. If not in direct
 communication, a service message must be used.

EXAMPLE 1: (Repetition of completed message required.)
AAA3USA DE AAA4USA INT ZDK
231426Z AUG 84 K

EXAMPLE 2: (Repetition of portion of message required.)
AAA4USA DE AAA5USA II R BT
INT ZDK 231426Z AUG 84 II WA
MY BT 1500Z K (AAA4USA replies):
AAA5USA DE AAA4USA ZDK
231426Z AUG 84 II WA MY II
222300Z AUG 84 K

e. Verifications.

(1) The prosign "J" means "VERIFY WITH ORIGINATOR AND REPEAT." Requests for verification are initiated by addressees only, and may be made in any of the forms described in paragraph 10-7.

(a) The prosign "J" may be used to request verification of messages or portions thereof using the identification data as shown in paragraph 13-12.

(b) Prosign "J" after a call without identifying data means "VERIFY WITH ORIGINATOR AND REPEAT YOUR LAST MESSAGE."

(c) Prosign "J" after a call and followed by identification data means "VERIFY WITH ORIGINATOR AND REPEAT MESSAGE OR PORTION THEREOF AS INDICATED."

(2) The answer to a request for a verification will be initiated by the originator of the original message, and will take the form of prosign "C," followed by the original identifying data (as shown in the request for verification) followed by the portion requested, except when a cancellation is initiated.

EXAMPLE: (AAA2USA requires a verification of the entire message shown in paragraph 13-11c(2). AAA2USA transmits):

WAR DE AAA2USA II T II R II
231550Z AUG 84 II FM
AAA2USA II TO AAA3USA BT
J 231426Z AUG 84 BT K

(NOTE: AAA2USA is not in direct communication with the originator, therefore, relay instructions must be included.)

(The response for AAA2USA as relayed by WAR would be):

AAA2USA DE WAR II R II 231604Z
AUG 84 II FM
AAA3USA II TO AAA2USA BT C
231426Z AUG 84 II R II 231426Z
AUG 84 II FM AAA3USA TO

(remainder of message as previously transmitted.)
(AAA2USA requires a verification of the time quoted in the text.)

WAR DE AAA2USA II T II R II
231607Z AUG 84 II FM
AAA2USA II TO AAA3USA BT J
231426Z AUG 84 II WA MY BT K
(Response from AAA3USA as relayed by WAR):

AAA2USA DE WAR II R II 231610Z
AUG 84 II FM AAA3USA II TO
AAA2USA BT C 231426Z AUG 84 II
WA MY II 222300Z AUG 84 BT K
(NOTE: Had the original message contained an error, a correction would be issued to all of the original addressees by the originator.)

13 - 11. Identification of messages.

a. Messages are identified by DTG. Additional data such as transmission identity, designator of the originating station, group count, preamble and partial or complete text may be used when absolutely necessary. In all cases, the data used to identify a message must be as brief as possible, consistent with positive identification.

b. The following rules for identifying parts or portions of messages will be followed:

(1) The heading—identify by:

- (a) All before prosign.
- (b) Prosign to prosign.
- (c) All after prosign.

(2) Plain language text—identify by:

- (a) BT to word.
- (b) Word to word.
- (c) Prosign WA and/or WB.
- (d) Prosign AA and/or AB.
- (e) BT to BT.
- (f) Word to BT.

(NOTE: When a word appears more than once, that word used as an identity applies to the first appearance. Reference to subsequent appearance of the word must be further identified by means of adjacent words.)

c. Sequence of dealing with queries. When a message is transmitted to a multiple or collective call, stations will answer in sequence. However should any station have any questions concerning the message, this will be dealt with by the transmitting station and a receipt obtained from the receiving station concerned before the next station in sequence transmits.

13-12. Acknowledgements.

a. An acknowledgement is a communication to the originator of a message indicating that the message to which it refers has been received and the purpose is understood by the addressee. An acknowledgement request is directed to the addressee and should not be confused with receipting following a message with the prosign "R."

b. Instructions to acknowledge mean "AN ACKNOWLEDGEMENT OF THIS MESSAGE (OR MESSAGE INDICATED), WHEN UNDERSTOOD, IS REQUIRED."

c. Message acknowledgement will be made only—

(1) When specifically requested by the word(s) "ACKNOWLEDGE" or "ACKNOWLEDGE IMMEDIATELY" appearing as the last word(s) of the text.

(2) When requested by a separate message. A separate message is used when the need for an acknowledgement is determined after release of the original message. Service messages will not be used for this purpose. Abbreviated service messages may be used in emergency nets to request acknowledgements:

EXAMPLE: AAA2UA DE WAR INT ZEV

1218Z K

(AAA2USA transmits):

DE AAA2USA AS AR

(Subsequently, AAA2USA acknowledges):

WAR DE AAA2USA ZEV 1218Z AR

(3) On other than emergency nets, the acknowledgement of a message will be composed as follows:

(a) The word "YOUR" or the address designator actually used to represent the originator.

(b) The message reference (DTG, reference number, and so forth).

(c) The word "ACKNOWLEDGE."

EXAMPLE: YOUR 121314Z OCT 84
ACKNOWLEDGED

13 - 13. Cancellations.

During transmission, cancellations are in accordance with paragraph 13-7k. To cancel a message after receipt has been obtained, a new message, authorized by the message originator, must be sent. Cancellations of transmissions may be accomplished by the transmitting station by use of the appropriate operating signals or prosigns. A station cancelling a transmission is

responsible for further handling of the contents of the transmission if further handling is required or notification to the message originator of its status, request for cancellation, or disposition.

13-14. Message group count.

a. In messages a sequence of characters not interrupted by a space is counted as one group.

b. Checking group count is accomplished as follows:

(1) The prosign "GR" preceded by the prosign "INT" and followed by a numeral means "IS THE NUMBER OF GROUPS AS INDICATED CORRECT?" When the number or groups received does not correspond with the group count transmitted, the receiving station will immediately question the transmitting station by using prosign "INT GR" followed by a numeral.

EXAMPLE: AAA6USA DE AAA9USA INT GR8 K

(After checking the message, the transmitting station finds that the receiving station is correct and transmits):

AAA9USA DE AAA6USA C K

(2) In all messages where groups are counted and the count is 50 or less, if the transmitting station considers the receiving station to be incorrect after it has questioned the group count, the transmitting station replies with the prosign "N," repeats the original group count followed by the prosign "BT," and then transmits the first character of each word or group in the text in succession, and then repeats the prosign "BT" in its proper position.

EXAMPLE: (AAA6USA transmits a message to

AAA9USA)

AAA9USA DE AAA6USA NR1 II R
II 272113Z OCT 84 GRIO BT

XTUMR BTRFH UBTXM
BCRST WHIHK

BBCOT WXCTI SRTXR OCUVC
WHISC

BT K

(AAA9USA questions the group count.)

AAA6USA DE AAA9USA INT
GR11 K

(AAA6USA checks and finds the group count correct as transmitted, then transmits):

AAA9USA DE AAA6USA II N II
GR10

B T X B U B W B W S O W BT
K

(3) In all messages with a group exceeding 50 groups, if the receiving station is considered to be incorrect, the transmitting station replies with prosign "N," repeats the original group count and transmits the identity of the 1st, 11th, and every subsequent 10th group followed by the letter of that group (The identity of the group will be separated from the initial letter of that group by a separative sign)

EXAMPLE: (AAA5USA transmits a message containing 76 groups to AAA3USA. AAA3USA questions the group count)
AAA5USA DE AAA3USA INT GR 75

(AAA5USA checks and finds the group count correct as transmitted, then transmits):

AAA3USA DE AAA5USA II N
II GR 76 BT I II D
11 II L 21 II H 31 II I 41 II Q
51 II M 61 II W 71 II A BT K

4) In the case of long messages sent in parts, the number of groups sent is to be checked at the end of each part. For parts of 100, see b(3) above; for final part if 50 or less, see b(2) above.

13-15 Break-in-procedure

a. Break-in procedure is the method whereby a receiving station may interrupt a transmission to request the transmitting station to wait, shift frequency repeat and so forth. Break-in procedure will not be used to obtain repetitions when more than one station is involved in the reception of a message.

b. The receiving station desiring to break-in on a transmission makes a series of dashes. When the transmitting operator hears these dashes, the operator stops transmitting to ascertain the reason for the break-in. If three attempts to break-in are unsuccessful, the receiving operator will cease attempts to break-in until the transmission in progress is completed.

c. The station breaking in to request an immediate repetition may omit the preliminary call before transmission of the last word or group received correctly when receiving conditions are good and no confusion will result. The transmitting station then commences transmission with the last word, group or prosign which the receiving station indicated it has correctly received.

EXAMPLE: AAA9USA is transmitting to AAA4USA
AND WILL PROCEED

IMMEDIATELY
(AAA4USA missed the word IMMEDIATELY. AAA4USA transmits

WA PROCEED
AAA9USA then transmits):
PROCEED IMMEDIATELY (and so forth)

d. A full or an abbreviated call must be employed when the interruption is for reason other than to request immediate repetition of a missed word.

EXAMPLE: (AAASUSA is transmitting to AAASUSA)

AND WILL PROCEED IMMEDIATELY

(AAASUSA has trouble with the receiver and desires AAA6IUSA to wait. He or she transmits):

AAA6USA DE AAA3USA

AAA6USA DE AAA3USA AS
(When ready to receive AAA3USA transmits): DE AAA3USA IMI AA PROCEED K

(AAA6USA then transmits): DE AA6USA AA PROCEED II IMMEDIATELY (and so forth)

e. Any station may break in on a transmission in order to transmit a message of higher precedence under the conditions specified in this paragraph. If in a directed net, NCS permission must be obtained before transmitting a message.

EXAMPLE: (WAR is transmitting along routine message to AAA6USA and has completed only a portion of the text. AAA2USA has been handed a priority message for transmission and desires to break in. WAR is transmitting): AND IS THE TIME FOR (AAA2USA transmits a series of dashes):

(Upon hearing the succession of dashes, WAR immediately ceases transmitting. AAA2USA transmits):

WAR DE AAA2USA P K

AAA2USA DE WAR K

AAA2USA send the message

(After AAA2USA has completed the higher precedence message WAR resumes transmitting to AAA6USA)

13-16. Method of synchronizing time.

a. If an accurate time check is desired, it will be requested by using the operational signal "ZUA" preceded by the prosign "INT" and followed by a four-digit time group. Time checks will always be given in Zulu time.

EXAMPLE: (AAASUSA desires time check at 2345 (Z)
(AAASUSA transmits):
WAR DE AAASUSA INT ZUA
2345ZK
(WAR transmits):
AAASUSA DE WAR ZUA 2345Z
(pauses) then sends 5-second
dash K (The end of the dash Is

the indicated time.)

WAR DE AAA5USA R AR

b. The intention to transmit a time signal may be conveyed by an abbreviated service message employing the operating signal "ZUJ," which means STANDBY followed by the operating signal "ZUA." Time will be synchronized based on the transmitted time signal.

EXAMPLE: (WAR informs AAA9USA that a time
signal will be transmitted at 0015Z)
A A A 9 U S A W A R Z U J Z U A
0015Z AR
(At the prescribed time he or she
transmits):
A A A 9 U S A D E W A R Z U A
(pauses) then sends 5-second dash
AR

CHAPTER 14

RADIOTELEPHONE PROCEDURES

14-1. General.

This chapter contains specific instructions for the conduct of communications employing radiotelephone. Adherence to prescribed procedures is mandatory. Unauthorized departures from or variations in prescribed procedure often create confusion and reduce reliability and speed. If the procedure prescribed herein does not cover a specific operating requirement, initiative and common sense should prevail. The basic rules essential to circuit discipline, good transmission practices, and practices to be avoided are outlined in paragraph 9-2. When authentication is required it will be accomplished in accordance with paragraph 9-22.

14-2. Reliability.

d. Reliability is the paramount consideration in military communication, always being more important than speed. Accuracy of transmission directly affects reliability and is enhanced by employing correct enunciation, pronunciation, and cadence.

e. Operators will transmit messages exactly as written. Prowords, operating signals, or abbreviations will not be substituted for text words or groups as written by the originator.

f. An operator who receipts for a message is responsible for the delivery or further relay of that message. Any delay or nondelivery of a message will be reported to the originator.

g. Speed of transmission may be attained primarily by employing standard phraseology and authorized prowords. Operators will transmit only as fast as the receiving operator can record the message. In net operations, the speed of the slowest operator will normally govern the speed of all transmissions in the net. This rule may be suspended by an NCS when practical reasons dictate higher speeds.

14-3. Prowords, operating signals, and phonetics.

a. Prowords. Prowords are pronounceable words or phrases which have been assigned meanings for the purpose of expediting message handling on nets where radiotelephone procedure is used. In no case will a proword or a combination of prowords be substituted for

any portion of message text. The list of prowords contained in appendix E are authorized for general radiotelephone use.

b. Operating signals. Operating signals "Q" and "Z" are not designed for radiotelephone transmissions. In radiotelephone procedure the operating information will be conveyed by concise phrases. When it is necessary to relay operating signals over radiotelephone nets, they will be transmitted by using their phonetic equivalent.

c. Phonetic alphabet. When necessary to identify any letter of the alphabet, the standard phonetic alphabet will be used. Difficult words or groups within the text of plain text messages may be spelled using the phonetic alphabet preceded by the proword "I SPELL." A single letter will be identified phonetically preceded by the proword "INITIAL." If the operator can pronounce the word to be spelled, he or she will do so before and after the spelling to identify the word. Where a text is composed of pronounceable words, they will be spoken as such.

EXAMPLE A: CATENARY - "I SPELL CHARLIE
ALFA NOVEMBER TANGO ECHO
NOVEMBER ALFA ROMEO
YANKEE - CATENARY."

EXAMPLE B: RENDEZVOUS POINT IS UNIMAK
- "I SPELL UNIFORM NOVEMBER
INDIA MIKE ALFA KILO-
UNIMAK."

EXAMPLE C: C W TRANSPORT IS - "INITIAL
CHARLIE INITIAL WHISKEY
TRANSPORT."

(1) Encrypted or enciphered groups will always be transmitted by the phonetic equivalents of the individual letters without using the proword, "I SPELL."

EXAMPLE: (The encrypted group LUXOW will be counted as one group.)

EXAMPLE: "LUXOW - LIMA UNIFORM X-RAY
OSCAR WHISKEY"

(2) The phonetic equivalents contained in table 9-1 are the only phonetics authorized for use in MARS operation. Pronunciation should be as indicated.

d. Pronunciation of numerals and words. To distinguish between numerals and words similarly pronounced, the proword "FIGURE(S)" must precede such numbers.

Numbers will be transmitted digit by digit except that exact multiples of hundreds or thousands may be spoken as such. The proword "PIGURE(S)" will precede the numerals to be transmitted except in conjunction with the prowords "NUMBER," "TIME," "GROUPS," and in the "HEADING OF THE MESSAGE" and in the phonetic pronunciation of a call sign. The numerals pronunciation chart is contained in table 9-1. The decimal point is to be spoken as "DAY-SEE-MAL." Dates will be spoken digit by digit, with the months in full. Roman numerals will be transmitted as the corresponding Arabic numerals preceded by the word "ROMAN."

EXAMPLE A: 123.4 is to be spoken as "WUN TOO THUH-REE DAY-SEE-MAL FOW-ER."

EXAMPLE B; "20 AUG" is spoken as "TOO ZE-RO AUGUST"

EXAMPLE C: "VII" is spoken as "ROMAN SEVEN"

e. Abbreviations in the text. Abbreviations in the text are transmitted as follows:

j. Initials used alone or in conjunction with short titles will be spoken phonetically each preceded by the proword "INITIAL."

k. Personal initials will be spoken phonetically prefixed by the word "INITIAL."

l. Abbreviations frequently used in normal speech may be used in the same manner when transmitted by voice.

EXAMPLE A: "Paragraph A" will be spoken as "Paragraph INITIAL Alfa"

EXAMPLE B: "ACP" will be spoken as "Alfa Charlie Papa"

EXAMPLE C: "G. M. SMITH" will be spoken as "INITIAL Golf INITIAL Mike Smith"

EXAMPLE D: "NATO" may be spoken as "NATO"
"CONUS may be spoken as
"CONUS"
"U.S. ARMY" may be spoken as
"U.S. Army"

14-4. The Call.

a. The preliminary call transmitted to establish communications between stations or to open a net may be a single, net (collective), or multiple call as outlined below. When net calls are used, the proword "EXEMPT" may be used to indicate station(s) not required to enter the net. Under difficult conditions the call signs may be transmitted twice. Two or more call signs transmitted as

part of the call or appearing in message components will normally be arranged in alphabetical, numerical order. Before transmitting, an operator will listen to ensure that his or her transmission will not cause interference to transmissions in progress. (NOTE: For purposes of illustration the examples in this paragraph will use full phonetic spelling. In succeeding paragraphs phonetic spelling has been omitted from all examples for simplicity of text.)

a. A single call consists of a call sign which represents the called station (AAR2USA), the proword "THIS IS," the call sign of the calling station (AAR3USA), and the proword "OVER."

EXAMPLE A: Conditions good:

ALFA ALFA ROMEO TWO
UNIFORM SIERRA ALFA THIS IS
ALFA
ALFA ROMEO THREE UNIFORM
SIERRA ALFA OVER.

EXAMPLE B: Conditions difficult:

WHISKEY ALFA ROMEO SLANT
ALFA WHISKEY ALFA ROMEO
SLANT ALFA THIS IS WHISKEY
ALFA ROMEO OVER

EXAMPLE C: Net call with exempted station:

WHISKEY ALFA ROMEO SLANT
ALFA EXEMPT ALFA ALFA
ROMEO TWO UNIFORM SIERRA
ALFA THIS IS WHISKEY ALFA
ROMEO OVER

b. A multiple call consists of the call signs of two or more called stations (AAR2USA, AAR3USA), and the proword "THIS IS," the call sign of the calling station (WAR), and the proword "OVER."

EXAMPLE: Using two called stations:

ALFA ALFA ROMEO TWO
UNIFORM SIERRA ALFA ALFA
ALFA
ROMEO THREE UNIFORM SI- E
R R A ALFA. THIS IS WHISKEY

" ALFA ROMEO OVER

14-5. Answering call,

a. An answer to a preliminary call consists of the call sign of the calling station, the proword "THIS IS," the call sign of the answering station, and the proword "OVER." When answering in response to a net or multiple call, stations will answer in alphanumerical order, in other NCS prearranged order, or in the order indicated in the multiple call.

EXAMPLE: Answer to a single call (AAR3USA has called AAR2USA):

AAR3USA THIS IS AAR2USA
OVER

Answer to a net or multiple call
(WAR has called net): WAR THIS IS
AAR2USA OVER WAR THIS IS
AAR3USA OVER WAR THIS IS
AAR4USA OVER

(a) When confusion will not result and when authorized by the NCS, the answer may be abbreviated by omitting the call sign of the calling station,

EXAMPLE: WAR has called net (abbreviated answer):

THIS IS AAR2USA OVER
THIS IS AAR3USA OVER
THIS IS AAR4USA OVER

(b) If any station fails to answer in proper sequence when a net or multiple call is employed, the next station waits 5 seconds and answers. The station which fails to answer in proper order must wait until all other stations have answered or have had time to answer.

(c) When a station hears a call without being certain that the call is intended for it, the station will not answer until the call has been repeated and is understood. When a station is called, but is uncertain of the call sign of the calling station, it will answer immediately by transmitting the proword "UNKNOWN STATION" followed by the proword "THIS IS," its own call sign, and the proword "OVER."

14—6. Transmission endings.

(a) When an answer is requested or expected, transmissions will end with the proword "OVER." When an answer is not requested or expected, transmissions will end with the proword "OUT" and without iteration of station's call sign (this is AAV9QB (OVER OR OUT)).

(b) Every transmission will end with only the proword "OVER" or "OUT." No other ending, such as, "GO," "GO AHEAD," "BACK TO YOU," and so forth, will be used to end MARS transmissions.

(c) The Prowords "OVER" and "OUT" are never used together to end a transmission.

14-7. Operating rules.

a. General. To use circuit time more efficiently, all messages or their substance should be written down prior to transmission. Those messages which must be delivered by the receiving operator to another person or which are preceded by the proword "MESSAGE FOLLOWS" will be written down. Transmission by

radiotelephone will be as concise as possible without sacrificing clarity. The use of standard phraseology enhances brevity. Transmissions over radiotelephone should be clear with natural emphasis on each word except the prescribed pronunciation of a numeral; transmissions should be spoken in natural phrases, not word by word. The operator will, during the transmission of a message, pause after each natural phrase and interrupt the transmission, momentarily, to another station to break in if necessary. To avoid interfering with other traffic, an operator will listen to make certain that a net is clear before making any transmissions.

(a) When it is necessary for a station to initiate test signals, either for the adjustment of a transmitter before making a call, or for the adjustment of a receiver, such signals will not continue for more than 10 seconds, and will be composed of spoken numerals (1, 2, 3, 4, and so forth) followed by the call sign of the station transmitting the signals and followed by the proword "OUT."

(b) After initial contact has been made between two stations, call signs will be used at the beginning of each transmission.

14-8. Establishing a net.

(a) The use of procedures prescribed below may be followed either when opening a net for the first time or when reopening a net. Proper control by the NCS and adherence to operating rules by stations enable the net to begin and maintain an exchange of traffic with minimum delay.

(b) Abbreviated calls will not be used in establishing communications. Before conducting regular traffic over radiotelephone nets, it may be necessary to make contact with the other stations involved to ascertain that communications is possible. Call signs in message headings will ordinarily be arranged in alphanumerical order in the format in which they are to be transmitted, that is, as shown in the TO and INFO lines.

(c) At the designated time or when ready to establish the net, the NCS transmits:

WAR/A This is WAR a long count and a 50-second clear period for adjusting equipment follows, WUN TOO THREE FOW-ER FI-YIV SIX SEVEN ATE NIN-ER ZE-RO "OUT" (50-second clear period - during this 50-second period net members will ensure their equipment is operating on frequency, within tolerances, and readjust their system on the air is necessary). Equipment should have been prewarmed at least 15 minutes before start of net and transmitted pretuned using a dummy antenna load. WAS transmits WAR/A THIS

IS WAR "OVER"

Each subordinate station then answers the call in alphanumeric order:

THIS IS AAK2U3A NO TRAFFIC OVER

THIS IS AAR3U8A NO TRAFFIC OVER

THIS IS AAR4USA NO TRAFFIC OVER

THIS IS AAR5USA NO TRAFFIC OVER

THIS IS AAR6USA NO TRAFFIC OVER

THIS IS AAE9USA NO TRAFFIC OVER

WAR now calls the net to inform all stations that their transmissions have been heard and that he or she has no traffic for them; WAR/A THIS IS WAR ROGER AAR2USA, A A R 3 U S A , A A R 4 U S A , A A R 5 U S A , AAR6USA, AAR9USA OUT

(1) Once the net has been established and the NCS desires to authorize the use of abbreviated call signs, the net control transmits:

WAR/A THIS IS WAR ABBREVIATED CALL SIGNS

AUTHORIZED OUT

(2) When a subordinate station is unable to answer a net call, the next station will wait five seconds and then transmit.

WAR transmits:

WAR/A THIS IS WAR OVER

Stations respond:

THIS IS AAR2IJS A NO TRAFFIC OVER

THIS IS AAR3USA NO TRAFFIC OVER

THIS IS AAR4USA NO TRAFFIC OVER

A A R 6 U S A hearing no answer from AAR5USA, waits approximately 5 seconds and then transmits:

THIS IS AAR6USA NO TRAFFIC OVER

THIS IS AAR9USA NO TRAFFIC OVER After receiving answers from all stations

except AAR5USA, WAR transmits:

WAR/A THIS IS WAR ROGER AAR2USA,

AAA3USA, AAA4USA, AAA6USA OUT

AAR5USA when able to transmit, calls the NCS to report into the net:

WAR THIS IS AAR5USE NO TRAFFIC OVER

WAR having no traffic for AAR5USA transmits:

AAR5USA THIS IS WAR ROGER OUT

14-9. Changing a net frequency.

a. WAR wishes to establish a net on a new frequency. At the time arranged for establishing

communications, WAS listens on the frequency and finding it clear transmits:

WAR/A THIS IS WAR LONG COUNT FOLLOWS
1-2-3-4-5-6-7-8-9-0 WAR/A THIS IS WAR "OUT."

b. After allowing time for stations to adjust their transmitters WAR will establish communications in the normal manner. If any station is off frequency, it will be given further tuning as follows:

AAR2USA THIS IS WAR YOU ARE ONE
KILOHERTZ LOW

ONE TWO THREE FOUR FIVE OVER

(AAR2USA adjusts frequency during count)

(AAR2USA transmits):

WAR THIS IS AAR2USA OVER

(WAR transmits):

AAR2USA THIS IS WAR OUT

14-10. Net operations.

a. Net operations will always be directed unless indicated otherwise by the NCS. When it is required to change a directed net to a free net, or vice versa, one of the following phrases will be used by the NCS: "THIS IS A DIRECTED NET" "THIS IS A FREE NET" "THE NET IS FREE" "THE NET IS INFORMAL,"

b. The following example illustrates the manner in which the NCS requests the amount and precedence of traffic to be transmitted:

EXAMPLE: WAR/A THIS IS WAR

WHAT PRECEDENCE AND FOR WHOM
ARE YOUR MESSAGES OVER

(Each subordinate station answers in numerical order of his or her full call sign indicating traffic onhand)

THIS IS AAR2USA ONE ROUTINE OVER

THIS IS AAR3USA NO TRAFFIC OVER

THIS IS AAR4USA ONE PRIORITY AAR

NINE USA OVER THIS IS AAR5USA

ONE ROUTINE AAR TWO USA OVER

THIS IS AAR6USA NO TRAFFIC OVER

THIS IS AAR9USA NO TRAFFIC OVER

WAR rogers the transmissions by saying:

WAR/A THIS IS WAR ROGER AAR TWO

USA ONE ROUTINE, AAR THREE USA,

AAR FOUR USA ONE PRIORITY AAR

NINE USA, AAR FIVE USA ONE

ROUTINE AAR TWO USA, AAR SIX

USA, AAR NINE USA OUT

(It is not necessary for WAR to say NO TRAFFIC for the stations which listed no traffic.)

c. WAR now commences to clear traffic in order of precedence.

THIS IS WAR, AAR FOUR USA
SEND ONE PRIORITY TO AAR
NINE USA OUT AAR9USA hearing
the previous transmission transmits:
THIS IS AAR NINE USA OVER
AAR4USA proceeds to send his
priority to AAR9USA. (Failure of
AAR 9USA making a preliminary
call to AAR4USA would necessitate
AAR4USA making a preliminary call
to AAR9USA). After AAR9USA
receipts for the traffic and transmits
the proword "OUT," WAR would
then authorize stations to transmit
their traffic in order of precedence
until traffic listed is cleared.

14-11. Radio checks, signal strength, and readability.

a. A station Is understood to have good signal strength and readability unless otherwise notified. Strength of signals and readability will not be exchanged unless one station cannot clearly hear another station.

b. A station that wishes to inform another of signal strength and readability will do so by means of a concise report of actual reception such as "Weak, but readable," "Loud, but distorted," "Weak, with interference," and so forth. Reports such as "Five by five," "Four by four," "Lima Charlie," will not be used to indicate strength and quality of reception

EXAMPLE: (Assuming abbreviated call signs are in use.)

(WAR desires a radio check with
WAR/A and transmits): WAR/A
THIS IS WAR RADIO CHECK
OVER

(All stations in the net hear WAR
loud and clear except R3USA and
R4USA. The replies of each station,
in order are): THIS IS R2USA
LOUD AND CLEAR OVER
THIS IS R3USA WEAK BUT
READABLE OVER THIS IS
R4USA WEAK WITH
INTERFERENCE OVER
THIS IS R5USA LOUD AND
CLEAR OVER
THIS IS R6USA LOUD AND
CLEAR OVER

THIS IS R9USA LOUD AND
CLEAR OVER

(WAR indicates his or her reception
of each of the called stations as loud
and clear, except for R3USA, who is
distorted and R4USA who was not
heard, by replying):

THIS IS WAR ROGER. R3USA
DISTORTED R4USA NOTHING
HEARD OUT

(In the event WAR heard all stations
loud and clear the reply would be):

THIS IS WAR ROGER OUT

14-12. Transmitting a message.

a. Communications good. When communication reception is satisfactory, message parts need be transmitted only once; preliminary calls are optional, Table 14-1 provides radiotelephone message format.

EXAMPLE: (WAR has one priority for AAR6USA

and transmits): R SIX USA THIS IS
WAR ONE PRIORITY OVER
(AAR6USA ready to receive traffic
transmits):

THIS IS R SIX USA OVER (WAR
transmits) THIS IS WAR NUMBER
ONE FIVE THREE PRIORITY
TIME ONE ONE TWO ZERO
THREE FIVE ZULU MAY EIGHT
FOUR FROM AAR4USA TO
AAR6USA GROUPS ONE FIVE
BREAK (TEXT) BREAK OVER
(AAR6USA missed the transmission,
transmits):

THIS IS R SIX USA SAY AGAIN
OVER

(WAR repeats the entire transmission
again.)

(AAR6USA having received the
transmission satisfactorily,
transmits):

THIS IS R SIX USA ROGER
NUMBER ONE FIVE THREE OUT

b. Communications difficult. When communications is difficult, call signs should be transmitted twice. Phrases, words, or groups may be transmitted twice and indicated by the use of the prowords "WORDS TWICE," Reception may be verified by use of the proword "READ BACK." Under such conditions preliminary calls are normally employed unless stations are on a directed net, in which case stations should have indicated they are ready to receive traffic.

EXAMPLE: (WAR transmits):

AAR2USA AAR2USA THIS IS
THIS IS WAR WAR ONE
PRIORITY ONE PRIORITY OVER
OVER
(AAR2USA transmits): WAR WAR
THIS IS THIS IS AAR2USA
AAR2USA ROGER ROGER OVER
OVER (WAR transmits): AAR2USA
AAR2USA THIS IS WAR WAR
WORDS TWICE WORDS TWICE
MESSAGE FOLLOWS MESSAGE
FOLLOWS NR101 NR101
PRIORITY PRIORITY CONVOY
HAS AR-
RIVED CONVOY HAS ARRIVED
TIME ONE SIX THREE ZERO

ZULU TIME ONE SIX THREE
ZERO ZULU OVER OVER
(AAR2USA transmits): WAR WAR
THIS IS AAR2USA AAR2USA
SAY AGAIN SAY AGAIN WORD
BEFORE HAS WORD BEFORE
HAS OVER OVER
(WAR transmits): AAR2USA
AAR2USA THIS IS WAR WAR I
SAY AGAIN I SAY AGAIN WORD
BEFORE HAS WORD BEFORE
HAS CONVOY CONVOY
OVER OVER
(AAR2USA transmits): WAR WAR
THIS IS THIS IS AAR2USA
AAR2USA ROGER NR101 ROGER
NR 101 OUT OUT

Table 14-1. Radiotelephone message format.

PART	COMPONENTS	ELEMENTS	FORMAT LINE	COMMENTS
H	Procedure	Called Stations	1	Not used.
			2	Stations call—Proword "EXEMPT," exempted calls.
E		Called Station	3	Proword "THIS IS"—station calling.
		b. Message follows		Proword "MESSAGE."
A		c. Transmission Identification	4	Proword "NUMBER" and station serial number
		d. Transmission Instructions		Prowords "RELAY TO"; "READ BACK"; "DO NOT ANSWER"; "WORDS TWICE": operating signals; call signs; plain language designators.
	Preamble	a. Precedence; DTG message instructions	5	Precedence designation; message instructions; proword "Time;" date and time expressed in digits and zone suffix; and operating signals.
I	Address	a. Originator's Sign; Originator	6	Proword "FROM." Sign; Originator Originator's address designator.
		b. Action Addressee sign	7	Proword "TO." Action addressee designator.
G		c. Information Addressee Sign; Information Addressee	8	Proword "INFO." Information addressee designators.
		d. Exempted Addressee Sign; Exempted Addressee	9	Proword "EXEMPT." Exempted addressee designators.
	Prefix	a. Accounting Information, group count	10	Accounting Symbol; group count; Proword "GROUPS (GROUP NO COUNT)."
SEPARATION			11	Proword "BREAK."
T E X T	TEXT	a. Subject Matter	12	CLEAR, UNCLASSIFIED, Proword "SERVICE," and/or internal instructions as appropriate; thoughts or ideas as expressed by the originator.

Table 14-1, Radiotelephone message format—continued.

PART	COMPONENTS	ELEMENTS	FORMAT LINE	COMMENTS
SEPARATION			13	Proword "BREAK."
E N DI N G	Procedure	a. Time Group	14	Proword "TIME." Hours and minute expressed in digit and zone suffix, when appropriate.
		b. Final Instructions	15	Prowords "WAIT, CORRECTION AUTHENTICATE, MORE TO FOLLOW," Station designators.
		c. Ending sign	16	Prowords "OVER, OUT."

14-13. Relay.

a. The proword "RELAY" used alone indicates that the station called is to relay the message to all addressees,

EXAMPLE: (Relay to an on-net station— WAR transmits):

AAR TWO USA THIS IS WAR
RELAY MESSAGE FOLLOWS NR
211 PRIORITY TIME ONE ONE
ONE THREE TWO TWO ZULU
MAY EIGHT FOUR FROM WAR
TO AAR NINE USA BREAK
(TEXT) BREAK OVER

(AAR2USA transmits): THIS IS
AAR TWO USA ROGER NR 211
OUT (AAR2USA relays the
message): AAR NINE USA THIS IS
AAR TWO USA MESSAGE FOL-
LOWS NR 421 PRIORITY TIME
ONE ONE ONE THREE TWO TWO
ZULU MAY EIGHT FOUR FROM
WAR TO AAR NINE USE BREAK
(TEXT) BREAK OVER

(AAR9USA transmits):
THIS IS AAR • NINE USA
ROGER NR 421 OUT

b. The proword "RELAY TO" followed by an address designator indicates that the station called is to relay the message to the stations indicated. When more than one station is called, the call sign of the station designated to perform

the relay will precede the proword "RELAY TO." At times it is necessary to relay a message originated on a radiotelephone circuit by some other means of communications. When such relay is necessary, it is the responsibility of the station relaying the message to place

the message in the proper format for the means of communications employed for relay. This includes use of full call signs for transmission instructions and address components.

EXAMPLE: (Relay to an off-net station— AAR3USA transmits): AAR FOUR USA AAR

SIX USA THIS IS AAR THREE
USA AAR FOUR USE RELAY
TO AAR9USA MESSAGE
FOLLOWS NR 624 ROUTINE
TIME TWO ONE ZERO EIGHT
ONE FOUR ZULU MAY EIGHT
FOUR FROM AAR3USA TO

A A R 1 USA INFO AAR4USA
AAR6USA BREAK (TEXT)
BREAK OVER (AAR4USA
transmits): THIS IS AAR FOUR
USA ROGER NR 624 OUT

(AAR6USA transmits): THIS IS R
SIX USA ROGER NR 624 OUT
(AAR4USA then leaves net and
enters net in which AAR9USA is a
member and after establishing
communications transmits):

AAR9USA THIS IS AAR4USA
MESSAGE FOLLOWS NR 146
ROUTINE TIME TWO ONE ZERO
EIGHT ONE FOUR ZULU MAY
EIGHT FOUR FROM AAR3USA
TO AAR9USA INFO AAR4USA
AAR6USA BREAK (TEXT)
BREAK OVER (AAR9USA
transmits): THIS IS AAR NINE USA
ROGER NR 148 OUT (Having
relayed the message AAR4USA
checks out of the net in which
AAR9USA is a member and reenters
his or her own net).

14-14. Repetitions.

a. When words are missed or are doubtful, repetitions will be requested by the receiving stations before receipting for the message. The proword "SAY AGAIN" used alone or in conjunction with "ALL BEFORE," "ALL AFTER," "BLANK TO BLANK," "WORD BEFORE," "WORD AFTER" will be used for this purpose. In complying with requests for repetitions, the transmitting station will identify that portion which is being repeated. In actual practice the requesting station should, to speed net operation, include in one transmission all requests for repetitions needed to complete the message.

EXAMPLE: (WAR calls the two stations for whom he or she has traffic): AAR THREE USA AAR FOUR USA THIS IS WAR MESSAGE FOLLOWS NR 268 ROUTINE TIME ZERO EIGHT ONE ZERO FIVE TWO ZULU MAY EIGHT FOUR BREAK AT (FIGURES) ONE EIGHT ZERO ZERO ZULU PROCEED HONG KONG TO ARRIVE (FIGURES) ONE SIX MAY FOR RETURN NINGPO I SPELL NOVEMBER INDIA NOVEMBER GOLF PAPA OSCAR NINGPO ACKNOWLEDGE BREAK OVER (AAR3USA having missed from "AT" to "PROCEED" transmits): THIS IS AAR THREE USA SAY AGAIN AT TO PROCEED OVER (WAR deals with the request for repetitions by AAR3USA before other stations request repetitions.) THIS IS WAR I SAY AGAIN AT TO PROCEED AT (FIGURES) ONE EIGHT ZERO ZERO ZULU PROCEED OVER (AAR3USA having now received the message satisfactorily transmits): THIS IS AAR THREE USA ROGER NR 268 OUT (Having heard AAR3USA give a receipt for the message, the next station asks for his or her repetitions. In this case AAR4USA missed from "AT"

to "PROCEED" and the word after "MAY". As AAR3USA has already asked for the phrase from "AT" to "PROCEED" and AAR4USA asks, therefore, for the word after "MAY" to be repeated).

THIS IS AAR FOUR USE SAY AGAIN WORD AFTER MAY OVER

(WAR repeats the word): THIS IS WAR I SAY AGAIN WORD AFTER MAY FOR OVER

(AAR4USA now having received the message satisfactorily transmits):

THIS IS AAR FOUR USA ROGER NR 268 OUT

b. In requesting repetitions of the heading of a message, a repetition may be requested of all that portion of the heading preceding or following a proword, or that portion of the heading between any two prowords. Requests for repetitions and replies thereto must include the nearest proword preceding and/or following the portion requested.

EXAMPLE A: (AAR5USA transmits):

AAR THREE USA THIS IS AAR FIVE USA MESSAGE FOLLOWS NR 281 PRIORITY TIME ONE EIGHT ONE SIX TWO FIVE ZULU MAY EIGHT FOUR FROM AAR5USA TO AAR3USA AAR6USA INFO AAR9USA BREAK (TEXT) BREAK OVER

(AAR3USA missed all before the addressee and transmits): THIS IS AAR THREE USA SAY AGAIN ALL BEFORE FROM OVER

(AAR5USA replies to AAR3USA):

AAR THREE USA THIS IS AAR FIVE USA I SAY AGAIN ALL BEFORE FROM AAR THREE USA THIS IS AAR FIVE USA PRIORITY TIME ONE EIGHT ONE SIX TWO FIVE ZULU MAY EIGHT FOUR OVER (AAR3USA transmits): THIS IS AAR THREE USA ROGER NR 281 OUT

EXAMPLE B: (AAR4USA transmits):

AAR NINE USA THIS IS AAR
FOUR USA MESSAGE FOLLOWS
NR 114 ROUTINE TIME TWO
FOUR ZERO NINER ONE TWO
ZULU MAY EIGHT FOUR FROM
AAR FOUR USA TO AAR NINE
USA INFO AAR SIX USA BREAK
CANCEL MY (FIGURES) TWO
THREE ONE FOUR TWO EIGHT
ZULU MAY (FIGURES) EIGHT
FOUR BREAK OVER

(AAR9USA having missed the
portion between the DTG and the
information addressee transmits):

THIS IS AAR NINE USA SAY
AGAIN TIME TO INFO OVER
(AAR4USA transmits): THIS IS
AAR FOUR USA I SAY AGAIN
TIME TO INFO TIME TWO FOUR
ZERO NINER ONE TWO ZULU
MAY EIGHT FOUR FROM AAR
FOUR USA TO AAR NINE USA
INFO OVER (AAR9USA having
received the repetition transmits):
THIS IS AAR NINE USA ROGER
NR 114 OUT

c. When it is necessary to ask for repetitions after a message has been receipted for, such requests and replies must identify the message being queried as well as the portion required. The DTG will be used for such identification.

14-15. Checking group count.

a. When a message is received and the number of groups actually received does not correspond with the group count of the message, the receiving station requests a check by transmitting the

phrase "ARE GROUPS, " stating the number of groups actually received.

b. When queried, the transmitting station will check and if the number of groups received is correct, the transmitting station will indicate this by use of the proword "CORRECT." The transmitting station must notate the group count accordingly.

EXAMPLE: (Group count incorrect. AAR3USA transmits): AAR SIX USA THIS IS AAR THREE USA MESSAGE FOLLOWS NR 864 ROUTINE TIME ZERO FOUR ONE TWO FIVE TWO ZULU MAY EIGHT FOUR GROUPS EIGHT BREAK

CJHRS HTWXX GHSA MORTI
GUVZI TUWXA UNOHM
BREAK OVER (AAR6USA noticing
that the group count and the number
of groups he or she has received are
different transmits): THIS IS AAR
SIX USE ARE GROUPS SEVEN
OVER (AAR3USA checks his or her
group count against the transmission
copy, discovers the error, and
transmits):

THIS IS AAR THREE USA
CORRECT OVER (AAR6USA
transmits): THIS IS AAR SIX USE
ROGER NR 864 OUT

c. If, after checking, the receiving station finds that the number of groups received differs from the group count that the groups have not been transmitted or received properly, the transmitting station repeats the group count followed by the initial letter of each group. This will enable the receiving station to discover which groups are missing and to ask for them.

EXAMPLE: (Group count correct—AAR4USA transmits):

AAR TWO USA THIS IS AAR
FOUR USA MESSAGE FOLLOWS
NR 125 ROUTINE TIME ONE
FOUR TWO ZERO THREE EIGHT
ZULU MAY EIGHT FOUR
GROUPS SIX BREAK IHNOV
RSTIJ MHORE JXMHR S7ZUH
VRJXM BREAK OVER (AAR2USA
having received 5 groups only,
transmits): THIS IS AAR TWO USA
ARE GROUPS FIVE OVER
(AAR4USA checks his or her group
count against the transmission copy,
finds it correct and transmits):
THIS IS AAR FOUR USA
NEGATIVE GROUPS SIX BREAK
INDIA ROMEO MIKE JULIETT
SIERRA VICTOR BREAK OVER
(AAR2USA can now see which
group he or she has missed and
transmits):

THIS IS AAR TWO USA SAY
AGAIN GROUP FOUR OVER
(AAR4USA transmits):
THIS IS AAR FOUR USA I SAY
AGAIN GROUP FOUR JXMHR
OVER

(AAR2USA transmits):
THIS IS AAR TWO USA
ROGER NR 125 OUT

14-16. Correction during transmission.

a. When an error is made by a transmitting operator, the proword "CORRECTION" will be transmitted followed by the proword "I SAY AGAIN" followed by the last word, group, proword, or phrase correctly transmitted. Transmission will then continue.

EXAMPLE A: (AAR6USA transmits):

AAR NINE USE THIS IS AAR SIX
USA MESSAGE FOLLOWS NR 962
ROUTINE TIME ONE FOUR TWO
ZERO THREE EIGHT ZULU
JANUARY EIGHT FOUR GROUPS
NOT COUNTED BREAK WILL
ARRIVE YOUR CITY (FIGURES)
ZERO NINE HUNDRED JANUARY
(FIGURES) ONE ONE
CORRECTION I SAY AGAIN
(FIGURES) ONE TWO REQUEST
TRANSPORTATION FROM TRAIN
STATION TO YOUR ACTIVITY
BREAK OVER

(AAR9USA transmits):

THIS IS AAR NINE USA
ROGER MR 962 OUT

EXAMPLE B: (AAR3USA transmits):

WAR THIS IS AAR THREE USA
MESSAGE FOLLOWS NR 691
ROUTING TIME TWO FOUR
ZERO SEVEN ONE TWO ZULU
MAY EIGHT FOUR GROUPS
NINER BREAK ZULU OSCAR
BRAVO ALFA XRAY XRAY INDIA
OSCAR CORRECTION I SAY
AGAIN ZULU OSCAR BRAVO
ALFA XRAY XRAY QUEBEC
INDIA ALFA KILO (and so
forth).....BREAK
OVER

(NOTE: In this example the text
is five-letter code groups—

ZOBAX, XQIAK, and so forth.)

(WAR transmits):

THIS IS WAR ROGER NR 691
OUT

EXAMPLE C: (AAR3USA transmits):

AAR SIX USA THIS IS AAR

THREE USA NR FOWER FIYIV
ROUTINE TIME ONE FOUR ZERO
SIX FOUR TWO ZULU JANUARY
EIGHT FOUR FROM AAR3USA
TO AAR4USA CORRECTION I
SAY AGAIN TO AAR6USA INFO
AAR5USA BREAK (TEXT)
BREAK OVER (AAR6USA
transmits): THIS IS AAR SIX USA
ROGER NR FOWER FIYIV OUT

b. When an error in transmission is made but not discovered immediately, a correction may be made in the final instructions element provided that the ending sign has not been transmitted. When making such a correction, the word, group, proword, or phrase must be properly identified.

EXAMPLE: (AAR4USA transmits):

AAR NINE USA THIS IS AAR
FOUR USA MESSAGE FOLLOWS
NR 946 PRIORITY TIME ONE
TWO ONE SIX TWO TWO ZULU
JANUARY EIGHT FOUR FROM
AAR4USA TO AAR9USA BREAK
RELIEF TEAMS WILL ARRIVE
YOUR CITY (FIGURES) ONE
FOUR THREE ZERO HOURS
THURSDAY BREAK
CORRECTION WORD AFTER
ONE FIVE OVER (AAR9USA,
uncertain as to correction, transmits):
THIS IS AAR NINE USA IS THE
WORD AFTER ONE FIVE OVER
(AAR4USA verifies correction,
transmits):

THIS IS AAR FOUR USA
CORRECT OVER (AAR9USA
transmits): THIS IS AAR NINE USA
ROGER NR 946 OUT

c. If it is necessary to make corrections after a receipt has been obtained for a message, an abbreviated service message, identifying the message and the portion to be corrected, should be transmitted.

EXAMPLE: (AAR6USA transmits):

c. AAR NINE USA THIS IS AAR SIX USA SVC
BREAK CORRECTION MY (FIGURES) ONE THREE
ONE FIVE ONE SIX ZULU JANUARY EIGHT FOUR
WORD AFTER ONE FIVE OVER

14-17. Cancelling messages.

a. During the transmission of a message and prior to the transmission of the ending proword "OVER" or "OUT," the transmission may be cancelled by use of the proword "DISREGARD THIS TRANSMISSION"—OUT.

EXAMPLE: (During the transmission of a message WAR realizes that the transmission is being sent in error and therefore cancels it): AAR SIX USA THIS IS WAR MESSAGE FOLLOWS NR 196 ROUTINE TIME ONE ZERO ZERO SIX ZERO ZULU MAY EIGHT FOUR FROM WAR TO AAR SIX USA BREAK BEGIN UNLOADING AT ONE ONE ONE TWO THREE ZERO ZULU PROCEED DISREGARD THIS TRANSMISSION OUT

b. A message which has been completely transmitted and receipted for can only be cancelled by a message from the originator.

EXAMPLE: (AAR5USA transmits):

AAR SIX THIS IS AAR FIVE USA MESSAGE FOLLOWS NR 46 ROUTING TIME ZERO SEVEN ONE TWO ZULU BREAK CANCEL MY (FIGURES) ONE ZERO ZERO SIX ZERO TWO ZULU MAY EIGHT FOUR BREAK OVER

14-18. Read back messages.

a. If it is desired that a message or portion thereof be read back, the proword "READ BACK" and identifying data will be transmitted immediately following the call.

EXAMPLE: (WAR transmits):

AAR SIX USA THIS IS WAR MESSAGE FOLLOWS NR 284 READ BACK BREAK CONVOY HAS ARRIVED TIME ONE SIX THREE ZERO ZULU BREAK OVER (AAR6USA transmits): THIS IS AAR SIX USA I READ BACK AAR SIX USA THIS IS WAR READ BACK MESSAGE FOLLOWS NR 284 BREAK CONVOY HAS ARRIVED TIME ONE SIX THREE ZERO ZULU BREAK OVER

(WAR transmits): THIS IS WAR CORRECT OUT (NOTE: When read back procedure is employed, the proword "ROGER" is not necessary to indicate receipt of the message.)

b. The proword "READ BACK," when not preceded by identifying call signs, means that all stations are to read back. If a net call is used, but only part of the stations represented in the call are required to read back, that station, or those stations, will be specified by transmitting the appropriate call signs preceding the proword "READ BACK." When the order to read back is given, only those stations directed to do so will read back; remaining stations called will keep silent unless directed by the calling station to receipt.

EXAMPLE A: (WAR transmits):

WAR SLANT ALFA THIS IS WAR AAR TWO USA READ BACK MESSAGE FOLLOWS NR 281 ROUTE TIME BREAK CONVOY HAS ARRIVED (FIGURES) ONE SIX THREE ZERO ZULU BREAK OVER (AAR2USA transmits): THIS IS AAR TWO USA I READ BACK WAR SLANT ALFA THIS IS WAR AAR TWO USA READ BACK MESSAGE FOLLOWS NR 281 ROUTINE TIME BREAK CONVOY HAS ARRIVED TIME ONE SIX THREE ZERO ZULU BREAK OVER (WAR transmits and directs AAR3USA to receipt): THIS IS WAR CORRECT AAR THREE USA OVER (AAR3USA transmits): THIS IS AAR THREE USA ROGER NR 281 OUT
EXAMPLE B: (If AAR2USA, in above example, had read back incorrectly, WAR would have called attention to the error by use of the proword "WRONG" followed by the correct version.) (WAR transmits): THIS IS WAR WRONG TIME ONE SIX THREE ZERO ZULU OVER

(AAR2USA transmits): THIS IS AAR TWO USA TIME ONE SIX THREE ZERO ZULU OVER

(WAR transmits):

THIS IS WAR CORRECT OUT

14-19. Receipt.

a. Receipt is employed in direct station-to- station traffic handling. No message is considered delivered until a receipt is obtained (see b below). A receipt may be effected as follows: the receiving station transmits a receipt after each message or string of messages by the proword "ROGER NO ___."

EXAMPLE: (AAR2USA transmits):

AAR SIX USA THIS IS AAR TWO USA
MESSAGE FOLLOWS NR 662 BREAK
SEND BOAT FOR MAIL BREAK
OVER

(AAR6USA transmits):

THIS IS AAR SIX USA ROGER
NR 662 OUT

b. When the transmitting station considers speed of handling a primary consideration, one station in the net may be directed to receipt for the message and no other stations may answer until instructed to do so. This does not prohibit any station from requesting repetition.

EXAMPLE: (WAR sends a message to the net call WAR/A, and only desires a receipt from AAR2USA, WAR transmits):

WAR SLANT ALFA THIS IS WAR
MESSAGE FOLLOWS NR 4
ROUTINE—BREAK SEND BOAT
FOR MAIL TIME ONE TWO ONE SIX
ZULU BREAK AAR TWO USA OVER
(AAR2USA transmits): THIS IS AAR
TWO USA ROGER NR 4 OUT

(AAR9USA having missed the word
"MAIL" transmits): WAR THIS IS AAR
NINE USA. SAY AGAIN WORD
AFTER FOR OVER (WAR transmits):
THIS IS WAR I SAY AGAIN WORD
AFTER FOR MAIL OVER
(AAR9USA transmits):
THIS IS AAR NINE USA
ROGER OUT

c. Should either the transmitting or receiving station wish to indicate that he or she has further traffic to transmit to the station that he or she is working, this will be done by the proword "MORE TO FOLLOW" in the message ending or receipt.

EXAMPLE: (AAR2USA, in

transmitting a message to AAR5USA,
wishes to indicate that he or she has
further traffic for AAR5USA,
transmits:

AAR FIVE USA THIS IS AAR TWO
USA BREAK (TEXT) BREAK MORE
TO FOLLOW OVER

(AAR5USA transmits):

THIS IS AAR FIVE USA
ROGER NR _ OVER

14-20. Acknowledgement of messages.

An acknowledgement should not be confused with a reply or receipt. A prompt reply referring to the message may serve in lieu of an acknowledgement to a message from any or all addressees of that message. If an acknowledgement is desired for a message, the request for acknowledgement normally is included in the text of that message. If the message has already been transmitted, the request for acknowledgement will constitute a new message. Acknowledgements are originated only by the addressee to whom the request for acknowledgement was made.

EXAMPLE A: (WAR transmits a message to

AAR4USA and desires an
acknowledgement. WAR transmits):

AAR FOUR USA THIS IS WAR
MESSAGE FOLLOWS NR 284
ROUTINE TIME BREAK
CONFERENCE START ONE
THREE ZERO ZERO FIVE
MARCH EIGHT FOUR
ACKNOWLEDGE BREAK OVER

(AAR4USA transmits): THIS IS
AAR FOUR USA ROGER NR 284
OUT (AAR4USA having delivered
the message to the addressee and
having been requested to
acknowledge the message by the
addressee transmits): WAR THIS IS
AAR FOUR USA YOUR ONE ONE
TWO ZERO ZULU
ACKNOWLEDGED TIME ONE
ONE TWO FIVE ZULU
OVER

14-21. Verifications.

a. When verification of a message or a portion thereof has been requested by an addressee, the originating station will verify with the originator and send the correct version.

EXAMPLE: (AAR9USA transmits):

WAR THIS IS AAR NINE USA
BREAK VERIFY YOUR ONE ZERO
ZERO ZERO EIGHT ZERO ONE
ZULU ALL BEFORE BREAK OVER
(WAR transmits): THIS IS WAR
ROGER OUT (WAR checks with the
originator, establishes that the heading
previously transmitted was correct
transmits):

AAR NINE USA THIS IS WAR
BREAK I VERIFY MY ONE ZERO
ZERO EIGHT ZERO ONE ZULU
ALL BEFORE BREAK AAR NINE
USA THIS IS WAR PRIORITY TIME
ONE ZERO ZERO EIGHT ZERO
ONE ZULU FROM WAR TO AAR
NINE USA INFO AAR SIX USA
GROUPS ONE SEVEN BREAK
OVER

(AAR8USA transmits):

THIS IS AAR NINE USA
ROGER OUT

b. When a message to a number of addressees is queried by one station and found to be incorrect, the corrected version must be sent to all addressees.

EXAMPLE: (AAR4USA transmits):

AAR SIX USA THIS IS AAR FOUR
USA BREAK VERIFY YOUR ZERO
EIGHT FOUR FIVE ZULU WORD
AFTER PROCEED HAIPHONG
BREAK OVER

(AAR6USA transmits):

THIS IS AAR SIX USDA
ROGER OUT

(AAR6USA checks the message with
the originator and finds that the word
after proceed should have been "Hong
Kong" instead of "Haiphong." He or
she, therefore, transmits a correction
to all the original addressees).

AAR FOUR USA AAR FIVE
USA AAR NINE USA THIS IS AAR
SIX USA CORRECTION MY ZERO
EIGHT FOUR FIVE ZULU WORD

AFTER PROCEED HONG KONG
OVER (Stations respond): THIS IS
AAR FOUR USA ROGER OUT
THIS IS AAR FIVE USA ROGER
OUT
THIS IS AAR NINE USA ROGER
OUT

14-22. Single side band break-in procedures.

a. The inherent break-in capability of a single side band (SSB) equipment makes it possible to adopt CW break-in methods for SSB voice transmissions. Consequently, the break-in procedure outlined here is basically the same as that for CW as contained in chapter 13,

b. To enhance the efficiency of break-in procedure during SSB transmission, it is suggested that the transmitting operator pause occasionally for 2 or 3 seconds to allow the receiving operator sufficient time to break in. During the transmission of a message these short pauses should occur between the heading and the text, at the end of phrases or sentences, and every 20 groups in a long message.

c. The receiving operator will break in, during pauses, using the proword "BREAK" followed by a request for repetition, WAIT, and so forth. If three attempts to break in are unsuccessful, the receiving operator will cease attempts to break in until the transmission in progress is completed. Stations directly concerned with the transmission of a message need not use station calls when employing break-in procedure.

EXAMPLE: AAR6USA is transmitting to AAR9USA. AAR9USA missed the word Thursday. AAR9USA transmits the proword BREAK, AAR6USA stops transmitting.

d. A station having a message of higher precedence than the transmission in progress may break in and thus suspend that transmission in the following circumstances: Priority may break in at once and pass the message. A preliminary call may be made before transmitting the message if necessary. On a directed net NCS approval to transmit the message must be obtained. In the case of priority messages only long routine messages should be interrupted.

e. The word "BREAK" spoken three times means "cease transmissions immediately." Silence will be maintained until the station breaking in has passed his or her message.

f. Break-in procedure for messages of priority precedence is illustrated in the following:

EXAMPLE: (AAR2USA is transmitting a long Routine message to AAR5USA when WAR is handed a Priority message for AAR4USA. When AAR2USA pauses, WAR transmits): BREAK BREAK BREAK (AAR2USA hearing the word "BREAK" spoken three times ceases his or her transmission, WAR then continues): AAR TWO USA THIS IS WAR ONE PRIORITY AAR FOUR USA OVER WAR THIS IS AAR TWO USA. SEND ONE PRIORITY AAR FOUR USA OUT. (WAR transmits.) (AAR4USA transmits): THIS IS AAR FOUR USA ROGER OUT (AAR2USA then continues his or her transmission): AAR FIVE USA THIS IS AAR TWO USA ALL AFTER (and so forth)

14-23. Synchronizing time.

a. When an accurate time check is desired, it will be requested by using the phrase "request time check." The time at which the check is required may be indicated by the addition of a four-figure group, normally Zulu time.

EXAMPLE: AAR2USA requires to check his or her clock and transmits: WAR THIS IS AAR2USA REQUEST TIME

CHECK OVER WAR transmits:

THIS IS WAR TIME CHECK ONE EIGHT ZERO TWO (pause) ONE FIVE SECONDS ONE ZERO SECONDS FIVE FOUR THREE TWO ONE TIME ONE EIGHT ZERO TWO OVER AAR2USA transmits: THIS IS AAR TWO USA ROGER OUT

b. Should the occasion arise when the NCS desires to give an accurate time check to all stations on the net, he or she will pause a sufficient period of time between his or her warning phrase and the commencement of the countdown for all receiving operators to prepare their watches. The NCS may announce his or her intention of transmitting a time check at a certain time, using the phrase "time check at"

EXAMPLE: WAR transmits:

WAR/A THIS IS WAR TIME CHECK AT ZERO NINE THREE ZERO (pause to allow operators to prepare) ONE FIVE SECONDS ONE ZERO SECONDS FIVE FOUR THREE TWO ONE ZERO NINE THREE ZERO OVER

14-24. Closing a net.

At the end of the allotted time, or when the net has served its purpose, the NCS will close the net. When the net is closed, no further transmissions will be made by stations in that net. EXAMPLE: WAR transmits:

WAR/A THIS IS WAR THE NET IS CLOSED OUT

CHAPTER 15

RADIOTELETYPEWRITER PROCEDURES

15-1. General.

a. This chapter contains specific instructions for the conduct of radioteletypewriter communications. Adherence to prescribed procedures is mandatory. Unauthorized departures from or variations in prescribed procedures often create confusion and reduce reliability and speed. If the procedures prescribed herein do not cover a specific operational requirement, initiative and common sense should prevail. The basic rules essential to circuit discipline and good transmission practices and practices to be avoided are outlined in paragraph 9-2. Message format preparation is listed in chapter 11.

b. In the many examples used in this chapter, extensive use has been made of MARS call signs in format lines 6 and 7 in the address portion of the message; for example, the FROM and TO lines. In normal message preparation and transmission, the FROM line will reflect the name, city, state or country of the originator; the TO line will be the address to whom the message is intended. The MARS call signs are used for simplicity and ease of construction. Examples used are for illustrative purposes only.

EXAMPLE: FM SGT RICHARD ROWE,
PIRMASENS GE/AEM1USA/GE
TO MRS. RICHARD ROWE 123
West 45th Street Baltimore MD
21001 Tel: (301) 567-8901

c. Machine functions are discussed below and in chapter 17.

(1) Shift. Operators must always depress the "LTRS" key when going from upper case (figures) to lower case (letters), and the "FIGS" key when going from lower case to upper case.

(2) Carriage return. The carriage return (CR) function is employed to reset the machine to the left margin of the paper. Two CR functions are used to ensure the proper return of the carriage.

(3) Line feed. The line feed (LF) function is employed to advance the paper vertically on the page teletypewriter.

(4) Space. The space (SP) function is employed to advance the typing unit laterally without printing a character on the page teletypewriter.

(5) Bell signal. The bell signal is used to attract the attention of the receiving operator when required and will be transmitted as a series of ten

characters, upper case "J" and "S" as follows: "(FIGS) JJJJSSSSS (LTRS)."

d. Specific machine functions are necessary to facilitate the handling of messages and to align receiving page teletypewriters.

(1) All transmissions must be preceded by five SP, 2CR, and LF.

(2) The end of line (EOL) function will be 2CR and LF.

(3) The separative function between pages of long messages is 2CR and 4LF.

(4) The end of message functions are 2CR, 4LF, the letter "N" repeated four times, and 12 letter keys,

(5) No line will exceed 69 characters, including spaces.

e. Examples of specific machine functions are discussed below.

(1) Calling. The basic procedure for call signs, calling, and answering are paragraph 13-7. The following examples show the relationship to teletypewriter procedure:

EXAMPLE: Net Call. Identifying all stations on that net.

(5 SPACES) (2CR) (LF)
WAR/A DE WAR K (2CR) (LF)

EXAMPLE: Single Station. Identifying a single station.

(5 SPACES) (2CR) (LF)
AAR5USA
DE WAR K (2CR) (LF)

EXAMPLE: Net call with exempted station identifying stations to be exempted from net.

(5 SPACES) (2CR) (LF)
WAR/A XMT
AAR3USA
DE WAR K (2CR) (LF)

EXAMPLE: Multiple Call. Identifying a predetermined group of stations.

(5 SPACES) (2CR) (LF)
AAR2USA AAR3USA
AAR4USA
DE WAR K (2CR) (LF)

(2) Answering, EXAMPLE:
Answering

(5 SPACES) (2CR) (LF)

WAR DE AAR5USA (2CR)
 R K (4LF)
 EXAMPLE: Abbreviated Answer
 (5 SPACES) (2CR) (LF)
 DE AAR5USA R K (2CR)
 (4LF)

(3) Testing. When a station wishes to test on a net or circuit, the following test will be used:

EXAMPLE: Testing
 (5 SPACES) (2CR) (LF)
 WAR DE AAR1USA (2CR)(LF)
 TEST THE QUICK
 BROWN FOX JUMPS
 OVER THE LAZY DOG
 1234567890 (2CR) (LF)
 RYRYRYRYRYRYRY
 RYRYRYRYRYRYRY
 RYRYRYRYRYR (2CR) (LF)
 *INT ZHN K (2CR)
 (4LF)

WAR determines that operation is satisfactory for the reception of traffic and answers:

(5 SPACES) (2CR) (LF)
 AAR1USA DE WAR (2CR) (LF)
 *ZHN1 K (2CR) (LF)
 *Or other appropriate operating signal.

PRESENT FREQUENCY: ZERO BEAT
 WITH MY TRANSMITTER TUNE
 AND/OR ADJUST YOUR
 EQUIPMENT."

This transmission will be followed by a 50-second clear period for stations to adjust to the marked frequency. WAR/A DE WAR K (NOTE: Member stations will tune their equipment during the 50-second clear period.)

c. In response to the initial net call, stations reporting into the net will indicate traffic by number, precedence, destination (ZBO, and so forth) or no traffic (QRU). Stations entering a net already in progress will include a traffic report at the time they enter the net.

d. Radioteletypewriter operation is discussed below.

(1) The procedures in paragraph 15-1e for calling and answering apply. Because of the instability of automated equipment and systems on marginal or unstable radio circuits, the following procedures should be followed when operating a radio station and encountering the situation described below. In carrying out these procedures, attention should be paid to the operating precautions prescribed in paragraphs 9-1 and 9-2.

(2) Action to be taken when in contact, but not transmitting traffic, where tape perforator and automatic sender (or transmitter-distributor) are available:

(a) If the signal shows signs of failing, but has not deteriorated to the point requiring a frequency change, a continuous test tape, constructed as indicated below, should be transmitted by each station if working on different frequencies, or by the NCS when all stations are working on the same frequency. CAUTION: Tape should be marked to prevent erroneous transmission.

EXAMPLE: (Sufficient LTRS to permit splicing into a continuous loop).

(5 SPACES) (2CR) (LF)
 THE QUICK BROWN FOX JUMPS
 OVER THE LAZY DOG
 1234567890
 DE AAR3USA (2CR) (LF)
 RYRYRYRYRYRYRY
 RYRYRYRYRYRYRY
 RYRYRYRYRYR (2CR) (LF)

(Sufficient LTRS to permit splicing into a continuous loop.)

(b) Stations not receiving above test satisfactorily will so inform the sending stations.

15-2. Opening a MARS radioteletype net.

a. The following example of a MARS teletypewriter net operation will be used as a guide for all MARS teletypewriter nets. It will be followed as closely as possible. This theoretical net operation is based upon the following:

(1) Net call WAR/A.

(2) NCS WAR.

(3) Alternate NCS (ANCS) AAR3USA.

(4) Alphabetical, numerical roster or reporting order of net stations, AAR2USA, AAR3USA, AAR4USA, AAR5USA, AAR6USA, AAR7USA, and AAR9USA.

b. An opening net call example is shown below.

EXAMPLE: WAR/A WAR/A WAR/A DE
 WAR WAR WAR
 ZRF - ZRC2 (10 second dash to zero beat) ZRC1 AR
 Meaning "I AM ABOUT TO SEND A
 MARK TUNING SIGNAL ON MY

(3) Action to be taken when contact has been lost:

The test tape above will be run; transmission of test will be interrupted at frequent intervals to manually transmit information required to regain contact (for example, call sign of station being listened for, call sign of station transmission is being received from, and so forth); this information may be perforated into the standard test tape if desired, but if done, use of the test tape for other situations is negated; action to be taken upon faulty reception of traffic is discussed in (a) through (d) below.

(a) When faulty reception of traffic is encountered, the transmitting station will be requested to run continuous tests until further orders.

(b) The standard test message is designed to facilitate the diagnosis and correction of faults and to standardize testing procedure on point-to-point radioteletypewriter channels.

(c) While the standard test message is being ran, all links in the circuit should be monitoring the test transmission and, since its exact composition will be known, the diagnosis and location of faults should be a simple matter subject to the intelligent monitoring and interpretation of results obtained, and good cooperation between all links positions concerned.

(d) When the test transmission contained in (d)(2)(a) above must be transmitted manually from the keyboard, the LTRS to permit tape splicing and the first line of "THE QUICK BROWN FOX" will be omitted.

15-3. Answering calls.

a. Answering calls by stations entering net:

EXAMPLE 1: WAR DE AAR2USA ZBO-1P-K

WAR DE AAR3USA

ZBO-1R-AAR6USA-K

AAR4USA fails to answer immediately. AAR5USA waits 5 seconds and transmits:

WAR DE AAR5USA ZBO-1R - IP
AAR2USA K

WAR DE AAR6USA ZBO-1R-K

EXAMPLE 2: WAR DE AAR9USA
QRU K

AAR4USA having missed his or her place in the callup waits until the roster has been completed and then transmits:

WAR DE AAR4USA QRU K If
AAR4USA had not responded, WAR
would have initiated a single call to
AAR4USA.

b. The NCS acknowledges the transmissions of the

net stations:

EXAMPLE: WAR/A DE WAR - R - AAR2USA
IP - AAR3USA-1R- AAR6USA -
AAR4USA QRU - AAR5USA-1R
- 1P-AAR2USA - AAR6USA-1R -
AAR9USA QRU AR

WAR then pauses 5 seconds to allow any other stations to call and to allow the ANCS or other net stations to inform the NCS of any errors or omissions in the callup or traffic reports.

c. WAR not hearing any station call proceeds to direct the clearance of traffic in order of precedence highest to lowest. EXAMPLE: WAR transmits:

AAR2USA DE WAR ZOG-P-K
(Meaning: "TRANSMIT YOUR
MESSAGE TO THIS STATION.")

(AAR2USA hearing WAR's
transmission, transmits):

WAR DE AAR2USA NR1 - P-
1.12025Z APR 84

FM AAR2USA NJ

TO WAR

GR NC

BT

TEXT

BT

K

(WAR having received the message,
transmits):

AAR2USA DE WAR-R-NR1 AR
(WAR pauses 5 seconds and not
hearing anything, transmits):

AAR5USA DE WAR ZOG-1P-
AAR2USA AR

(AAR2USA hearing WAR's
transmission, transmits): AAR5USA
DE AAR2USA K (If AAR2USA had
not called AAR5USA, AAR5USA
would have had to call AAR2USA to
see if AAR2USA was ready to copy
his or her message). (AAR5USA
receiving AAR2USA invitation,
transmits): AAR2USA DE

AAR5USA NR5

P-012345Z APR 84

FM TO

GR NC

BT

TEXT BT

K
 (AAR2USA after requesting and receiving missed words, transmits):
 AAR5USA DE AAR2USA-R-NR5 AR
 (After a 5-second pause, WAR transmits):
 AAR6USA DE WAR ZOG 1R K
 (Meaning "TRANSMIT YOUR MESSAGE TO THIS STATION")
 (AAR6USA transmits):
 WAR DE AAR6USA NR7 - R-
 120456Z APR 84
 FM
 TO
 GR NO
 BT
 TEXT
 BT
 (Upon completion of AAR6USA transmission WAR transmits):
 AAR6USA DE WAR-R-NR7 AR
 (After a 5-second pause, WAR proceeds):
 AAR3USA DE WAR ZOG-1R-
 AAR6USA AR
 (AAR6USA hearing WAR, transmits):
 AAR3USA DE AAR6USA K
 (AAR3USA hearing AAR6USA invitation transmits):
 AAR6USA DE AAR3USA NR4 - R-
 050607Z APR 84
 FM
 TO
 GR NC
 BT
 TEXT
 BT
 (AAR6USA having received the message transmits):
 AAR3USA DE AAR6USA-R-NR4 AR
 (WAR pauses for 5 seconds and AAR9USA having received an original priority message for transmission, transmits):
 WAR DE AAR9USA ZBO-1P-
 AAR6USA K
 (WAR transmits):

AAR9USA DE WAR ZOG IP
 AAR6USA AR
 (AAR6USA hearing the last two transmissions, transmits):
 AAR9USA DE AAR6USA K
 (AAR9USA hearing AAR6USA invitation, transmits):
 AAR6USA DE AAR9USA NR8 - P-
 121450Z APR 84 FM TO
 GR NO
 BT
 TEXT
 BT
 (AAR6USA having requested and received missed words, transmits):
 AAR9USA DE AAR6USA-R-NR8 AR
 (WAR pauses 5 seconds then proceeds):
 AAR5USA DE WAR ZOG-1R-
 AAR2USA AR
 (AAR2USA hearing WAR transmits):
 AAR5USA DE AAR2USA K
 (AAR5USA hearing AAR2USA invitation, transmits):
 AAR2USA DE AAR5USA NR3 - R-
 120340Z APR 84 FM TO
 GR NO
 BT
 TEXT
 BT
 (Having requested and received missed words transmits): AAR5USA DE
 AAR2USA R NR3 AR
 (Traffic is now cleared and the NCS will proceed to other net business.)

15-4. Message transmission.

a. Messages will be transmitted exactly as written. Abbreviations will not be substituted for full words or full words for abbreviations without the approval of the originator. A station receiving a message is responsible for the message until a receipt is obtained from another station or delivery is accomplished. Instructions to take no further action regarding a message relieves that station of any further forwarding responsibility.

(1) Transmission of plaindress messages,

(a) A single address plaindress message:

EXAMPLE: (5 SPACES) (2CR) (LF) (LTRS)
 (Line 2) AAR2USA (2CR) (LF)

(Line 3) DE WAR NR (2CR) (LF)
 114 (LTRS)
 (Line 5) R 151412Z (2CR) (LF)
 JAN 78 (LTRS)
 (Line 6) FM
 AAR2USA (2CR) (LF)
 (Line 7) TO
 AAR2USA (2CR) (LF)
 (Line 10) GRNC (2CR) (LF)
 (Line 11) BT (2CR) (LF)
 (Line 12) PLAIN SINGLE
 ADDRESS (2CR) (LF)
 MESSAGES WILL BE
 TRANSMITTED
 OVER (2CR) (LF)
 TELETYPEWRITER
 CIRCUITS AS (2CR) (LF)
 INDICATED IN THIS
 EXAMPLE (2CR) (LF)
 (Line 13) BT (2CR) (LF)
 (Line 15) C WA
 OVER TELETYPE-
 WRITER (2CR) (LF)
 (Line 16) K (2CR)
 (4LF)
 (NNNN) (12 LTRS)

(b) A single address abbreviated plaindress message:
 EXAMPLE: (5 SPACES) (2CR) (LF) (LTRS);
 (Line 2, 3) AAR2USA
 DE WAR (2CR) (LF)
 NR120 (LTRS)
 (Line 5) R 022320Z (2CR) (LF)
 Jan 84 (LTRS)
 (Line 6) FM
 AAR3USA (2CR) (LF)
 (Line 7) TO
 AAR2USA (2CR) (LF)
 (Line 11) BT (2CR) (LF)
 (Line 12) THIS IS AN
 ABBREVIATED (2CR)
 (LF)
 PLAINDRESS MES-
 SAGE IN (2CR) (LF)
 WHICH THE CALL
 DOES NOT (2CR) (LF)
 SERVE AS THE ORIGIN-
 ATOR (2CR) (LF)
 (Line 13) BT (2CR) (LF)
 (Line 16) K (2CR)
 (4LF)

(NNNN) (12 LTRS)
 c) An abbreviated plaindress message with the call
 serving as the address:
 EXAMPLE: (5 SPACES) (2CR) (LF) (LTRS)
 (Line 2,3) AAR2USA
 DE WAR (2CR) (LF)
 (Line 11) THIS IS AN
 ABBREVIATED (2CR) (LF)
 MESSAGE WHERE
 THE CALL (2CR) (LF)
 SERVES AS ADDRESS (2CR) (LF)
 (Line 13) BT (2CR) (LF)
 (FIGS)
 (Line 14) 0730Z (2CR) (LF)
 (Line 16) K (2CR)
 (4LF)
 (NNNN) (12 LTRS)
 (NOTE: Line 14. "0730Z" is the time group
 used in place of DTG in line 5. (The filing
 time.)
 (d) A multiple address plaindress message:
 EXAMPLE: (5 SPACES) (2CR) (LF) (LTRS)
 (Line 2,3) AAR2USA
 DE WAR (2CR) (LF)
 NR325 (LTRS)
 (Line 5) P 150903Z (2CR) (LF)
 Jan 84 (LTRS)
 (Line 6) FM HQ 7TH
 SIG COMD (2CR) (LF)
 FT RITCHIE
 MD (2CR) (LF)
 (Line 7) TO CDR
 USAISC FT (2CR) (LF)
 HUACHUCA AZ //AS-
 OPS- OM// (2CR) (LF)
 (Line 8) INFO HQ 7TH SIG
 COMD (2CR) (LF)
 FT LEWIS
 WA (2CR) (LF)
 SMD TX
 SAN
 ANTONIO
 TX (2CR) (LF)
 SMD ME
 WISCASSTE ME
 (2CR) (LF)
 (Line 11) BT (2CR) (LF)
 (Line 12) THIS IS A PLAINDRESS
 (2CR) (LF)

MULTIPLE ADDRESS

MESSAGE (2CR) (LF)

(Line 13) BT (2CR) (LF)

(Line 16) K (2CR) (LF)

(NNNN) (12 LTRS)

(2) Multiple page messages. Messages containing more than 20 lines of heading and text will be divided into pages for transmission as follows:

(a) The number of pages of message text in any transmission will not exceed five.

(b) The first page will consist of not more than 20 lines and will begin with formal line 5, counting from format line 5 of message heading and continuing for a total of 20 lines.

(c) All succeeding pages will contain 20 lines of text, except the last page, which may contain less.

(d) The abbreviation "CFN" (confirmation.) or "C" may be used at the end of each page as necessary.

(e) The second and succeeding pages will be identified by the page number, the routing indicator or designator, and the station serial number of the originating station. When the message text is transmitted in plain language, the security classification or the abbreviation "UNCLAS" will also be included as part of the page identification on each page. Page identification will appear on a separate line and will not be included in the line count.

(f) The machine functions between pages will be 2CR and 4LF as follows:

EXAMPLE OF PAGING:

PAGE 2 MDE 027 (2CR) (LF)

UNCLAS (LTRS)

TEXT(20 lines) (2CR) (LF)

(LTRS)

PAGE 3 MDE 027

UNCLAS (2CR) (LF)

TEXT ... (Remaining (2CR) (LF)

lines) (LTRS)

BT (2CR) (LF)

(FIGS)

29/1405Z (2CR) (LF)

NNNN (12 LTRS)

(g) Transmission of long messages. Messages which exceed 5 teletypewriter pages are considered to be long messages. Such messages will be divided into transmission sections. A transmission section will not exceed 5 teletypewriter pages.

(h) Since long messages monopolize circuit time when transmitted in their entirety, it is sometimes advisable to separate them into transmission sections

even though they may be below the prescribed length. They must be divided into transmission sections if they exceed the prescribed length.

(i) Messages to be forwarded in transmission sections will be divided at a convenient point, but not beyond the maximum number of pages described, and the text will be separated at the end of a sentence; prior to the text, or following the security classification. Each section will be numbered and the section number will be inserted at the beginning of the text. Example: "SECTION ONE OF " Each additional transmission section will be preceded by an identical message heading except it will contain a different SSN and group count (if used) for that particular transmission section. At the beginning of the text of each transmission, the security classification will be inserted in plain language. The final transmission section is identified "FINAL SECTION OF The section number inserted at the beginning of the text will differ for each transmission section. The first transmission of a message separated into two sections would appear: "SECTION ONE OF TWO." The following examples show the manner in which a 1,600-word message may be separated into pages and into two transmission sections:

EXAMPLE: (5 SPACES) (2CR) (LF) (LTRS)

(Line 2,3) AAR6USA (2CR) (LF)

DE WAR (LTRS)

(Line 4) TNR 6 (2CR) (LF)

(Line 5) R 030400Z (2CR) (LF)

APR 84 (LTRS)

(Line 6) FM WAR (2CR) (LF)

(Line 7) TO

AAR6AAA (2CR) (LF)

AAR6AAB (2CR) (LF)

AAR6AAC (2CR) (LF)

(Line 8) INFO

AAR6USA (2CR) (LF)

(Line 10) GR NC (2CR) (LF)

(Line 11) BT (2CR) (LF)

(Line 12) UNCLAS (2CR) (LF)

SECTION ONE OF TWO

(EIGHT LINES OF TEXT)

(2CR) (LF)

C WA UNCLAS SECTION .

(2CR) (LF)

(Line 12) PAGE TWO

WAR NR 6

UNCLAS (2CR) (LF)

(TWENTY LINES OF

TEXT) (2CR) (LF)

(NOTE: Succeeding pages of this transmission section would appear as shown above. Final page is shown below:

(Line 12) PAGE FIVE WAR
NR6
UNCLAS (2CR) (LF)
(FINAL LINES
OF TEXT OF SECTION
ONE) (2CR) (LF)
(Line 13) BT (2CR) (LF)
(Line 15)
C PAGE 3 LINE 2 (2CR)
WA (4LF)
(LTRS)
ESTABLISHED
FAULT
K (4 N's)
(12 LTRS)

(The second and final transmission section would appear):

EXAMPLE: (5 SPACES) (2CR) (LF) (LTRS)
(Line 2,3) AAR6USA
DE WAR (2CR) (LF)
NR6 (LTRS)
(Line 4) T (2CR) (LF)
(Line 5) R 030500Z (2CR) (LF)
APR 84 (LTRS)
(Line 6) FM WAR (2CR) (LF)
(Line 7) TO
AAR6AA (2CR) (LF)
AAR8AB (2CR) (LF)
AAR6AC (2CR) (LF)
(Line 8) INFO
AAR6USA (2CR) (LF)
(Line 10) GRNC (2CR) (LF)
(Line 11) BT (2CR) (LF)
(Line 12) UNCLAS (2CR) (LF)
FINAL SECTION OF
TWO. . . . (2CR) (LF)
(EIGHT LINES
OF TEXT) (2CR) (LF)
(Line 12) PAGE TWO
WAR NR6
UNCLAS (2CR) (LF)
TWENTY LINES OF
TEXT (2CR) (LF)

(NOTE: Succeeding pages of this transmission section would appear as shown above with the appropriate page number. The final page would appear the same as in section one above,

b. Checking group count.

(1) The prosign "GR" preceded by the prosign "INT" and followed by a numeral means "IS THE NUMBER OF GROUPS AS INDICATED?" When the number of groups received does not correspond with group count transmitted, the receiving station will immediately question the transmitting station by using the prosign "INT GR" followed by a numeral.

EXAMPLE: (5 SPACES) (2CR) (LF) (LTRS)
WAR DE AAR2USA (2CR) (LF)
INT GR (LTRS)

If after rechecking the message the transmitting station finds that the receiving station is correct, the transmitting station sends abbreviation "C" for confirmation.

EXAMPLE: (5 SPACES) (2CR) (LF) (LTRS)
AAR2USA DE
WAR C K (2CR) (LF)

(2) When a group count is used for messages containing less than 50 groups and the receiving station is considered to be incorrect, the transmitting station repeats the original group count and transmits the first character of each group in the text in succession.

EXAMPLE: (5 SPACES) (2CR) (LF) (LTRS)
AAR2USADE WAR (2CR) (LF)
NR9 (LTRS)
R 272113Z JAN 84 (2CR) (LF)
(LTRS)
GR10 (2CR) (LF)
(LTRS)
BT (2CR) (LF)
XTUMR BTRFII
UBTXM BCRST
WHIIK
BBCDT WKCTI
SRTXR OCUVC
WHISC (2CR) (LF)
BT (2CR) (LF)
K (2CR) (LF)
NNNN (12 LTRS)
AAR2USA questions the group count):
(5 SPACES) (2CR) (LF) (LTRS) WAR
DE AAR2USA
INT GR11 K (2CR) (LF)
(LTRS)
WAR checks and finds the group
correct as transmitted and transmits:
(5 SPACES) (2CR) (LF)
AAR2USA DE WAR
GR10 BT X B U B W
B W S O W (2CR) (LF)

c. The prosign "T" used alone indicates that the station called is to relay the message to all addressees.

EXAMPLE: (Relay to an on-net station - WAR

transmits):

(5 SPACES) (2CR) (LF) (LTRS)

AAR2USA DE WAR (2CR) (LF)

NR (LTRS)

TP 111322Z MAY 84 (2CR) (LF)

(LTRS)

FM WAR (2CR) (LF)

TO AAR9USA (2CR) (LF)

GR NC (2CR) (LF)

BT (2CR) (LF)

TEXT (2CR) (LF)

BT (2CR) (LF)

K (2CR) (LF)

(AAR2USA transmits):

(5 SPACES) (2CR) (LF) (LTRS)

DE AAR2USA R

NR__AR (2CR) (LF)

(AAR2USA relays the message):

(5 SPACES) (2CR) (LF) (LTRS)

AAR2USA DE WAR

NR__ (2CR) (LF)

P 111322Z MAY 84 (2CR) (LF)

FM WAR (2CR) (LF)

TO AAR9USA (2CR) (LF)

GR NC (2CR) (LF)

BT (2CR) (LF)

TEXT (2CR) (LF)

BT (2CR) (LF)

K (2CR) (LF)

(AAR9USA transmits):

DE AAR9USA R AR (2CR) (LF)

(1) The prosign "T" followed by an address designator indicates that the station called is to relay the message to the stations indicated. When more than one station is called, the call sign of the station designated to perform the relay will precede the prosign "T." At times it is necessary to relay a message originated on a radiotelephone circuit by some other means of communications. When such relay is necessary, it is the responsibility of the station relaying the message to place the message in the proper format for the means of communications employed for relay,

EXAMPLE: (Relay to an off-net station - AAR3USA transmits):

(5 SPACES) (2CR) (LF) (LTRS)

AAR4USA AAR2USA

DE AAR3USA

AAR4USA NR (2CR) (LF)

(LTRS)

T AAR9USA R

210814Z MAY 84 (2CR) (LF)

(LTRS)

FM AAR3USA (2CR) (LF)

TO AAR1USA (2CR) (LF)

INFO AAR5USA (2CR) (LF)

AAR6USA (2CR) (LF)

GR NC (2CR) (LF)

BT (2CR) (LF)

TEXT (2CR) (LF)

BT (2CR) (LF)

K (2CR) (LF)

(AAR4USA transmits):

(5 SPACES) (2CR) (LF) (LTRS)

DE AAR4USA R

NR__AR (2CR) (LF)

(AAR6USA transmits):

(5 SPACES) (2CR) (LF) (LTRS)

DE AAR6USA R

NR__AR (2CR) (LF)

(AAR4USA then leaves the net and enters net in which AAR9USA is a member and after establishing communications, transmits):

(5 SPACES) (2CR) (LF) (LTRS)

AAR9USA DE

AAR4USA NR (2CR) (LF)

(LTRS)

R 210814Z MAY 84 (2CR) (LF)

(LTRS)

FM AAR3USA (2CR) (LF)

TO AAR9USA (2CR) (LF)

INFO AAR4USA (2CR) (LF)

AAR6USA (2CR) (LF)

GR NC (2CR) (LF)

BT (2CR) (LF)

TEXT (2CR) (LF)

BT (2CR) (LF)

K (2CR) (LF)

AAR9USA transmits):

(5 SPACES) (2CR) (LF) (LTRS)

DE AAR9USA R

NR__AR (2CR) (LF)

(Having relayed the message AAR4USA checks out of the net in which AAR9USA is a member and reenters his or her own net.)

d. Cancellation of messages. A message may be cancelled only by the originator.

(1) Cancellation of a transmission.

(a) While it is in progress, a transmission may be cancelled by use of the error prosign "EEEEEEEE" and the ending prosign "AR." Each letter of the error sign must be separated by a space.

EXAMPLE: NOW IS THE (2CR) (LF) (LTRS)

E E E E E E E E A R

(b) The prosigns "EEEEEEEE" and "AR" will not be used to cancel a transmission after a receipt has been given. A service message containing operating signals or a service message must be used for this purpose.

(2) A station cancelling a transmission is responsible for any further action necessary in connection with the message involved.

e. Read back.

(1) If it is desired that a message or portion thereof be read back, the prosign "G" and identifying data will be transmitted immediately following the call.

EXAMPLE: (WAR transmits):

(5 SPACES) (2CR) (LF) (LTRS)

AAR6USA DE WAR

NR__ (2CR) (LF) (LTRS)

GR 210814Z MAY 84 (2CR) (LF) (LTRS)

GR NC (2CR) (LF)

BT (2CR) (LF)

CONVOY HAS ARRIVED TIME

ONE SIX (2CR) (LF)

THREE ZERO ZULU (2CR) (LF)

BT (2CR) (LF)

K (2CR) (LF)

(AAR6USA transmits):

(5 SPACES) (2CR)

(LF) (LTRS)

DE AAR6USA R NR (2CR) (LF) (LTRS)

R 210814Z MAY 48 (2CR) (LF) (LTRS)

GR NC (2CR) (LF)

BT (2CR) (LF)

CONVOY HAS ARRIVED TIME

ONE SIX (2CR) (LF)

THREE ZERO ZULU (2CR) (LF)

BT (2CR) (LF)

K (2CR) (LF)

(WAR transmits):

(5 SPACES) (2CR) (LF) (LTRS)

DE WAR CORRECT

AR (2CR) (LF)

(NOTE: When "READ BACK" procedure is employed, the prosign "R" is not necessary to indicate receipt of the message.)

(2) The prosign "G" when not preceded by identifying call signs means that all stations are to read back. If a net call is used, but only part of the stations represented in the call are required to read back, that station or those stations will be specified by transmitting the appropriate call signs preceding the prosign "G." When the order to read back is given, only those stations directed to do so will read back; remaining stations called will keep silent unless directed by the calling station to receipt.

EXAMPLE A. (WAR transmits):

(5 SPACES) (2CR) (LF) (LTRS)

WAR/A DE WAR (2CR) (LF)

NR_ (LTRS)

AAR2USA G R (2CR) (LF)

210814Z MAY 84 (LTRS)

GR NC (2CR) (LF)

BT (2CR) (LF)

CONVOY HAS

ARRIVED TIME

ONE SIX (2CR) (LF)

THREE ZERO ZULU (2CR) (LF)

BT (2CR) (LF)

K (2CR) (LF)

(AAR2USA transmits):

(5 SPACES) (2CR) (LF) (LTRS)

DE AAR2USA NR_ (2CR) (LF) (LTRS)

R 21Q814Z MAY 84 (2CR) (LF) (LTRS)

GR NC (2CR) (LF)

BT (2CR) (LF)

CONVOY HAS

ARRIVED TIME

ONE SIX (2CR) (LF)

THREE ZERO ZULU (2CR) (LF)

BT (2CR) (LF)

K (2CR) (LF)

(WAR transmits and directs AAR3USA to receipt):

(5 SPACES) (2CR) (LF) (LTRS)

DE WAR CORRECT

AAR2USA K (2CR) (LF)
(LTRS)
(AAR2USA transmits):
(5 SPACES) (2CR) (LF) (LTRS) DE
AAR2USA R
NR_AR (2CR) (LF)
(LTRS)
EXAMPLE B: (If AAR2USA, in
above example, had read back
incorrectly, WAR could have called
attention to the error by use of the
prosign "EEEEEEEE" followed by
the correct version.) (WAR
transmits): DE WAR EEEEEEEE AA
ARRVED TIME ONE SIX THREE
ZERO ZULU K
(AAR2USA transmits): DE
AAR2USA ONE SIX THREE ZERO
ZULU K
(WAR transmits): DE WAR
CORRECT AR

f. Obtaining corrections and repetitions.

(1) When used for repetitions and corrections, service messages may be assigned the precedence considered necessary to accomplish their purpose.

(2) Corrections involving discrepancies in SSNs, mutilations, or garbles caused by mechanical difficulties are usually obtained from the station from which the transmission was received.

(3) Corrections or requests for corrections should contain transmission identification, and such other data as necessary to prevent confusion. If a word or group occurring more than once in a message is used to identify a part of that message, it is to be assumed that the first occurrence of that word or group is implied.

(4) When the transmission identification and the DTG are in error, the message should be identified by quoting the message heading and a portion of the text.

(5) If a message contains more than one error, all discrepancies in the same message should be cited in the same correction request.

(6) In requesting repetitions of the heading of a message, if portions of more than one component are desired, a repetition of the entire heading must be requested. If portions of more than one element of the address component are desired, a repetition of the entire component must be requested. The answers to such requests must include the entire element and the prosign of the next element.

EXAMPLE A: Original transmission by WAR:

(5 SPACES) (2CR) (LF) (LTRS)

AAR2USA DE WAR (2CR) (LF)
NR8 (LTRS)
R 281545Z JAN 84 (2CR) (LF)

(LTRS)
FN AAR3USA (2CR) (LF)
TO AAR2USA (2CR) (LF)
GRNC

(2CR) (LF)
BT (2CR) (LF)

RECEIVED PACKAGE
AND LETER

THANK ALL WHO

HELPED OUT (2CR) (LF)

BT (2CR) (LF)

K (2CR)

(4LF)

NNNN (12 LTRS)

EXAMPLE B: AAR2USA requests WAR repeat last transmission:

(5 SPACES) (2CR) (LF) (LTRS) WAR
DE AAR2USA

IMI K (2CR) (LF)

WAR answers:

(5 SPACES) (2CR) (LF)

AAR2USA DE WAR (2CR) (LF)

NR8 (LTRS)

R 281545Z Jan 84 (2CR) (LF)

(LTRS)

FM AAR3USA (2CR) (LF)

TO AAR2USA (2CR) (LF)

GRNC (2CR) (LF)

BT (2CR) (LF)

RECEIVED PACK-
AGE AND LETTER

THANK ALL WHO

HELPED OUT (2CR) (LF)

BT (2CR) (LF)

K (2CR)

(4LF)

NNNN (12 LTRS)

EXAMPLE C: AAR2USA requests WAR repeat all before the text of the message

(5 SPACES) (2CR) (LF) (LTRS)

WAR DE AAR2USA

IMI AB BT K (2CR) (LF)

(LTRS)

WAR answers:

(5 SPACES) (2CR) (LF) (LTRS)

AAR2USA DE WAR (2CR) (LF)

AB BT

AAR2USA DE WAR (2CR) (LF)
 NR8 (LTRS)
 R 281545Z JAN 84 (2CR)(LF)
 (LTRS)
 FM AAR3USA (2CR) (LF)
 TO AAR2USA (2CR) (LF)
 GRNC (2CR) (LF)
 BT K (2CR) (LF)

EXAMPLE D: AAR2USA requests WAR to repeat all between TO and GRNC:

(5 SPACES) (2CR) (LF) (LTRS)
 WAR DE AAR2USA
 IMI TO TO GRNC (2CR) (LF)
 WAR answers:
 (5 SPACES) (2CR) (LF) (LTRS)
 AAR2USA W DE
 WAR (2CR) (LF)
 TO TO GRNC (2CR) (LF)
 TO AAR2USA (2CR) (LF)
 GRNC K (2CR) (LF)

EXAMPLE E: AAR2USA requests WAR to repeat all after helped:

(5 SPACES) (2CR) (LF) (LTRS)
 WAR DE AAR2USA
 IMI AA HELPED 8 K (2CR) (LF)
 WAR answers
 (5 SPACES) (2CR) (LF) (LTRS)
 AAR2USA DE WAR
 AA HELPED OUT
 BT K (2CR) (LF)

(7) Messages forwarded subject to correction,

(a) If corrections and repetitions cannot be obtained immediately, messages received with portions missing or portions of doubtful accuracy will be delivered or forwarded subject to correction if the precedence is immediate or higher. Discretion should be exercised in forwarding or delivering messages that are so garbled as to be of no value.

(b) A station delivering or forwarding a message "subject to correction" is still responsible for obtaining and forwarding corrections.

g. Receipt.

(1) Receipt is employed in direct station-to- station traffic handling. No message is considered delivered until a receipt is obtained. A receipt may be effected as follows: The receiving station transmits a receipt after each message or string of messages by the prosign "R NR_____, NR____, and so forth."

EXAMPLE: (AAR2USA transmits):

(5 SPACES) (2CR) (LF) (LTRS)
 AAR6USA DE
 AAR2USA NR 4 (2CR) (LF)
 (LTRS)
 R 210730Z MAY 84 (2CR) (LF)
 (LTRS)
 GRNC (2CR) (LF)
 BT (2CR) (LF)
 SEND BOAT FOR MAIL (2CR) (LF)
 BT (2CR) (LF)
 K (2CR) (LF)
 (AAR6USA transmits):
 (5 SPACES) (2CR) (LF) (LTRS)
 DE AAR6USA R NR
 4 AR (2CR) (LF)

(2) When the transmitting station considers speed of handling a primary consideration, one station in the net may be directed to receipt for the message and no other stations may answer until instructed to do so. This does not prohibit any station from requesting repetition.

EXAMPLE: (WAR sends a message to the net call WAR/A, and only desires a receipt from AAR2USA, WAR transmits):

(5 SPACES) (2CR) (LF) (LTRS)
 WAR/A DE WAR (2CR) (LF)
 NR 4 (LTRS)
 R 210730Z MAY 84 (2CR) (LF)
 (LTRS)
 GRNC (2CR) (LF)
 BT (2CR) (LF)
 SEND BOAT FOR MAIL (2CR) (LF)
 BT (2CR) (LF)
 K (2CR) (LF)

(AAR2USA transmits):
 (5 SPACES) (2CR) (LF) (LTRS)
 DE AAR2USA R NO (2CR) (LF)
 4 AR (LTRS)

(AAR9USA having missed the word "MAIL" transmits):

(5 SPACES) (2CR) (LF) (LTRS)
 WAR DE AAR9USA
 IMI WA FOR (2CR) (LF)
 K (2CR) (LF)
 (WAR transmits):
 (5 SPACES) (2CR) (LF) (LTRS)
 DE WAR IMI WA
 FOR MAIL (2CR) (LF)

K (2CR) (LF)
 (AAK9USA
 transmits):
 (5 SPACES) (2CR) (LF) (LTRS)
 DE R9USA E N R 4
 AR (2CR) (LF)

(3) Should either the transmitting or receiving station wish to indicate that he or she has further traffic to transmit to the station that he or she is working, this will be done by the prosign "B" in the message ending or receipt.

EXAMPLE A: (AAR2USA, in transmitting a message to AAR5USA wishes to indicate that he or she has further traffic for AAR5USA and transmits):

TEXT
 BT
 B
 K
 (AAR5USA transmits):
 DE AAR5USA R K

EXAMPLE B; (WAR in transmitting a message to WAR/A, wishes to indicate that he or she has further traffic for AAR2USA and AAR9USA, transmits):

WAR/A DE WAR (TEXT) B FOR
 AAR2USA AND AAR9USA K
 (Net responds):
 DE AAR2USA R K
 DE AAR3USA R AR
 DE AAR4USA R AR
 DE AAR5USA R AR
 DE AAR6USA R AR
 DE AAR9USA R K

h. Verifications,

(1) Requests for verifications are initiated by address only and may be made by service or regular message.

(2) The prosign "J" may be used to represent verification of the heading, or portions of the heading of any message; the entire text of any message, or a portion of the text of a plain language message. (See para 13-10d.)

i. Confirmation.

(1) When less than 50 percent of a message consists of figures, unusual letter combinations, or figures and letter combinations, such groups may be confirmed. Unless for operational reasons, consequent additional transmission (such as operator's notes) should not be made and are prohibited.

2) It is the responsibility of the station called to compare the confirmation with the text to ensure correctness prior to delivery or refile.

3) The following confirms a portion of the text:

EXAMPLE A: Station WAR confirms figures and figures and letter combination.

(5 SPACES) (2CR) (LF) (LTRS)
 AAR5USA DE WAR
 NR12 (2CR) (LF)
 (LTRS)
 R 181545Z MAR 85 (2CR) (LF)
 (LTRS)

FM CDR USAISC
 CHIEF ARMY MARS (2CR) (LF)
 TO CDR 7TH SIG
 COMD (2CR) (LF)
 GRNC
 BT
 FORWARD 15 BOXES PAPER
 WEIGHT 200W
 NEXT COURIER
 BT
 CFN 15 200E (2CR) (LF)
 K (2CR)
 (4LF)
 NNNN (12 LTRS)

EXAMPLE B: Before acknowledging receipt.

The operator at AAR5USA discovers a discrepancy between the text and confirmation. AAR5USA requests confirmation of the word after the word "weight."

(5 SPACES) (2CR) (LF) (LTRS)
 DE AAR5USA ZDL2
 INT WA WEIGHT K (2CR) (LF)
 (WAR transmits):
 DE WAR ZDL2 WA
 15 200W K (2CR) (LF)

EXAMPLE C: After acknowledging receipt, the operator at AAR5USA finds a discrepancy between the text and confirmation. AAR5USA services WAR for the correct portion:

(5 SPACES) (2CR) (LF) (LTRS)
 WAR DE AAR5USA (2CR) (LF)
 SVC BT ZUI 181545Z
 MAR 84 (2CR) (LF)
 (LTRS)
 ZDL2 INT WA
 WEIGHT BT K (2CR) (LF)

15-5. Correction.

a. Keyboard transmission. When an error is detected during transmission, the operator transmits the error prosign "EEEEEEEE" (eight E's) after the error and resumes transmission by repeating the last word or group transmitted correctly.

EXAMPLE: "IN ACCORDANCE WITH
MESGAEEEEEEEE WITH
MESSAGE SENT:

Errors made in a message heading will be corrected by use of the error prosign followed by the last prosign or operating signal correctly sent.

b. Tape preparation. Errors made in preparing chad tapes will be corrected by backspacing the tape and rubbing out or lettering out the error by means of the LTRS key. Errors made in preparing chadless tapes will be corrected by use of the error prosign except that when errors occur in a message heading, a new tape will be prepared (see chap 17).

c. End of month (EOM). If the transmitting operator discovers an error has been made which was not corrected as indicated above, the error may be corrected at the end of the message. Such corrections will be separated from the prosign "BT" by 2CR 1LF and will be preceded by the prosign "C."

d. Corrections. The called station is responsible for making any corrections indicated in the correction line prior to delivery or refile. In instances where the time element is such that immediate delivery is warranted without all indicated corrections, the addressee should be apprised of the corrections).

e. Characters and spaces. The number of characters and spaces typed on one line should not exceed 69 (para 15-1(d)(5)), Sending operators should perforate or type the message in such a manner that a long word falling at the end of a line will not be hyphenated. In order to prevent hyphenating a word at the end of the line, strict attention must be paid to the EOL indicators (warning light or margin bell, depending upon the type of transmitting equipment used).

15-6. Break-in procedure.

a. A station having a message of higher precedence than the transmission in, progress may break-in at the end of the transmission and thus suspend further transmission in the following circumstances: Priority may break-in at the end of the transmission and pass the message. A preliminary call may be made before transmitting the message if necessary. On a directed net, NCS

approval to transmit the message must be obtained.

b. The prosign "BT" transmitted three times means "CEASE TRANSMISSION IMMEDIATELY." Silence

will be maintained until the station breaking in has passed his or her message.

c. Break-in procedure for message of priority precedence is illustrated as follows:

EXAMPLE: AAR2USA is transmitting a long Routine message to AAR5USA when WAR is handed a Priority message for AAR4USA. When AAR2US A pauses, WAR transmits):

BT BT BT

(AAR2USA hearing the prosign "BT" three times does not resume transmission, WAR then transmits:

AAR2USA DE WAR PRIORITY
AAR4USA K

WAR DE AAR2USA SEND IP
AAR4USA AR

(WAR transmits Priority message):

DE AAR4USA R AR (AAR2USA then continues his or her transmission):

AAR5USA DE AAR2USA AA

15-7. Return to directed net.

When a net is informal, any station entering the net initially, or a station listing traffic, will automatically convert an informal net to a directed net. The NCS can also return the informal net to a directed net.

15-8. Checking out of a net.

Stations having to leave the net early will notify the NCS when they leave the net by asking permission to close station.

15-9. Reentering a net.

Stations that may have had to leave the net early and wishing to reenter must do so as if this is their initial check-in.

15-10. Closing a net.

The NCS will close the net at the scheduled time by announcing the net call and his or her own call, and stating the net is closed and the prosign "AR."

15-11. Packet radio operations.

Packet radio operations also apply to HF radio systems but the baud rate must be reduced to keep the transmission within specific bandwidths. See paragraph 16-5 for further details on packet radio operations.

CHAPTER 16

VERY HIGH FREQUENCY OPERATIONS

16-1. General.

This chapter contains specific instructions for the conduct of communications employing the various signaling modes that apply to VHF radio operations. Adherence to prescribed procedures is mandatory. Unauthorized departures from or variations in prescribed procedure often create confusion and reduce reliability and speed. If the procedure prescribed herein does not cover a specific operating requirement, initiative and common sense should prevail. The basic rules essential to circuit discipline and good transmission practices are outlined in paragraph 9-2. When authentication is required, it will be accomplished in accordance with paragraph 9-22. Because of line-of-sight propagation VHF radio operations are characterized by the relatively short range or local nature of such operations, and by the frequent use of repeaters to enhance the power and range of individual stations.

16-2. Very high frequency procedures.

Normal day-to-day MARS VHF net operations employ the voice mode and standard radiotelephone procedures. The procedures outlined in chapter 14 apply to VHF radiotelephone operations, with the sole exception that net opening procedure is simplified in accordance with that indicated in paragraph 16-3.

16-3. Establishing a very high frequency radiotelephone net.

- a. The use of procedures prescribed below will be followed when opening a net for the first time or when reopening a net. Proper control by the NCS and adherence to operating rules by the stations within the net enable the net to begin and maintain operations with minimum delay.
- b. Complete call signs will be given phonetically when establishing initial communications.
- c. At the designated time or when ready to establish the net, the NCS transmits:

WAR/C THIS IS WAR OVER Each subordinate station then answers the call in such order as designated by the NCS. THIS IS AAR3ABC NO TRAFFIC OVER THIS IS AAR3BCD NO

TRAFFIC OVER

THIS IS AAR3USA NO TRAFFIC OVER
THIS IS AAR3USB NO TRAFFIC OVER
THIS IS AAR3USC NO TRAFFIC OVER

NOTE: In order that local stations entering the net understand their position in the alphabetical order, it will be necessary that such stations be specifically assigned to the net. Following completion of net roster, unassigned stations may request and receive NCS permission to enter net.

WAR now calls the net to inform all stations that their transmissions have been heard and advises if he or she has traffic for them: WAR/C THIS IS WAR ROGER AAR3ABC, AAR3BCD, AAR3USA, AAR3USB, AAR3USC OUT

- (1) When a subordinate station is unable to answer a net call, the next station in the alphabetical reporting order will wait 5 seconds and then transmit.

(WAR transmits):

WAR/C THIS IS WAR OVER

(Stations respond):

THIS IS AAR3ABC NO TRAFFIC OVER THIS IS AAR3BCD NO TRAFFIC OVER (AAR3USB hearing no answer from AAR3USA waits approximately 5 seconds and then transmits):

THIS IS AAR3USB NO TRAFFIC OVER THIS IS AAR3USC NO TRAFFIC OVER (After receiving answers from all stations except AAR3USA, WAR transmits): THIS IS WAR ROGER

AAR3ABC, AAR3BCD, AAR3USA, AAR3USB, AAR3USC OUT

(AAR3USA, when able to transmit, calls the NCS to report into the net):

THIS IS AAR3USA NO TRAFFIC OVER

(WAR having no traffic for AAR3USA transmits):

THIS IS WAR ROGER AAR3USA OUT

- (2) Once the net has been established and the NCS desires to authorize the use of abbreviated call signs, the NCS transmits:

WAR/C THIS IS WAR ABBREVIATED CALL SIGNS AUTHORIZED THIS NET OUT

16-4. Operations.

MARS VHF operations may be conducted as simplex, single repeater, or multiple repeater operations.

a. In operating simplex, transmission and reception occur on an established frequency and no repeater is involved. In view of the effect of terrain on line-of-sight propagation, simplex operation often involves a complex series of relays in order that the net may be made an operational entity. While more time consuming than a repeater-linked net, simplex operation provides excellent training in tactical-type field operations and an understanding of the effect of terrain in a given local situation. During an emergency situation, often involving power outages affecting both repeaters and individual fixed stations, MARS operators are forced to employ simplex operation using battery power in order to maintain communications. A thorough understanding of and experience with simplex operation in the local situation is therefore important,

b. In operating VHF with a single repeater, the repeater acts as an amplifier for individual stations in the local net. Separate frequencies are employed for transmission (up frequency) and reception (down frequency) through the repeater. The repeater antenna is normally located high in order to provide maximum line-of-sight contact with individual stations in a given area, thus overcoming many of the difficulties associated with terrain. In operating with a single repeater, however, the MARS operator is faced with the problem of properly locating his or her antenna to fully use the amplifying properties of the repeater.

c. Ideal conditions for VHF operations in a local situation are approached when it is possible to employ two or more repeaters in widely spaced locations. By judicious placement, multiple repeaters can virtually eliminate the adverse effect of terrain on line-of-sight propagation. In itself, each repeater acts as a single repeater using frequencies in common with other repeaters in the group and simultaneously triggering the latter to accomplish maximum coverage of the area.

d. Many repeaters incorporate a built-in timer which cuts off the receiver-transmitter function after the repeater has been in use by an individual station for 2 or 3 minutes. Before this occurs, the transmitting operator must cease transmitting, release his or her key (XMT button), and allow time for the repeater to reset. He or she may then continue the transmission. The prescribed procedure for resetting the repeater timer is as follows: WAR is transmitting a long message and determines that he or she is approaching the time limit of the repeater. At the end of a sentence he or she pauses, transmits the proword "WAIT," then unkeys the transmitter for 2 or 3 seconds to allow the repeater to reset, after which he or she continues with the message. This procedure must be accomplished during each increment of transmission

time allowed by the repeater.

e. The easy portability and reduced power requirement of VHF equipment make it ideal for mobile use. Such mobility is invaluable in the conduct of emergency net operations, both real and simulated, and in providing communications for military and civic events,

f. The characteristics of VHF operations make it an ideal means for establishing emergency net operations, nets devoted to training in radiotelephone procedures, nets devoted to MARS administration, technical discussions, and other subjects related to the integration of the local MARS membership. It is the policy of Chief, Army MARS that VHF operations be established wherever practicable in the support and discharge of the MARS mission at the local level.

g. Adaptation of existing VHF equipment is possible to permit its use in the transmission of modulated signals using the international Morse code, as well as the audio tones (baudot or ASCII) required for radioteletype, computer, and packet radio operations.

h. The system known as autopatch employs VHF signaling equipment and special repeater equipment to permit the dialing of a commercial telephone number from the individual radio station, whether fixed or mobile. A touchtone pad is used to create coded tones at the transmitter. These tones key the repeater causing it to access the local commercial telephone system. The commercial telephone number is then dialed in the usual manner using the touchtone pad. Care must be taken not to exceed the transmission time allowed by the repeater timer. Procedures for the use of the local autopatch circuits are established by the area MARS director having jurisdiction over the autopatch repeater. Prescribed transmission practices and circuit discipline must be observed at all times when using autopatch.

16-5. Packet radio operations.

a. Packet radio operations use specialized circuitry in conjunction with VHF radio transceivers to accomplish high speed data burst transmission and reception. Error correcting circuitry is normally incorporated into the system. It is considered ideal for the handling of record traffic as well as computer data transfer using either single or multiple linked repeaters. Packet radio repeaters usually will accommodate several interference-free transmissions simultaneously.

b. Where packet radio operations are employed in passing record traffic, the procedures for formatting and transmitting messages will conform to the standardized procedures indicated in paragraph 15-4.

c. Where packet radio operations are employed during a local emergency situation, real or simulated, transmissions will conform to the standard MARS message formatting procedures.

16-6. Participation credit.

It is the policy of Chief, Army MARS that VHF operations will not be conducted to the exclusion of HF operations by individual MARS stations. At least 6 of

the 12 hours of participation required per calendar quarter for the maintenance of Army MARS membership will be on MARS frequencies in the HF portion of the radio spectrum (3 to 30 MHz).

CHAPTER 17

COMPUTER OPERATIONS

17-1. General

a. This chapter contains guidance and information on the use of personal computers in Army MARS. The inclination of technically oriented MARS and amateur radio operations to obtain and use personal computers has led to the widespread application to CW and RATT operations, in addition to the standard data processing modes (word processing, data bases, calculating, and so forth). Interface equipment and the necessary software for communications are readily available. Personal computers can bring to the MARS program the same rise in productivity and reliability that they have provided for the business and administrative segments of society.

b. Adherence to prescribed standards and procedures is mandatory. Unauthorized departures from these will result in confusion or incompatibility and interrupting or impeding MARS operations rather than enhancing them.

17-2. Applications.

a. Personal computers can be used in the following types of applications:

(1) CW communications—the use of machine-generated international Morse code produces signals that are easier to copy than hand-keyed signals, thus raising the speed and reliability of message handling. High-speed CW operations (100-150 WPM) are possible on CW nets where all members are suitably equipped. Hard-copy of received and transmitted traffic provides more accuracy for relaying and forwarding traffic.

(2) RATT communications—the use of magnetic storage media (tape and disk) is faster and easier than paper-tape operations. The capability to receive, edit, and relay traffic in minimal time provides for greater speed, accuracy, and reliability in message handling. Higher speed operation (75-1200 baud) is possible where all net members are suitably equipped and the baud rate of transmission can be accommodated within specified bandwidths of assigned frequencies.

(3) Station management—computer data bases can be used for equipment inventories, net member rosters, net lists (frequencies, times, and modes), and similar lists, thus reducing paper products and associated storage, filing, and updating.

(4) MARS administration—although word processing, with its obvious advantages in saving time and effort, is the most extensive application to administration, the computer can provide automated products for member rosters, participation reporting, training records, member status, and so forth, which can greatly reduce the paperwork in the administration of the Army MARS program.

(5) Training—computer programs will permit training on MARS procedures, on an individual basis, at the member's own rate, and at times determined by the member. These programs allow for repetition of the training as required, can provide expanded material on the subject (where on-the-air training nets have restricted time), and can incorporate interactive examples.

(6) Bulletin board systems (BBS)—these store- and-forward message systems can be used to provide WAR broadcasts, area broadcasts, and State director messages to the members at other than established net times. Messages, notices, and bulletins can be disseminated to members quickly.

17-3. Continuous wave operations.

a. The primary mode of CW operations in MARS is manual international Morse code. Any CW net with members using computers (unless specified and established as a high-speed computer CW net) will operate at the speed of the slowest operator, in accordance with MARS policy and procedures.

b. Some computer CW software is not reliable in copying hand-keyed international Morse code due to operator changes in character speed and spacing. If this occurs on a manual CW net, the member using the computer will revert to regular manual procedures to copy the traffic. Traffic cannot be refused due to computer incompatibility. If the computer software will not permit- reducing speed to that of the slowest operator on a manual CW net, the computer will not be used in transmission of traffic.

c. High-speed, computerized, CW nets can be established by area directors for efficiency in traffic handling. Net members will be limited to those with automated capability. Net operating speed will be limited to the slowest operating system used by any member of the net.

Highspeed CW nets will not be established to eliminate manual CW nets, but will be additional nets. A sufficient number of manual CW nets must still be active to permit CW participation by the majority of MARS members operating in the manual code.

d. All CW operations will be in accordance with standard MARS CW procedures and policies.

17-4. Radioteletypewriter operations.

a. The primary mode of MARS RATT operation is 60 WPM and baudot 5-level code, 850 Hz frequency shift. All RATT nets (except those specifically authorized as high-speed data nets) must be capable of operating in the primary mode to permit participation by members using electromechanical teletype equipment. The primary speed can be 100 WPM when authorized by the area director.

b. High-speed computerized data nets can be established by area directors for increased efficiency in traffic handling. Each such net will operate in a specified mode of operation. The standard modes are—

(1) ASCII code, 110 baud, 170 Hz frequency shift keying.

(2) ASCII code, 300 baud, industry standard 103 tone pairs, frequency shift keying.

(3) ASCII code, 1200 baud, industry standard 212 tone pairs, frequency shift keying.

c. Transmission of coded or encrypted text is prohibited except when specifically authorized by the area director.

d. All RATT operations will be in accordance with standard MARS operating procedures and policies (chap 15).

e. Transmission of computer programs (software) is not permitted on data nets unless specifically authorized by the area director.

17-5. Station management.

a. Development of special computer programs tailored to MARS requirements is encouraged. The availability of such programs should be made known to other members and the area director. Area projects have been initiated to establish lists and libraries of MARS software, to permit widespread distribution among MARS members.

b. The use of commercial, copyrighted programs, or modified versions of such programs is limited to the original purchaser of the program, unless the licensing agreement specifically permits distribution. This provision also applies to programs obtained from computer books or magazines.

c. Recordkeeping or data base programs which contain information on MARS members fall under the purview of the Federal Privacy Act. Such information (roster lists with addresses and telephone numbers, net member listings, and so forth) must be protected from accessibility by third parties and cannot be published or distributed without written permission from the member whose data is contained in the files.

17-6. MARS administration.

a. Development of special computer programs tailored to MARS requirements is encouraged. The availability of such programs should be made known to other MARS officers and the area director. Area projects have been initiated to establish lists and libraries of MARS software, to permit widespread distribution among MARS officers.

b. The use of commercial, copyrighted programs, or modified versions of such programs is limited to the original purchaser of the program, unless the licensing agreement specifically permits distribution. This provision also applies to programs obtained from computer books or magazines.

c. Recordkeeping or data base programs which contain information on MARS members fall under the purview of the Federal Privacy Act. Such information (roster lists with addresses and telephone numbers, State and area officer listings, and so forth) must be protected from accessibility by third parties and cannot be published or distributed without written permission from the member whose data is contained in the files. Authorization for release of certain types of records will be provided by Chief, Army MARS.

17-7. Training.

a. Development of computer programs for MARS training is encouraged. Emphasis should be placed on originating interactive, hands-on programs with examples and tests, rather than just narrative text presentations. The availability of such programs should be made known to training officers and the area director. Area projects have been initiated to establish lists and libraries of MARS software, to permit widespread distribution among MARS training officers.

b. The use of commercial, copyrighted programs, or modified versions of such programs is limited to the original purchaser of the program, unless the licensing agreement specifically permits distribution. This provision also applies to programs obtained from computer books or magazines. Many copyrighted holders of programs

which permit the development of training packages by using the master program will permit distribution of the derived program upon written application. This is particularly true for cases where the resulting program is distributed with no financial gain to the distributor. Some licensing agreements give the purchaser the right to distribute derived programs, while restricting use of the master program to the purchaser. MARS members are responsible for compliance with licensing agreements and copyright laws applying to their computer programs.

17-8. Bulletin board system operations.

a. The establishment of MARS computer bulletin boards is encouraged. Programs are readily available to provide this service either over wire (telephone lines) or radio (HF or VHF).

b. Computer bulletin boards are considered to be a form of RATT operations and, as such, will comply with MARS RATT procedures and policies contained in chapter 15. The standard speed and mode of operation will be 60 WPM baudot code, 850 Hz shift for radio. Wireline standards will be 300 baud ASCII, industry standard 103-tone pairs.

c. Radio operation will include provisions for station identification, use of call signs, and all MARS operating procedures which are compatible with program operation. Frequencies and time periods for BBS operation will be authorized by the area director. A control station or system operator must monitor the operation at all times, to permit closing down the operation in the event of interference to other nets or equipment malfunction or posting of prohibited material. An alternate control station will be appointed to assume

the duties of the system operator when the primary operator is not available for duty.

d. Data on the bulletin boards, such as messages, files, or bulletins, will be restricted to official MARS matters. Commercial or political messages are prohibited. The BBS operator is responsible for monitoring the data on the bulletin board to ensure compliance.

e. Coded or encrypted data is prohibited on MARS bulletin board systems. The use of coded log-on passwords is permitted.

f. MARS operating logs will be kept on a monthly basis for all BBSs. System operating logs (user lists, stored messages, and so forth) will be made available for inspection by the State or area directors, if required. File copies of formal MARS messages entered into the system will be retained by the system operator for 90 days.

g. Storage and distribution of commercial copyrighted computer programs through upload or download functions of the bulletin board is prohibited. The BBS operator is responsible for ensuring that permission has been obtained for distribution of any computer program files.

h. Participation report credit will be given only for formal MARS messages entered into or retrieved from the system. The member entering a formal MARS message into a BBS for relay or delivery is still responsible for the message until it has been retrieved and rogered for by another member. The member retrieving a formal MARS message for delivery or further relay will leave a message for all members rogering for the message. The system operator is responsible for deleting rogered messages to prevent duplication by later retrieval by another member.

APPENDIX A

REFERENCES

Section I. Required Publications

AR 25-6 U.S. Army Military Affiliate
Radio System (MARS)

Section II. Related Publications

AR 105-70	Amateur Radio Operations	DODD 4650.2	Military Affiliate Radio
AR 140-1	Mission, Organization, and Training	NGB 105-75	System (MARS)
AR 140-185	Training and Retirement	NGB 640-1	Military Affiliate Radio
	Point Credits and Unit Level		System
	Strength Accounting Records		Retirement for Members of
AR 340-3	Official Mail Cost Control	FCC Rules &	the Army National Guard
	Program	Regulations	Part 97, Amateur Radio
AR 340-18	The Army Functional Files	American Radio	Service
	System	Relay League	
FM 24-18	Tactical Single-Channel	(ARRL)	
	Radio Communications	ACP 121 (E)	Communication Instructions
	Techniques		General
DODD4160.21M	Defense Disposal Manual		

Section III. Required Forms

DA Form 630	Application for Membership in Military Affiliate Radio System (MARS)	DA Form DDDD-R	Monthly Phone Patch Report
DA Form 4016	Telecommunication Center	DA Form IEEE-R	MARS Radiogram
	Originating Message Register	DA Form FFFF-R	Net Control Station Report
DA Form 4341	Radio Station Log	DA Form GGGG-R	Interference Report
DA Form 4342	Radio Wire Integration Log	DA Form HHHH-R	Army MARS Questionnaire
DA Form AAAA-R	MARS Traffic and Participation Report	DA Form IIII-R	MARS Membership
		DA Form JJJJ-R	Termination Request
			Army MARS Equipment Request

Section IV. Related Forms

DA Form BBBB-R	MARS Member Information Sheet	DA Form 3328	Supply Actions
DA Form CCCC-R	State Consolidated	DD Form 1348-1	Property Book Page
	Participation Report		DOD Single Line Item
DA Form KKKK-R	Screened by U.S. Army	DD Form 2350	Release/Receipt Document
	MARS Tag		DOD Military Affiliate Radio
DA Form 145	Army Correspondence	PG Form 17GX	System (MARS) Official Mail
	Course Enrollment		Stamp Requisition Form
	Application	GSA Form 2946	Official Mail Stamp
DA Form 1149	Requisition and Invoice/ Shipping Document	DPDS Form 130	Requisition Form
			Screeners Identification Card
DA Form 1687	Delegation of Authority		Screeners Tally Request to
DA Form 2064	Document Register for		Freeze Excess/Surplus Property

APPENDIX B

OPERATING SIGNALS (Q AND Z)

B-1. Operating signals are a concise code designed primarily for use by communications personnel in exchanging information incident to the handling of messages or in establishing communications. They are also used in service messages and other forms of messages between communication personnel.

B-2. Operating signals provide no security and therefore must be regarded as the equivalent of plain language.

B-3. The Z signals shown in table B-1 are designed to cover military requirements and should be used whenever necessary in military communications. The Q signals shown in table B-2 may be used in military communications where no suitable Z signal exists. Only Q signals will be used in nonmilitary communications.

B-4. The meaning of Q. and Z operating signals may be amplified or completed by the addition of appropriate call signs, time groups, complementary groups, and so forth. Call signs used to complement an operating signal normally follow the signal, but under certain conditions,

such as to achieve clarity or to effect separation, they may be placed ahead of the operating signal. Plain language is prohibited except when no other method is provided to complete the meaning. B-5. When desired, an operating signal may be given an interrogative sense—
h. When communicating with military stations by inserting the prosign "INT" before the Q and Z signal.
i. When communicating with nonmilitary stations, by inserting the prosign "IMF" after the Q signal and data used with it.

B-6. Operating signals should not normally be used in radiotelephone procedures. Instead, the operating information will be conveyed by concise phrases. When it is necessary to relay operating signals over voice circuits, that is, in messages, they are transmitted by their phonetic equivalents.

B-7. Blank spaces in the meaning of Q and Z signals will be completed in the order in which they appear; however, blank spaces enclosed in parentheses normally will be completed on an optional basis only.

Table B-1. Z Signals.

Signal	Question	Answer, Advice, or Order
ZAA		You are not observing proper circuit discipline.
ZAB		Your speed key is improperly adjusted.
ZAC		Cease using speed key
ZAD		Your operating signal (made at) received as is— b. Not understood. c. Not held,
ZAE		I am unable to receive you. Receipt through
ZAF	Will you patch me through to ____ (Via you ____ or ____).	I am about to patch you through to ____).
ZAG		Break; b. Go ahead with next tape. c. Go back ____ feet. d. Advance your key tape to reference mark number and repeat last transmission for transmission ____).
ZAH		Unable to relay message in present form: (a) Not in prescribed format. (b) Format lines incorrect. (c) No online facility available. (d) Call signs not encrypted. (e) Text not encrypted. We file. Transmit correctly prepared message to all addressees (or to ").

Table B1 - Z signals Continued

Signal	Question	Answer, Advice, or Order
ZAI		Run— 1. Caller. 2. Test tape. 3. Synchronizing tape. 4. Traffic tape. 5. Marking signals. 6. Spacing signals. 7. Reversals. 8. Run Teletypewriter space bar signals
ZAJ		I have for has) been unable to break vou.
ZAK		Transmission on kHz (or MHz) suspended until ____ (or was suspender! at 1 on account, of electrical hazards (or).
ZAL		I am closing down (until) due to
ZAN		Transmit only messages of and above precedence ____
ZAO		I cannot understand your speech. Use radiotelegraph.
ZAP	Shall I work 1. Simplex? 2. Duplex? 3. Diplex? 4. Multiplex? 5. Single sideband? 6. With automatic error correction system? 7. Without automatic error correction system?	Work— 1. Simplex. 2. Duplex. 3. Diplex. 4. Multiplex. 5. Single sideband. 6. With automatic error correction system. 7. Without automatic error correction system.
ZAQ		The last word (or group)— 1. Received from vou was 2. Transmitted to vou was
ZAR		This is my ____ request (or reply). 1. First. 2. Second. 3. Third, and so forth.
ZAS		Rerun all tapes run on since— 1. Your present frequency. 2. kHz (or MHz). 3. Call sign 4. This channel or channel).
ZAT		Am preparing traffic (perforating tape) for transmission
ZAU	What is teletype range taken on my signal(s)?	Teletype range taken on your signal(s) is to
ZAV		Transmit traffic for me (or for) by broadcast method until further directed (or until ____)
ZAW	Shall I transmit 12 inches of blank tape at short intervals at ____ WPM?	Transmit 12 inches of blank tape at short intervals at WPM.
ZAX		You are— 1. Causing interference. Listen before sending. 2. Causing interference by inattention to order to wait. 3. Sending at the same time as (call signal). 4. Causing delay by slowness in answering my service or procedure messages. 5. Answering out of turn
ZAY		Transmit traffic to me (or to ____ on kHz (or MHz)) without waiting for receipts. I (or ____) will receipt for traffic later (on kHz (or MHz)').
ZAZ		One or more transmitters, simultaneously keying on this broadcast are defective but transmission will continue on remaining transmitters. Traffic so transmitted will be repeated when full service is restored.
ZBA	What is cause of delay (or of bad transmission)?	Delay (or bad transmission) due to

Table B1 - Z signals Continued

Signal	Question	Answer, Advice, or Order
ZBB		For following message you will require a total of copies.
ZBC		You are transmitting a continuous— 1. Mark. 2. Space.
ZBD	Will you repeat what you (or ___) sent (at ___)?	Following is what I (or ___) sent, (at ___) 1
ZBE		Retransmit message ___ to ___ (for ___) 1, Action, 2. Information.
ZBG		You are transmitting in upper case.
ZBH		Make preliminary call before transmitting traffic.
ZBI		Listen for radiotelephony.
ZBK	Are you receiving my traffic clear?	I am receiving your traffic— 1. Clear. 2. Garbled.
ZBL		I am unable to receive you while I am transmitting.
ZBM		Do not use break-in procedure. Place on watch on this frequency. 1. A qualified speed key operator. 2. A competent operator.
ZBN		Your is reversed. 1. Tape 2. Mark and space.
ZBO	Of what precedence(s)?	I have (or has) (numeral followed by precedence prosign for each precedence message(s) for you and for).
ZBP		Your— 1. Characters are indistinct 2. Spacing is bad.
ZBQ	When and on what frequency was message received?	Message was received at on ___ kHz (or MHz).
ZBR	Shall I send by— 3) Direct (R) method? 2. Broadcast (F) method? 3. Intercept (I) method? 4. Repeat back (G) method?	Send by— 1. Direct (R) method. 2. Broadcast (F) method. 3. Intercept (I) method. 4. Repeat back (G) method.
ZBT	How do you count following text group(s) ___ ?	Text, group(s) should be counted group(s).
ZBU		Report when you are in radio communication with .
ZBV		Answer me (or ___) on ___ kHz (or ___ MHz).
ZBW	Will you shift, for ask to transmit, on kHz or MHz)?	I am (or ___ is) shifting to transmit, on ___ kHz (or MHz),
ZBX	Will you shift for ask to receive on ___ kHz (or MHz)?	I am (or ___ is) shifting to receive on ___ kHz (or ___ MHz).
ZBZ	What is the printing accuracy of my signals for those of ___ ?	The printing accuracy of your signals (or those of) is— 1. Totally corrupt 2. Very corrupt 3. Partly corrupt rendering traffic unacceptable 4. Occasionally corrupt but traffic acceptable. 5. Perfect—no corruption.
ZDA		I have a formal message for you (precedence is).
ZDB		Expedite reply(ies) to my— 1. Previous operating signal. 2. Request for repetition and correction. c. Service message
ZDE		Message undelivered— 1. Station closed until 2. Attempting to alert station to reopen (for Flash or Immediate precedence only). 3. Will continue efforts to effect disposal.

Table B-1 Z Signals - Continued

Signal	Question	Answer, Advice, or Order
		4. Advise disposition.
		5. Will not continue further efforts. Request cancel and file.
		6. Give more complete address.
ZDF		Message was— 1. Received by (addressee(s) designation) at ____ Z. 2. Received by (addressee(s) designation communications center) at ____ Z. 3. Delivered to (addressee(s) designation) by broadcast at ____ Z. 4. Forwarded to (addressee(s) designation) by commercial means at ____ Z. 5. Mailed to (addressee(s) designation) at ____ Z
ZDG		Accuracy of following message(s) or (message) ____ is doubtful. Correction or confirmation will be forthcoming.
ZDH		Request corrected copy of message be forwarded to ____.
ZDJ	How many groups does your message contain?	I have a message containing ____ groups to transmit to you (or to ____).
ZDK	Will you repeat message (or portion ____) (or will you rerun number ____)?	Following repetition of is made in accordance with your request
ZDL		Confirmation— 1. Was omitted 2. Differs from text.
ZDM		I am holding your message
ZDN		Report disposal of message your station with any reason for delay.
ZDO		I could not send message to ____
ZDP		Hold my message until correctness is confirmed.
ZDQ		Message __ was relayed to __ at __ by (on __ kHz? (or MHz)).
ZDR		This is a multiple address or book message tape containing routing indicator in the routing line which is to be routed in accordance with the established doctrine.
ZDS		Message (or message) which you have (or has) just forwarded was incorrectly transmitted. Correct version of message (or part or portion) is
ZDT		Exercise messages are not to be sent until further orders for until ____)
ZDX		Messages up to and including serial (or circuit) number have been previously transmitted.
ZDZ		On (date) message(s) was (were) mailed to ____ (Command(s)) bearing serial numbers .
ZEC	Have you received message ?	Message 1. Not received 2. Unidentified. Give better identifying data.
ZED		The following confirmatory information as received is at variance with the text.
ZEE		Request message be transmitted.
ZEH		Accuracy of 1. Heading 2. Text. 3. Group to portion of following message (of message) is doubtful. Correction or confirmation will be forwarded when received:

Table B-1 Z Signals - Continued

Signal	Question	Answer, Advice, or Order
ZEI		Accuracy is doubtful of heading of message received as follows . Check to station of origin if necessary and repeat.
ZEJ		Replies to this message (of message ...) are to be sent now (or at) .
ZEK	Is message a correction to _____ which was previously transmitted with doubtful or missing groups (words)?	No answer is required.
ZEL		This message is a correction (to message transmitted hv). Note: May only be used in conjunction with ZDG.
ZEN		This message has been delivered by a separate transmission or by other means (1) messenger or courier or (2) mail to the addressee(s) immediately following this operating signal.
ZEP		This message (or message) was incompletely received. Each word or group missed, which is indicated by position of ZEP in the message, will be forwarded as soon as obtained.
ZEQ		Message has been— 1. Missent to this station. Received at Z. Message protected. 2. Missent. to this station and unable to protect delivery due to . Request von retransmit to the appropriate called station(s). 3. Misrouted to this station and has been rerouted to at Z. Recommend corrective action. 4. Misrouted to this station and unable to protect delivery (for) due to Request your station protect delivery. 5. Routed to this station in format line 2, but without delivery responsibility in format lines 7 or 8. Advise.
ZEE		This is a multiple address or book message tape containing routing indicators in the routing line for which the station called in the pilot is responsible.
ZES		Your message has been received (1) incomplete or (2) garbled. Request retransmission.
ZET		Message has been protected and no further action by _____ is required,
ZEU		Exercise (drill) message.
ZEV	Request you acknowledge message	Message (or message) is acknowledged.
ZEW		Your attention is invited to message which is in your files for— 1) Action, 2) Information.
ZEX		This is a book message and may be delivered as a single address message to addressees for whom you are responsible.
ZEY		This message (or message) has been (or will be) put on the broadcast schedules (serial number) .
ZEZ		When this book message is delivered to the addressee by commercial means, or when copy is forwarded to confirm message previously delivered by telephone, it is to be sent as a single address message. (May only be used in conjunction with ZEX.)
ZFA		Following message has been read (received or intercepted).
ZFB		Pass this message — 1. On arrival. 2. On return to base.
ZFC	Have you received (or sent) the executive signal ("Execute") for message ?	Executive signal ("Execute") for last message (or following message) has been made (or was made at _____).

Table B-1 Z Signals - Continued

Signal	Question	Answer, Advice, or Order
ZFD		This message is a suspected duplicate.
ZFE		Pass message (which is in your files) to addressee(s) for whom you are responsible using the indicated supplementary heading.
ZFF		Inform me when this message (or message) has been received by (addressee designation) or by— 1.Action addressee(s). 2.Information addressee(s). 3.All addressees. 4.Action addressees or addressees message center. 5.Information addressees or addressees message center. NOTE: Not authorized for use on general messages.
ZFG		This message is an exact duplicate of a message previously transmitted.
ZFH		This message (or message____) is being (or has been) passed to you (or____) for— 1.Action. 2.Information. 3.Comment, (at the request of _____).
ZFI	Is there any reply to message ?	There is no reply to message
ZFJ		Message(s) number(s) not transmitted on this schedule (or on schedule) are no longer needed.
ZFK	Does message (or SSN concern me for (or)?	Message____ (or SSN____). 1. Does not concern you (or____). 2. No longer concern you (or____).
ZFL	Was there any traffic addressed to me on broadcast schedule serial numbers and ?	Following traffic was addressed to you on broadcast schedule between serial numbers and .
ZFM		Message(s) serial numbers____ (to____) which was (were) transmitted during unserviceability of one or more transmitters, simultaneously keyed, will now be repeated.
ZFQ		Two messages (and) received under channel number (or SSN). . Both released.
ZFR		Cancel transmission ... (made under channel or SSN
ZFS		Make message same channel or station serial number at this procedure message.
ZFT		Message received without channel number (or SSN) following message bearing channel number (or SSN) . Message released.
ZGA	What is my call sign for use on this circuit only?	Your call sign for use on this circuit only is _____.
ZGB		Send (answer)— 1. In alphabetical sequence of call signs, each station to make its call signs once (or _____ times). 2. In the following sequence of call signs.
ZGE		Send your call sign(s) once (or time(s) on this frequency (or on kHz (or MHz)).
ZGF		Make call signs more distinctly.
ZGH		I am using this transmitter to answer calls on two or more frequencies. Answer to calls may be delayed.
ZGI		I am or have been (or is or has been) calling you (on kHz (or MHz)).
ZGJ	When will you call me again on present frequency (or on) kHz (or MHz)	I will call you again as soon as possible (or as____) on present frequency (or____ on kHz for____ MHz)).
ZGK	When shall I call you again on present frequency (or on kHz for MM)?	Call me again at on present frequency (or on kHz (or MHz)).

Table B-1 Z Signals - Continued

Signal	Question	Answer, Advice, or Order
ZGM	Did you (or _____) hear _____ (at _____)?	I have (or _____ has) been unable to communicate with _____ (since _____).
ZGN	When was I for) last heard?	Nothing heard from you (or _____) (since _____).
ZGO	What is my number and sequence in answering?	Your number is _____. Answer after number _____. (Numbers to be separated by separative sign.)
ZGP		Answer calls for me on present frequency (or on _____ kHz or _____ MHz)).
ZHA	Shall I decrease frequency very slightly for kHz to clear interference?	Decrease frequency very slightly (for _____ kHz to clear interference.
ZHB	Shall I increase frequency very slightly for kHz to clear interference?	Increase frequency very slightly for _____ kHz) to clear interference.
ZHN	How do you receive my automatic transmission?	Your automatic transmissions are— 1. Good. 2. Fair. 3. Unreadable.
ZHO	What is your speed of automatic transmission in— 1. Revolutions per minute 2. WPM? 3. Bauds?	My speed of automatic transmission in (1) revolutions per minute. f2) WPM. and f3) bauds is _____.
ZHP	What is preventing automatic reception?	Automatic reception is prevented by _____.
ZHQ		Please listen for me on kHz (or MHz), and transmit to me on kHz (or MHz).
ZHR	Is my speed of automatic transmission correct?	Your speed of automatic transmission is— 1. Fast. 2. Slow. 3. Erratic. 4. Correct.
ZIF		I (or _____) did not use serial numbers).
ZIG	Is (are) number(s) _____ to _____) blank?	Number(s) _____ (to _____) is (are) blank.
ZIH		Repeat headings of message Number _____ to _____ transmitted by your (or) to check serial numbers.
ZII	What was _____ of your or (_____ 's) number ? 1. DTG? 2. Filing time?	My (or _____ 's) number had following: 1. DTG _____ 2. Filing time _____
ZIJ		I am now changing my channel number or letter. Last number sent in this series is that of this message. Repeat back each group of the text of this message as it is transmitted.
ZJO		
ZKA	Who is controlling station (net station) on this frequency (or on kHz (or MHz))?	I am (or _____) controlling (NCS) on this frequency for (on _____ kHz (_____ or MHz)).
ZKB	Is it necessary to obtain permission of the controlling station (NCS) before transmitting messages?	It is necessary to obtain permission of the controlling station (NCS) before transmitting messages.
ZKD	Shall I take control of net (for _____) (until _____)?	Take control of net (for _____) (until _____).
ZKE		I (or _____) report(s) into circuit (net).
ZKF		Station leaves net temporarily (or for _____ minutes) (to communicate with _____) (will be on _____ kHz (or _____ MHz)).
ZKG		Observe for direct to observe) schedule with on _____ kHz (_____ or MHz)) at _____).
ZKH	Did you (or _____) observe schedule with _____ (at _____)?	I (or _____) observed schedule with (at _____).
ZKI		Set watch on _____ kHz (or _____ MHz)— 1. Continuous. 2. Until further notice.

Table B-1 Z Signals - Continued

Signal	Question	Answer, Advice, or Order
ZKJ	Mav I close down (until ____)?	(1) Close down until ____). (2) I am closing down (until ____)
ZKL		Resume normal radio communication now (or at ____).
ZKM		Take guard (for > (on ____ kHz for ____ MHz)).
ZKN		I have taken over guard on ____ kHz for ____ MHz.
ZKO		I have handed over guard(to ____) (on ____ kHz (or ____ MHz)) serial number of last message received was ____
ZKP	Are vou (or is) radio guard for (on ____ kHz (for ____ MHz))?(or ____ MHz))?	I am (or ____ is) radio guard (for ____ (on ____ kHz for ____ MHz)).
ZKB	On what frequencies are you (or) maintaining watch?	I am (or ____ is) maintaining watch on ____ kHz (or ____ MHz).
ZKS	What stations are keeping watch on kHz (or MHz) (or are in net)?	Following stations are keeping watch on ____ MHz (or ____ MHz) (or are in net).
ZKV		I am (or ____ is) standing splitphone watch on ____ and ____ kHz (or ____ MHz).
ZNB	What is authentication of— 1. Message? 2. Last transmission? 3. ____)?	Authentication (of) is 1. Message. 2. Last transmission, 3. ____).
ZNC		All transmissions will be authenticated— 1. On all circuits. 2. On this circuit. 3. On freauency.
ZND		You are using authenticator incorrectly: 1. Verify authenticator system key. 2. Check authentication of your last transmission.
ZOB		Take (I will take) no further action regarding forwarding message
ZOC		Station(s) called relay this message to addressees for whom you are responsible.
ZOD		Act as radio link (relaying station) between me and (or between and).
ZOE	Can vou accept message for ? 1. Online? 2. Offline?	Give me your message. I will dispose of it— 1. Online. 2. Offline.
ZOP		Relay (pass) this message for message) to now (or at).
ZOG		Transmit (pass) this message for message) to (for) 1. Action. 2. Information.
ZOH		Send message for on kHz for MHz) by- 1. Receipt method. 2. Broadcast method. 3. Intercept method.
ZOK		Relay this message via
ZON		Place this message (or message _) on— 1. CW broadcast. 2. Radio teletype broadcast (indicated by following specific broadcast designator(s)).
ZOQ		Deliver this message for message)to all broadcast areas (or to the following specific broadcast area(s)).
ZOU	How should traffic for be routed?	Route traffic for through fon kHz (or ____ MHz)).
ZOV		Station Designation preceeding this operating signal is the correct routing for this message rerouted by ____.

Table B-1 Z Signals - Continued

Signal	Question	Answer, Advice, or Order
ZOY		Relay this message only to the station(s) whose designations) precede this operating signal.
ZPA		Your speech is distorted.
ZPB		Your transmitter has strong radiation while idling.
ZPC		Your signals are— 1.Fading badly. 2.Fading slightly. 3.Good for ____ WPM. 4.Getting stronger. 5.Getting weaker.
ZPD	Is your signal as checked by monitor satisfactory— 1, Locally? 2. As radiated?	My signal as checked by monitor ____ is satisfactory— 1.Locally. 2.As radiated.
ZPE		Maximum power is not being radiated.
ZPF	What is the readability of the signals of the group (net) for of)?	The readability of the signals of the group (net) (or of) is (1 to 5).
ZPO		The text of this message is to be relayed in precisely the same format as that in which it is received. No characters or machine functions are to be added, inserted, or deleted and the relative positions of the group are to be retained.
ZPR		I have ____ message (precedence) prepared in tape relay format for you for _____. (Consideration should be given to receiving this message directly on to a printing perforator or typing reperforator).
ZPW		This message canceled at time indicated. File without further transmission.
ZRA	How does my frequency check?	Your frequency is— 1.Correct. 2.Slightly (or ____ hertz (or kHz)) high. 3.Slightly (or ____ hertz (or MHz)) low. 4.Stable on steady mark. 5.Stable on steady pace. 6.Unstable. 7.Erratic.
ZRB		Check your for 'si frequency on this circuit) or on ____ kHz (or ____ MHz)).
ZRC	Shall I tune my transmitter to— 1.Proper frequency? 2.Zero beat with your (or ____) transmitter)?	Tune your transmitter to— 1.Proper frequency. 2.Zero beat with my for transmitter).
ZRE	On what frequency do you hear me best?	I hear you best on ____ kHz (or ____ MHz).
ZRF	Will you send tuning signal on your present frequency or on ____ kHz (or ____ MHz)).	Am about to send tuning signal on my present frequency (or on ____ kHz (or ____ MHz)).
ZRG	When will a change of frequency (to ____ kHz (or ____ MHz)) be necessary?	A change in frequency (to ____ kHz (or ____ MHz)) will be necessary at approximately
ZRH	Is my frequency shift correct?	Your frequency shift is— 1.Too wide. 2.Too narrow, 3.Not linear. 4.Correct (by hertz).
ZRJ	Will you check your ?	I will check my
ZRM	Can you receive both sidebands of my independent sideband transmissions?	I can receive— 1.The upper sideband. 2.The lower sideband. 3.Both sidebands.
ZRN	Do you intend to transmit on— 1.The upper sideband? 2.The lower sideband? 3.Both sidebands independently?	I intend to transmit on— 1.The upper sideband. 2.The lower sideband. 3.Both sidebands independently.
ZRQ		Change to other sideband.
ZTA		Change to— 1.Radio. 2.Visual. 3.Landline.

Table B-1 Z Signals - Continued

Signal	Question	Answer, Advice, or Order
ZTB		4. Teletypewriter (teleprinter). 5. Telephone. 6. Automatic. 7. Facsimile. 8. Shore telephone. 9. Radio teletypewriter. Change to— 1. Transmitter-distributor. 2. Autohead. 3. Perforator. 4. Reperforator. 5. Printer. 6. Undulator. 7. Keyboard. 8. Frequency shift keyer. 9. Multiplexing carrier base. 10. Online crypto device.
ZTC	1. Does my _____ appear to be faulty? 2. Is your _____ faulty?	1. Your _____ appears to be faulty. 2. My _____ is faulty.
ZTD	Shall I use _____ ?	Use _____
ATE	Are you (or is _____) able to use ?	I am (or _____ is) able to use _____
ZTF	Are repairs completed?	Repairs completed.
ZTI		Change to— 1. Receiver. 2. Transmitter. 3. Power supply. 4. Antenna system. 5. Radio direction finder.
ZTJ		Cease using _____
ZTK	Are you for is _____ connected to ?	I am for is _____ connected to _____
ZTL	Are you (or is _____) about to disconnect?	I am (or is) about to disconnect.
ZTM		I am (or is) unable to use _____
ZTN	What is my bias distortion?	Your bias distortion is— 1. Excessive. 2. _____ percent (marking). 3. _____ percent (spacing). 4. _____ percent (total distortion).
ZTG	Is my character information correct?	Your character information is— 1. Correct. 2. Defective in the start element. 3. Defective in the first unit. 4. Defective in the second unit. 5. Defective in the third unit. 6. Defective in the fourth unit. 7. Defective in the fifth unit. 8. Defective in the stop element.
ZTS	Is the line satisfactory?	The line is— 1. Satisfactory. 2. Unsatisfactory.
ZUA	Request a timing signal now for at _____).	Timing signal will be transmitted now (or at _____), The numerals _____ indicating time will be followed by a 5-second dash terminating exactly at the time indicated.
ZUB		At _____
ZUC		From to _____
ZUD		Until further orders for until _____).
ZUE		Affirmative (yes).
ZUG		Negative (no).
ZUH		Unable to comply.
ZUI		Your attention is invited to _____
ZIJ		Stand by.
ZVA		Station called is responsible for relay or delivery to all stations in line two or to stations indicated.

Table B-1 Z Signals Continued.

Signal	Question	Answer, Advice, or Order
ZVB		Retransmit this message (or message) at once to all subordinate stations.
ZWF		Incorrect.
ZWG		You are correct.
ZWH		Try again.
ZWI		Answer last question for question ____
ZWK		Following is answer to the last question (or question ____).
ZWL		No forwarding action to the designation(s) immediately following is required.
ZWM		Correct answer to the last question (or question ____) is ____ (or will be found in ____).
ZWN		Correct version of the past of the last message (or message ____) which was sent incorrectly is (or will be found in).
ZXC		This transmission has been interrupted. A retransmission will follow without request. (Must be followed immediately by valid end of message sequence.)
ZXK		Station called relay this message (or message ____) to in addition to predetermined responsibilities.
ZXO		Request you obtain retransmission of messages from station
ZXW		This message has been delivered to all action addressees whose designations follow this operating signal.
ZXX		This message has been delivered to all information addresses whose designations follow this operating signal.
ZXY		Transmit this message to the addressee(s) indicated by the numeral(s) following . All addressees are to be counted consecutively as they appear (numbers to be separated by the separative sign).

Table B-2. Q - Signals.

Signal	Question	Answer, Advice, or Order
QAP	Shall I listen for von for for) nn kHz (or MHz)? NOTE: If the frequency is given in megahertz, the abbreviation "MHz" is to be used. (See also signal "QSX.")	Listen for me (or for on kHz (-....MHz). NOTE: If the frequency is given in megahertz, the abbreviation "MHz" is to be used. (See also signal "QSX.")
QAR	May I stop listening on the watch frequency for minutes?	You may stop listening on the watch frequency for minutes.
QBM	Has sent, arre message. for me?	Here is the message sent by at hours.
QCB		Delay is being caused by— 1. Your transmitting out of turn. 2. Your slowness in answering. 3. Lack of your reply to my
QCS		Mv reception on frequency has broken down.
QCX	What is your full call sign?	My full call sign is , If advice or oder, use your call sign until further notice.
QDB	Have vou sent message to ?	I have sent message to
QDP	Will you accept control (or responsibility) of (for) now (or at hours)?	I will accept control (or responsibility of (for) now (or at hours).
QIC	May I establish communication with radio station on kHz (or MHz) now (or at, hours)?	Establish communications with radio station on KHz for MHz") now for at hours). If advice or order. I will establish communication with radio station ____ on kHz for MHz) now (or at _ hours).
QIF	What frequency is using?	_____ is using kHz (or MHz).

Table B-2 Q Signals Continued

Signal	Question	Answer, Advice, or Order
QJA	Is my ___ reversed? 1. Tape.	Your (is/is not) reversed. 1. Tape.
QJB	2. Mark and space. Will you use— 1. Radio? 2. Cable? 3. Telegraph? 4. Teletypewriter? 5. Telephone? 6. Receiver? 7. Transmitter? 8. Reperforator?	2. Mark and space. I will use— 1. Radio 2. Cable. 3. Telegraph. 4. Teletypewriter. 5. Telephone. 6. Receiver. 7. Transmitter, 8. Reperforator.
QJC	Will you check your— 1. Transmitter distributor? 2. Autohead? 3. Perforator? 4. Reperforator? 5. Printer? 6. Printer motor? 7. Keyboard? 8. Antenna system?	I will check my— 1. Transmitter distributor. 2. Autohead. 3. Perforator. 4. Reperforator. 5. Printer. 6. Printer motor. 7. Keyboard. 8. Antenna system.
QJD	Am I transmitting— 1. In letters? 2. In figures?	You are transmitting— 1. In letters. 2. In figures.
QJE	Is my frequency shift— 1. Too wide? 2. Too narrow? 3. Correct?	Your frequency shift is— 1. Too wide. 2. Too narrow. 3. Correct.
QJF		Mv signal as checked by monitor is satisfactory 1. Locally. 2. As radiated.
QJH	Shall I run— 1. My test tape? 2. A test sentence?	Run— 1. Your test tape. 2. A test sentence.
QJI	Will your transmit a continuous— 1. Mark? 2. Space?	I am transmitting a continuous— 1. Mark. 2. Space.
QJK	Are you receiving— 1. A continuous mark? 2. A continuous space? 3. A mark bias? 4. A space bias?	I am receiving— 1. A continuous mark. 2. A continuous space. 3. A mark bias. 4. A space bias.
QLB	Will you monitor station and report regarding range, quality, and so forth?	I have monitored station and report (briefly) as follows
QLH	Will you use simultaneous keying on frequency and frequency?	I will now key simultaneously on frequency and frequency.
QMH		Shift to transmit and receive on ___ kHz (or MHz); if communication is not established within 5 minutes, revert to present frequency.
QRA	What is the name of your station?	The name of my station is
QRB	How far approximately are you from my station?	The approximate distance between our station is nautical miles (or kilometers).
QRG	Will you tell me my exact frequency (or that of)?	Your exact frequency for that of) is kHz (or MHz).
QRH	Does my frequency vary?	Your frequency varies.
QRI	How is the tone of my transmission?	The tone of your transmission is— 1. Good. 2. Variable. 3. Bad.
QHJ	How many radio telephone calls have you to book?	I have _ radiotelephone calls to book.
QRK	What is the intelligibility of my signals (or those of)?	The intelligibility of your signals (or those of is— 1. Bad. 2. Poor.

Table B-2. Q Signals—continued.

Signal	Question	Answer, Advice, or Order
		3.Fair. 4.Good. 5.Excellent.
QRI	Are you busy?	I am busy (or I am busy with). Please do not interfere.
QRM	Are you being interfered with?	I am being interfered with— 1.Nil. 2.Slightly. 3.Moderately. 4.Severely. 5.Extremely.
QRN	Are you troubled by static?	I am being troubled by static— 1.Nil. 2.Slightly. 3.Moderately. 4.Severely. 5.Extremely.
QRO	Shall I increase transmitter power?	Increase transmitter power.
QRP	Shall I decrease transmitter power?	Decrease transmitter power.
QRQ	Shall I send faster?	Send faster at _____ WPM.
QRR	Are you ready for automatic operation?	I am ready for automatic operation. Send at _____ WPM.
QRS	Shall I send more slowly?	Send more slowly at _____ WPM.
QRT	Shall I stop sending?	Stop sending.
QRU	Have you anything for me?	I have nothing for you.
QRV	Are you ready?	I am ready.
QRW	Shall I inform that you are calling on _____ kHz (or _____ MHz)?	Please inform that I am calling on _____ kHz (or _____ MHz).
QRX	When will you call me again?	I will call you again at _____ hours on _____ kHz (or _____ MHz),
QRY	What is my turn? (Relates to communication.)	Your turn is number _____ (or according to any other indication). (Relates to communication.)
QRZ	Who is calling me?	You are being called by _____ on kHz (or _____ MHz).
QSA	What is the strength of my signals (or those of)?	The strength of your signals for those of) is— 1.Scarcely perceptible. 2.Weak. 3.Fairly good. 4.Good. 5.Very good.
QSB	Are my signals fading?	Your signals are fading.
QSD	Is my keying defective?	Your keying is defective.
QSI		I have been unable to break in on your transmission, or will you inform (call sign) that I have been unable to break in on his or her transmission on _____ kHz (or _____ MHz).
QSK	Can you hear me between your signals and if so can I break in on your transmission?	I can hear you between my signals; break in on my transmission.
QSL	Can you acknowledge receipt?	I am acknowledging receipt.
QSM	Shall I repeat the last telegram which I sent you (or some previous telegram)?	Repeat the last telegram which you sent me (or telegrams) numbers(s) _____
QSN	Did you hear me (or) (call sign) on kHz (or _____ MHz)?	I did hear you or () (call sign) on kHz (or _____ MHz),
QSO	Can you communicate with direct or by relay?	I can communicate with direct (or by relay through).
QSR	Shall I repeat the call on the calling frequency?	Repeat your call on the calling frequency; did not hear you (or have interference).
QSS	What working frequency will you use?	I will use the working frequency _____ kHz (normally only the last three figures of the frequency need be given).
QSU	Shall I send or reply on this frequency or on _____ kHz (or _____ MHz) (with emissions of class)?	Send or reply on this frequency or on _____ kHz (or _____ MHz) (with emissions of _____)

Table B-2 Q Signals Continued

Signal	Question	Answer, Advice, or Order
QSV	Shall I send a series of V's on this frequency or on _____ kHz (or _____ MHz)?	Send a series of V's on this frequency or on _____ kHz (or _____ MHz),
QSW	Will you send on this frequency or on _____ kHz (or MHz) (with emissions of class 1)?	I am going to send on this frequency or on _____ kHz (or MHz) with emissions of class 1.
QSX	Will you listen to (call sign) on kHz (or _____ MHz)?	I am listening to (call sign) on kHz (or _____ MHz),
QSY	Shall I change to transmission or another frequency?	Change to transmission on another frequency (or on _____ kHz (or _____ MHz).
QSZ	Shall I send each word or group more than once?	Send each word or group twice (or _____ times).
QTA	Shall I cancel telegram number?	Cancel telephone number .
QRB	Do you agree with my counting of words?	I do not agree with your counting of words. I will repeat the first letter or digit of each word or group.
QTC	How many telegrams have you to send?	I have telegrams for you for ().
QTH	What is your position in latitude and longitude (or according to any other indication)?	My position is latitude (or according to any other indication).
QTR	What is the correct time?	The correct time is hours.
QRT	Will you send your call sign for tuning purposes or so that your frequency can be measured (or at _____ hours) on _____ kHz (or _____ MHz)?	I will send my call sign for tuning purposes or so that my frequency may be measured now (or at hours) on kHz (or _____ MHz).
QTU	What are the hours during which your station is	My station is open from _____ to _____ hours.
QTV	Shall I stand guard for you on the frequency of kHz (or _____ MHz) (from to hours)?	Stand guard for me on the frequency of _____ kHz (or MHz) from to hours).
QTX	Will you keep your station open for further communication with me until further for until hours)?	I will keep my station open for further communication with you until further notice for until ().
QUA	Have you news of (call sign)?	Here is news of (call sign).
QUO	What is the number (or other indication) of the last message you received from me (or from (call sign))?	The number (or other indication) of the last message I received from you for from (call sign) is .
QUF	Have you received the distress signal sent by (call sign of mobile station)?	I have received the distress signal sent by (call sign of mobile station) at hours.
QUM	May I resume normal working?	Normal working may be resumed.

APPENDIX C

AUTHORIZED PROSIGNS AND PROWORDS

C-1 Prosigns are procedure signs consisting of thereof. They are used to facilitate rapid communication by conveying, in condensed standard form, certain frequently used orders, instructions, requests, reports, and information related to communications

in radio telephone procedure, one or more letters or characters or combinations. Operating personnel will not under any circumstances, substitute prosigns, prowords, or combinations thereof for the textual component of message received for transmission without the consent and approval of the originator. Table C-1 explains the meaning of the prosigns and prowords.

C-2. Prowords are word equivalents of prosigns, for use

Table C-1. Prosigns and Prowords.

Prosign	Meaning	Prowords
AA	Unknown station.....	UNKNOWN STATION
AA	All after.....	ALL AFTER
AB	All before.....	ALL BEFORE
AR	End of transmission, no receipt required.....	OUT
AS	I must pause for a few seconds.....	WAIT
AS AR	I must pause longer than a few seconds, will call you back..	WAIT OUT
B	More to follow.....	MORE TO FOLLOW
BT	Break.....	BREAK
C	Correct.....	THIS IS CORRECT or CORRECT
DE	That which immediately follows is the time or DTG of this message.....	THIS IS TIME
EEEEEEEE		CORRECTION
EEEEEEEE AR	This message is in error, disregard it.....	DISREGARD THIS TRANSMISSION
F	Do not answer.....	DO NOT ANSWER
FM	Originator's sign.....	FROM
G.....	Repeat this entire transmission back to me.....	READ BACK I READ BACK
	The following is my response to your instructon to read back....	
GR (numerals)	Group count.....	GROUPS (Numerals)
GRNC	The groups in this message have not been counted.....	GROUP NO COUNT
HM HM HM	Emergency silence.....	SILENCE
	Resume normal transmissions, station imposing it or by higher authority.....	SILENCE LIFTED
	When an authentication system is in force, transmissions lifting silence are to be authenticated.	
	Your transmission is too fast speed.....	SPEAK SLOWER
	Reduce speed of transmission.	
II	Separative sign,	
IMI	Repeat.....	SAY AGAIN
	I am repeating transmission or portion indicated.....	I SAY AGAIN
	I will spell the next word(s) phonetically.....	I SPELL
	Communication is difficult.	WORDS TWICE
	Transmitting) each phrase (or each code group) twice. (This proword may be used as an order, request, or as information.)	
	Your last transmission is in error.....	INCORRECT
INFO	The address designations immediately following are addressed for information.....	INFO
INT	Interrogative. Used to indicate single alpha letters.....	INITIAL
IX	Action on the message or signal which follows is to be carried out upon receipt of "EXECUTE."	EXECUTE TO FOLLOW
IX (o-sec dash)	Carry out now, the purpose of the message or signal, to which this applies.....	EXECUTE
J	Verify with operator and repeat. Message or portion indicated been verified.....	VERIFY I VERIFY

Table C-1. Prosigns and Prowords—continued.

Prosign	Meaning	Prowords
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K	Go ahead, or this is the end of my transmission to you and a response is necessary.....	OVER
	Message requiring recording follows.....	MESSAGE FOLLOWS
NR	Station serial number. Numerals or numbers follows.....	FIGURES
0	Immediate precedence.....	IMMEDIATE
P	Priority precedence.....	PRIORITY
R	I have received your last transmission satisfactorily.....	ROGER
R	Routine precedence.....	ROUTINE
SVC	Service (group count of service message),,,.....	SERVICE
T	Transmit this message to all addressees or to the address designations immediately following.....	RELAY TO
TO	Action addressee.....	TO
WA	Word after.....	WORD AFTER
WB	Word before.....	WORD BEFORE
XMT	Exempt.....	EXEMPT
Z	Flash precedence.....	FLASH

APPENDIX D MISCELLANEOUS ABBREVIATIONS

D-1. Two-letter state and territory abbreviations.

Alabama	AL
Alaska	AK
Arizona	AZ
Arkansas	AR
California	CA
Canal Zone	CZ
Colorado	CO
Connecticut	CT
Delaware	DE
District of Columbia	DC
Florida	FL
Georgia	GA
Guam	GU
Hawaii	HI
Idaho	ID
Illinois	IL
Indiana	IN
Iowa	IA
Kansas	KA
Kentucky	KY
Louisiana	LA
Maine	ME
Maryland	MD
Massachusetts	MA
Michigan	MI
Minnesota	MN
Mississippi	MI
Missouri .	MO
Montana	MT
Nebraska	NE
Nevada	NV
New Hampshire	NH
New Jersey	NJ
New Mexico	NM
New York	NY
North Carolina	NC
North Dakota	ND
Ohio	OH
Oklahoma	OK
Oregon.	OR
Pennsylvania	PA
Puerto Rico	PR
Rhode Island	RI
South Carolina	SC
South Dakota	SD
Tennessee	TN
Texas	TX
Utah	UT
Vermont	VT
Virginia	VA

Virgin Islands

Washington

West Virginia

Wisconsin

VI

WA

WV

WI

D-2. Three-letter month abbreviations.

January	JAN
February	FEB
March	MAR
April	APR
May	MAY
June	JUN
July	JUL
August	AUG
September	SEP
October	OCT
November	NOV
December	DEC

D-3. Three-letter day abbreviations.

Monday	MON
Tuesday	TUE
Wednesday	WED
Thursday.	THU
Friday	FRI
Saturday	SAT
Sunday	SUN

D-4. Punctuation abbreviations.

Punctuation	Symbol	Abbreviation	Morse	Spoken
Apostrophe	'	APOS	.-.-.-.	Apostrophe
Ampersand	&			And
Colon	:	CLN	- - -...	Colon
Comma	>	CMM		Comma
Dash	-		- - . - .	Dash
			-	
Dollar Sign	\$			Dollars
Hyphen	- -		- . . . -	Hyphen
Paragraph		PARA		Paragraph
Parenthesis (left)	(	PAREN	-.--.	Paren
Parenthesis (right)	)	UNPAREN	-.--.	Unparen
Period		PD	.-.-.-	Period
Question Mark	?	QUES	.-.-.-.	
Quotation marks	"	QUOTE/ UNQUOTE	-.--.	Question Mark
Slant sign	/	SLANT	-.--.	SLANT

APPENDIX E

INTERNATIONAL PREFIXES

AAA-ALZ	United States of America	IAA-IZZ	Italy and Mandated Territories
AMA-AOZ	Spain	JAA-JSZ	Japan
APA-ASZ	Pakistan	JTA-JVZ	Mongolian People's Republic
ATA-AWZ	India	JWA-JXZ	Norway
AXA-AXZ	Commonwealth of Australia	JYA-JYZ	Jordan
AYA-AZZ	Argentine Republic	JZA-JZZ	West New Guinea
BAA-BZZ	China	KAA-KZZ	United States of America
CAA-CEZ	Chile	LAA-LNZ	Norway
CFA-CKZ	Canada	LOA-LWZ	Argentine Republic
CLA-CMZ	Cuba	LXA-LXZ	Luxembourg
CNA-CMZ	Morocco	LYA-LYZ	Lithuania
COA-COZ	Duba	LZA-LZZ	People's Republic of Bulgaria
CPA-CPZ	Bolivia	MAA-MZZ	Great Britain and North Ireland
CQA-CRZ	Portuguese Overseas Provinces	NAA-NZZ	United States of America
CSA-XUZ	Portugal	OAA-OCZ	Peru
CVA-CXZ	Uruguay	ODA-ODZ	Lebanon
CYA-CZZ	Canada	OEA-OEZ	Austria
DAA-DTZ	Germany	OFA-OJZ	Finland
DUA-DZZ	Republic of the Philippines	OKA-OMZ	Czechoslovakia
EAA-EHZ	Spain	ONA-OTZ	Belgium
EIA-EJZ	Ireland	OUA-OZZ	Denmark
EKA-EKZ	Union of Soviet Socialist Republics	PAA-PIZ	Netherlands
ELA-ELZ	Liberia	PJA-PJZ	Netherlands Antilles
EMA-EOZ	Union of Soviet Socialist Republics	PKA-POZ	Republic of Indonesia
EPA-EQZ	Iran	PPA-PYZ	Brazil
ERA-ERZ	Union of Soviet Socialist Republics	PZA-PZZ	Surinam
ESA-ESZ	Estonia	QAA-QZZ	(Service Abbreviations)
ETA-ETZ	Ethiopia	RAA-RZZ	Union of Soviet Socialist Republics
EUA-EWZ	Bielorussian Soviet Socialist Republic	SAA-SMZ	Sweden
EXA-EZZ	Union of Soviet Socialist Republics	SNA-SRZ	People's Republic of Poland
FAA-FZZ	France and French Community	SSA-SSM	United Arab Republic
GAA-GZZ	United Kingdom	SSN-STZ	Sudan
HAA-HAZ	Hungarian People's Republic	SUA-SUZ	United Arab Republic
HBA-HBZ	Switzerland	SVA-SZZ	Greece
HCA-HDZ	Ecuador	TAA-TCZ	Turkey
HEA-HEZ	Switzerland	TGA-TDZ	Guatemala
HFA-HFZ	People's Republic of Poland	TEA-TEZ	Costa Rica
HGA-HGZ	Hungarian People's Republic	TFA-TFZ	Iceland
HHA-HHZ	Repblic of Haiti	TGA-TGZ	Guatemala
HIA-HIZ	Dominican Republic	THA -THZ	France and French Community
HJA-HKZ	Republic of Colombia	TIA-TIZ	Costa Rica
HLA-HNZ	Korea	TJA-TJZ	Republic of Cameroon
HMA-HNZ	Iraq	TKA-TKZ	France and French Community
HOA-HPZ	Republic of Panama	TLA-TLZ	Central African Republic
HQA-HRZ	Republic of Honduras	TMA-TMZ	France and French Community
HSA-HSZ	Thailand	TNA-TNZ	Republic of Congo (Brazzaville)
HTA-HTZ	Nicaragua	TOA-TQZ	France and French Community
HUA-HUZ	Republic of El Salvador	TRA-TRZ	Republic of Gabon
HVA-HVZ	Vatican City State	TSA-TSZ	Tunisia
HWA-HYZ	France and French Community	TTA-TTZ	Republic of Chad
HZA-HZZ	Saudi Arabia	TUA-TUZ	Republic of the Ivory Coast
		TVA-TXZ	France and French Community
		TYA-TYZ	Republic of Dehomey

TZA-TZZ Republic of Mali
UAA-UQZ Union of Soviet Socialist
Republics
URA-UTZ Ukrainian Soviet Socialist Republic
UUA-UZZ Union of Soviet Socialist Republics
VAA-VGZ Canada
VHA-VNZ Commonwealth of Australia
VOA-VOZ Canada
VPA-VSZ Great Britain and North Ireland
VTA-VWZ India
VXA-VYZ Canada
VZA-VZZ Commonwealth of Australia
WAA-WZZ United States of America
XAA-XIZ Mexico
XJA-XOZ Canada
XPA-XPZ Denmark
XQA-XRZ Chile
XSA-XSZ China
XTA-XTZ Republic of the Upper Volta
XUA-XUZ Cambodia
XVA XVZ Vietnam XWA-XWZ Laos
XXA-XXZ Portuguese Overseas Provinces

3YA-3YZ	Norway
3ZA-3ZZ	People's Republic of Poland
4AA-4CZ	Mexico
4DA-4IZ	Republic of the Philippines
4JA-4LZ	Union of Soviet Socialist Republics
4MA-4MZ	Venezuela
4NA-4OZ	Yugoslavia
4PA-4SZ	Ceylon
4TA-4TZ	Peru
4UA-4UZ	United Nations
4VA-4VZ	Republic of Haiti
4WA-4WZ	Yemen
4XA-4XZ	State of Israel
4YA-4YZ	International Civil Aviation Organization
4ZA-4ZZ	State of Israel
5AA-5AZ	Libya
5BA-5BZ	Republic of Cyprus
5CA-5GZ	Morocco
5HA-5IZ	Tanzania
5HA-5KZ	Colombia
5LA-5MZ	Liberia
5NA-5OZ	Nigeria
5PA-5QZ	Denmark
5RA-5SZ	Malagasy Republic
5TA-5TZ	Islamic Republic of Mauretania
5UA-5UZ	Republic of the Niger
5VA-5VZ	Togolese Republic
5WA-5WZ	Western Samoa
5XA-5XZ	Uganda
5YA-5ZZ	Kenya
6AA-6BZ	United Arab Republic
6CA-6CZ	Syria
6DA-6JZ	Mexico
6KA-6NZ	Korea
6OA-6OZ	Somalia
6PA-6SZ	Pakistan
6TA-6UZ	Sudan
6VA-6WZ	Republic of the Senegal
6XA-6XZ	Malagasy Republic
6YA-6YZ	Jamaica
6ZA-6ZZ	Liberia
7AA-7IZ	Indonesia
7JA-7NZ	Japan
7OA-7OZ	South Yemen Popular Republic
7PA-7PZ	Lesotho
7QA-7QZ	Malawi
7RA-7RZ	Algeria
7SA-7SZ	Sweden
7TA-7YZ	Algeria
7ZA-7ZZ	Saudi Arabia
8AA-8IZ	Indonesia
8JA-8NZ	Japan
8OA-8OZ	Botswana
8PA-8PZ	Barbados
8QA-8QZ	Haldire Islands

APPENDIX F

STANDARD TIME ZONES AND TIME CONVERSION CHARTS

F-1. There are 24 worldwide zones based upon longitude. By international agreement, the Central cross hair of an historic instrument called the Airy Transit Circle, in Greenwich, England marks the prime Meridian at Zero degree of longitude. The prime Meridian is an imaginary great circle, crossing both geographic poles. Standard time zones are established at intervals of 15 degrees of longitude East and West of the prime Meridian. The exact boundaries of the time zones are modified by political and geographic considerations, but generally form about seven and one half degrees, on either side of the zone reference Meridian.

F-2. The United States is divided into seven standard time zones (fig F-1). The time of each zone is an integral number of hours different from

(earlier than) Universal Time Standard (UTS) based on Greenwich, United Kingdom. The zones are identified as Eastern Standard, 5 hours earlier than UTS; central standard, 6 hours earlier than UTS; Mountain Standard, 7 hours earlier than UTS; and Pacific standard, 8 hours earlier than UTS.

F-3. To ensure communications facilities are functioning properly, many governments have activated radio stations which provide very accurate time signals and standard tones for equipment calibration (table F-1). Depending on geographic location and propagation conditions, one of the stations listed in the following charts should be available for use to calibrate equipment. The receiver station must account for local area variation such as Daylight Savings or Summer Time.

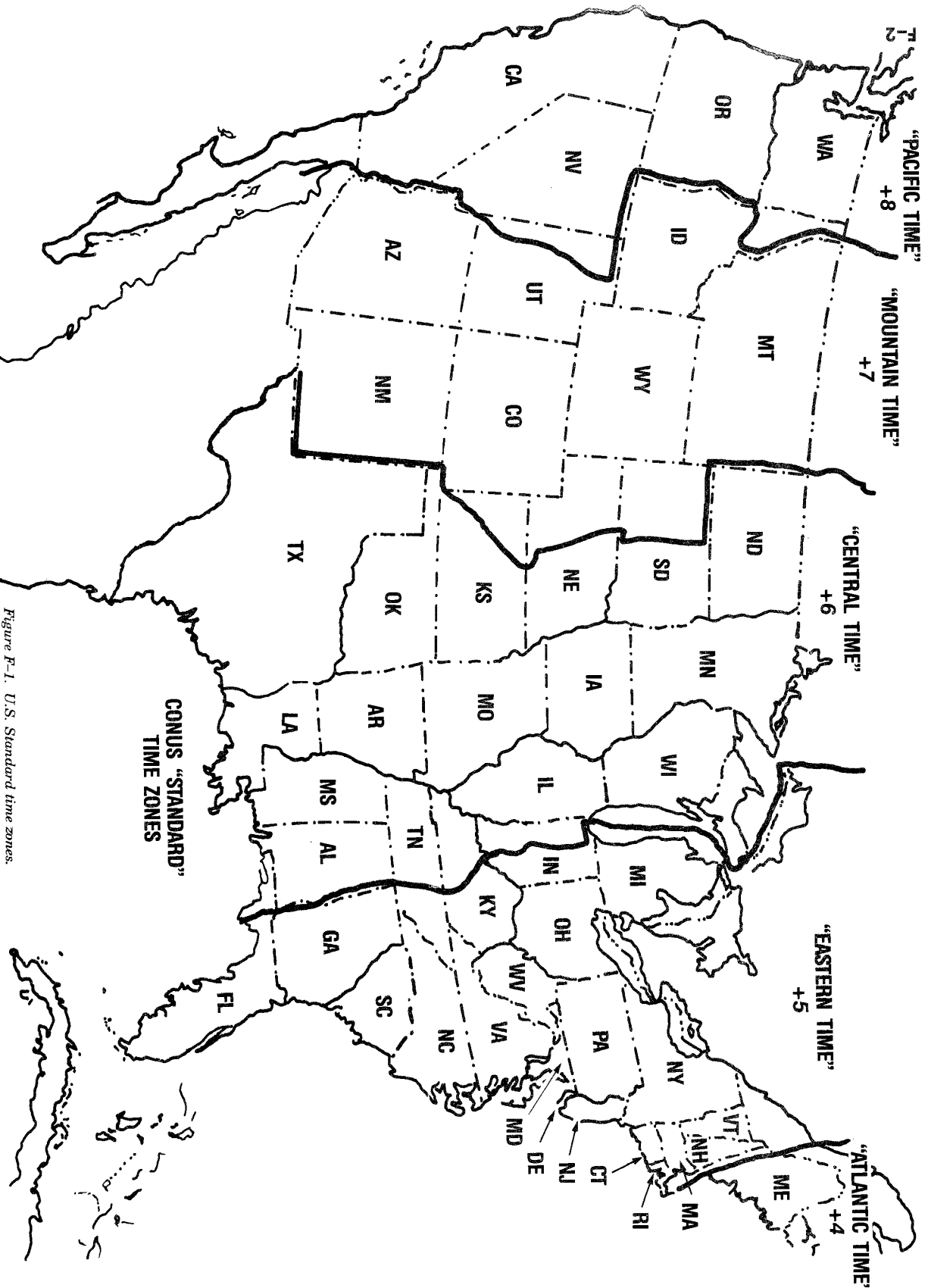


Figure F-1. U.S. Standard time zones.

Table F-1. Time-Signals Emitted in the UTC System

Station	Location Latitude Longitude	Frequency (kHz)	Schedule (UT)	Form of the Time Signal
BSF	Taiwan Rep. of China	5000	Between min 00-05, 10-15, 20-25, 30-35, 40-45, 50-55 from 0100-0900	Second pulses of 5 ms duration. Minute marker is pulse of 300 ms duration. During 29th and 59th min., Morse code and Chinese voice announcement of time. Second markers for DUTI are pulses of 100 ms.
CHU	Ottawa Canada ♦45°18' ♦75°45'	3330 7335 14670	continuous	Second pulses of 300 cycles of a 1 kHz modulation. Minute pulses are 0.5 s long. A bilingual (Fr.- Eng.) announcement of time is made each minute. DUTI: CCIR code by split pulses
DAM	Elmshorn Germany, F.R. ♦53° 46' - 9°40'	8638.5 1698.4 4625 8638.5 6475.5 12763.5	11 h 55 m to 12 h 6m 23 h 55 m to 24 h 6m from 21 Sept. to 20 March 23 h 55 m to 24 h 6 m from 21 March to 20 Sept.	New international system, then Second pulses from minutes 0.5 to 6.0 (Minute pulses prolonged). A1 type. DUTI: CCIR code by doubling after Minute pulses 1 to 5
DAN	Osterloog Germany, F.R. +53°38' - 7° 12'	2614	11 h 55 m to 12 h 6 m 23 h 55 m to 24 h 6 ra	As DAM (see above)
DAO	Kiel Germany, F.R. ♦54 "26' -10° 8'	2775	11 h 55 m to 12 h 6 m 23 h 55 m to 24 h 6 m	As DAM (see above)
DCF77	Mainflingen German, F.R. +50° 1' - 9° 0'	77.5	continuous, except second Tuesday of every month from 4 h to 8 h	The Second marks are reduction to 1/4 of the carrier's amplitude of 0.1 s duration; the reference point is the beginning of the pulse modulation. The second 59 marker is omitted. DUTI: CCIR code by lengthening to 0.2 s
DGI	Oranienburg Germ. Dem. Rep. +52°48' -13°24'	185	5 h 59 m 30 s to 6 h 00 m 11h 59 m 30 s to 12 h 00 m 17 h 55 m 30 s to 18 h 00 m	A2 type Second pulses of 0.1 s duration for seconds 30-40, 45-50, 55-60. The last pulse is prolonged.
DIZ	Nauen Germ. Dem. Rep. +52°39' -12°55'	4525	continuous except from 8 h 15 m to 9 h 45 m for maintenance if necessary	A1 type Second pulses of 0.1 s duration. Minute pulses prolonged to 0.5 s. Hour pulses marked by prolonged pulses for seconds 58, 59, 60. DUTI: CCIR code by double pulse.
FFH	Chevannes France +48°32' - 2°27'	2500	continuous from 8 h to 16 h 25 except Saturday and Sunday	Second pulses of 5 cycles of 1 kHz modulation. Minute pulses prolonged to 0.5 s. DUTI: CCIR code by lengthening to 0.1 s.
FTA91	Saint-Andre-de- Corcy France +45°55' - 4°55'	91.15	at 8 h, 9 h, 9 h 30 m, 13 h, 20 h, 21 h, 22 h 30 m.	A1 type Second pulses during the 5 minutes preceding the indicated times. Minute pulses are prolonged, DUTI: in Morse code.
FTH42 FTK77 PTN87	Pontolse France ♦40° 4' - 2° 7'	7428 10775 13873	at 9 h and 21 h at 8 h and 20 h at 9 h 30 m, 13 h, 22 h, 30 m.	A1 type Second pulses during the 5 minutes preceding the Indicated times. Minute pulses are' prolonged. DUTI: in Morse code.

Table F-1. Time-Signals Emitted in the UTC System (Continued)

Station	Location Latitude Longitude	Frequency (kHz)	Schedule (UT)	Form of the Time Signal*
GBR	Rugby United Kingdom +52°22' + 1°11'	16	at 3 h. 9 h. 15 h, 21 h	A1 type Second pulses during the 5 minutes preceding the indicated times. DUTI: CCIR code by double pulse
HBG	Prangins Switzerland +46°24' - 6° 15'	75	Continuous	Interruption of the carrier at the beginning of each second, during 100 ms. The minutes are identified by a double pulse, the hours by a triple pulse. No transmission of DUTI.
I AM	Rome Italy +41°52' -12°21'	5000	10 m every 15 m from 7 h 30 m to 8 h 30 m and from 13 h to 14 h except Saturday afternoon and Sunday Advanced by 1-hour in summer.	Second pulses of 5 cycles of 1 kHz modulation. Minute pulses of 20 cycles (Announcements and 1 kHz modulation, 5 m before the emission of time signals).
IBF	Torino Italy +45° 2' - 7°42'	5000	During 15 m preceding 7 h, 9 h, 10 h, 11 h, 12 h, 13 h, 14h, 15 h, 16 h, 17 h, 18 h. Advanced by 1-hour in summer.	Second pulses of 5 cycles of 1 kHz modulation. These pulses are repeated 7 times at the minute. Voice announcement at the beginning and end of each emission. DUTI: CCIR code by double pulse.
JG2AE	Koganei Japan +35°42' -139° 31'	8000	from 20 h 59 m to 10 h 59 m.	Second pulses of 1600 Hz modulation. Minute pulses are preceded by a 600 Hz modulation. DUTI: CCIR code by lengthening
JG2AS	Chiba Japan + 35°38' -140° 4'	40	from 23 h 30 m to 8 h (exc. Sunday) and from 8 h to 23 h 30 on Monday. Interruptions during communications.	A1 type Second pulses of 0.5 sec. duration. Second 59 is omitted. No DUTI code.
JJY	Koganei Japan * +35°42' -139°31'	2500 5000 10000 15000	continuous, except interruptions between minutes 25 and 34.	Second pulses of 8 cycles of 1600 Hz modulation. Minute pulses are preceded by a 600 Hz modulation. DUTI: CCIR code by lengthening
LOL1	Buenos-Aires Argentina -34° 37' +58°21'	5000 10000 10000	11 h to 12 h, 14 h to 15 h, 17 h to 18 h, 20 h to 21 h 23 h to 24 h	Second pulses of 5 cycles of 1000 Hz modulation. Second 59 is omitted. Announcement of hours and minutes every 5 minutes, followed by 3 m of 1000 Hz and 440 Hz modulation. DUTI: CCIR code by lengthening
LOL2 LOL3	Buenos-Aires Argentina -34° 37' +58°21'	8030 17180	1 h, 13 h, 21 h	A1 Second pulses during the 5 minutes preceding the indicated times. Minute pulses are prolonged. DUTI: CCIR code by lengthening
LQB9 LQC20	Planta Gral Pacheco Argentina -34°26' +58°37'	8167.5 17551.5	22 h 5 m, 23 h 50 m 10 h am, 11 h 50 m	A1 Second pulses during the 5 minutes preceding the indicated times. Second 59 is omitted, second 60 is prolonged. After the emission, OK is transmitted if the emission is correct, NV if not correct. DUTI: CCIR code by omission of second markers.

Table F-1. Time-Signals Emitted in the UTC System (Continued)

Station	Location Latitude Longitude	Frequency (kHz)	Schedule (UT)	Form of the Time Signal*
OLB5	Podebrady Czechoslovakia +50° 9' -15° «'	3170	continuous except from 5 h to 11 h on the first Wednesday of every month	A1 type, Second pulses No transmission of DUTI
OMA	Liblice Czechoslovakia +50° 4' -14°53'	50 2500	continuous except from 5 h to 11 h on the first Wednesday of every month between minutes 5 and 15 25 and 30, 35 and 40, 50 and 60 of every hour except from 5 h to 11 h on the first Wednesday of every month	Interruption of the carrier of 100 ms at the beginning of every second, of 500 ms at the beginning of every minute. The precise time is given by the beginning of the interruption. Pulses of 5 cycles of 1 kHz modulation (prolonged for the minutes). The first pulse of the 5th minute is prolonged to 500 cycles. No transmission of DUTI.
PPE	Rio de Janeiro Brazil -22°54' +43°13*	8721	0 h 30 m 11 h 30 m, 13 h 30 m, 19 h 30 m, 20 h 30 m, 23 h 30 m	Second ticks, of A1 type, during the 5 minutes preceding the indicated hours. The minute ticks are longer. DUTI: CCIR Code by double pulse
PPR	Rio de Janeiro Brazil -22°59' +43° 11'	435 8634 13105 17194.4	01 h 30 m, 14 h 30 m, 21 h 30 m	Second ticks, A1 type, during the 5 minutes preceding the indicated hours. The minute ticks are longer
RAT	Moscow USSR +55° 19' -38°41'	2500 5000	between minutes 30 and 35, 41 and 45, 50 and 60 from 17 h 50 m to 24 h between minutes 30 and 35, 41 and 45, 50 and 60 from 1 h 30 m to 17 h	Second pulses* at the beginning of the minute are prolonged to 0.5 s. DUTI ♦ dUTI by Morse Code each hour between minutes 11 and 12.
RBU	Moscow USSR +55°19' -38°41'	66-2/3	between minutes 0 and 5 from 0 h to 22 h 5 m	A1 type. Second pulses*. The pulses at beginning of the minute are prolonged to 0.5 s. DUTI + dUTI: by Morse Code each hour between minutes 6 and 7.
RCH	Tashkent USSR +41°19' -69° 15'	2500	between minutes 15 and 20; 25 and 30, 35 and 40, 45 and 50 from 0 h to 3 h 50 m from 5h35mto9h30m from 10 h 15 m to 13 h 30 m from 14 h 15 m to 24 h	Second pulses*. The pulses at the beginning of the minute are prolonged to 0.5 s. DUTI + dUTI: by Morse Code each hour between minutes 51 and 52.
RID	Irkutsk USSR 52°46' -103°39'	5004 10004	between minutes 5 and 10, 15 and 20, 25 and 30, 51 and 60 from 0 h to 1 h 10 m from 13 h 51 m to 24 h between minutes 5 and 10, 15 and 20, 25 and 30, 51 and 60 from 1 h 51 m to 13 h 10 m	Second pulses*. The pulses at the beginning of the minute are prolonged to 0.5 s. DUTI + dUTI: by Morse Code each hour between minutes 31 and 32.

The information about the value and the sign of the DUTI + dUTI difference is transmitted after each minute signal by the marking of the corresponding second signals by additional impulses. In addition, it is transmitted in Morse Code as indicated.

Table F-1. Time-Signals Emitted in the UTC System (Continued)

Station	Location Latitude Longitude	Frequency (kHz)	Schedule (UT)	Form of the Time Signal
RIM	Tashkent USSR +41°19' -69° 15'	5000 10000	between minutes 15 and 20, 25 and 30, 35 and 40, 45 and 50 from 0 h to 1 h 30 m from 2 h 15 m to 3 h 50 m from 18 h 15 m to 24 h between minutes 15 and 20, 25 and 30, 35 and 40, 45 and 50 from 5h35-mto9h30m from 10 h 15 m to 13 h 30 m from 14 h 15 m to 17 h 30 m	Second pulses'. The pulses at the beginning of the minute are prolonged to 0.5 s. DUT1 + dUT1: by Morse Code each hour between minutes 51 and 52.
RKM	Irkutsk USSR + 52°46' -103°39'	10004 15004	between minutes 5 and 10, 15 and 20, 25 and 30, 51 and 60 from 0 h to 1 h 10 m. from 13 h 51 m to 24 h between minutes 5 and 10, 15 and 20, 25 and 30, 51 and 60 from 1 h 51 m to 13 h 10 m	Second pulses'*. The pulses at the beginning of the minute are prolonged to 0.5 s. DUT1 + dUT1: by Morse Code each hour between minutes 31 and 32.
RTA	Novossibirsk USSR +55°04' -82°58'	4996 9996 14996	between minutes 5 and 10, 15 and 20, 25 and 29, 35 and 39 from 0 h to 1 h 29 m from 18 h 5 m to 24 h between minutes 5 and 10, 15 and 20, 25 and 29, 35 and 39 from 3h5mto4h39m from 14 h 5 m to 17 h 29 m between minutes 5 and 10, 15 and 20, 25 and 29, 35 and 39 from 5h35mto9h29m f4om 10 h 5 m to 13 h 29 m	Second pulses*. The pulses at the beginning of the minute are prolonged. DUT1 + dUT1: by Morse Code each hour between minutes 45 and 46.
RWM	Moscow USSR ♦55° 19' -38°41'	10000 15000	between minutes 30 and 35, 41 and 45, 50 and 60 from 1 h 30 m to 3 h from 17 h 50 m to 24 h between minutes 30 and 35, 41 and 45, 50 and 60 from 3 h 50 m to 17 h	Second pulses*. The pulses at the beginning of the minute are prolonged to 0.5 s. DUT1 + dUT1 by Morse Code each hour between minutes 11 and 12.
RTZ	Irkutsk USSR ♦ 52° 18' -104°18'	50	between minutes 0 and 5 from 0 h to 22 h 5 m	All type second pulses*. The pulses at the beginning of the minute are prolonged. DUT1 * dUT1: by Morse Code each hour between minutes 6 and 7.

The information about the value and the sign of the DUT1 + dUT1 difference is transmitted after each Minute signal by marking the corresponding Second signals with additional impulses. In addition, it is transmitted in Morse Code as indicated.

Table F-1. Time-Signals Emitted in the UTC System (Continued)

Station	Location Latitude Longitude	Frequency (kHz)	Schedule (UT)	Form of the Time Signal*
VNG	Lyndhurst Australia 38° -3' -145°16'	4500 7500 12000	9 h 45 m to 21 h 30 m continuous except 22 h 30 m to 22 h 45 m 21 h 45 m to 9 h 30 m	Seconds markers of 50 cycles of 1 kHz modulation; 5 cycles only for Seconds markers 55 to 58; Seconds marker 59 ie omitted; 500 cycles for Minute markers. During the 5th, 10th, 15th, etc...minutes, 5 cycles for Seconds markers 50 to 58. Identification by voice announcement during 15th, 30th 45th, and 60th minutes. DUT1: CCIR code by 45 cycles of 900 Hz modulation immediately following the normal Seconds markers.
wwv	Fort-Collins USA + 40°41' +105° 2'	2500 5000 10000 15000 20000 25000	continuous	Pulses of 5 cycles of 1 kHz modulation. 59th and 29th second pulse omitted. Hour is identified by 0.8 second long, 1500 Hz tone. Beginning of each minute identified by 0.8 second long, 1000 Hi tone. DUT1: CCIR code by double pulse. Additional information on corrections.
WWVB	Fort-Collins USA + 4G°40' +105° 3'	60	continuous	Second pulses given by reduction of the amplitude of the carrier. Coded announcement of the date and time and of the correction to obtain UT1, No CCIR code.
WWVH	Kauai USA + 21°59' +159°46'	2500 5000 10000 15000 20000	continuous	Pulses of 6 cycles of 1200 Hz modulation. 59th and 29th seconds pulse omitted. Hour identified by 0.8 second long 1500 Hz tone. Beginning of each minute identified by 0.8 second long, 1200 Hz tone. DUT1: CCIR code by double pulse. Additional information on UT1 corrections.
YVTO	Caracas Venezuela +10°30' +66°56'	6100	12 h to 20 h 0 h 30 m to 1 h 30 m	Second pulses of 1 kHz modulation with 0.1 s duration. The minute is identified by a 800 Hz tone and a 0.5 s duration Between seconds 52 and 57 of each minute, voice announcement of hour, minute, and second.
ZUO	Olifantsfontein South Africa -25°58' -28°14'	2500 5000 100000	18 h to 4 h continuous continuous	Pulses of 5 cycles of 1 kHz modulation. Second 0 is prolonged. DUT1: CCIR code by lengthening

OTHER TIME SIGNALS: HPV, XSG, Shanghai, China, P R . Latitude: +31° 12', Longitude: -121° 26'. Characteristics and schedule not known.

Table G-3. Anniversaries—continued.

Transmission	Message
THREE FIVE	Wish we could be with you.
THREE SIX	Wish you could be with us.
THREE EIGHT	Congratulations and best wishes on your retirement.

Table G-4. Weddings and birthdays.

Transmission	Message
FOUR ZERO	Congratulations and best wishes for health, happiness, and prosperity.
FOUR ONE	Heartly congratulations and wishes for your happiness together.
FOUR SIX	Greetings on your birthday and best wishes for many more to come.
FOUR EIGHT	Congratulations on the new arrival. Hope mother and child are well.

Table G-B. Greetings and seasonal messages.

Transmission	Message
FIVE ZERO	Greetings by amateur radio.
FIVE TWO	Love and best wishes, I am thinking of you most affectionately on this day.
FIVE THREE	Best wishes for a Thanksgiving Day full of cheer and happiness.
FIVE FOUR	Victory of defeat, our best wishes are with you. Hope you win.
FIVE SIX	All the best for a Merry Christmas.
FIVE EIGHT	Wishing you a very Merry Christmas and a Happy New Year.
FIVE NINER	Merry Christinas from all of us. Wish you could be here.
SIX ZERO	All the best in the New Year.
SIX ONE	Love and best wishes for a Merry Christmas and a Happy New Year.
SIX TWO	The heartiest of seasons greetings.
SIX THREE	Most sincere wishes for health, happiness, and prosperity.
SIX SEVEN	Most sincere wishes for health and happiness on this Easter Day,
SEVEN ONE	Have not heard from you in some time. Please write or answer by amateur radio, through the station delivering this message.
SEVEN TWO	Wishing you and yours the best this holiday season.
SEVEN FOUR	Hope you have a wonderful vacation. See you when you get back.
SEVEN FIVE	Having a wonderful time. See you when I return.
SEVEN SIX	This message is sent free of charge by amateur radio from an exhibit at the {name of fair, exposition, hobby show, and so forth).
SEVEN SEVEN	Having a wonderful time at (name of fair, exposition, exhibit, and so forth). Wish you were here.

Table G-6. Miscellany.

Transmission	Message
SEVEN EIGHT	This acknowledges receipt of your letter.
SEVEN NINER	This acknowledges receipt of your message.
EIGHT ZERO	This acknowledges receipt of your recent communication.
EIGHT ONE	Please acknowledge receipt of this message by return radiogram.
EIGHT TWO	Sorry to hear you are ill. Best wishes for a speedy recovery.
EIGHT SIX	Everything well here. Hope all well there. Will write later.
EIGHT SEVEN	Send. .dollars as soon as possible.
EIGHT NINER	Am sending. dollars immediately.
NINER ZERO	Arrived at. (place) safely. Will write as soon as possible.
NINER ONE	I am in the hospital and receiving excellent care. Will write soon. Don't worry.
NINER TWO	Please arrange quarters for davs.
NTNER THREE	Have been reassigned toWill write as soon as possible.
NINER FOUR	Will travel on temporary duty orders to.....(place) on (date).
NINER FIVE	Will arrive. (place) on.....(date).
MINER SIX	Send no further mail. Will send new address soon.
NINER SEVEN	Leave granted. Should arrive on (date).
NINER EIGHT	Leave denied. Will write later.
NINER NINER	Please arrange transportation to meet me.

APPENDIX H

EMISSION DESIGNATIONS AND FREQUENCY TOLERANCE

H-1. Designation of emissions.

a. Emissions are designated according to their classification and the width of the frequency band occupied. Classification is according to type of modulation, mode of transmission, and supplementary characteristics. The designations in table H-1 are in accordance with international agreements.

Table H-1. Designation of emissions.

Modulation Types:	Designator
Amplitude	A
Frequency	F
Pulse	P
Modulation (transmission mode):	Designator
None	0
Telegraphy (keyed EF carrier)	1
Telegraphy (tone)	2
Telephone	3
Facsimile	4
Television	5
Four-channel duplex telegraphy	6
Multichannel voice frequency	7
Telegraphy	9
Complex forms	Designator
Supplemental Characteristics	None Single sideband
Double sideband	A
-reduced carrier*	H
-full carrier	J Two independent sidebands,
-suppressed carrier	suppressed
carrier	B
Vestigial sideband	C Pulse
-amplitude modulated	D
-width modulated	E
-phase modulated	F
-code modulated	G

* Sometimes a lower case letter instead of a capital.

b. Examples of specific emission designations.

(1) In the specific emission designations in table H-2 refer to the occupied bandwidth. The numerals preceding the first letter signify the bandwidth; the first letter signifies the modulation type; the next numeral is the modulation or transmission mode; the final letter is the supplemental characteristic (sometimes a lower case letter):

(2) In each of the cases in table H-2, the assigned frequency is at the center of the bandwidth indicated.

(3) In the case of 6A3B, the individual

voice channels are used only for voice telephony- 'Jilie individual voice channels of these emissions may be used one at a time, or simultaneously. If multichannel telegraphy is employed in one or more voice channels, the emission designations become 6A9B, signifying complex waveforms. Emission designations 6A9B, 9A9B, and 12A9B are in general use in the military and commercial point-to-point services,

c. Examples of emissions:

(1) 3A3J emission (upper single sideband).

Assigned frequency — 4031.5 kHz

Mode — 3 A3 J upper single sideband

Bandwidth — 3 kHz

Reference frequency — 4030.0 kHz
(suppressed carrier)

This emission consists of one 3 kHz voice in upper sideband mode, with suppressed carrier

(2) 3A3J emission (lower single sideband)

Assigned frequency — 4028.5 kHz

Mode — 3A3J lower single sideband

Bandwidth - 3 kHz

Reference frequency — 4030.0 kHz
(suppressed carrier) — 4030.0 kHz

This emission consists of one 3 kHz voice channel, in lower sideband mode, with suppressed carrier.

Table H-2. Specific emission designations.

Designator	Explanation
100HA1A	CW, 15-25 WPM, telegraphy
1.08F1	RATT, 60 WPM, single channel
1.24F1	RATT, 100 WPM, single channel
	RATT, 120 WPM, single channel Four channel mux
3A3J	Single sideband; one 3 kHz channel with voice, with suppressed carrier offset + 1.5 kHz from assigned frequency
4F4	Facsimile
	Amplitude modulated telephony

Table H-2. Specific emission designations—continued.

Designator	Explanation
6A3B	Two independent sidebands; two 3 kHz channels with voice telephony in each, with suppressed carrier on the assigned frequency
9A9B	Two independent sidebands; three 3 kHz channels with voice or multichannel telegraphy in each, with voice or multichannel telegraphy in each, with suppressed carrier offset + 1.5 kHz from assigned frequency
12A9B	Two independent sidebands; four 3 kHz channels with voice or multichannel telegraphy in each with suppressed carrier on the assigned frequency
16F3	Frequency modulated telephony. Television (video signal - 525 lines at 15750 lines/sec). Television (aural signal).

H-2. The new worldwide radio emission codes are considerably more complicated than those emission codes that have been used for the past 25 or so years. It would take several pages to go into each column's possible definitions. Therefore, only those emission codes that are normally authorized on Army MARS frequencies, crossed as close as possible to the old emission codes, are shown in table H-3.

Table H-3. Frequency range and tolerance (in parts per million unless stated otherwise).

Fixed Stations	Parts	Mobile Stations	Parts
Band 10 to 535 kHz		Band 10 to 535 kHz	
No power limitation	200	No power limitation	100
Band 1605 to 5 MHz		Band 1605 to 5 MHz	
200W or less	100	No power limitation	200
Above 200W	50		
SSB/ISB in excess of 1 KW is 20 Hz			
Band 4 to 29.7 MHz		Band 4 to 29.7 MHz	
200W or less	50	No power limitation	1000
Above 500W	15		
SSB/ISB in excess of 1 KW is 10 Hz			
Band 29.7 to 100 MHz		Band 29.7 to 100 MHz	
200W or less	50	5W or less	100
Above 200W	30	30 above 5W	50
SSB/ISB in excess of 1KW is 20 Hz			
Band 100 to 470 MHz		Band 100 to 470 MHz	
5W or less	50	5W or less	50
Above 5W	20	Above 5W	20

H-3. Frequency tolerances.

a. The tolerance levels for Army fixed and mobile stations are shown in table H-4.

EXAMPLE:

If the assigned frequency for a fixed station is 4031.5 kHz and the power being used is 400 W the tolerance can be determined by multiplying 4015 kHz x

50 x 106 which equates to 201.595 hertz. The tolerance for this operation is, therefore, plus or minus 201.575 hertz from 4015 kHz.

b. The tolerance for OB.0720 maintenance apply to the entire military service and must be maintained by MARS stations. Communications security activities monitor Army frequencies to ensure maintenance of the standards in table H-3, and to report off-frequency transmission.

c. The FCC also monitors Army frequencies and forwards reports of OB.0730 violations to DA. Reports involving MARS stations are referred to the Chief, Army MARS, who forwards them through channels to the station involved directing that corrective actions be taken.

d. If experience during actual use reveals that the frequencies authorized are unsatisfactory or inadequate because of OB.0800 interference, the Chief, Army MARS will advise DA with full details. MARS frequencies must be used in such a manner as to avoid interference to other Army communications, which take precedence. If such interference develops and the Army station affected requests that the MARS station cease operation on the frequency in use, such action will be taken immediately, until the frequency is clear.

e. Facts pertaining to interference on MARS frequencies should be reported in accordance with ACP 121(E) to the Chief, Army MARS through channels providing the following specific information via message or DA Form 5528-R.

(1) Station causing interference:

A. Call sign or other identification.

B. Measured frequency.

C. Type of mission and type of traffic being transmitted.

D. Measured bandwidth of interfering signal. (Indicate lowest and highest frequencies. Give type of measuring equipment.)

E. Signal strength. (If field meter not available, use scale 1 to 5.)

F. Severity of interference. (Indicate percentage of copy or intelligence lost due to interference.)

(2) Transmitter station experiencing interference:

G. Call sign or name of station.

H. Assigned frequency.

I. Measured frequency.

J. Type of emission and traffic being transmitted.

K. Measured bandwidth,

L. Signal strength.

(3) Receiving station experiencing interference.

M. Call sign or name of station.

N. Location. (Give coordinates in latitude

and longitude, if possible. Indicate nearest major city, state, or country.)

O. Date and time of interference, and duration in minutes. (Example: 251030Z (22), 261143Z (5), and so forth.)

P. Other particulars.

Q. Requested action.

NOTE:

The letter "X" will be used after any of the above letters if no information on that particular item is reported.

Table H-4. Worldwide radio emission codes

WARC	FREQ TOL	OLD CODE	MEANING
100HA1A	500w-30	0.1A1	Continuous wave (CW)
	500w-10		
1K10F1B	15	1.1F1	Frequency shift keying
2K50F3F	20	3A5j	Slow scan television
3K000J2B	20Hz	3A2j	Audio frequency shift keying
			Radioteletype
3K00J2C	20	3A3j	Facsimile
3K00J3E	20	3A7j	DATA
6K00A3E	20	6A3	AM voice
6K00B9W	20	6A7b	Multichannel voice-AFSK RATT
16K0P2B	5	16F2	FM AFSK radioteletype
16K0F3E	8	16F3	Frequency modulation (FM) Voice
16K0F7D	5	16F9	FM DATA

Units for frequency tolerance are (+ or -) parts per million (ppm) unless otherwise stated in Hz. PSK transmitter operating on a frequency of 10MHz. Example; $10,000,000 = 10 \times 15 = 150\text{Hz}$ 1,000,000

Glossary

SECTION I. ABBREVIATIONS

AARS	Army Amateur Radio System	MARS	Military Affiliate Radio System
ACP	Allied Communications Publications	MF	medium frequency
ANCS	alternate net control station	MPM	MARS Property Manager
APO	Army Post Office	MUF	maximum usable frequency
ARNG	Army National Guard	NC	no contact
ARRL	American radio relay league	NCS	net control station
BBS	bulletin board systems	NFT	no file time
C-E	communications-electronics	NSN	national stock number
CFN	confirmation	OCONUS	outside continental United States
CK	check	OMS	official mail stamp
CONUS	continental United States	PDO	property disposal officer
CR	carriage return	P.d.t.	Pacific daylight time
CW	continuous wave	PMS	penalty mail stamp
DA	Department of the Army	ppm	parts per million
DCSLOG	Deputy Chief of Staff for Logistics	Prosigns	procedure signs
D/F	direction finding	RATT	radioteletype
DLSC	Defense Logistics Services Center	RCS	requirement control symbol
DOD	Department of Defense	ROTC	Reserve Officer Training Corps
DODAAC	Department of Defense activity address code	RWI	radio wire integration
DRMO	Defense Reutilization and Marketing Office	SDO	Stamp Distribution Offices
DRMS	Defense Reutilization and Marketing Service	SGD	signed
DTG	date-time group	SHF	super high frequency
EOL	end of line	SOP	standard operating procedure
EOM	end of month	SP	space
EPPL	excess personal property list	SSB	single sideband
e.s.t.	eastern standard time	SSN	station serial number
FCC	Federal Communications Commission	TDA	tables of distribution and allowances
FM	frequency modulated	TOE	table of organization and equipment
FPO	Fleet Post Office	TOF	time of file
FSC	Federal Supply Classes	UCT	Universal Coordinated Time
G.m.t.	Greenwich mean time	UHF	ultrahigh frequency
GSA	General Services Administration	U.S.	United States
HF	high frequency	USAISC-	USAISC-Western Command
HQDA	Headquarters, Department of the Army	WESTCOM	
I.T.U.	International Telecommunications Union	USAR	U.S. Army Reserve
JCS	Joint Chiefs of Staff	USAREUR	U.S. Army Europe
LF	line feed/low frequency	USPS	U.S. Postal Service
LTRS	letters	VHF	very high frequency
		VLF	very low frequency
		WAR	radio station call letters
		WPM	words per minute

Section II. TERMS

Address designator

A plain language name (full or abbreviated), routing indicator, call sign, or address group of a

unit, activity, or other authority used to indicate the originator and/or addressee(s).

Address group, collective

An address group which represents two or more commands, authorities, activities, units, or any combination thereof, and includes the commander of the organization or group and all subordinate commanders therein.

Affiliate radio station

A privately owned HF radio station (licensed by the FCC and/or host authority), operated by an individual who is affiliated with and licensed by Army MARS.

Allocated

In reference to a circuit or channel indicates the exclusive use by one or more specified authorities.

Allocated band

A frequency band allocated to the service concerned in the Table of Frequency Allocations of the International Telecommunications Union (I.T.U.) Radio regulations.

Area MARS director

An individual appointed by the command MARS director and assigned a specific geographical area of responsibility for the management, direction, and operation of the Army MARS and Army amateur radio program in accordance with DOD Directive 4605.2.

Audio frequency

A frequency which can be detected as a sound by the human ear. The range of audio frequencies extends from approximately 10 to 20,000 hertz.

Automatic frequency control

A circuit or device which maintains a receiver or transmitter on a desired frequency.

Automatic gain control

A circuit or device which regulates the gain of a receiver or amplifier so that the output tends to remain constant although the incoming signal may vary in strength.

Azimuth

- a. A direction expressed as a horizontal angle usually in degrees or miles and measured clockwise from north. Thus, azimuth will be true azimuth, grid azimuth, or magnetic azimuth depending upon which north is used.
- b. The angle at the zenith between the observer's celestial

meridian and the vertical circle through a heavenly body.

Bandwidth, necessary

For a given class of emission, the minimum value of the occupied bandwidth sufficient to ensure the transmission of information at the rate and with the quality required for the system employed under specified conditions. Emissions, useful for the good functioning of the receiving equipment as, for example, the emission corresponding to the carrier or reduced carrier systems, will be included in necessary bandwidth.

Bandwidth, occupied

The bandwidth occupied by an emission is the band of frequencies comprising 99 percent of the total radiated power extended to include any discrete frequency on which the power is at least 0.25 percent of the total radiated power.

Bauds

The unit of modulation rate. It corresponds to a rate of one unit interval per second. Example: If the duration of the unit interval is 20 milliseconds, the modulation rate is 50 bauds.

Beam (radio)

A radio transmission concentrated into a sector narrow in either azimuth or elevation, or in both.

Beam width

The angle between the directions, on either side of the axis, at which the intensity of the radio frequency field drops to one-half the value it has on the axis.

Carrier frequency

The frequency of the carrier wave.

Carrier suppressed

Carrier restricted to a power level more than 32db below the peak envelope power and preferably 40db or more below the peak envelope power.

Carrier wave

A wave, usually sinusoidal, which is modulated to transmit signals. The frequency of the wave is called the "carrier frequency." The carrier wave is not transmitted in some types of modulation.

Channel (radio)

A specific portion of the radio frequency band usually associated with a specific telecommunications function.

Chief, Army MARS

An individual appointed by Commander, USAISC, serving as the executive agent for the Chief of Staff, Army, and responsible for the management and direction of the worldwide Army MARS and Army amateur radio program in accordance DOD Directive 4650.2,

Clear text

See plain language (plain text).

Coaxial Cable

A transmission line consisting of one conductor, usually a small copper tube or wire, within and insulated from another conductor of larger diameter, usually copper tubing or copper braid. The outer conductor may or may not be grounded. Coaxial cable is sometimes called concentric line or coaxial line.

Code

Any system of communication in which arbitrary groups of symbols represent units of plain text of varying length. Codes are provided primarily for one of three purposes:

- a. In the broadcast sense, coding is a means of converting information into a form suitable for communication and/or encryption; for example, international Morse code, teletypewriter (teleprinter) codes, and so forth.
- b. Brevity codes are used to reduce the length of time necessary to transmit information; for example, long stereotyped sentences may be reduced to a few characters which are transmitted. (See code, brevity.)
- c. Security codes are used to provide some degree of cryptographic protection for the information being transmitted.

Code, brevity

A code which provides no security but which has as its sole purpose the shortening of messages rather than the concealment of their content.

Code, International Morse

A code in which letters and numbers are represented by specific groupings of dots and/or dashes. The international Morse code is used especially in radiotelegraph and visual communications.

Command MARS director

An individual appointed by commanders of USAISC' subordinate commands (5th Signal Command, 7th Signal Command, 1st

Signal Brigade

(USAISC), USAISC-Japan, and USAISC- WESTCOM) responsible for the management, direction, and operation of the Army MARS and Army amateur radio programs within the major subordinate command's geographical area of responsibility in accordance with DOD Directive 4650.2.

Communications security

The protection resulting from all measures designed to deny to unauthorized persons information of value which might be derived from the possession and study of telecommunications, or to mislead unauthorized persons in their interpretation of the results of such a study. COMSEC includes—

- a. Transmission security.
- b. Cryptosecurity.
- c. Physical security of COMSEC materials and information.

Conditional gift

Money or other intangible personal property (stocks and bonds) offered with specified limitations on its ownership and use; or real property or tangible personal property offered on condition that it would be used in a manner, or for a purpose, or in a place that is not considered normal use or placement; or real property or tangible personal property offered on condition that it would be used by specific organizations other than those that use such property.

Counterpoise

A conductor or system of conductors used as a substitute for earth or ground in an antenna system.

Crystal (xtal)

- a. A slice of a substance such as quartz or tourmaline which exhibits piezoelectric characteristics. It has the property of responding markedly to a given frequency when cut to a given thickness.
- b. A detector using the asymmetrical conducting properties of certain crystal-crystal or crystal- metal contacts.

Crystal control

Control of the frequency or an oscillator by means of a piezoelectric crystal.

Cycle

One complete positive and one complete negative alternation of a current or voltage.

Date-time group

Depending upon national requirements the DTG may indicate either the date and time when the message was officially released by the releasing officer, or the date and time the message was handed into a communication facility for transmission. The DTG is expressed as six digits followed by a zone suffix and the month expressed by the first three letters. The last two digits of the year of origin may be added if required by national authorities.

Deviation ratio

The ratio of the maximum frequency (of a frequency modulated (FM) wave) deviation to the maximum modulation frequency.

Dipole

An antenna consisting of two elements each approximately one-fourth wave length in length and fed with radio-frequency energy of opposing polarity at adjacent ends of the elements.

Distortion

The amount by which the output wave form differs from the input wave form. Distortion may exist in amplitude, frequency, or phase modulation.

Electronic

A generic term used to describe that branch of electrical science and technology which treats the behavior of free electrons in vacuous or gaseous space and in semiconductors and the circuitry associated therewith.

End of transmission

- a. A standardized uninterrupted sequence of character and machine functions used to terminate a transmission and disconnect the circuit and transmitting equipment.
- b. A prosign used in manual systems to signify the completion of a transmission.

Ending

The part of a message containing all components following the text.

Equipment review board

A committee appointed by the Commander, USAISC to evaluate all requests for equipment from MARS affiliate members, consisting of Chief, Army MARS, representative from DCSLOG, HQ, USAISC; and the MPM.

Exercise, communications

Any transmission or reception of information directed specifically to evaluate the efficiency of communications facilities, procedures, and training.

Facsimile

A system of telecommunications for the transmission of fixed images with a view to their reception in a permanent form.

Fading

A variation in strength of received signals due to variation with time in the conditions of propagation.

Feedback

A transfer of energy from the output circuit of a device back to its input.

Filing time or time handed in

The date and time a message is received from an originator by the communications center for transmission (shown as date, hours, and minutes expressed in digits followed by zone suffix; for example, 111245Z). The filing time for refile messages is the date and time the message is received by a communication center for refile.

Final modulation

Final modulation of a carrier wave (carrier wave of radio link transmitter) is modulation applied to the transmitted carrier. A transmitter may be modulated either directly by multiplex telephony, or by signals from premodulation. There are two types of final modulation: amplitude modulation (from all or nothing in the case of pulses) and frequency modulation (or phase modulation).

Forced issue

Items of supply or equipment that are issued without a requirement or request having been submitted.

Freeze authority

The authority to reserve a specific item of equipment being offered for reutilization pending receipt of requisition documents.

Frequencies, distress

Frequencies allotted to distress calls by international agreement.

Frequency

The number of recurrences of a periodic phenomenon in a unit of time. In specifying the electrical frequency, the unit of time is the second; for example, the frequency is normally expressed in kHz at and below 30,000 kilohertz, and in MHz above this frequency.

Frequency assigned

This assigned frequency is the frequency coinciding with the center of the frequency band in which the station is authorized to work. This frequency does not necessarily correspond to any frequency in an emission.

Frequency, authorized

A portion of the radio spectrum the width of which is the necessary bandwidth of emission plus twice the prescribed frequency tolerance.

Frequency, lowest usable

The lowest high frequency effective at a specified time for ionospheric propagation of radio waves between two specified points.

Frequency, maximum usege

The upper limit of the frequencies that can be used at a specified time for radio transmission between two points and involving propagation by reflection from the regular ionized layers of the ionosphere.

Frequency, optimum transmission

The most effective frequency at a specified time for ionospheric propagation of radio waves beetween two specified points (commonly taken as 85 percent of the monthly median value of maximum usable frequency (MUF) for the specified time and path).

Frequency, primary

A frequency assigned for normal use on a particular circuit.

Frequency, radio

The nomenclature of radio frequencies is as follows:

Frequency Subdivision	Frequency Range	Metric Subdivision
"Very low frequency (VLF)	3 to 30 kHz	Myrimetric waves
Low frequency (LF)	20 to 200 kHz	Kilometric waves

Medium frequency (MF) HF	300 to 3000 kHz	Hectometric waves
VHF	3 to 30 MHz	Decametric waves
Ultrahigh frequency (UHF)	30 to 300 MHz	Metric waves
Super high frequency (SHF)	300 to 3000 MHz	Decametric waves
	3 to 30 GHz	Centimetric waves

Frequency, secondary

A frequency assigned for use on a particular radio circuit when primary frequency becomes unusable for any reason.

Frequency stability

The ability of an oscillator to maintain its operation on a constant frequency.

Frequency tolerates

The maximum permissible deviation, with respect to the reference frequency, of the corresponding characteristic frequency of an emission; the reference frequency may differ from the frequency assigned to a station by a fixed and specified amount. Frequency tolerance is expressed as a percentage or in hertz.

Function key

A term associated with specific keys on a teletypewriter (for example, OR, LF, FIGS, LTHS, and so forth), which when operated cause the teletypewriter to perform mechanical functions in order that a message may be received in proper form.

Garble

An error in transmission reception or encryption which renders the message or a portion thereof incorrect or undecryptable.

Gift

A contribution, donation,, device, or bequest of real or personal property.

Half-duplex operation

Operating method using a circuit designed for duplex operation, but which on account of the terminal equipment can be operated alternately only.

Harmonic

An integral multiple of a fundamental frequency.

Hertz

A unit of frequency equivalent to one cycle per second.

Interface

A boundary or point common to two or more systems or other entities across which useful information flow takes place. (The boundary implies that useful information flow requires the specification of the interconnection of the systems which enables them to interoperate.)

Interference

Any electrical disturbance which causes undesirable response in electronic equipment.

Ionosphere

The region of the atmosphere extending from roughly 40 to 250 miles altitude, in which there is appreciable ionization. The presence of charged particles in this region profoundly affects the propagation of electromagnetic radiations of long wavelengths (radio and radar waves).

Ionospheric prediction

The forecasting of ionospheric conditions and the preparation of radio propagation data derived therefrom.

Jamming

The deliberate radiation, reradiation, or reflection of electromagnetic signals with the object of impairing the use of electronic devices by the enemy.

Joint operations center

A jointly manned facility of a joint force commander's headquarters established for planning, monitoring, and guiding the execution of the commander's decisions.

Land line

A general term applied to metallic conductors used for conveyance of intelligence.

Limiter

A device which limits the amplitude of a signal to a specified value.

Line of sight

In communications, a direct propagation path that does not go below the radio horizon.

Link

A general term used to indicate the existence of communication facilities between two points.

Log, operator's

A chronological record of events relating to the operation of a particular circuit.

Log, station

A chronological record of station events; for example, entries relating to message handling, equipment difficulties, personnel, and so forth.

Long lines

Includes all forms of physical conductors used for communication purposes such as open wire systems, underground and overhead cables, and submarine cables, but do not include local circuits. They also may contain radio relay systems when they are integrated with the wire system.

Mark (marking signal)

The signal corresponding to the inactive condition in a teleprinter. Normally MARK is the signalling condition which produces a STOP signal.

MARS call sign

A set of combined letters and a numeral assigned to an individual MARS member or MARS station that are used as identification when transmitting radio signals.

MARS club station

A radio station properly licensed, established, operated, and maintained by volunteers under the auspices of a properly constituted military and/or civilian amateur radio club.

MARS gateway station

A USAISC operation and maintenance supported, HF radio station serving a major geographic area or command through which incoming and outgoing record traffic (MARS) is processed.

MARS Joint-Service panel

A DOD chartered panel consisting of a primary member and an alternate member from the U.S. Army, U.S. Navy/Marine Corps, and the U.S. Air Force to address mutual problems; improve overall MARS operations; develop common standards, procedures, and practices; institute plans for greater affiliate participation, and review overall effectiveness and need for management awareness.

Message

Any thought or idea expressed briefly in plain or secret language, prepared in a form suitable for transmission by any means of communication.

Message, book

A book message which is destined for two or more addressees and is of such nature that the originator considers that no addressees need to be informed of any other addressees. Each addressee must be indicated as action or information.

Message, drill

Messages intended for training communications personnel.
Message, exercise
Messages sent during and relating to training exercises, command post exercises, tactical exercises, and maneuvers.

Message, multiple address

A multiple address message is one which is destined for two or more addressees each of whom is informed of all addressees. Each addressee must be indicated as action of information.

Message, service

A brief, concise message between operating or supervisory personnel at communication or signal centers or relay stations pertaining to any phase of traffic handling, status of communication facilities, circuit conditions, or other matters affecting communication operation.

Message, single address

A message destined for only one addressee.

Message heading

The part of a message containing all components preceding the text.

Military community MARS station

A MARS licensed radio station sponsored and supported by the military community and has as a minimum one full-time operator. The operator must have an FCC license, host nation amateur radio license, and a MARS station license.

Military activity or unit MARS station

An HF radio station located at ARNG, USAR, or Reserve Officer Training Corps (ROTC) facilities, or an active Army unit, activity, or agency which may or may not be located on a military installation that has a secondary mission to

support the MARS.

Military community station

A MARS licensed HF radio station designated to support the local military community as authorized by the community commander.

Military installation MARS station.

- a. An HF command and control or contingency radio station installed on a post, camp, or station, operated and maintained by USAISC that has a secondary mission to support the MARS.
- b. An HF radio station installed on a military installation, approved, operated, and maintained by the installation commander, with the primary mission of supporting the MARS.

Minimize

A condition wherein normal message and telephone traffic is drastically reduced in order that messages connected with an actual or simulated emergency will not be delayed.

Mobile communications equipment

Communications equipment which is installed in or on a vehicle and can be operated while the vehicle is in motion. Some member nations apply this term to similarly installed equipment which can only be operated while the vehicle is stationary.

Mobile service

A service or radio communications between mobile and land stations, or between mobile stations.

Modulation

The process in which the amplitude, frequency, or phase of a carrier wave is varied with time in accordance with the wave form of superimposed intelligence.

Multiplex

Denotes the simultaneous use of a number of channels on a single circuit.

Net (communications)

An organization of stations capable of direct communications on a common channel or frequency.

Net control station

A station designated to control traffic and enforce circuit discipline within a given net.

Network

- a. In communications, an organization of stations capable of intercommunication but not necessarily on the same channel.
- b. In engineering, two or more interrelated circuits.

Number, originator's reference

The number assigned to a message by an originator to provide a means of reference.

Number,, station serial

A message reference number assigned within a communication or signal center. It will normally consist of a number allotted in sequence. However, in those instances where SSNs are allotted at more than one position, as prescribed by installation procedure, a single letter designator follows each number, for example, 107A, 119B.

One-way operation

Communications between two points in one direction only. (One-way operations is preferred to the term "half duplex," which has various interpretations.)

Originator

The command by whose authority a message is sent. The originator is responsible for the functions of the drafter and releasing officer.

Oscillator

A device which produces an electrical signal of relatively constant frequency and amplitude.

Oscilloscope

An instrument for showing visual and graphical representations of the waveforms encountered in electrical circuits.

Page copy

A message in page form which is the result of a transmission.

Parabolic antenna

An antenna provided with a reflector having the characteristic that radio frequency waves emitted from a focal point will be reflected into space along parallel paths thus creating a narrow beam.

Personal sign

Signs composed of one or more letters (normally initials) used when endorsing station records and messages to indicate individual responsibility of operating and supervisory personnel.

Phonetic alphabet

A list of standard words used to identify letters of the alphabet in communications.

Plain language (plain text)

Text or language which conveys and intelligence text underlying encrypted text.

Plaindress

A type of message in which the originator and addressee designations are indicated externally of the text.

Polarization

The direction of the electrical field component of radiated energy.

Polarization diversify

The use or availability of various polarizations such as horizontal, vertical, cross, circular, and ecliptical either simultaneously or singly.

Power (of a radio transmitter)

When not otherwise specified, the definition of "peak power" of a radio transmitter will be used.

Power, carrier

The average power supplied to the antenna transmission line by a transmitter during one radio frequency cycle under conditions of no modulation. This definition does not apply to pulse modulated emissions.

Power, effective radiated

The power supplied to the antenna multiplied by the relative gain of the antenna.

Power, mean

The power supplied to the antenna transmission line by a transmitter during normal operations averaged over a time sufficiently long, compared with the period of the lowest frequency encountered in the modulator. A time of one-tenth second during which the mean power is greatest will be selected normally.

Power, peak envelope

The mean power supplied to the antenna during one radio frequency cycle at the highest crest of modulation envelope taken under conditions of normal operation.

Preamble

One of the components contained in the heading of the message whose elements include the degree of precedence, the DTG, and message instructions.

Precedence

A designation assigned to a message by the originator to indicate to communication personnel the relative order of handling and to the addressee the order in which the message is to be noted.

Precedence, dual

Messages containing two precedence designations the higher one for all action addressees and the lower one for all information addressees.

Prefix

One of the components contained in the heading of a message whose elements may include the accounting information, group count, and service.

Procedure messages

A message in which the text contains only prosigns, operating signals, addressee designation(s), identification of messages, parts of messages, and amplifying data, as necessary.

Propagation

The manner in which an electromagnetic emission travels outward from its source.

Pulse

A variation in the value of an electrical quantity as a function of time such that the value departs from a given datum for a time interval and then returns to this datum for a much longer time interval.

Radiation (in radio communication)

- a. The outward flow of radio frequency energy from a source.
- b. Energy flowing in a medium in the form of radio waves.

Radio

A descriptive term applied to the use of electromagnetic waves between 10 kHz and 3,000,000 MHz. (It is used principally as an adjective.)

Radio relay

The reception and retransmission by a radio station of signals received from another radio station from the line portion of an integrated line and radio communications system.

Radio silence

A period during which all or certain radio equipment capable of radiation is kept inoperative. (In combined or U.S. joint or intraservice communications the frequency bands and/or types of equipment effected will be specified.)

Radio station

Installation, assembly of equipment, and location where radio transmissions are sent and/or received,

Range

The distance between specified radio stations over which effective communications can be provided.

Readability

The ability to be understood, for example, the readability of signals sent by any means of telecommunications.

Recipient of an offer

The commander, or his or her designated representative, of an organization chosen to receive a gift.

Refile

The processing of a message into appropriate procedure for transfer to another system which involves alteration of the message format, for example, tape relay or radiotelegraph, point-to-point telegraph network to tape relay network, military to civil or commercial.

Rerun

The repetition of a message which was previously sent by tape, card, or similar stored mode of communications.

Retransmission

The repetition of a message which was previously transmitted by any mode of communications.

Routing

The process of determining and prescribing the path or method to be used in forwarding messages.

Routing alternative

A process whereby substitute routes are used for transmitting messages when circuit failures occur or primary transmission paths or backlogs develop.

Saturation

The overwhelming of a receiver by an excessively high input signal such as jamming.

Screening

The examination of surplus property being offered for reutilization to determine usability of a specific item.

Selectivity

That characteristic of a radio receiver which determines its ability to discriminate between a designed signal and coexistent undesired signals at other frequencies.

Sensitivity

That characteristic of a radio receiver which determines the minimum strength of signal input capable of producing a desired strength of signal output.

Service, amateur

A service of self-training, intercommunication, and technical investigations carried on by amateurs; that is, by duly authorized persons interested in radio techniques solely with a personal aim and without pecuniary interest.

Sideband

A sideband is the frequency band, above or below the carrier, produced by the process of modulation.

Signals

- a. As applied to electronics, any transmitted electrical impulse.
- b. Operationally, a type of message, the text of which consists of one or more letters, words, characters, signal flags, visual display, or special sounds, with prearranged meanings, and which is conveyed or transmitted by visual, acoustical, or electrical means.

Signal-to-noise ratio

The ratio, at a selected point in the circuit, of signal power to total circuit-noise power.

Simplex operation

Simplex operation refers to communication between two points in both directions, but not simultaneously.

Single-sideband transmission

That system of carrier transmission in which one sideband is transmitted and the other wideband is suppressed. The carrier wave may be either transmitted or suppressed.

Skip distance

The minimum distance between the transmitting station and the point of return to the earth of the transmitted wave reflected from the ionosphere.

Skip zone

The space or region wherein a transmission is not received between the farthest point reached by the ground wave and the nearest point at which the refracted sky waves come back to earth.

Sky wave

That portion of a radiated wave which travels in space and is returned to earth by refraction in the ionosphere.

Space (spacing signal)

The signal corresponding to the active condition in a teleprinter. Normally "SPACE" is the signalling condition which produces a "START" signal.

Space diversity (spaced aerial diversity)

A term used to designate any method of transmission and/or reception for combatting effects of flat or selective fading which employs antennae having common polarization and spatial separations.

State MARS director

An individual member, appointed by the area MARS director, who serves on a volunteer basis, without remuneration, under the direct supervision of the area MARS director and is responsible for a specific geographic area or state.

Station

A separate transmitter or receiver or a combination of transmitters and receivers including the accessory equipment required for carrying on a definite radio communication of the service in which it operates permanently or temporarily.

Station, amateur

A station in the amateur service.

Switchboard

A manually operated apparatus at an exchange on which the various circuits from subscribers and other exchanges are terminated to enable operators to establish communication either between two subscribers on the same exchange, or between subscribers on different exchanges.

System

An overall term used to describe communication facilities from an engineering aspect including all the associated equipment.

Telecommunications

Any transmission, emission, or reception of signals, signs, writing, images, and sounds, or intelligence of any nature by wire, radio, visual, or other electromagnetic systems.

Test, communications

Any transmission or reception specifically intended to evaluate the efficiency of communications media or facilities.

Text

That part of a message which contains the thought or idea which the originator desires to be communicated.

Time, Greenwich Mean Time (GMT) or Universal Coordinated Time (UTC)

Mean solar time at the meridian of Greenwich.

Time of delivery

The date and time at which a message is delivered to an addressee.

Time of dispatch

The date and time at which a communication is dispatched to an addressee or communication agency. (Not to be used in connection with messages transmitted by telecommunications.)

Time of receipt

The date and time at which a communication agency completes reception of a message transmitted to it by another communication agency.

Time zone

The description and designation of letters assigned to time zones of the world.

Traffic (communications)

All transmitted and received messages.

Tuning

The process of adjusting a circuit so that it resonates at a desired frequency.

Unconditional gift

Money or real or personal property offered with no limitation on its ownership or use. If a donor does specify that a gift (real or personal property) be used in a certain place, manner, or for a certain purpose, but the condition is for normal use, the gift is to be considered unconditional.

Wave length

A wave length is the distance traveled in one period or cycle by a periodic disturbance. It is the distance between corresponding phases of two consecutive waves or a wave train. A wave length is the quotient of velocity divided by frequency.

By Order of the Secretary of the Army:

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